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Teradata Vantage optimizing the Analytics Database through Workload Management

version 17.10.2

96789
student guide

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Module 0: Course introduction

Vantage performance optimization curriculum	0-2
Conferencing tool controls.....	0-3
Audio or video issues?.....	0-4
Virtual classroom ground rules	0-5
Course description and objectives.....	0-6
Workshop Pre-Work.....	0-7
Workshop modules and collaterals	0-8
Workshop modules schedule (VILT)	0-9

Module 1: Workload management overview

Objectives	1-2
Topics	1-3
Current topic: Mixed workload overview	1-4
What is a mixed workload?.....	1-5
What's purpose for having mixed workloads?	1-6
Current topic: Workload management overview	1-7
What is workload management?	1-8
Workload management benefits.....	1-9
Workload management offerings	1-10
Current topic: Mixed workload features	1-11
Workload management key features	1-12
Workload classification criteria types	1-13
Virtual partitions	1-14
Workload management methods	1-15
Pre-Execution controls: Filters.....	1-16
Arrival rate meter.....	1-17
Pre-Execution controls: Throttles.....	1-18
State matrix.....	1-19
During execution controls: Exceptions	1-20
Exception actions	1-21
Workload management levels.....	1-22
Example: Query management architecture	1-23
Workload management: Workloads and rules	1-24
Workload management offering comparison	1-25
Current topic: Viewpoint portlets	1-26
Workload management: Administration.....	1-27

Workload management: Monitoring and reporting	1-28
Summary	1-29

Module 2: Case study

Objectives	2-2
Topics	2-3
The case study	2-4
Case study characteristics	2-5
Simulation workloads	2-6
Simulation hardware	2-7
Data model	2-8
Vantage Analytics Database environment	2-9
Service level goals	2-10
Workload users	2-11
Workload profiles	2-12
Case study summary	2-13

Module 3: Introduction to Workload Designer

Objectives	3-2
Topics	3-3
Current topic: Working with Workload Designer	3-4
About Workload Designer	3-5
Workload Designer	3-6
Workload Designer: Ready rulesets	3-7
Workload Designer: Working rulesets	3-8
Workload Designer: Working rulesets: View/edit	3-9
Workload Designer: Working rulesets: Show all	3-10
Workload Designer: Working rulesets: Unlock	3-11
Workload Designer: Working rulesets: Clone	3-12
Workload Designer: Working rulesets: Export	3-13
Workload Designer: Working rulesets: Delete	3-14
Workload Designer: Local: Import a ruleset	3-15
Workload Designer: Local: Create a new ruleset	3-16
Workload Designer: TIWM	3-17
Workload Designer: TASM	3-18
Case study: Differences between VOWM_starting_ruleset and FirstConfig rulesets	3-19
Current topic: Lab instructions	3-20
Lab: IP address for your team's Linux server	3-21
Lab: Run QryPerfMetrics.sql in Teradata Studio	3-22
Lab: Record workload simulation query results in VOWM simulation results spreadsheet ..	3-23
Lab: Find the load jobs information	3-24
Lab: Record workload simulation load-job results in VOWM simulation results	3-25
Lab: Record the simulation results	3-26
Summary	3-27

Module 4: Monitoring portlets

Objectives	4-2
Topics	4-3
Current topic: Overview of workload monitoring portlets	4-4
About workload health and monitor	4-5
About the dashboard	4-6
Current topic: Workload health	4-7
The workload health portlet	4-8
Workload health: Health states	4-9
Workload health: Filters	4-10
Workload health: Summary information	4-11
Workload health: Detailed display	4-12
Current topic: Workload monitor	4-13
Workload monitor: Dynamic pipe display	4-14
Workload monitor: Time interval	4-16
Workload monitor: Current state	4-17
Workload monitor: Workload status	4-18
Workload monitor: Workload details	4-19
Workload monitor: Active requests	4-20
Workload monitor: Active requests details	4-21
Workload monitor: Deferred requests	4-22
Workload monitor: Deferred request details	4-23
Workload monitor: Delayed requests	4-24
Workload monitor: Delayed request details	4-25
Workload monitor: Static pipe display	4-26
Workload monitor: Distribution view: CPU consumption	4-27
Workload monitor: Distribution view: I/O consumption	4-28
Workload monitor: Distribution highlights	4-29
Workload monitor: Distribution details	4-31
Current topic: Dashboard	4-32
The viewpoint dashboard	4-33
Dashboard: System health	4-34
Dashboard: Workloads	4-35
Dashboard: Queries	4-36
Summary	4-37

Module 5: Workload Designer: General settings

Objectives	5-2
Topics	5-3
Current topic: General settings: General, bypass, limits/reserves	5-4
General button: General tab	5-5
General button: Bypass tab	5-6
General button: Limits/reserves tab	5-7
Current topic: General settings: Other settings	5-8
General settings: Other tab	5-9
Other tab: Intervals	5-10
Logging interval relationships	5-11
Logging tables	5-12
Multi-Request transaction example	5-13

Other tab: Blocker	5-14
Other tab: Other settings.....	5-15
Workload priority order.....	5-17
Other tab: Utility limits	5-18
Current topic: AMP work tasks	5-19
Before we discuss the last option on the other tab	5-20
What are AMP worker tasks?	5-21
Reserved pools of AWTs	5-22
Work types.....	5-23
AMP message queues	5-24
BYNET retry queue.....	5-25
Other tab: Define 'available AWTs' as.....	5-26
AWTs available for the WorkNew (work00) work type.....	5-27
AWTs available in the unreserved pool for use by any work type.....	5-28
Summary	5-29

Module 6: Workload Designer: Classifications

Objectives	6-2
Topics	6-3
Current topic: Classification overview	6-4
Workload management levels: Classify	6-5
Classification criteria	6-6
Classification criteria options.....	6-7
Classification criteria exactness.....	6-9
Classification criteria recommendations	6-10
Current topic: Classification tab	6-11
Where you'll find the classification tab	6-12
Request source criteria.....	6-13
Target criteria	6-14
Target Sub-Criteria	6-15
Query characteristics criteria.....	6-16
Queryband criteria	6-18
Utility criteria.....	6-19
Multiple request source criteria	6-20
Data block selectivity.....	6-21
Multi-Statement request	6-22
Multi-Statement request classification settings	6-23
Estimated memory usage.....	6-24
Where to define values for estimated memory.....	6-25
Incremental planning and execution	6-26
Summary	6-27

Module 7: Workload Designer: State matrix

Objectives	7-2
Topics	7-3
Current topic: State matrix overview	7-4
About the state matrix.....	7-5
State matrix.....	7-6

Event actions	7-7
Event notifications.....	7-8
Event notifications: Alert setup.....	7-9
Event notifications: Alert action set	7-10
Event notifications: Run program and post to Qtable	7-11
State transitions	7-12
Rule sets and working values	7-13
Displaying working values.....	7-15
Default state matrix.....	7-17
Current topic: Setup wizard: Planned environments.....	7-18
Setup wizard: Getting started.....	7-19
Setup wizard: Planned environments	7-20
Creating planned environments	7-21
Setup wizard: Planned events	7-22
Creating period events	7-23
Creating user defined events	7-24
Creating event combinations.....	7-25
Assigning planned events	7-26
Current topic: Setup wizard: Health conditions	7-27
Setup wizard: Health conditions.....	7-28
Creating health conditions	7-29
Setup wizard: Unplanned events.....	7-30
Creating system events.....	7-31
System event types: Component down events.....	7-32
Event qualification time	7-33
AMP activity level events: Available AWTs	7-35
AMP activity level events: Flow control	7-36
System level events: CPU utilization.....	7-37
System level events: CPU skew	7-38
Creating workload events	7-39
Workload event types.....	7-40
Unplanned event guidelines.....	7-42
Assigning unplanned events	7-43
Current topic: Setup wizard: States	7-44
Setup wizard: States.....	7-45
State guidelines.....	7-46
Creating states	7-47
Assigning states	7-48
Completed state matrix	7-49
Current topic: Lab activity.....	7-50
Summary	7-51

Module 8: Workload Designer: Session control

Objectives	8-2
Topics	8-3
Current topic: Sessions overview	8-4
Workload management levels: Session limits	8-5
Query session throttles	8-6
Sessions view	8-7
Session control	8-8
Current topic: Query sessions	8-9
Creating query sessions	8-10
Session limit rule types	8-11
Collective and members example	8-12
Collective and individual example	8-13
Request source classification criteria	8-14
State specific settings	8-15
Query sessions by state	8-17
Current topic: Utility limits	8-18
Creating utility limits	8-19
Utility limits classification	8-20
Supported utility protocols	8-21
State specific settings	8-22
Utility limits by state	8-24
Current topic: Utility sessions	8-25
Utility sessions	8-26
Utility protocols	8-27
Default utility sessions	8-28
Default utility session rules	8-29
Creating utility sessions	8-30
Create utility session: UtilityDataSize	8-31
Create utility session: Classification	8-32
Utility sessions evaluation order	8-33
Summary	8-34

Module 9: Workload Designer: System filters

Objectives	9-2
Topics	9-3
Workload management levels: Filter	9-4
Pre-Execution controls: Filters	9-5
Bypass filters	9-6
Creating filters	9-7
Warning only	9-8
Classification criteria	9-9
State specific settings	9-10
Enabled by state	9-12
Using filters	9-13
Summary	9-14

Module 10: Workload Designer: Arrival rate meter

Objectives	10-2
Topics	10-3
Current topic: Arrival rate meter overview	10-4
Workload management levels: Arrival rate meter	10-5
Arrival rate meter	10-6
Deferring requests	10-7
Arrival rate meter: Use case 1	10-8
Arrival rate meter: Use case 2	10-9
Arrival rate meter: Rule attributes	10-10
Arrival rate meter: Current arrival rate: Example	10-11
The arrival rate meters view	10-12
Current topic: Creating arrival rate meters	10-13
ARM general tab	10-14
Collective rule type with classification of profile	10-15
Disable manual release or abort	10-16
Classification criteria	10-17
State specific settings	10-18
Meter rates by state	10-20
Summary	10-21

Module 11: Workload Designer: System throttles

Objectives	11-2
Topics	11-3
Current topic: Throttling overview	11-4
Workload management levels: System throttle	11-5
Throttling levels	11-6
Throttling requests	11-7
Why use throttles?	11-8
Delay queue order	11-9
Bypass system throttles	11-10
Current topic: The throttles tab	11-11
Creating throttles	11-12
Creating system throttles	11-13
System throttle rule types	11-14
Collective and members example	11-15
Collective and individual example	11-16
System throttle scenario: Collective and member	11-17
Disable manual release or abort	11-18
Classification criteria	11-19
State specific settings	11-20
Creating virtual partition throttles	11-22
State specific settings	11-23
Throttle limits by state	11-25
Current topic: Throttling examples and recommendations	11-26
Overlapping associations	11-27
Average response time example	11-28
Throttle recommendations	11-30
Current topic: The AWT resource limits tab	11-31

AWT resource limits	11-32
Creating AWT resource limits	11-33
Classification criteria	11-34
State specific settings	11-35
Resource limits by state	11-37
Current topic: Lab activity	11-38
Summary	11-39

Module 12: Workload Designer: Workloads and exceptions

Objectives	12-2
Topics	12-3
Current topic: Workloads overview	12-4
Workload management levels: Workloads	12-5
What is a workload?	12-6
Advantages of workloads	12-7
Default workload	12-8
Current topic: The workloads view	12-9
Creating a new workload	12-10
Workload tabs	12-12
Classification criteria	12-13
Throttles state specific settings	12-14
Current topic: Flex throttles	12-16
What are flex throttles?	12-17
Characteristics of flex throttles	12-18
Enabling the flex throttles feature	12-19
Flex throttles example	12-20
Current topic: Workload group throttles	12-21
Workload throttles delay queue problem	12-22
Workload throttles delay queue solution	12-23
Creating workload group throttles	12-24
State specific settings	12-26
Workload group throttles and demotions	12-27
Current topic: Additional options for workloads	12-28
Workload service levels goals	12-29
Establishing service level goals	12-30
Hold query responses	12-31
Minimum response time	12-32
Current topic: Workload exceptions	12-33
Workload management levels: Exceptions	12-34
Exceptions	12-35
Creating exceptions	12-36
Unqualified exception thresholds	12-38
Qualified exception conditions	12-39
Qualification time	12-40
Exceptions example	12-41
Exception monitoring	12-42
Asynchronous exception monitoring example	12-43
CPU disk ratio	12-44
Skew detection	12-45
Skew impact	12-47

False skew	12-48
Exception actions	12-49
Change workload exception action.....	12-50
Abort exception action	12-51
Exception action conflicts	12-52
Exception notifications	12-53
Enabling exceptions by planned environment	12-54
Enabling exceptions by workload.....	12-55
Enabling exceptions by exceptions.....	12-56
Tactical workload exception	12-57
Tactical exception	12-58
Current topic: Additional workload tabs	12-59
SLG summary.....	12-60
Workload evaluation order	12-61
Console utilities	12-62
Summary	12-63

Module 13: Refining workload definitions

Objectives	13-2
Topics	13-3
Current topic: Capturing the workload performance metrics	13-4
Run SQL in Teradata Studio	13-5
Export SQL result set	13-6
Select the CSV file format.....	13-7
Current topic: Creating a pivot table.....	13-8
Select pivot table	13-9
Create pivot table.....	13-10
Populating the pivot table	13-11
Changing value field settings.....	13-13
Formatting data cells	13-14
Current topic: Using the pivot table to analyze workload performance metrics	13-15
Create a pivot chart	13-16
Select a style for your pivot chart	13-17
Filter the queries displayed in the pivot chart.....	13-18
Identify the outlier query (or queries)	13-20
Current topic: Refining the workloads	13-21
Workload refinement	13-22
Create a new workload.....	13-23
Adjust the workload evaluation order	13-24
Workloads: Refinement.....	13-25
Current topic: Lab activity.....	13-26
Summary	13-27

Module 14: Workload Designer: Mapping workloads to priority levels

Objectives	14-2
Topics	14-3

Current topic: The completely fair scheduler.....	14-4
Linux SLES 11 scheduler	14-5
Control groups.....	14-6
Resource shares.....	14-7
Virtual runtime	14-8
Current topic: Priority scheduler overview.....	14-10
Teradata SLES 11 priority scheduler	14-11
Hierarchy of control groups	14-12
TDAT control group.....	14-14
Virtual partitions	14-15
Preemption.....	14-16
Remaining control group	14-17
Current topic: Tactical workload management method.....	14-18
What is the tactical workload management method?	14-19
Tactical workload exceptions	14-20
Reserving AMP worker tasks	14-21
Guidelines for reserving AWTs	14-23
Current topic: SLG tier workload management method	14-24
SLG tier workload management method	14-25
SLG workload share percent.....	14-26
SLG tier target share percent	14-28
Current topic: Timeshare workload management method	14-29
What is the timeshare workload management method?.....	14-30
Timeshare access rates.....	14-31
Timeshare access rates concurrency	14-32
Automatic decay option.....	14-33
Automatic decay characteristics	14-34
Current topic: Priority scheduler recommendations	14-35
Managing resources	14-36
I/O prioritization.....	14-37
Tactical recommendations	14-38
SLG tier recommendations.....	14-39
Timeshare recommendations	14-40
Current topic: Assigning workloads to workload management methods	14-41
Virtual partitions	14-42
Adding virtual partitions	14-43
Partition resources	14-44
Workload distribution.....	14-45
System workload report	14-47
Penalty box workload	14-48
Current topic: Lab activity.....	14-49
Summary	14-50

Module 15: Summary

Objectives	15-2
Topics	15-3
Current topic: Mixed workload review	15-4
Mixed workloads	15-5
What is workload management?	15-6
What are the advantages of workload definitions?	15-7
Current topic: A final solution	15-8
Workload management solution	15-9
Baseline lab exercise results.....	15-18
Filters and throttles lab exercise results.....	15-19
Refine workloads and exceptions lab exercise results	15-20
Workload management final lab exercise results	15-21
Course summary	15-22

Appendix A: Viewpoint Configuration

Appendix B: Viewpoint Portlets

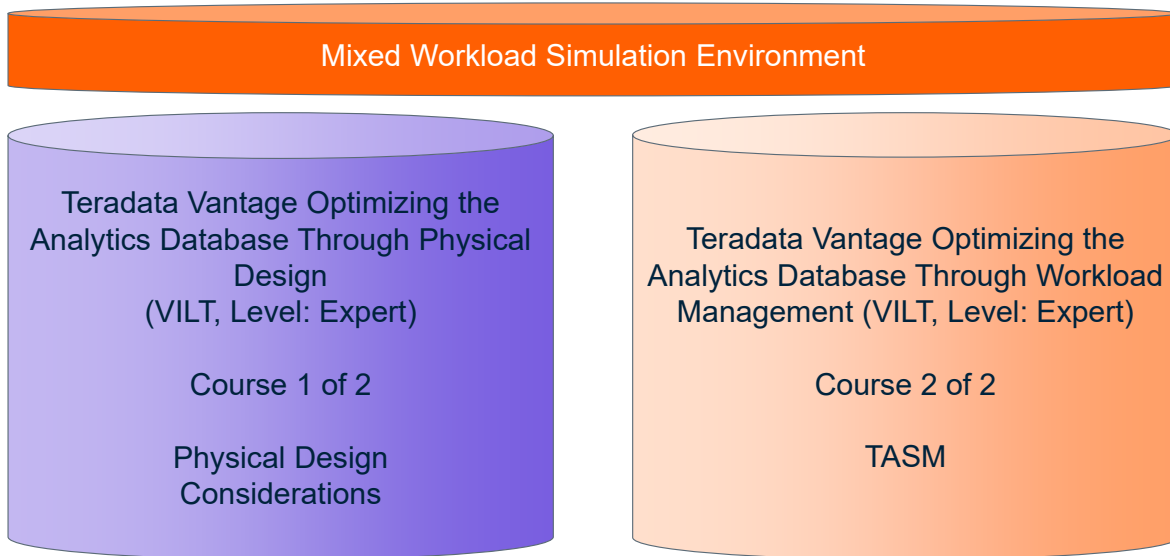
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Vantage performance optimization curriculum



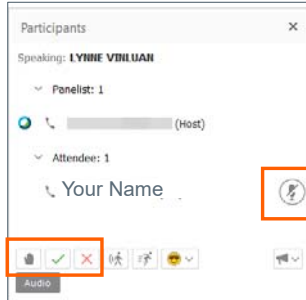
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Conferencing tool controls

(Upper Edge of Webex Window)



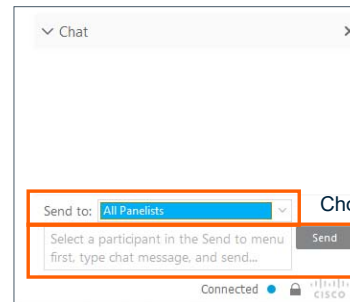
Participant Panel



Click the icon to Mute/Unmute

- Raise your hand
- Green checkmark
- Red X

Chat Panel



Choose **All Participants**

Type and click **Send**

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Let's take a few moments to get familiar with the features of the conferencing tool we will be using today. To see your Control Panel, move your cursor to the top of your Webex Training Center screen, and it will pull down.

The Participants icon is a toggle. When it is highlighted in blue, you can see the names of everyone in this session somewhere on your screen. If you have two monitors, it may be on your other monitor. Point your cursor to your name in the Participants panel, and you will see a red or gray microphone next to it. This is another toggle, where you can mute and unmute yourself. Red is muted, click again to make it gray to unmute and speak up.

At the bottom of the Participants panel, below everyone's name, is a set of icons that your instructor may be asking you to use. You can raise and lower your hand, you can give it a try now. During the labs, we will be asking you to click either the green checkmark if you are done with that lab, or the red X if you need more time. This is helpful to us so that we know if we can move ahead to the next section.

Some of us have been using the Chat panel already. When you have a question, use the drop-down box by the Send To: Prompt and choose "All Participants" so that our instructor does not have to repeat the question. Then type your message in the box below it and click Send.

Audio or video issues?



If you lose **Audio**, click the Audio icon to reconnect.
You do not have to leave the Webex session.

If your screen "freezes," send a chat to your Host.

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Depending on your internet connection, the state of Webex Training Center, or other factors, we have found that some attendees have trouble with audio. If that is the case, click on the Audio icon on the Control Panel to reconnect your audio. This lets you stay connected in Webex and not have to leave the session completely to get back in.

On rare occasions, your screen may seem to freeze up and get out of synch. If that is the case, send a chat message to me, your Host. We can address it then.

Virtual classroom ground rules

- Join class on time – attendance will be taken (after 15 minutes you will be considered a No-Show)
- For course credit, you must attend the entire class
- Mute your line (in WebEx or on your phone) while listening
- Do not put this call on hold
- To get the most out of this class, participate in activities and labs



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Now to make this the best possible learning experience for everyone, here are a few ground rules for this session.

- Attendance is taken, after 15 minutes you will be considered a no-show.
- To get course credit, you need to attend the entire class.
- Quick reminder to ensure your line is muted when you are listening to others and unmute to contribute a question or comment.
- Also, do not put this call on hold, to avoid having elevator music that some providers use.
- If you already have a second monitor set up, you may find it helpful in this training, especially when doing the labs.
- And most of all, please participate in the activities and labs to get the most out of your time in this class.

Course description and objectives

The purpose of the *Vantage: Optimizing the Analytics Database through Workload Management* (VOWM) workshop is to provide students the opportunity to optimize performance of a Teradata Vantage Analytics Database utilizing Teradata Active System Management (TASM). Using TASM's Workload Management ruleset, students are tasked with applying various workload management rules, in a simulated mixed workload environment, to achieve a set of response time and throughput service level goals.

During this hands-on workshop, students will start with an existing Active Data Warehouse that is not meeting performance objectives and perform the following tasks:

- Execute a mixed workload simulation
- Use Viewpoint's Workload Monitor portlet to monitor the performance of an active mixed workload environment
- Analyze performance metrics captured in Database Query Logging (DBQL)
- Use Viewpoint's Workload Designer portlet to apply various rules, such as Filters, Throttles, and Exceptions

After completing this workshop, the students should be able to:

- Establish a Baseline measurement
- Leverage workload management tools to monitor and analyze the mixed workload environment
- Refine workload management to achieve given service level goals

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Workshop Pre-Work

Prior to attending the VOWM Workshop, the students will be expected to complete the following pre-work or have 1-2 years of experience as a Teradata Vantage Administrator:

- [96021](#) - Teradata Vantage Administration Intermediate
- [95044](#) - Teradata Vantage MasterClass
- [95918](#) - Teradata Vantage Administration Basics Curriculum

Recommended Readings:

- Teradata Vantage - Workload Management User Guide
- Orange Books:
 - AMP Worker Tasks and ResUsage
 - Teradata Priority Scheduler for Linux SLES 11
 - Teradata Active Systems Management

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Workshop modules and collaterals

Course Modules

1. Workload Management Overview
2. Case Study
3. Introduction to Workload Designer
4. Monitoring Portlets
5. Workload Designer – General Settings
6. Workload Designer – Classifications
7. Workload Designer – State Matrix
8. Workload Designer – Session Control
9. Workload Designer – Filters
10. Workload Designer – Meters
11. Workload Designer – Throttles
12. Workload Designer – Workloads
13. Refining Workload Definitions
14. Workload Designer – Mapping Workloads to Priority Levels
15. Summary

Amazon Workspace

SSH Software

Putty

TD Client Software

TTU16.20

Teradata Studio TD17

Web Browser

Firefox

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Workshop modules schedule (VILT)

Day 1

1. Workload Management Overview
2. Case Study
3. Introduction to Workload Designer
4. Monitoring Portlets

Day 2

- Workload Designer:
5. General Settings
 6. Classifications
 7. State Matrix

Day 3

- Workload Designer:
8. Sessions
 9. Filters
 10. Meters
 11. Throttles

Day 4

12. Workload Designer – Workloads & Exceptions
13. Refining Workload Definitions

Day 5

14. Workload Designer – Mapping Workloads to Priority Levels
15. Summary

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Module 1: Workload management overview

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Discuss the characteristics of a mixed workload
- Discuss the different types of decision-making mixed workloads support
- Discuss the concepts and features of Workload Management
- Identify the differences between Teradata Integrated Workload Management (TIWM) and Teradata Active System (TASM)

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Topics

- Mixed workload overview
- Workload management overview
- Workload management features
- Overview of workload management portlets in viewpoint

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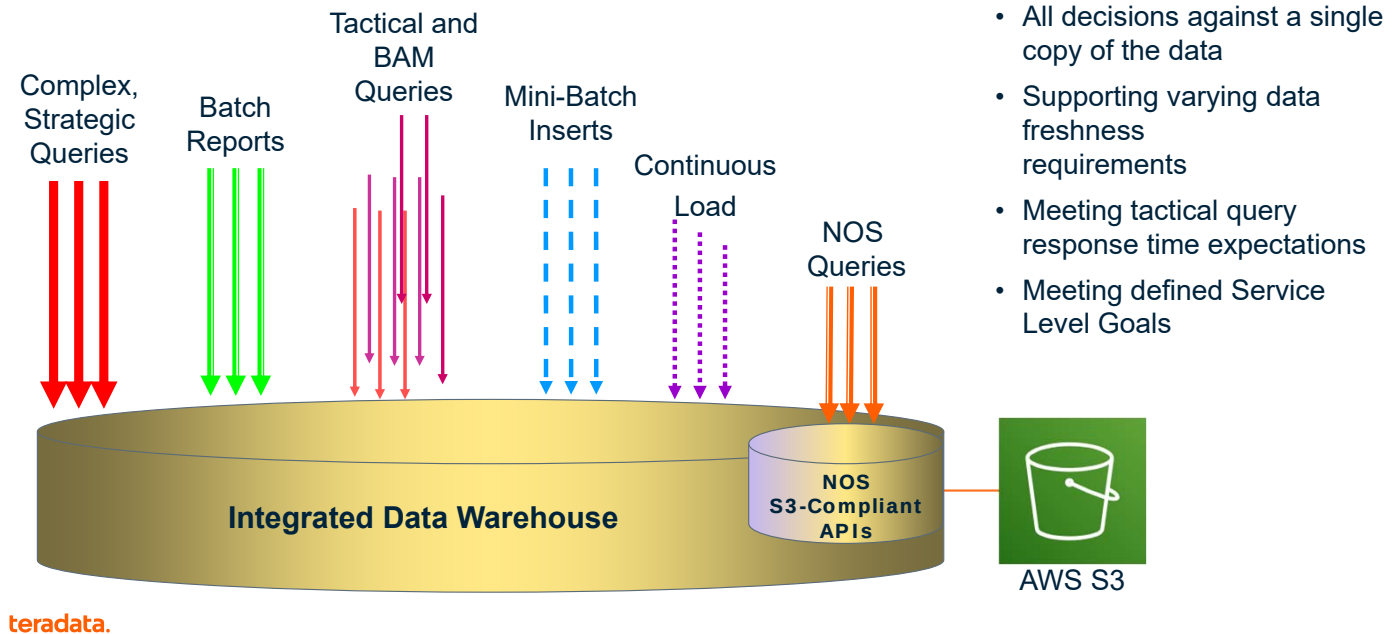
Current topic: Mixed workload overview

- **Mixed workload overview**
 - **What is a mixed workload?**
 - **What's the purpose for having mixed workloads?**
- Workload management overview
- Workload management features
- Viewpoint portlets

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What is a mixed workload?



Traditionally, data warehouse workloads have been based on drawing strategic advantage from the data. Strategic queries are often complex, sometimes long-running, and usually broad in scope. The parallel architecture of Vantage Analytics Database supports these types of queries by spreading the work across all of the parallel units and nodes in the configuration.

Today, data warehouses are being asked to support a diverse set of workloads. These range from the traditional complex strategic queries and batch reporting, which are usually all AMP requests requiring large amounts of I/O and CPU, to tactical queries, which are similar to the traditional OLTP characteristics of single or few AMPs requiring little I/O and CPU.

In addition, the traditional batch window processes of loading data are being replaced with more real-time data freshness requirements.

The ability to support these diverse workloads, with different service level goals, on a single data warehouse is the vision of Teradata's Active DW. However, the challenge for the PS consultants is to implement, manage and monitor an effective mixed workload environment.

What's purpose for having mixed workloads?

Mixed workloads are required to support different decision-making requirements.

- **Tactical** decision-making
 - **Short term decisions** focused on a narrow topic
 - Requires more **aggressive data freshness** service levels
- **Strategic** decision-making
 - **Long range decisions** covering multiple subject domains
 - Requires data that is **integrated** with **historical data**
 - **Time lags** in the data freshness service levels are **acceptable**
- **Event** based decision-making
 - Decisions made as a result of an event
 - Set of actions are **performed automatically** when a specified operation on a specified table is performed
 - Useful to **enforce business rules**
- **Near real time** data loading
 - Loading data **continuously**

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Mixed workloads are required to support different decision-making requirements.

- Tactic decision-making
 - Short term decisions focused on a narrow topic
 - Requires more aggressive data freshness service levels
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 - Decisions made as a result of an event
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 - Useful to al enforce business rules

Current topic: Workload management overview

- Mixed workload overview
- **Workload management overview**
 - **What is a workload management?**
 - **Workload management benefits**
 - **Workload management offering comparison**
- Workload management features
- Overview of workload management portlets in viewpoint

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What is workload management?

Workload Management is a goal-oriented, workload-centric, automated management of mixed workloads.

With Workload Management users create rules that can:

- Limit sessions, utilities, and requests – delaying or rejecting them when limits are reached
- Divide system resources among different types of requests
- Send alerts or change the workload of active requests that exceed limits
- Monitor time of day and system resources and react in predefined ways



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The Workload infrastructure is a Goal-Oriented, Automatic Management and Advisement technology in support of performance tuning, workload management, capacity planning, configuration and system health management. Workload Management features greatly improves system management capabilities, with a key focus being to reduce the effort required by DBAs, application developers and support engineers through automation. In addition, Workload Management provides many more system management monitoring and analysis capabilities than previously available to Teradata users. Business-driven Service Level Goals can be specified and monitored against for a quick-and-easy evaluation of performance when using the monitoring capabilities. Users of Workload Management will realize improved response time consistency and predictability of their workloads.

Workload management benefits

Workload Management provides the following benefits:

- **Support business** operations priority decisions
- **Stabilize** response times of the critical work
- **Increase throughput**
- **Protect** known, proven work from impact by unknown, adhoc unpredictable queries
- **Give priority** to proven, good performing queries
- **Automatically manage prioritization** based on processing periods or system health conditions

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Benefits of Workload Management include:

- Support business operations priority decisions
- Stabilize response times of the critical work
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- Protect known, proven work from impact by unknown, adhoc unpredictable queries
- Give priority to proven, good performing queries
- Automatically manage prioritization based on processing periods or system health conditions

Workload management offerings

Teradata Active System Management (TASM)

- Performs full workload management
- Gives administrators the ability to prioritize workloads, tune performance, and monitor and manage workload and system health
- Automates tasks that were previously labor-intensive for application DBAs and operational DBAs

Teradata Integrated Workload Management (TIWM)

- Provides basic workload management capabilities to customers without full TASM. TIWM offers a subset of TASM's capabilities



There are two Workload Management Offerings:

- Teradata Active System Management (TASM)
- Teradata Integrated Workload Management (TIWM)

With TASM licensing agreement Customers have all Workload Management features and capabilities. TIWM offers a subset of TASM's capabilities.

Current topic: Mixed workload features

- Mixed workload overview
- Workload management overview
- **Workload management features**
 - **Classification**
 - **Virtual partitions**
 - **Workload management methods: TIWM/TASM**
 - **Filters**
 - **Arrival rate meter**
 - **Throttles**
 - **State matrix**
 - **Exceptions**
 - **Query management architecture**
- Overview of workload management portlets in viewpoint

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Workload management key features

Key Features	
	Workload Classification
	Virtual Partitions
	Prioritization
	Resource Management
	Filters, Meters, & Throttles
	State Matrix
	Exceptions

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Teradata offers Workload Management bundled with all the platforms and it is called Teradata Integrated Workload Management. Teradata also offers advanced Workload Management on some platforms and it is called TASM.

Here are the differences between the offerings for starting from Release 14.0 for SLES11.

- All platforms now get full workload classification
- All platforms default to one virtual partition and TASM offers up to 10 virtual partitions
- Prioritization methods have been enhanced with all platforms utilizing Tactical and Timeshare methods while TASM adding SLG Tiers method
- All platforms utilize now also utilize workload throttles in addition to system throttles and filters
- TASM offers sophisticated operating period and health conditions state matrix
- Finally, all platforms utilize Tactical exceptions and Timeshare Decay and TASM adds Workload Exceptions

Teradata Integrated Workload Management ([TIWM](#)) has limited capabilities versus Teradata Active Systems Management ([TASM](#))

- [SLG Tiers](#) are not available
 - [Priority weighting](#) is automatic and not configurable
 - [Reserved AWTs and expedited status](#) is only available for Tactical Workloads
- [Exceptions](#) are only available for Tactical Workload automatic exceptions and for Timeshare automatic decay option can be enabled
- [Limited State Matrix](#) configuration:
 - Can configure additional Planned Environments and User Defined or Period Events
 - Cannot configure additional Health Conditions or Unplanned Events
- Cannot add additional [Virtual Partitions](#)
- Cannot add [Arrival Rate Meter](#) rules

- Only **Tactical** and **Timeshare** workload management methods are available, not **SLG Tiers**
- **Flex Throttles** are not available in TIWM

Workload classification criteria types

- With Workload Management, queries can be classified using **multiple classification criteria**
 - Classification **determines priority** and other workload **management activities**
- Multiple classification criteria **can be combined** together

Request Source Criteria

User
Account String
Account Name
Client ID
Client IP Address
Profile
Application

Target Criteria

Databases
Tables
Views
Macros
Stored Procedures
Functions
Methods

Query Characteristics Criteria

Join types
Full table scans
AMP limits
Statement type
Estimated Row counts
Estimated processing time
Memory Usage
Incremental Planning

QueryBand Criteria

Name/Value pairs

Utility Criteria

Fastload
Multiload
Fastexport
Archive/Restore

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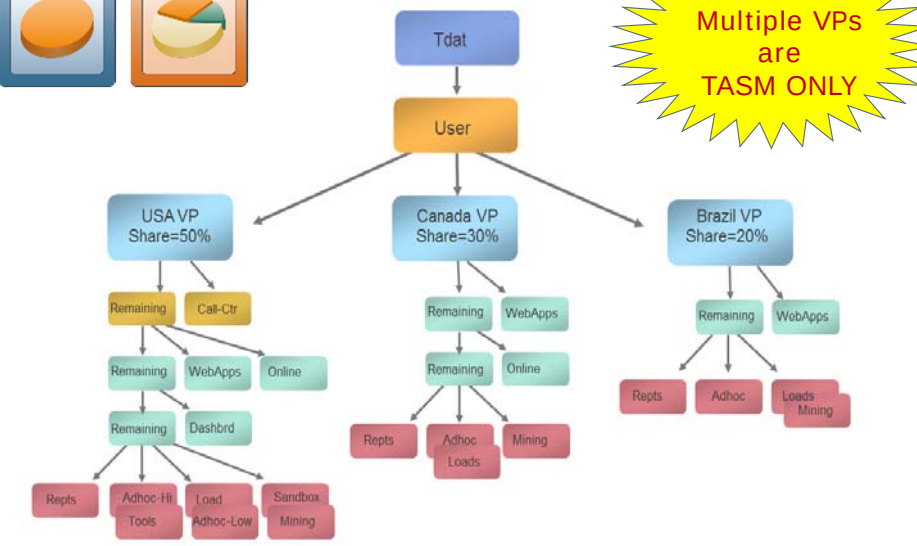


Workload Management allows queries to be classified using multiple criteria, whereas in earlier versions of Teradata only Account ID was used.

We will start with workload classification or classification criteria which consists of the following:

- **Request Source** – username, account name, account string, profile, application, client IP, client ID
- **Target** – database, table, macro, view, or stored procedure
Subcriteria: Full Table Scan, Join Type, Min Step Row Count, Max Step Row Count, or Min Step Time
- **Query Characteristics** – Statement type, AMP Limits, Step Row Count, Final Row Count, Estimated Processing Time, Min Step Time, Join Type, or Full Table Scan
- **QueryBand** – User-define metadata about the query
- **Utility** – FastLoad, FastExport, MultiLoad, or Backup Utilities

Virtual partitions



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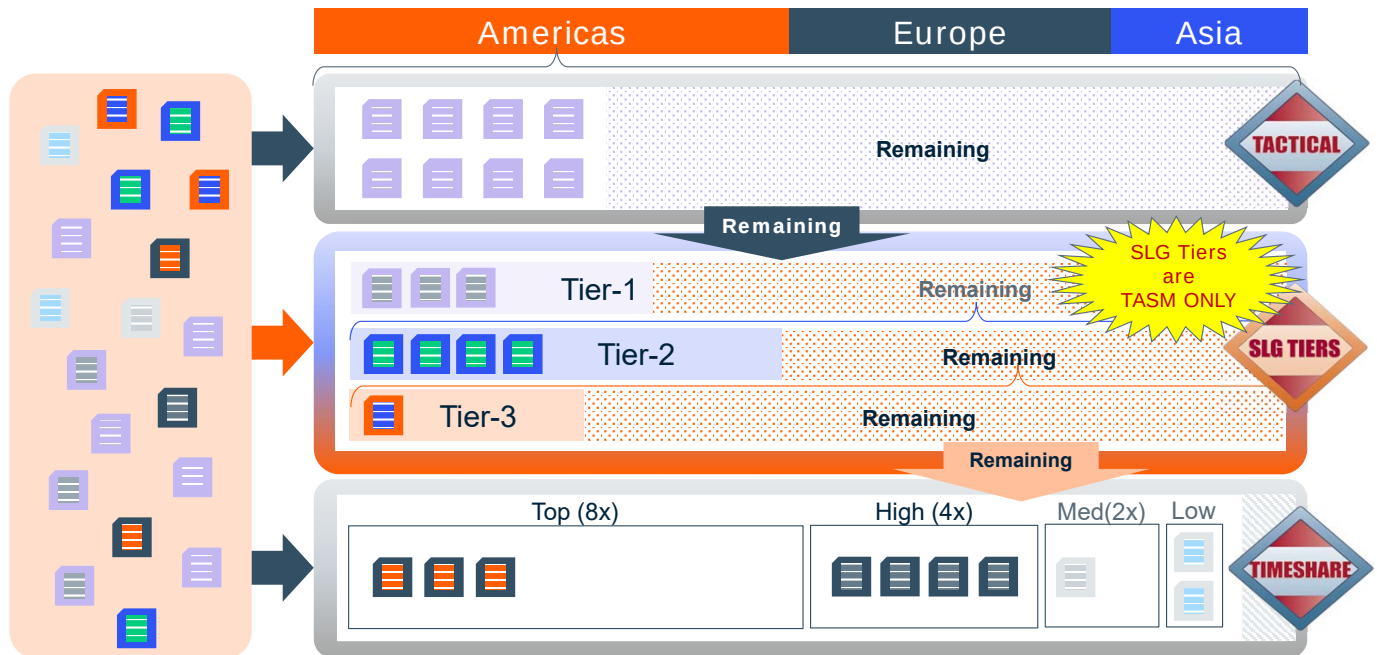
- Virtual Partitions are intended for sites supporting multiple **geographic entities** or **business units**
- The share percent assigned to a Virtual Partition will **determine** how the **CPU** is initially **allocated** across multiple Virtual Partitions
- If there are **spare resources** **not** able to be **consumed** within one Virtual Partition, another Virtual Partition will be able to consume more than it's assigned share percent unless **hard limits** are specified
- Recommendation is to **start with one** Virtual Partition

The first level in the priority hierarchy that the administrator can interact with is the virtual partition level. All platforms default to one virtual partition, and TASM offers up to 10 virtual partitions.

A virtual partition represents a collection of workloads. A single virtual partition exists for user work by default, but up to 10 can be defined with TASM.

A single virtual partition is expected to be adequate to support most priority setups. Multiple virtual partitions are intended for platforms supporting several distinct business units or geographic entities that require strict separation.

Workload management methods



TASM allows for workloads utilizing the Tactical, SLG and Timeshare methods. This diagram provides an example of a hypothetical TASM environment.

Note that SQL Tiers are only available with TASM.

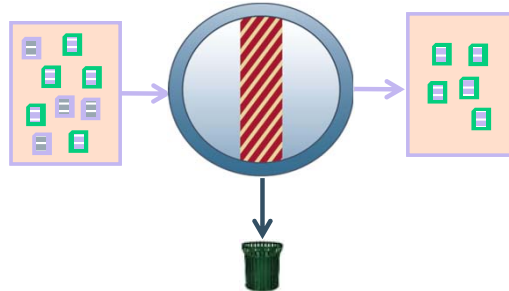
So, with TASM we can have multiple partitions. As shown in this example the Americas Virtual Partition has workloads assigned to all three Workload Management Methods: Tactical, SLG Tier and Timeshare. And you would have similar configurations for the other Virtual Partitions.

All the Americas resources are available to the first method which is the Tactical method, and as work arrives, the **Tactical Workload consumes all resources needed and the remaining resources are passed to the SLG and Timeshare Levels below.**

Work gets assigned to the various methods and remaining resources are passed down the next method, from Tactical to SLG-Tiers, and from SLG-Tiers to Timeshare.

Pre-Execution controls: Filters

- Filters are applied system-wide and **reject a query** before the query starts running based on the classification criteria
- **Classification criteria** include:
 - Source, Target, Query Characteristics, and QueryBand
- The “**Warning Only**” option can be used for testing
- **System Bypass** privileges can be applied based on username, account name, account string, or profile, allowing queries from **Bypass** users to skip the check for Filter rules



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Filters are system-wide and allows the DBA to reject queries before they begin running based on the classification criteria. If it is determined that a certain type of request should never run during the day, for example, system filter rules are able to enforce that. Filters are very useful for rejecting problematic query requests (e.g., unconstrained product joins against large tables) before they consume resources and are eventually cancelled due to exception criteria

In order to restrict the impact and scope of a filter to a selective number of deserving queries, the administrator can apply a variety of qualifying criteria. These are the same criteria choices that can be used for workload classification purposes.

Filter rules need to be used with caution and forethought, and applied very selectively, as rejecting queries is a strong action to take and may be considered as inappropriate in an ad hoc environment. “Warning Only” option can be used for testing as queries are logged but not rejected (can be used for testing new filters).

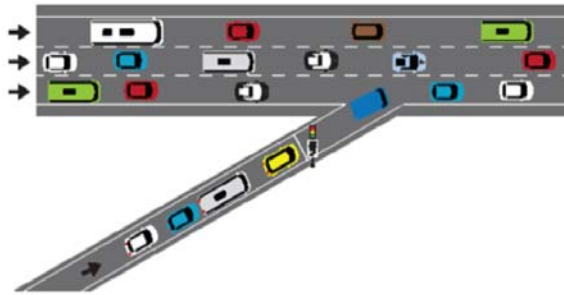
System Bypass privileges can be applied based on username, account name, account string, or profile, **allowing queries from Bypass users to skip the check for Filter rules.**

Arrival rate meter

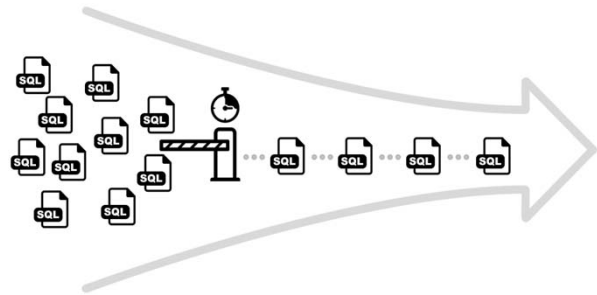
- An Arrival Rate Meter (ARM) is a new TASM rule type
- ARM rules provide a means to regulate the flow of SQL requests admitted by TASM into the system
- An ARM rule specifies the max number of SQL requests per time unit that TASM will admit into the system. Qualification criteria can be used to target specific types of requests (e.g., 3 queries per hour for queries with estimated processing time of more than 1 hour)



Freeway with Ramp Metering



TASM Arrival Rate Meter



Arrival Rate Meters (ARMs) are TASM rules that provide a way to regulate the flow of requests admitted by TASM into the system. An ARM rule specifies the maximum number of requests per time unit that TASM will admit into the system. Qualification criteria can be used to target specific types of requests.

TASM's filter and throttle processing sequence for a SQL request is enhanced to check for ARM. New Workload Management ruleset APIs allow you to automate the creation of a system throttle or arrival rate meter without using Viewpoint Workload Designer.

Benefits

- Prevents a predictable request surge from flooding the TASM delay queue and impacting service time of subsequent queries.
- Limits resource-intensive requests to a specific rate.
- Prevents you from submitting many extremely complex queries frequently.

Considerations

- Use arrival rate meters carefully after an evaluation period in warning mode.

Pre-Execution controls: Throttles

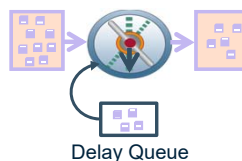
- Throttles can be used to **limit** the number of **active queries**
- **System Bypass** privileges can be applied based on username, account name, account string, or profile, allowing queries from **Bypass** users to skip the check for System Throttle rules

System Throttle

- ☐ **Session** throttles limit active sessions and reject new sessions
- ☐ **Query** throttles limit concurrent queries and reject / delay new queries
- ☐ **Utility** throttles limit concurrent utility jobs and reject / delay new jobs

Workload Throttle

- ☐ **Limit Workload** the number of concurrent queries for the workload
- ☐ **Reject/Delay** new queries
- ☐ **Flex Throttle** feature attempts to utilize unused resources by automatically releasing work from the Delay queue based on triggering conditions which DBA defines



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Controlling the number of concurrent request is by far the most popular use of throttles at Vantage sites today. When a throttle rule is active, a counter is used to keep track of the number of requests that are active at any point in time among the queries under control of that rule. When a new query is ready to begin execution, the counter is compared against the limit specified within the rule. If the counter is below the limit, the query runs immediately; if the counter is equal to or above the limit, the query is either rejected or placed in a delay queue. Most often throttles are set up to delay queries, rather than reject them.

Once a query which has been delayed is released from the delay queue and begins running, it can never be returned to the delay queue. Throttles exhibit control before a query begins to execute, and there is no mechanism in place to pull back a query after it has been released from the delay queue.

Starting in Teradata 15.10 there is a new option to order the delay queue by workload priority. A priority value is calculated for each workload based on the workload management method assigned to the workload. Requests in the delay queue are ordered from high to low based on the workload value. Ties are ordered by start time. If the option to order the delay queue by workload priority is not selected, the queue is ordered by query start time. In that case queries are released from the delay queue in first-in-first-out (FIFO) order if all applicable throttles are within limits.

Two types of throttles are available; system and workload throttles.

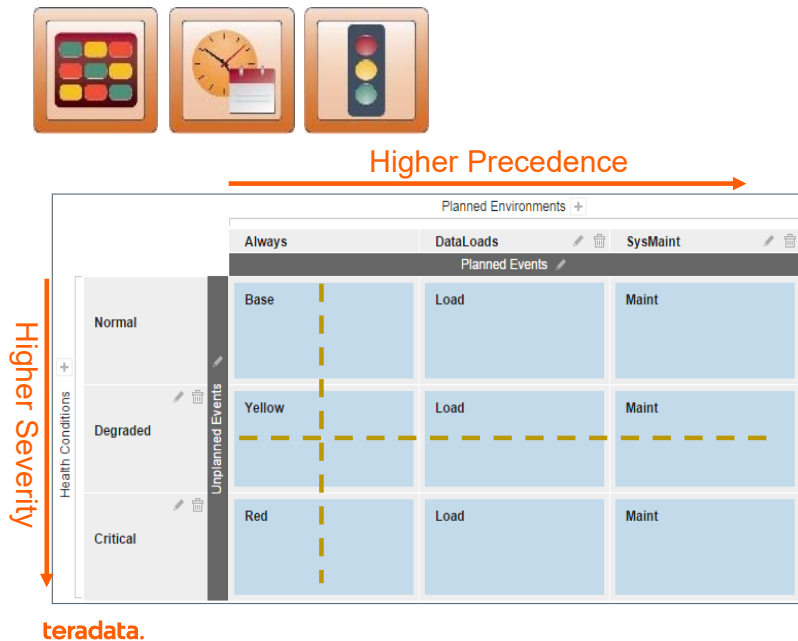
System throttles include session throttles which are used to limit the number of active sessions and reject any new sessions and the user must resubmit the query. Query throttles limit the number of concurrent queries and will reject or delay new queries. And last utility throttles limit the number of concurrent utility jobs and will reject or delay new utility jobs.

The other type of throttles is the workload throttle which is used to limit the number of concurrent queries for the workload and will reject or delay any new queries for that workload.

And the same Bypass privileges discussed with filters can also be applied to throttles.

The Flex Throttle feature attempts to utilize unused resources by automatically releasing work from the Delay queue based on triggering conditions a DBA defines.

State matrix



The State Matrix consists of two dimensions:

- **Health Condition** – (TASM ONLY) the condition or health of the system. Health Conditions are **unplanned events** that include system performance and availability considerations, such as number of AMPs in flow control or percent of nodes down at system startup
- **Planned Environment** – the kind of work the system is expected to perform. Usually indicative of **planned time periods or operating windows** when particular critical applications, such as load or month-end, are running
- **State** – identifies a set of **Working Values** and can be associated with one or more intersections of a Health Condition and Planned Environment
- **Current State** – the **intersection** of the current Health Condition and Planned Environment

Generally, workloads do not generate consistent demand, nor do they maintain the same level of importance throughout the day/week/month/year. For example, suppose there are two workloads: A query workload and a load workload. Perhaps the load workload is more important during the night and the query workload is more important during the day. Or perhaps there are tactical workloads and strategic workloads, and when the system is somehow degraded, it is more important to assure tactical workload demands are met, at the expense of the strategic work. Or finally, a year-end accounting workload may take precedence over all other workloads when present. The State Matrix allows a transition to a different working value set to support these changing needs.

The State Matrix allows a simple way to enforce gross-level management rules amidst these types of situations. In TASM it is a two-dimensional matrix, with Operating Environments and System Conditions represented, with the intersection of any Operating Environment and System Condition pair being associated with a State with different rule set working values. Multiple Operating Environment and System Condition pairs can be associated with a single State

During execution controls: Exceptions

Exception Rules are

- Used to **detect inappropriate queries** in a workload
- Automated for Tactical Workloads
- Available for Non-Tactical Workloads (**TASM ONLY**)



Take Action **NOW**

Unqualified Criteria

☐ Must exist for a period of time

- I/O Count & Physical Bytes
- Sum & Node CPU Time
- Spool Usage Bytes
- Maximum Spool Rows
- Blocked & Elapsed Time
- Number of Amps

WAIT Before Action

Qualified Criteria

☐ Must exist for a period of time

- ☐ **Length of time** (measured in CPU seconds) must exist before an action is triggered
- I/O Skew Difference & Percent
- CPU Skew Difference & Percent
- CPU Disk Ratio

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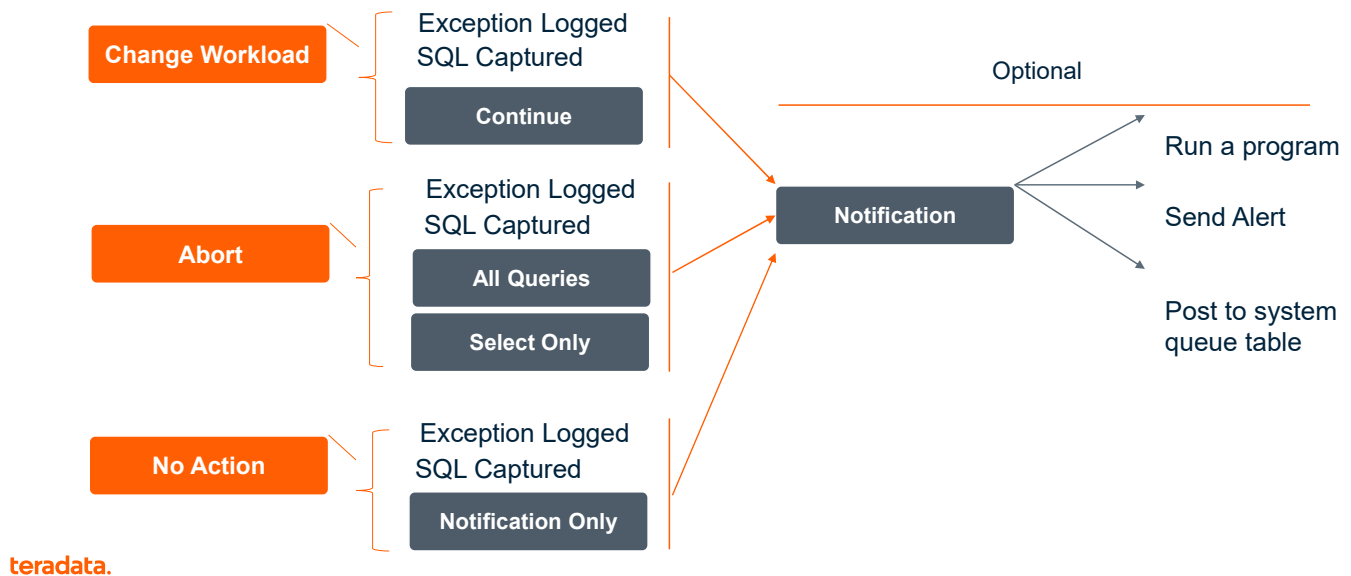
Workload Management has additional functionality to monitor queries during execution for adherence to specified criteria. Prior to Workload Management only CPU accumulations was able to be monitored.

It allows TASM to recognize atypical query processing conditions not intended for that workload so the priority scheduler can perform actions. The qualification criteria prevents false triggers.

The atypical query processing exception is defined via exceptions criteria such as max spool rows, I/O count, spool size, block time, response time, number of amps, cpu time, tactical cpu usage threshold (per node), tactical I/O physical bytes (per node), and/or I/O physical bytes.

Exception actions

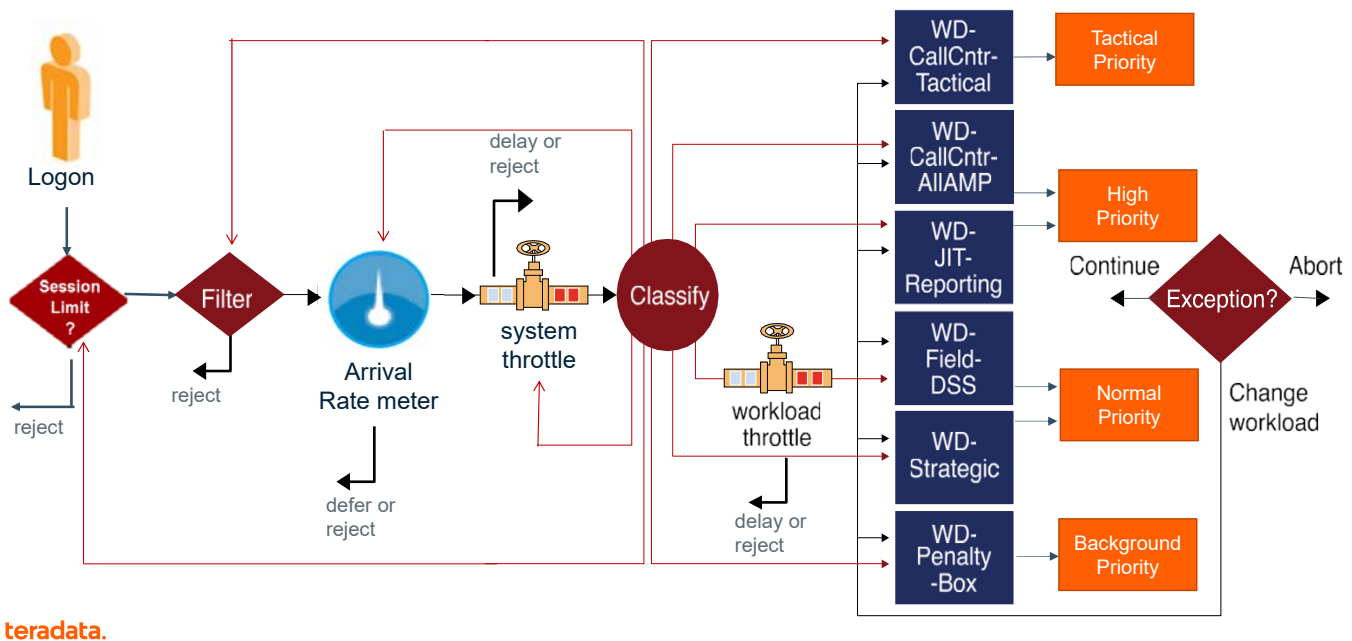
An Exception Action specifies **what to do** when an Exception condition is detected



Workload Management now allows a variety of actions to be taken when an exception condition is detected. Prior to Workload Management only demotion to another allocation group was possible.

The actions that can be taken by the priority scheduler include: abort or stop the query, abort select only queries, change the workload definition to a different work load, send a notification only as an alert or run a program or post an entry in the Qtable for processing by other applications.

Workload management levels



There are seven different methods of management offered, as illustrated below:

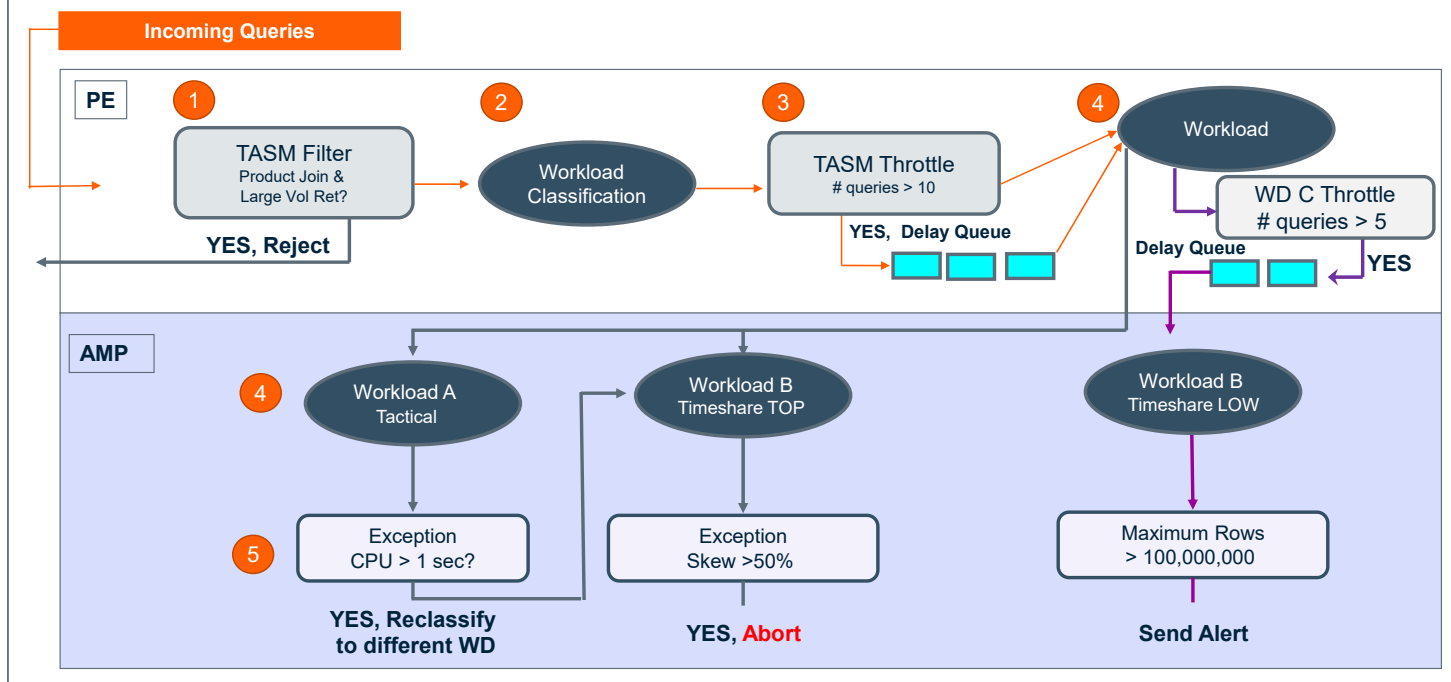
Methods regulated prior to the query beginning execution

- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level.
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level

Methods regulated during query execution

- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

Example: Query management architecture



The Workload Management rules you create are stored in tables in the TDWM user database. Unless otherwise specified, every logon and every query in every Analytics Database session is checked against the enabled Workload Management rules. That includes SQL queries from any supported Analytics Database interface, such as BTEQ, CLIV2, ODBC, and JDBC.

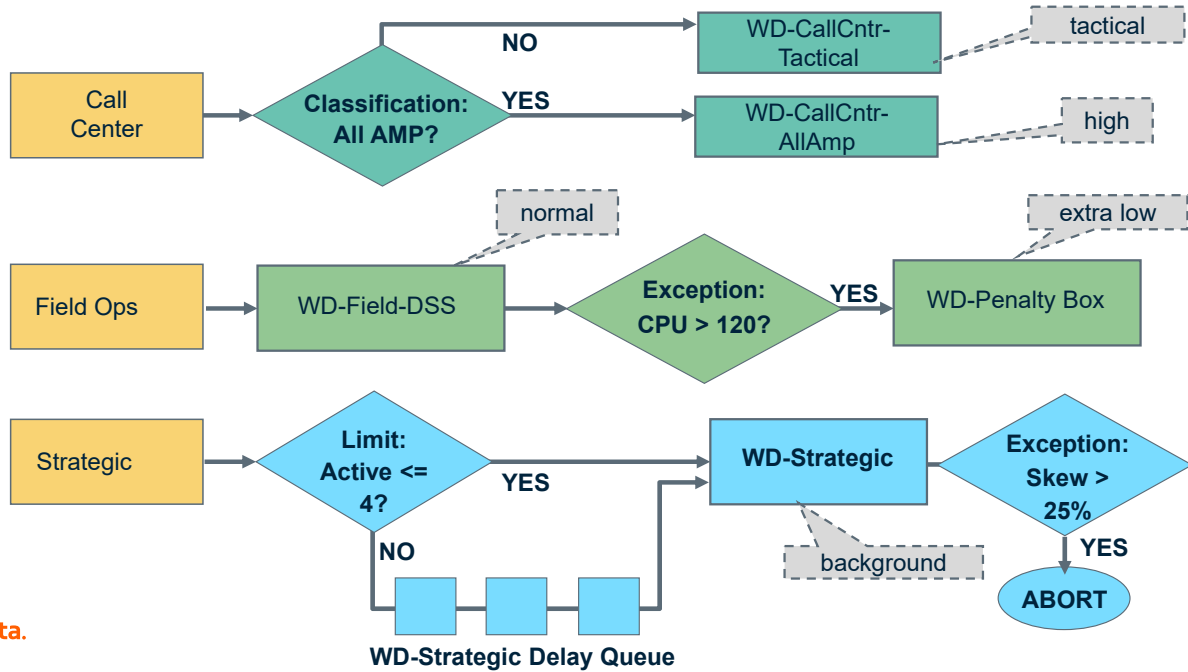
Workload Management rules are loaded into the Dispatcher components of the Analytics Database. When a Vantage client application issues a request to the Analytics Database, the request is examined and checked by TDWM functions in the Dispatcher before being forwarded to the AMPs to execute the request against the user database.

Query Management analyzes the incoming requests and compares the requests against the active rules to see if the requests should be accepted, rejected, or delayed.

- Queries that do not pass Filter Rules are rejected
- Queries that pass Filter rules are classified into a Workload Definition.
- Queries that do not pass Throttle Rules can be delayed or rejected. Additional throttles can also be applied at the Workload Definition level.

As queries execute within their assigned workload, they will be monitored against any exception rules. Violations of exception rules can invoke several actions from changing workloads, abort the query, send alert or run a program.

Workload management: Workloads and rules



This diagram illustrates an example of creating five workload definitions to handle a mix of queries.

Workload management offering comparison

Key Features	TIWM	TASM
 Workload Classification	Source, Target, Query Characteristics, QueryBand, Utility	Source, Target, Query Characteristics, QueryBand, Utility
 Virtual Partitions	One Partition	Multiple Virtual Partitions
 Prioritization	Tactical and Timeshare	Tactical, SLG Tiers , and Timeshare
 Resource Management	CPU and I/O	CPU and I/O
 Meters, Filters & Throttles	Filters, Workload and System Throttles	Filters, Arrival Rate Meters , Workload and System Throttles; Flex Throttles
 State Matrix	Planned Environments	by Planned Environment and by Health Conditions
 Exceptions	Tactical and Timeshare Decay	Tactical, Timeshare Decay, and Workload Exceptions

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 - Cannot configure additional Health Conditions or Unplanned Events
- Cannot add additional [Virtual Partitions](#)
- Cannot add [Arrival Rate Meter](#) rules

- Only **Tactical** and **Timeshare** workload management methods are available, not **SLG Tiers**
- **Flex Throttles** are not available in TIWM

Current topic: Viewpoint portlets

- Mixed workload overview
- Workload management overview
- Workload management features
- **Overview of workload management portlets in viewpoint**
 - **Workload management: Administration**
 - **Workload management: Monitoring and reporting**

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Workload management: Administration

The screenshot displays the 'Workload Designer' interface for 'Team1'. The top navigation bar includes tabs for 'General', 'States', 'Sessions', 'Filters', 'Meters', 'Throttles', 'Workloads', and 'Exceptions'. The 'General' tab is selected, showing sub-tabs for 'GENERAL', 'BYPASS', 'LIMITS/RESERVES', and 'OTHER'. The 'GENERAL' sub-tab is active, displaying a form with the following fields:

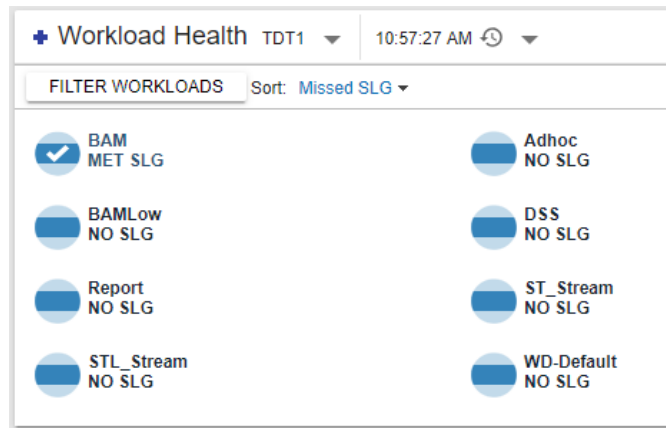
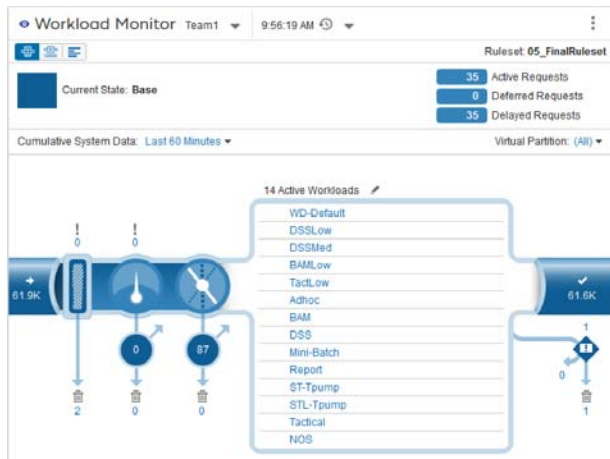
- Name: *** FirstConfig
- Description:** (empty text area)

Workload Management Administration

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Administration of Workload Management is administered using the Viewpoint Porlet Workload Designer.

Workload management: Monitoring and reporting



Workloads can be monitored on a real-time basis or reported at a summary and historical level.

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Workload Management provides for monitoring and reporting by workloads.

Summary

- Mixed Workloads consist of various types of queries with different performance requirements
- Mixed Workloads must also support different levels of data freshness requirements
- Mixed Workloads must support different types of decision making with different resource requirements
- Mixed Workloads must be managed to control the distribution of limited resources based on the priority
- Workload Management consists of a set of Goal Oriented, Workload Centric, and Automated supporting tools

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- Mixed Workloads consist of various types of queries with different performance requirements
- Mixed Workloads must also support different levels of data freshness requirements
- Mixed Workloads must support different types of decision making within the business
- Mixed Workloads must be managed to control the distribution of limited resources
- Analyzing Mixed Workloads is done at two levels:
 - System
 - Workload
- Workload Management consists of a set of Goal Oriented, Workload Centric, and Automated supporting products

Module 2: Case study

Teradata Vantage optimizing the Analytics Database through Workload Management

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Objectives

After completing this module, you will be able to:

- Discuss the business requirements of the case study
- Describe the simulation environment that will be used
- Identify the Service Levels Goals that need to be achieved

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Topics

- Your case study customer's environment
 - Case study characteristics
 - Simulation workloads
 - Hardware used for your workloads
 - Data model
 - Vantage Analytics Database environment
 - Service level goals
 - Workload users and profiles

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The case study

- The case uses a portion of Retail LDM to model a Retail business where customers purchase products at different stores
- The customer has implemented a Data Warehouse environment to support a set of mixed workloads consisting of Tactical, Business Activity Monitoring, Decision Support, Batch Reporting and Adhoc queries, as well as Native Object Store (NOS) queries, along with several real-time load workloads
- A comprehensive set of mixed queries and load components will be used to simulate various business requirements that will need to be addressed utilizing the various workload management choices
- Simulation has been designed to maximize critical resources such as AWTs, CPU and I/O
- You will be part of team, and will be assigned a Vantage Analytics Database in which you will implement your team's workload management choices
- After implementing your workload management choices, you will execute a mixed workload simulation and measure your performance gains or losses
- At the end of the class, each team will be measured on their ability to meet defined Service Level Goals (SLGs)

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The Mixed Workload Optimization case study models a retail business where customers purchase products at different stores. You will be working to teams applying various design options with a goal of meeting defined Service Level Goals.

Case study characteristics

- Your Data Warehouse environment consists of 30 tables using approximately 210GB of permanent space
- The tables have Secondary and Join Indexes implemented as necessary and collected statistics on all indexes as well as all non-indexed columns used for Value, Join or Range access
- The mixed workload simulation consists of a set of workload scripts that will randomly submit a set of queries or execute a utility
 - Tactical
 - BAM
 - DSS
 - Reporting
 - Adhoc
 - NOS
 - Continuous load
 - Mini-Batch load
- Each team will work through a process of implementing workload management choices, executing a mixed workload simulation and analyzing performance at the system and workload levels

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- BAM
- DSS
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- Adhoc
- Continuous load
- Mini-Batch load

Each team will work through a process of implementing workload management choices, executing a mixed workload simulation and analyzing performance at the system and workload levels

Simulation workloads



There are 8 distinct workloads that will be executed in the Mixed Workload simulation.

Tactical workload – executed in 25 streams and consists of 12 queries executed as macros. These queries will be submitted randomly across all 25 sessions every 2 seconds.

Business Activity Monitoring (BAM) workload – executed in 5 streams and consists of 5 queries executed as macros. These queries will be submitted every 5 minutes across 5 sessions.

Complex (DSS) workload – executed in 30 streams and consists of 25 queries. These queries will be submitted randomly across all 30 sessions every 2 seconds.

Batch Reports workload – executed in 5 streams and consists of 5 queries submitted. These queries will be submitted every 30 minutes.

Native Object Store (Nos) workload – executed in 5 streams and consists of 3 queries submitted. The queries will be submitted every 10 minutes.

Adhoc workload – executed in 10 streams and consists of 10 queries submitted. These queries will be submitted every 10 minutes.

Batch Reports workload – executed in 3 streams and consists of 3 queries submitted. These queries will be submitted every 30 minutes. The queries access an AWS S3 bucket that contains websales data in CSV format.

Mini-Batch workload – executed as a series of FastLoad jobs inserting 20,000 records into a staging table followed by a BTEQ Insert/Select into the Item_Inventory table. At the completion of the mini-batch job, there will be a 5 minute delay and another mini-batch job will be submitted.

Continuous (TPump) workload – There are 2 T pump workloads. One is inserting 40,000 rows continuously into the Sales_transaction_Line, the other is inserting 20,000 records into the Sales_Transaction table. This execution will continue until the simulation is stopped.

Simulation hardware

Each team will have their own Vantage Analytics Database running in AWS.

AWS m5.4xLarge
2 PEs
24 AMPs
16GB Memory
16 vCPU

Linux SLES12

DBS 17.10



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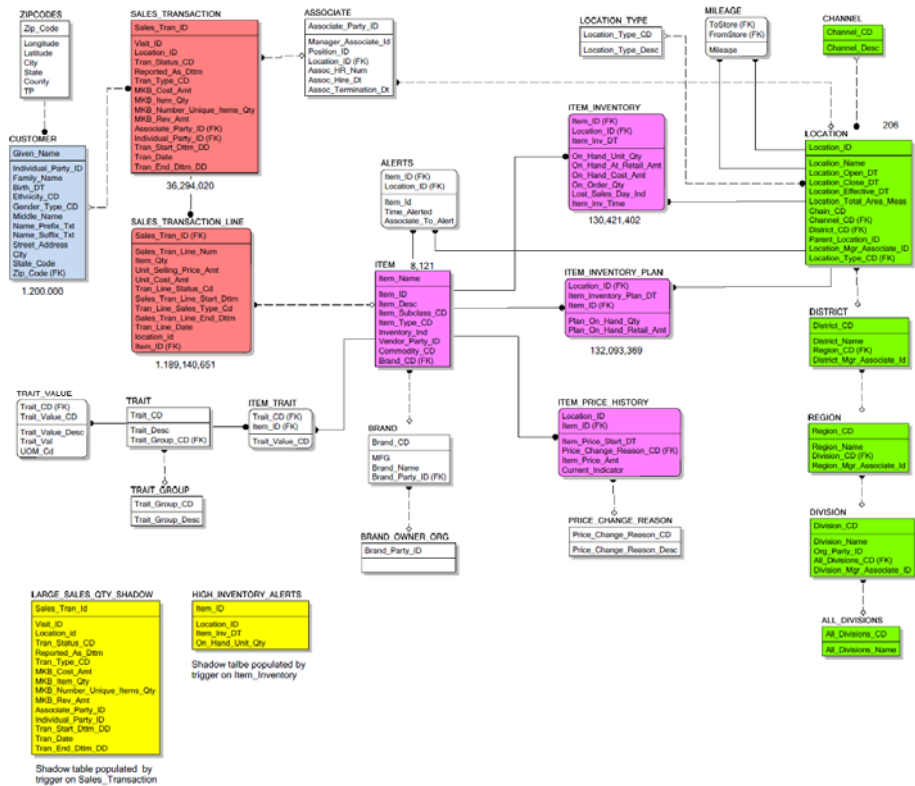
The lab environment for each team also includes a Viewpoint server running in AWS.

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Our lab systems run in the AWS cloud.

Data model

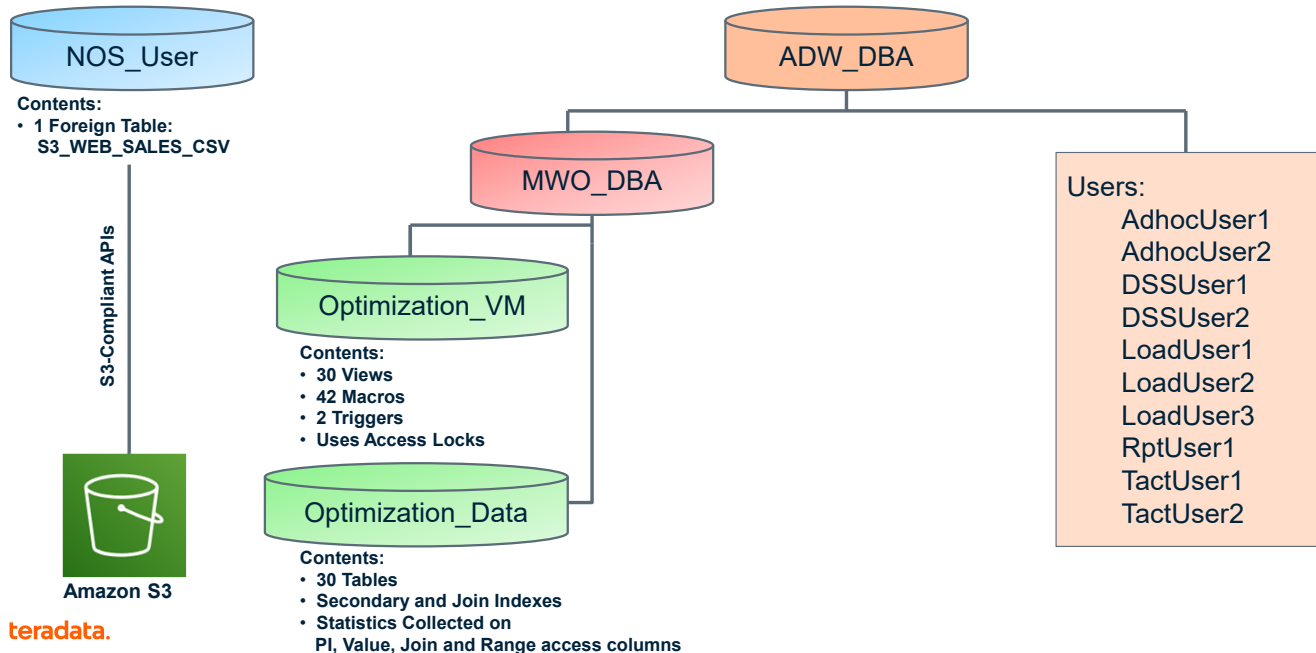
Note: This model is also provided as a PDF file in your VOWM Collaterals folder.



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A retail-oriented database has been designed to support this workshop. The physical data is on the facing page.

Vantage Analytics Database environment



The Teradata software will be Vantage Analytics Database Version 17.10. All logon passwords will be the same as the UserID. The hierarchical user structure is as follows:

ADW_DBA is the Parent user database for **MWO_DBA**, **Optimization_VM** and **Optimization_Data**:

MWO_DBA is the parent User for all of the databases used in this course.

Optimization_VM – This Database contains all views, macros and Triggers with references to the **Optimization_Data** database. All Views and Macros use access locks.

Optimization_Data – This contains all of the tables referenced by **Optimization_VM** as well as Join Indexes. Statistics have been collected on all indexes as well as all value, join and range access columns.

Following Users that submit queries:

AdhocUser1	RptUser1
AdhocUser2	TactUser1
DSSUser1	TactUser2
DSSUser2	
LoadUser1	
LoadUser2	
LoadUser3	

The **NOS_User** User database contains one Foreign Table that points to a bucket in AWS S3 which contains Web Sales data. The **NOS_User** submits queries against its **S3_WEB_SALES_CSV** Foreign table.

Service level goals

Service Percent Goals

Tactical Queries	Avg Resp Time <= 1 sec
BAM Queries	Avg Resp Time <= 6 sec
Known DSS Queries	Avg Resp Time <= 120 sec

Throughput Goals

Tactical Queries	45,000 per hour
BAM Queries	60 per hour
DSS Queries	1000 per hour
Item Inventory Mini-Batch	60 Inserts/sec
Sales_Trans Stream	90 Inserts/sec
Sales_Trans_Line Stream	200 Inserts/sec

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The Service Level Goals (SLGs) is the goal state that we are working towards. These performance and throughput numbers are determined and agreed upon by both the Customer and the Teradata PS Representative.

Workload users

There are 11 Users that are used to establish sessions and submit queries.

User Name	Default Profile	Workload Type
AdhocUser1	ADHOC_Profile	User submits one of 10 adhoc queries
AdhocUser2	ADHOC_Profile	User submits one of 10 adhoc queries
DSSUser1	DSS_Profile	User submits one of 25 decision support queries
DSSUser2	DSS_Profile	User submits one of 25 decision support queries
LoadUser1	LOAD_Profile	User executes Mini_Batch job (Item_Inventory)
LoadUser2	STREAM1_Profile	User executes T pump job (Sales_Transaction)
LoadUser3	STREAM2_Profile	User executes T pump job (Sales_Transaction_Line)
RPTUser1	REPORT_Profile	User executes one of 5 batch reports
NOSUser	NOS_Profile	User executes one of 3 NOS queries
TACTUser1	TACTICAL_Profile	User executes one of 12 tactical queries
TACTUser2	BAM_Profile	User executes one of 5 business activity monitoring queries

The following DBQL Logging statement is used for DSS and Tact Users:

```
Begin Query Logging with All on User_Name;
```

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There workload users used to establish sessions and submit queries are on the facing page.

Workload profiles

There are nine Profiles that have been assigned to Users submitting queries

Profile	Workload Characteristics
Adhoc_Profile	Adhoc queries, AdhocUser1 and AdhocUser2
BAM_Profile	BAM queries, TactUser2
DSS_Profile	DSS Queries, DSSUser1 and DSSUser2
Load_Profile	Mini-batch into Item_Inventory, LoadUser1
Report_Profile	Report queries, RptUser1
NOS_Profile	NOS queries, NOSUser
Stream1_Profile	Tpump into Sales_Transaction, LoadUser2
Stream2_Profile	Tpump into Sales_Transaction_Line, LoadUser3
Tactical_Profile	Tactical queries, TactUser1

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There are 8 distinct workload profiles as described on the facing page.

Case study summary

- The case study is based on the Retail LDM where customers purchase products at different store
- A comprehensive set of mixed queries and load components will be used to simulate various business requirements that will need to be addressed utilizing the various workload management choices
- Each team will work through a process of implementing workload management choices, executing a mixed workload simulation and analyzing performance by workload
- At the end of the class, each team will be required to meet defined Service Level Goals (SLGs)

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Module 3: Introduction to Workload Designer

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Discuss the characteristics and purpose of Viewpoint's Workload Designer portlet
- Show, edit, copy and activate Rulesets
- Explain how Workload Designer is used within a mixed workload environment

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Topics

- Working with the Workload Designer portlet
- Lab instructions

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Current topic: Working with Workload Designer

- **Working with Workload Designer**
 - **An overview of Workload Designer**
 - **Workload Designer for TIWM**
 - **Workload Designer for TASM**
 - **Managing rulesets with Workload Designer**
- Lab instructions

teradata.



About Workload Designer

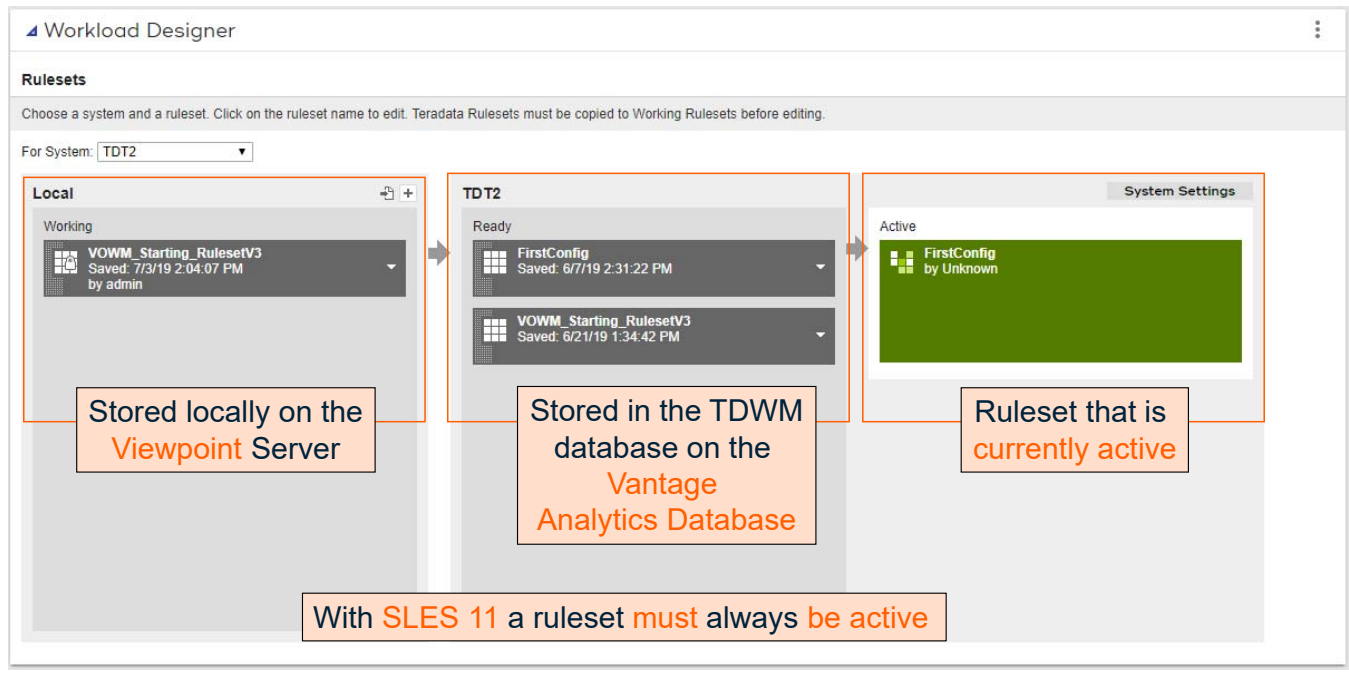
- It is a tool to provide **Query Management** capability
- It uses a set of **user-defined “rules”** to determine if query requests will be accepted by the database for execution or delayed for later execution
- Rules can consider **characteristics** and **expected resource** usage of each query when it executes and determine the number of queries or utilities that will be allowed to execute
- Rules can be applied as **system** rules or **workload** specific rules
- They can provide for **more predictable** response times for **high-priority**, low resource consuming, queries by **limiting** excessive interference from **lower priority**, higher resource consuming, queries
- **Avoid exhaustion** of uncontrolled system resources, such as AMP Worker Tasks
- **Protect** against poorly formulated queries that may require an unreasonable share of resources
- Utilizes the **“TDWM” User** to store its tables, macros and stored procedures
- TDWM User is created via a DIP program (**DIPTDWM**)

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Key features of Workload Designer include:

- It is a tool to provide Query Management capability.
- It uses a set of user-defined “rules” to determine if query requests will be accepted by the database for execution or delayed for later execution.
- Rules can consider the characteristics of each query or utility and control which ones be allowed to execute the priority they will be allowed to execute under and how many will be allowed to execute.
- Provide more predictable response times for high-priority queries by limiting excessive interference from lower priority queries.
- Avoid exhaustion of system resources, such as AMP Worker Tasks.
- Protect against poorly formulated queries that may require an unreasonable share of resources.
- Utilizes the “TDWM” User to store its tables and macros.

Workload Designer



The Workload Designer view shows high-level information about rulesets. Items in the options list depend on if you are the ruleset owner. If a ruleset is locked by someone else, you have fewer options than if you are the ruleset owner. Different options are available in **Working**, **Ready**, and **Active**.

- **Working** Names and descriptions of rulesets not yet moved to the production system. In **Working**, you can create and import rulesets. Rulesets in **Ready** can be copied to **Working** for editing. Rulesets in **Working** can also appear in **Ready** and **Active**.
- **Ready** Rulesets that have been saved to the production system, but are not active. A ruleset must be in **Ready** before it can be moved to **Active**. The **Active** ruleset cannot be deleted from **Ready**.
- **Active** Active ruleset on the production system. The only option available in the options list, if you have permissions, is to deactivate the ruleset.

Workload Designer Portlet Actions

- Rulesets are created and edited only in the *Working* section of Viewpoint
- Rulesets can be locked to prevent updates by others
- Clone an existing ruleset to utilize it as the basis for any modifications, to simplify changes, and to maintain consistency
- Use *Show All* to prepare a printable listing of the details of the ruleset
- Making a ruleset *ready* copies an inactive ruleset from the Viewpoint *Working* section to the target Teradata Vantage system
- Making a ruleset *active* makes the ruleset *ready*, if it is not, and then makes it the current ruleset in use for the Teradata Vantage system

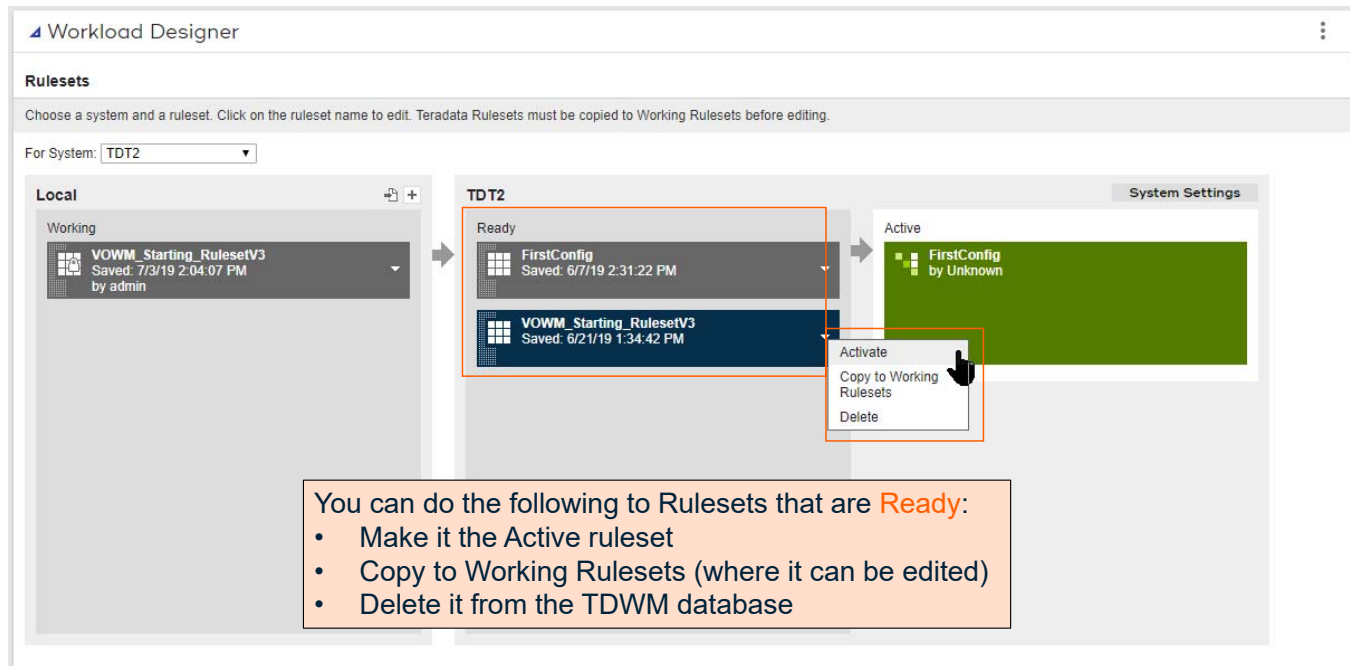
Creating a Ruleset

A ruleset is a complete collection of related filters, throttles, events, states, and workload rules. You can create multiple rulesets, but only one ruleset can be active on the production server.

After creating a ruleset, you can specify settings, such as states, sessions, and workloads, using the toolbar buttons. New rulesets are automatically locked so only the owner can edit the ruleset.

1. From the Rulesets view, select a system from the list.
2. Click + button

Workload Designer: Ready rulesets



The Workload Designer view shows high-level information about rulesets. Items in the options list depend on if you are the ruleset owner. If a ruleset is locked by someone else, you have fewer options than if you are the ruleset owner. Different options are available in **Working**, **Ready**, and **Active**.

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1. From the Rulesets view, select a system from the list.
2. Click + button

Workload Designer: Working rulesets

Workload Designer

Rulesets

Choose a system and a ruleset. Click on the ruleset name to edit. Teradata Rulesets must be copied to Working Rulesets before editing.

For System: TDT2

Local

Working

VOWM_Starting_RulesetV3
Saved: 7/3/19 2:04:07 PM
by admin

TDT2

Ready

FirstConfig
Saved: 6/7/19 2:31:22 PM

VOWM_Starting_RulesetV3
Saved: 6/21/19 1:34:42 PM

Active

FirstConfig
by Unknown

View/Edit

Show All

Unlock

Clone

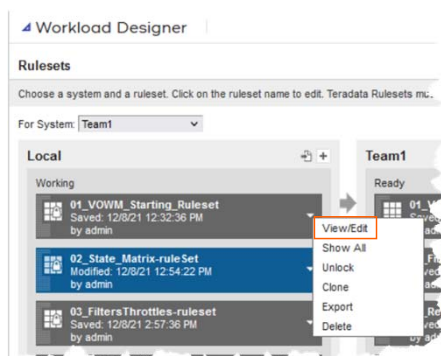
Export

Delete

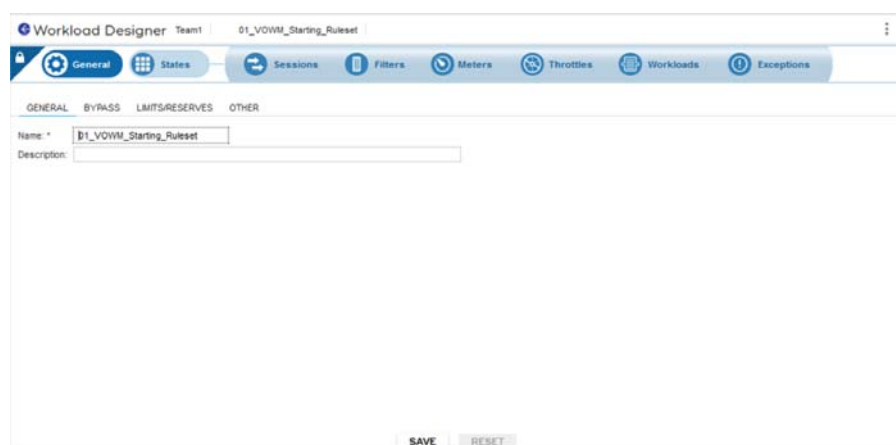
You can do the following to Rulesets that are Working:

- View and edit the rules of the Ruleset
- Display a summary of all rules and settings made for the Ruleset
- Copy (Clone) the Ruleset
- Export the Ruleset to an XML file
- Delete the Ruleset

Workload Designer: Working rulesets: View/edit

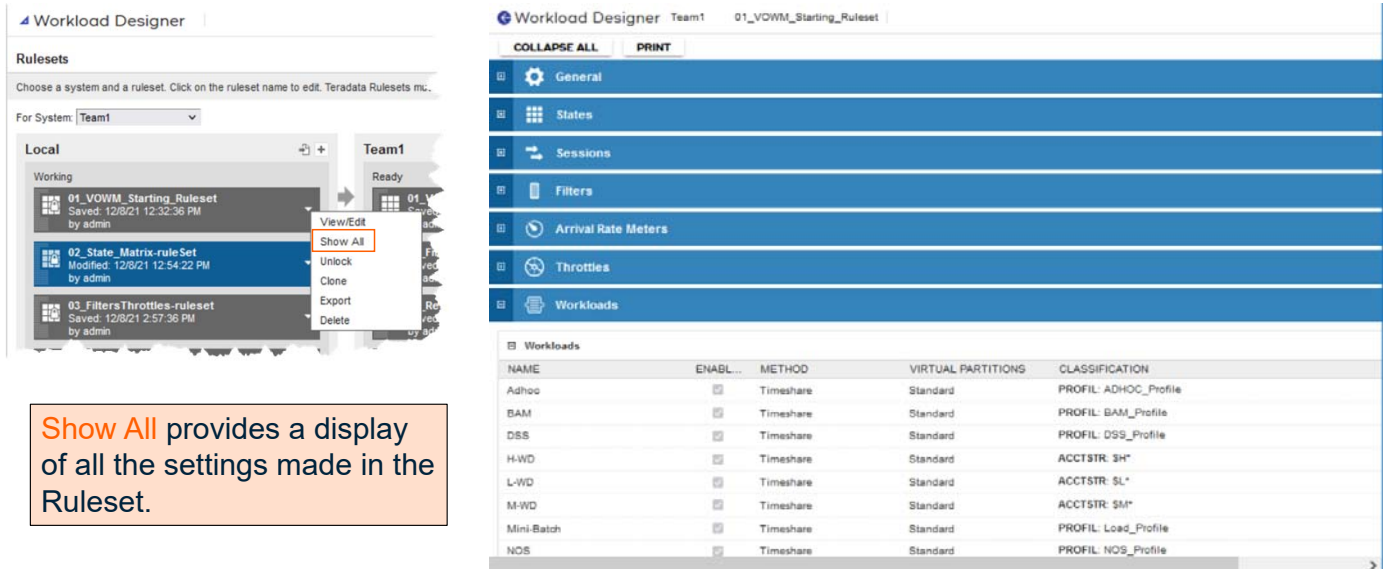


View/Edit opens the Ruleset where you can define its rules



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Workload Designer: Working rulesets: Show all



Rulesets

Choose a system and a ruleset. Click on the ruleset name to edit. Teradata Rulesets mc.

For System: **Team1**

Local

Working

- 01_VOWM_Starting_Ruleset
Saved: 12/8/21 12:32:36 PM
by admin
- 02_State_Matrix-ruleSet
Modified: 12/8/21 12:54:22 PM
by admin
- 03_FiltersThrottles-ruleset
Saved: 12/8/21 2:57:36 PM
by admin

Team1

Ready

View/Edit
Show All
Unlock
Clone
Export
Delete

Workload Designer Team1 01_VOWM_Starting_Ruleset

COLLAPSE ALL **PRINT**

- General
- States
- Sessions
- Filters
- Arrival Rate Meters
- Throttles
- Workloads

Workloads

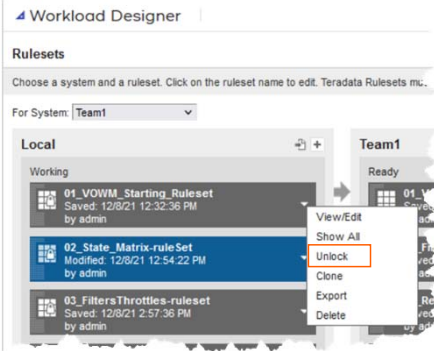
NAME	ENABL...	METHOD	VIRTUAL PARTITIONS	CLASSIFICATION
Adhoc	☒	Timeshare	Standard	PROFIL: ADHOC_Profile
BAM	☒	Timeshare	Standard	PROFIL: BAM_Profile
DSS	☒	Timeshare	Standard	PROFIL: DSS_Profile
H-WD	☒	Timeshare	Standard	ACCTSTR: 3H*
L-WD	☒	Timeshare	Standard	ACCTSTR: 5L*
M-WD	☒	Timeshare	Standard	ACCTSTR: 5M*
Mini-Batch	☒	Timeshare	Standard	PROFIL: Load_Profile
NOS	☒	Timeshare	Standard	PROFIL: NOS_Profile

Show All provides a display of all the settings made in the Ruleset.

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Lists all ruleset attributes on one page.

Workload Designer: Working rulesets: Unlock



Working in teams, only one person can have the ruleset locked.

The current lock status of a ruleset as shown in the Rulesets view

Option	Description
	Ruleset is locked.
	Ruleset is unlocked.

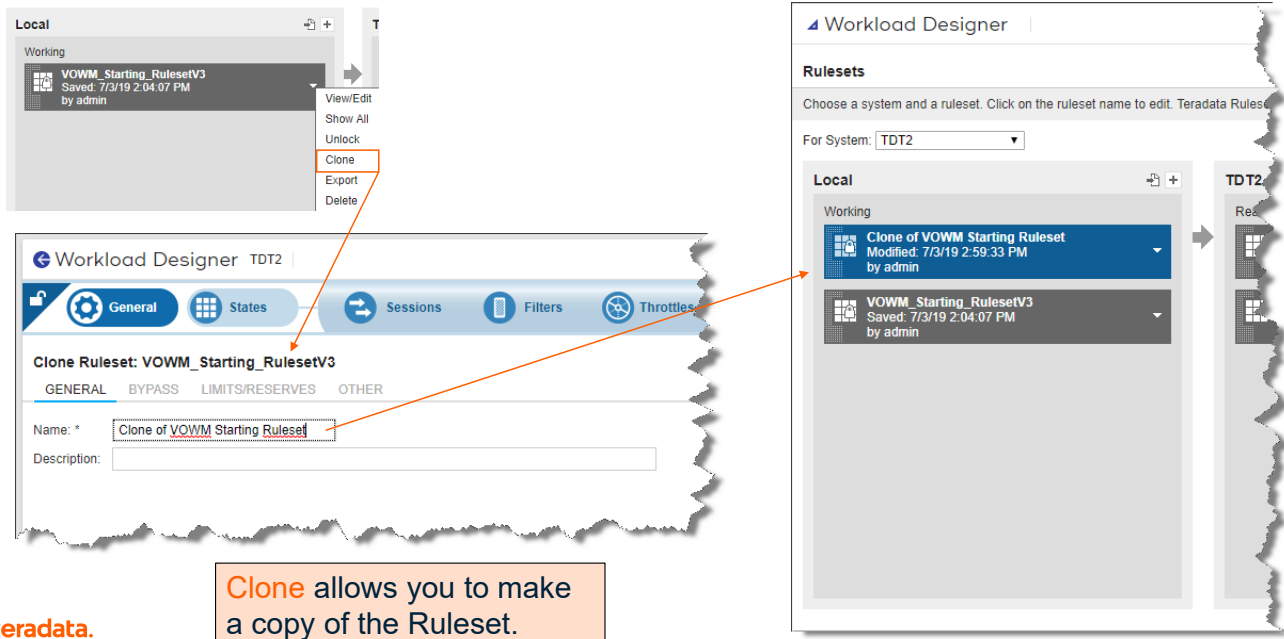
The current lock status of a ruleset as shown in the Rulesets Toolbar view

Option	Description
	Ruleset is unlocked.
	Ruleset is locked by the current user.
	Ruleset is locked by another user. Position the cursor over the icon to see the name of the other user.

teradata.

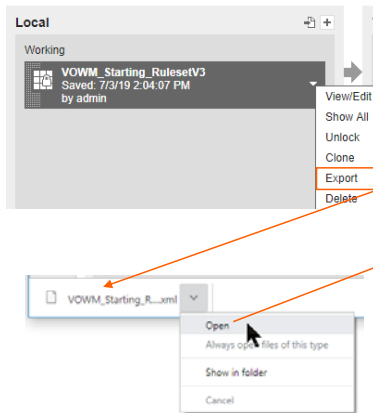
An exclusive lock can be placed on a ruleset so that the ruleset cannot be edited, deleted, or otherwise modified except by the owner of the lock. A ruleset is automatically locked when it is created and each time changes to the ruleset are saved. Use the **Workload Designer** view to lock and unlock rulesets. The Teradata Viewpoint Administrator must grant your role permission to edit rulesets so you can complete this action. The Teradata Viewpoint Administrator can also grant your role permission to unlock any ruleset.

Workload Designer: Working rulesets: Clone



Creates a copy of the ruleset. This option is useful if you want to use an existing ruleset as a base or template to create a ruleset.

Workload Designer: Working rulesets: Export



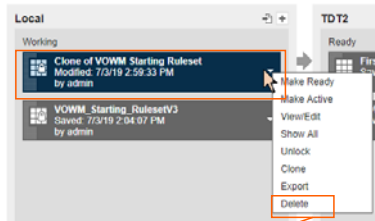
Export creates an XML file of the Ruleset. This XML file can be used to import the Ruleset into a Workload Designer portlet on another Viewpoint server.

[illegible]

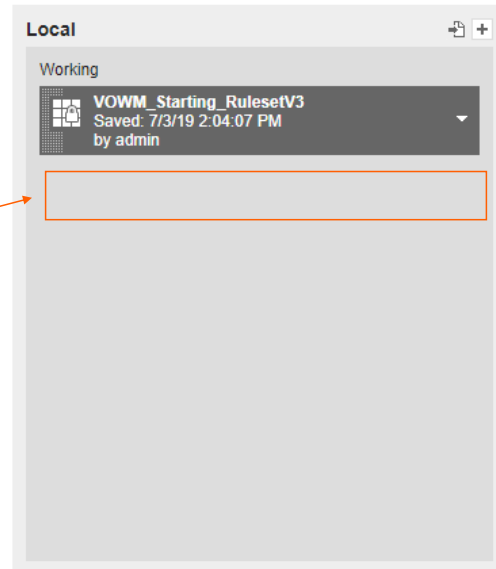
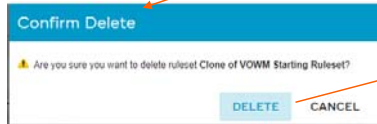
teradata.

Exports the ruleset as an XML file. Use with the **Import** button to copy a ruleset from one system to another.

Workload Designer: Working rulesets: Delete



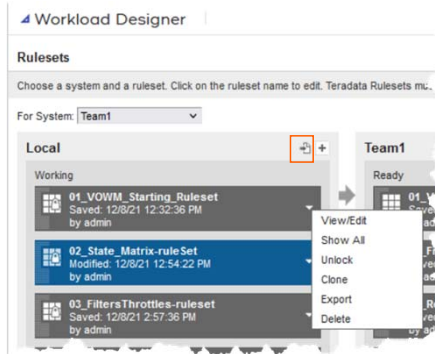
Delete allows you remove the Ruleset from Viewpoint.



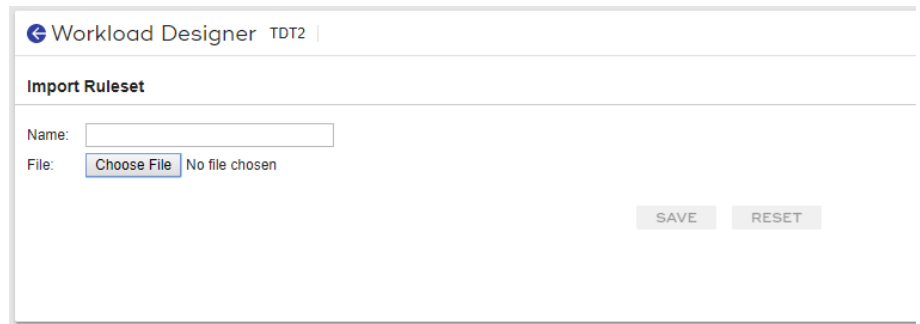
teradata.

Removes the ruleset from the **Working** section.

Workload Designer: Local: Import a ruleset



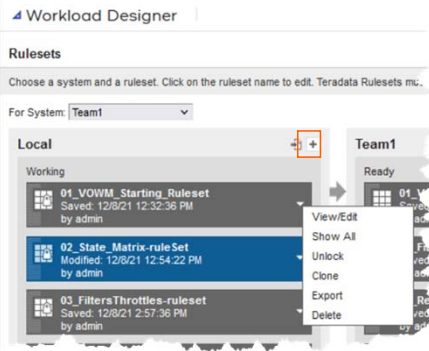
Import allows you to select an XML file and import it as a new Ruleset.



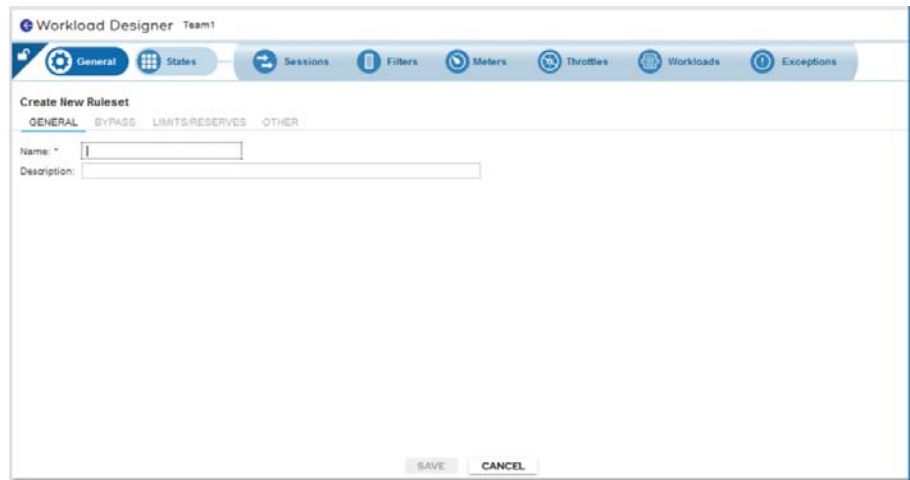
teradata.

The import and export options can be used to copy a ruleset from one Viewpoint system to another. The Teradata Viewpoint Administrator must grant your role permission to edit rulesets so you can complete this action. Only rulesets exported from **Workload Designer** and a database of the same release can be imported.

Workload Designer: Local: Create a new ruleset



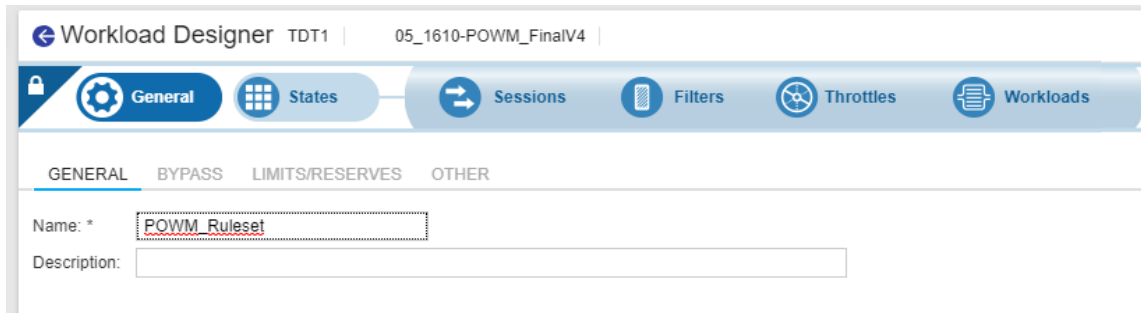
Create a new Ruleset allows you to make a brand-new Ruleset that's initialized with only the default rules and settings.



teradata.

Create a new Ruleset allows you to make a brand new Ruleset that's initialized with only the default rules and settings.

Workload Designer: TIWM



Enhanced TASM Functions



Enable this option if your Teradata system has license entitlement to TASM

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This example displays the main Workload Designer interface for non-TASM licensed (TIWM) systems. Notice that the Workload Designer interface for TIWM does not include the Meters and Exceptions button. When the *Enhanced TASM Functions* option is not selected in the *Administration Monitored Systems* portlet, then the Workload Designer screen will not include TASM-only Views, such as Meters and Exceptions.

Teradata offers Workload Management bundled with all the platforms and it is called Teradata Integrated Workload Management. Teradata also offers advanced Workload Management on some platforms and it is called TASM.

Here are the differences between the offerings for starting from Release 14.0 for SLES11.

- All platforms now get full workload classification
- All platforms default to one virtual partition and TASM offers up to 10 virtual partitions
- Prioritization methods have been enhanced with all platforms utilizing Tactical and Timeshare methods while TASM adding SLG Tiers method
- All platforms utilize now also utilize workload throttles in addition to system throttles and filters
- TASM offers sophisticated operating period and health conditions state matrix
- Finally, all platforms utilize Tactical exceptions and Timeshare Decay and TASM adds Workload Exceptions

Teradata Integrated Workload Management (**TIWM**) has limited capabilities versus Teradata Active Systems Management (**TASM**)

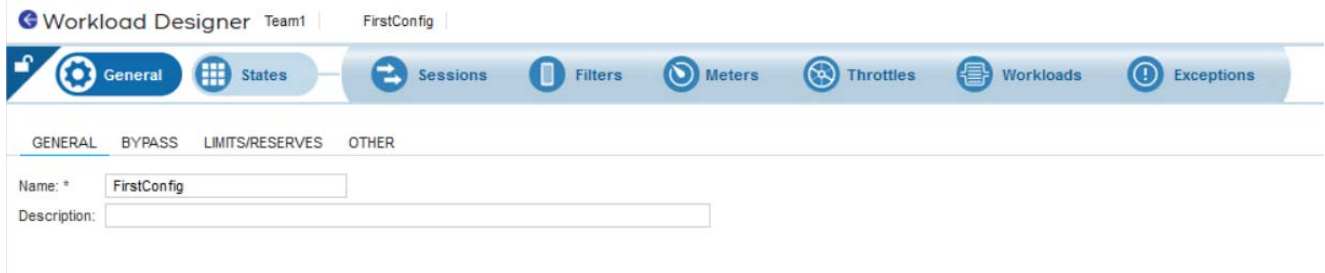
- **SLG Tiers** are not available
 - **Priority weighting** is automatic and not configurable
 - **Reserved AWTs and expedited status** is only available for Tactical Workloads
- **Exceptions** are only available for Tactical Workload automatic exceptions and for Timeshare

teradata.

automatic decay option can be enabled

- [Limited State Matrix](#) configuration:
 - Can configure additional Planned Environments and User Defined or Period Events
 - Cannot configure additional Health Conditions or Unplanned Events
- Cannot add additional [Virtual Partitions](#)
- Cannot add [Arrival Rate Meter](#) rules
- Only [Tactical](#) and [Timeshare](#) workload management methods are available, not [SLG](#) Tiers
- [Flex Throttles](#) are not available in TIWM

Workload Designer: TASM



Workload Designer Team1 | FirstConfig

General States Sessions Filters Meters Throttles Workloads Exceptions

GENERAL BYPASS LIMITS/RESERVES OTHER

Name: * FirstConfig

Description:

Enhanced TASM Functions

☒ Enable this option if your Teradata system has license entitlement to TASM

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The facing page displays the main Workload Designer interface for TASM licensed systems. Notice that the Workload Designer interface for TASM includes the Meters and Exceptions button.

Case study: Differences between VOWM_starting_ruleset and FirstConfig rulesets

FirstConfig default workloads

Workloads				
Workloads				
NAME	ENABL	METHOD	VIRTUAL PARTITIONS	CLASSIFICATION
H-WD	☒	Timeshare	Standard	ACCTSTR: SH*
L-WD	☒	Timeshare	Standard	ACCTSTR: SL*
M-WD	☒	Timeshare	Standard	ACCTSTR: SM*
T-WD	☒	Timeshare	Standard	ACCTSTR: SR*
WD-Default	☒	Timeshare	Standard	
WD-MapsMover	☒	Timeshare	Standard	WDClassif... MoveTableToMap

VOWM_Starting_Ruleset workloads

Workloads				
Workloads				
NAME	ENABLED	METHOD	VIRTUAL PARTITIONS	CLASSIFICATION
Adhoc	☒	Timeshare	Standard	PROFIL: ADHOC_Profile
BAM	☒	Timeshare	Standard	PROFIL: BAM_Profile
DSS	☒	Timeshare	Standard	PROFIL: DSS_Profile
H-WD	☒	Timeshare	Standard	ACCTSTR: SH*
L-WD	☒	Timeshare	Standard	ACCTSTR: SL*
M-WD	☒	Timeshare	Standard	ACCTSTR: SM*
Mini-Batch	☒	Timeshare	Standard	PROFIL: Load_Profile
NOS	☒	Timeshare	Standard	PROFIL: NOS_Profile
Report	☒	Timeshare	Standard	PROFIL: Report_Profile
ST-Tpump	☒	Timeshare	Standard	PROFIL: Stream1_Profile
STL-Tpump	☒	Timeshare	Standard	PROFIL: Stream2_Profile
T-WD	☒	Timeshare	Standard	ACCTSTR: SR*
Tactical	☒	Tactical	Standard	PROFIL: Tactical_Profile

Note: The only difference between FirstConfig Ruleset and the VOWM_Starting_Ruleset is that we added a new workload for each of the Profiles from our Case Study. Each new workload is mapped to Timeshare Medium except for Tactical which is mapped to the Tactical Workload Management Method.

We also added a Workload Throttle for the NOS workload.

Workload Throttles		
Flex Throttle: Disabled		
Evaluation Mode: Disabled		
ENABLED	NAME	CLASSIFICATION
☒	NOS	PROFIL: NOS_Profile
NAME	DEFAULT SETTING	BASE
NOS	1, DELAY	(Use Default)

FirstConfig is the default ruleset.

Current topic: Lab instructions

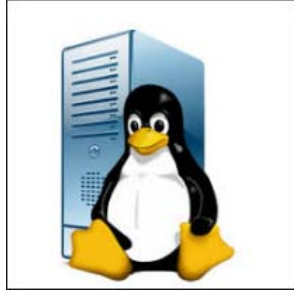
- Working with Workload Designer
- **Lab instructions**
 - **Setup a connection profile in Teradata Studio (refer to day 1: Lab guide)**
 - **Setup an SSH connection in PuTTY (refer to day 1: Lab guide)**
 - **Run the QryPerMetrics.sql in Teradata Studio**
 - **Update the VOWM simulation results spreadsheet**
 - **Find the load jobs information**
 - **Record the load-job results in the VOWM simulation results spreadsheet**

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Lab: IP address for your team's Linux server



Each team has its own Vantage Analytics Database running on a Linux server in AWS.



Your instructor will provide you with the IP address for your team's Linux server.

Refer to the Day 1 – Lab Guide, to setup:

1. Connection Profile in Teradata Studio
2. SSH Connection in PuTTY

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Each team has its own Analytics Database system running on a Linux server in AWS. Your instructor will provide you with the IP address for your team's Linux server.

Refer to the Day 1 – Lab Guide, to setup:

1. Connection Profile in Teradata Studio
2. SSH Connection in PuTTY

Your instructor will give you 20 minutes to complete these two steps. When you are done with these setup steps, the remaining lab activities in this module will be done together with your instructor.



Lab: Run QryPerfMetrics.sql in Teradata Studio

Run the **QryPerfMetrics.sql** script to capture the Average Response Time and Throughput metrics for the BAM, Tactical, and DSS workloads

The screenshot shows the Teradata Studio interface. The top pane is the SQL Editor, displaying a query that selects workload metrics. The bottom pane is the Teradata Result Set Viewer, showing the results of the query in a table format.

```
SELECT "ProfileName" AS "Workload",
       "Avg_ResponseTime" AS "Average Response Time",
       2 * "QueryCount" AS "Throughput/hour"
FROM "tdwm"."Workload_Candidate_Profile"
WHERE "ProfileName" in ('DSS_Profile', 'BAM_Profile', 'Tactical_Profile');
```

	Workload	Average Respon...	Throughput/hour
1	BAM_Profile	89.21	68
2	Tactical_Profile	7.89	7280
3	DSS_Profile	151.44	672

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After you have setup a Connection Profile in Teradata Studio, you can run this query to capture the Average Response Time and Throughput metrics for the BAM, Tactical, and DSS workloads

```
SELECT "ProfileName" AS "Workload",
       "Avg_ResponseTime" AS "Average Response Time",
       2 * "QueryCount" AS "Throughput/hour"
FROM "tdwm"."Workload_Candidate_Profile"
WHERE "ProfileName" in ('DSS_Profile', 'BAM_Profile', 'Tactical_Profile');
```



Lab: Record workload simulation query results in VOWM simulation results spreadsheet

After each simulation, capture:

- Average Response Time and Throughput per hour for:
 - Tactical Queries
 - BAM Queries
 - DSS Queries

	Workload	Average Respon...	Througput/hour
1	BAM_Profile	89.21	68
2	Tactical_Profile	7.89	7280
3	DSS_Profile	151.44	672

	VOWM Simulation Results											
	Tactical Queries		BAM Queries		DSS Queries		Item Inventory Loads		Sales Transaction Loads		Sales Transaction Line Loads	
	Average Response Time	Throughput per/hour	Average Response Time	Throughput per/hour	Average Response Time	Throughput per/hour	Rows Inserted	Inserts / Second	Rows Inserted	Inserts / Second	Rows Inserted	Inserts / Second
FirstConfig	7.89	7280	89.21	68	151.44	7280	0	0	0	0	0	0
Baseline							0	0	0	0	0	0
Test 1							0	0	0	0	0	0
Test 2							0	0	0	0	0	0
Test 3												
Test 4												
Test 5												
Test 6												
Test 7												
Test 8												
Test 9							0	0	0	0	0	0

Remember the Workload Simulation is run for 30 minutes, so the Query Count number needs to be doubled to determine the Throughput per hour

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Your instructor will provide you with the *VOWM Simulation Results Spreadsheet*. Prior to the start of class, your instructor ran the Workload Simulation script for your team with the default ruleset, *FirstConfig*, activated.

In this step you will record the *Average Response Time* and *Throughput* metrics for:

- Tactical Queries
- BAM Queries
- DSS Queries

Note: You will record these metrics after each running of the Workload Simulation.



Lab: Find the load jobs information

1. SSH into the Linux server and change to the **MWO** home directory:
`cd /home/ADW_Lab/MWO`
2. Open the **post_job.log** to get the number of rows inserted during the load jobs

```
10.0.2.141 - PuTTY
select 'sales_transaction_line', count(*)
from optimization_vm.sales_transaction_line
where tran_line_date = '2010-07-01'
UNION
select 'item_inventory', count(*)
from optimization_vm.item_inventory
where item_inv_dt = '2010-07-01'
;

*** Query completed. 3 rows found. 2 columns returned.
*** Total elapsed time was 18 seconds.

Updated Table          # of rows inserted
-----
item_inventory          60000
sales_transaction       60000
sales_transaction_line 120000

+-----+-----+-----+-----+-----+-----+
.quit;
*** You are now logged off from the DBC.
*** Exiting BTEQ...
** RC (return code) = 0

S1,2 Bot
```

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Once you have setup an SSH Connection in PuTTY, you can view the post_job.log file contains the number of rows insert for the load jobs. This example shows a portion of the post_job.log file which contains the summary of the load job information.

Lab: Record workload simulation load-job results in VOWM simulation results spreadsheet



After each simulation, capture:

- Inserts per Second for tables:
 - Item Inventory
 - Sales Transaction
 - Sales Transaction Line

```
Updated Table                                     # of rows inserted
-----
item_inventory                                     60000
sales_transaction                                 60000
sales_transaction_line                           120000
```

	VOWM Simulation Results											
	Tactical Queries		BAM Queries		DSS Queries		Item Inventory Loads		Sales Transaction Loads		Sales Transaction Line Loads	
	Average Response Time	Throughput per/hour	Average Response Time	Throughput per/hour	Average Response Time	Throughput per/hour	Rows Inserted	Inserts / Second	Rows Inserted	Inserts / Second	Rows Inserted	Inserts / Second
FirstConfig	7.89	7280	89.21	68	151.44	7280	60000	33.33333333	60000	33.33333333	120000	66.66666667
Baseline							0	0	0	0	0	0
Test 1							0	0	0	0	0	0
Test 2							0	0	0	0	0	0
Test 3							0	0	0	0	0	0
Test 4							0	0	0	0	0	0
Test 5							0	0	0	0	0	0
Test 6							0	0	0	0	0	0
Test 7							0	0	0	0	0	0
Test 8							0	0	0	0	0	0
Test 9							0	0	0	0	0	0
Test 10							0	0	0	0	0	0

Enter the values into the Rows Inserted column will automatically calculate the Inserts/Second value

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Your instructor will provide you with the *VOWM Simulation Results Spreadsheet*. Prior to the start of class, your instructor ran the Workload Simulation script for your team with the default ruleset, FirstConfig, activated.

In this step you will record the *Rows Inserted* for:

- Item Inventory
- Sales Transaction
- Sales Transaction Line

Note: The Spreadsheet will automatically calculate the Inserts/Second column, and you will record these metrics after each running of the Workload Simulation.



Lab: Record the simulation results

	VOWM Simulation Results											
	Tactical Queries		BAM Queries		DSS Queries		Item Inventory Loads		Sales Transaction Loads		Sales Transaction Line Loads	
	Average Response Time	Throughput per/hour	Average Response Time	Throughput per/hour	Average Response Time	Throughput per/hour	Rows Inserted	Inserts / Second	Rows Inserted	Inserts / Second	Rows Inserted	Inserts / Second
FirstConfig	7.89	7280	69.21	68	151.44	7280	60000	33.33333333	60000	33.33333333	120000	66.66666667
Baseline							0	0	0	0	0	0
Test 1							0	0	0	0	0	0
Test 2							0	0	0	0	0	0
Test 3							0	0	0	0	0	0
Test 4							0	0	0	0	0	0
Test 5							0	0	0	0	0	0
Test 6							0	0	0	0	0	0
Test 7							0	0	0	0	0	0
Test 8							0	0	0	0	0	0
Test 9							0	0	0	0	0	0
Test 10							0	0	0	0	0	0

We will continue to update the **VOWM Simulation Results** spreadsheet for each Workload Simulation that we run.

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We will continue to update the **VOWM Simulation Results** spreadsheet for each Workload Simulation that we run.

Summary

In this module we covered how to:

- Discuss the characteristics and purpose of Viewpoint Workload Designer portlet
- Show, edit, copy and activate Rulesets
- Explain how Workload Designer is used within a mixed workload environment

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Module 4: Monitoring portlets

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Use the Viewpoint Workload Health portlet to monitor workload health as compared to their service level goals
- Use the Viewpoint Workload Monitor portlet to monitor the workload performance and activity
- Use the Viewpoint Dashboard to monitor key metrics related to system health and workload activity

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Topics

- Overview of workload monitoring portlets
- Workload health
- Workload monitor
- Dashboard

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Current topic: Overview of workload monitoring portlets

- **Overview of workload monitoring portlets**
 - **Workload health**
 - **Workload monitor**
 - **Dashboard**
- Workload health
- Workload monitor
- Dashboard

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About workload health and monitor

- The WORKLOAD HEALTH portlet displays **workload health information** and provides Filter and Sort menus allowing the customization of the displayed data:
 - Data in the WORKLOAD HEALTH portlet is **refreshed every minute** to provide near-real-time reporting
 - The WORKLOAD HEALTH portlet displays workloads that:
 - Have **completed** processing according to their **Service Level Goals**
 - Have **missed** their Service Level Goals
 - Are **inactive** or **disabled**
 - Have **no** defined **Service Level Goals**
- The WORKLOAD MONITOR portlet allows you to monitor **workload activity**, management method and session data, and it provides:
 - Multiple **summary and details** views for presenting information
 - A state matrix icon that displays the **current state** of the Analytics Database
 - A choice of data **sampling periods**
 - The ability to **filter workloads** and sort columns

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The WORKLOAD HEALTH portlet displays workload health information and provides Filter and Sort menus allow you to customize the displayed data

Data in the WORKLOAD HEALTH portlet is refreshed every minute to provide near-real-time reporting

The WORKLOAD HEALTH portlet displays workloads that:

- Have completed processing according to their Service Level Goals
- Have missed their Service Level Goals
- Are inactive or disabled
- Have no defined Service Level Goals

The WORKLOAD MONITOR portlet allows you to monitor workload activity, management method, session data

The WORKLOAD MONITOR provides:

- Multiple summary and details views for presenting information
- A state matrix icon that displays the status of the Analytics Database
- A choice of data sampling periods
- The ability to filter workloads and sort columns

About the dashboard

The **DASHBOARD** provides access to the most commonly used information about a system including: **System Health**, **Workloads**, **Queries**, and **Alerts**.

When expanded, the Dashboard initially shows an overview for the selected system. For this at-a-glance system overview, there are **5 main content areas**:

1. **Trend graphs** for key metrics
2. **System Health** metrics that have exceeded thresholds
3. **Workload details** such as the current ruleset, state, and top active workloads
4. **Query details** showing counts of queries in each state and the **top 5 lists for queries** including:
 - Highest **Request CPU**
 - Highest **CPU Skew** Overhead
 - Longest **Duration**
 - Longest **Delayed**
5. **Alert details** showing counts of alerts in each state

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The **DASHBOARD** provides access to the most commonly used information about a system including: System Health, Workloads, Queries, and Alerts.

When expanded, the Dashboard initially shows an overview for the selected system. For this at-a-glance system overview, there are 5 main content areas:

1. Trend graphs for key metrics
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4. Query details showing counts of queries in each state and the top 5 lists for queries including
 - Highest Request CPU
 - Highest CPU Skew Overhead
 - Longest Duration
 - Longest Delayed

Current topic: Workload health

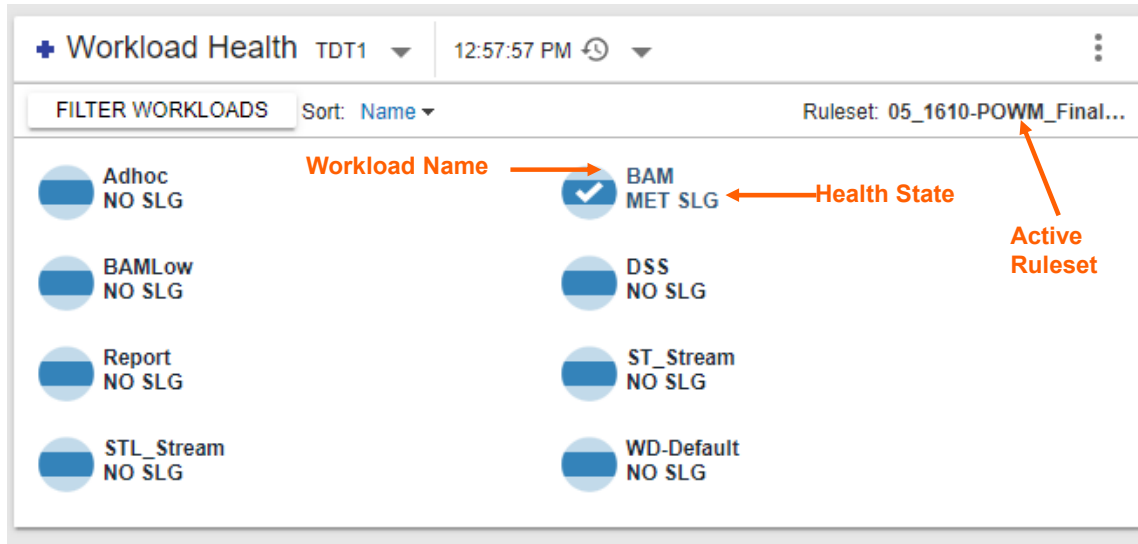
- Overview of workload monitoring portlets
- **Workload health**
 - **The workload health portlet**
 - **Health states**
 - **Filtering which workloads display**
 - **Summary display of workloads**
 - **Workload detailed information display**
- Workload monitor
- Dashboard

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The workload health portlet

The Workload Health portlet displays the **health status** of one or more workloads.



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Use the **WORKLOAD HEALTH** view to display the health status of one or more workloads. Workload health is determined in relation to a Service Level Goal (SLG).

The following list describes the features in this view:

- System name in the portlet frame, color-coded to red if a workload has missed its SLG
- Active ruleset name (the ruleset currently enabled on the Analytics Database)
- Workload names
- Workload health, presented using color, icons, and predefined states
- Workload sort and filter capabilities
- Portlet rewind and share capabilities

Workload health: Health states

Workload health is described using a set of icons and predefined states.

Icon	Text Color	State	Description
	Red	Missed SLG	Workload missed its response-time SLG for a period of time. Time periods for tier/access levels in SLES 11: • Tactical: last minute • SLG Tiers: last 30 minutes • Timeshare: last 60 minutes
	Blue	Met SLG	Workload met its enforcement priority SLGs in SLES 10 or lower, or tier/access level SLGs in SLES 11.
	Blue	No SLG	No response time or percent SLGs have been set for this workload.
	Gray	No Data	No data is available for this workload.
	Gray	Disabled	Workload has been disabled.
	Gray	Inactive	Workload is inactive.

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The facing page describes the various workload health states.

Workload health: Filters

Portlet: **Workload Health** > Button: **Filter Workloads**

The 'Filter Workloads' dialog box has a blue header. Below the header are two tabs: 'GROUPING FILTERS' (selected) and 'NAME FILTERS'. The 'GROUPING FILTERS' tab contains a list of checkboxes for filtering workloads. The list is organized into three sections: 'All Virtual Partitions', 'All Tier Types/Access Levels', and 'All Health States'. Each section has a master checkbox and several sub-items. At the bottom right are 'OK' and 'CANCEL' buttons.

Section	Item	Selected
All Virtual Partitions	All Virtual Partitions	☒
	DataLoads	☒
	Standard	☒
All Tier Types/Access Levels	All Tier Types/Access Levels	☒
	Tactical	☒
	SLG Tier	☒
	Top	☒
	High	☒
	Medium	☒
All Health States	All Health States	☐
	Missed SLG	☒
	Met SLG	☒

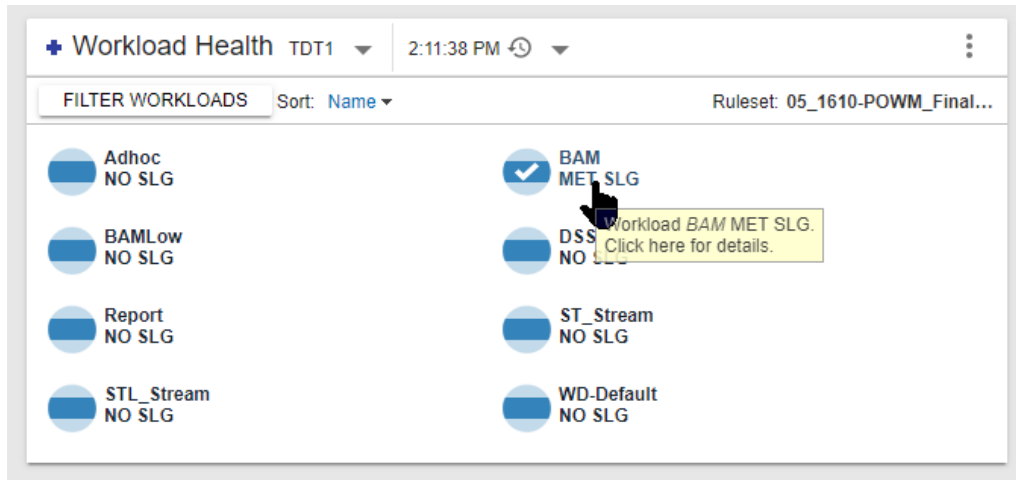
The 'Filter Workloads' dialog box has a blue header. Below the header are two tabs: 'GROUPING FILTERS' and 'NAME FILTERS' (selected). The 'NAME FILTERS' tab contains a section titled 'Deselect workloads to hide:' followed by a list of checkboxes for specific workload names. At the bottom right are 'OK' and 'CANCEL' buttons.

Item	Selected
Mini-Batch	☒
ST_Stream	☒
STL_Stream	☒
Adhoc	☒
BAM	☒
BAMLow	☒
DSS	☒
DSSLow	☒
H-WD	☒
L-WD	☒
M-WD	☒
Report	☐

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From the toolbar, you can choose to apply any filters. You can also choose to sort by workload name.

Workload health: Summary information



Moving the cursor over the Workload will display an **information balloon**
Selecting the Workload will drill down to a detailed metric display

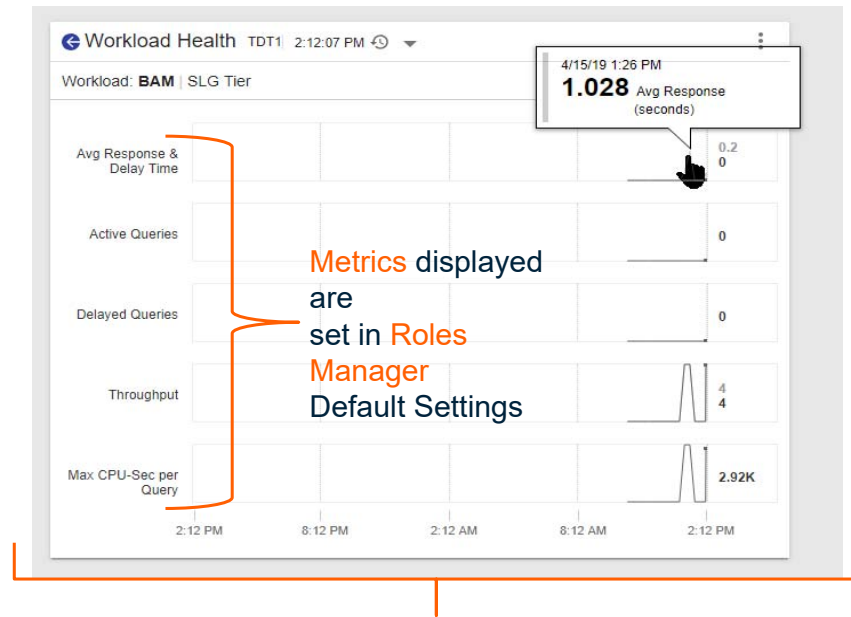
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Moving the cursor over a workload will display detailed information in an information balloon.
Selecting the workload will drill down to a detailed metric display.

Workload health: Detailed display

The **Workload Health details** view displays **metrics** for a **single workload**.

Moving the cursor over the Workload will display an **information balloon**.



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The **Trend Interval** is set in **Roles Manager Default Settings**

The Workload Health details view displays metrics for a single workload. Use the Settings view to select the metrics. This details view appears after you click the workload icon or name for a workload in the Workload Health view. The Workload Health details view is not available for workloads with a health state of NO DATA.

Current topic: Workload monitor

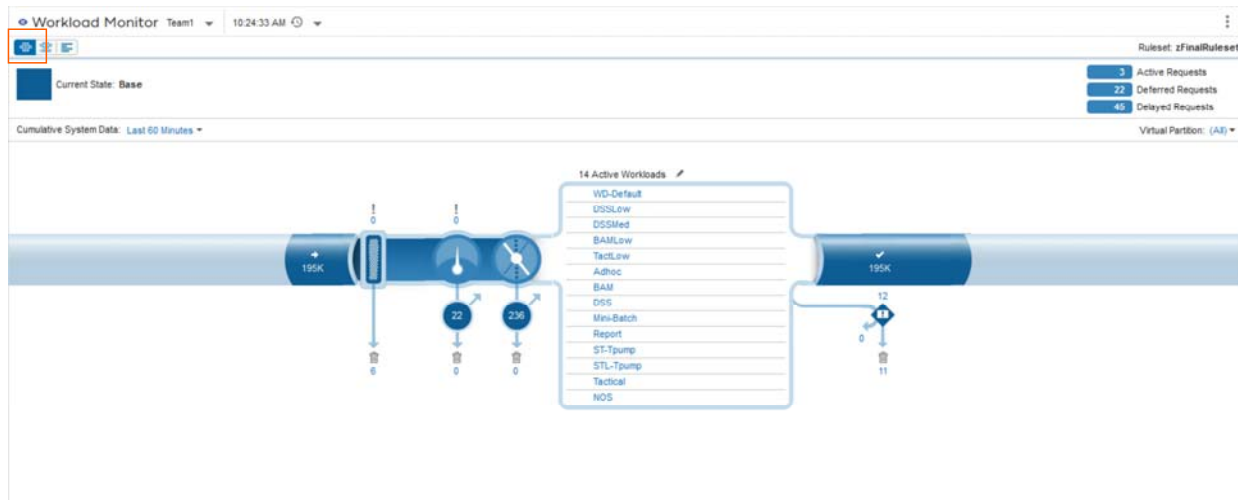
- Overview of workload monitoring portlets
- Workload health
- **Workload monitor**
 - **Dynamic pipe display**
 - **Static pipe display**
 - **Distribution view**
- Dashboard

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Workload monitor: Dynamic pipe display (1 of 2)

Portlet: **Workload Monitor** > Button: **Dynamic Pipes**



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The **WORKLOAD MONITOR** portlet allows you to monitor **detailed Workload activity data**.

The **WORKLOAD MONITOR** portlet allows you to monitor workload activity, allocation group, and session data in the Analytics Database.

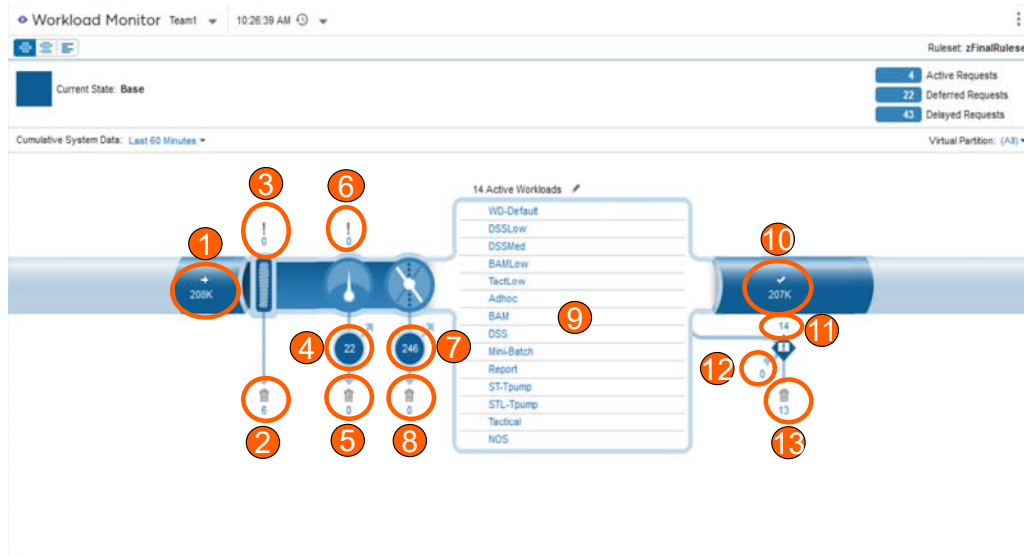
Use the Dynamic Pipes view to analyze workload data in near-real time at each system management point of control. You can choose the data sampling period and workload filter criteria. Workloads can be displayed within their enforcement priority (EP).

The **WORKLOAD MONITOR** provides:

- Multiple summary and details views for presenting information
- A state matrix icon that displays the status of the Analytics Database
- A choice of data sampling periods

The ability to filter workloads and sort columns

Workload monitor: Dynamic pipe display (2 of 2)



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Selecting any of the above areas on the display will drill down to detailed information

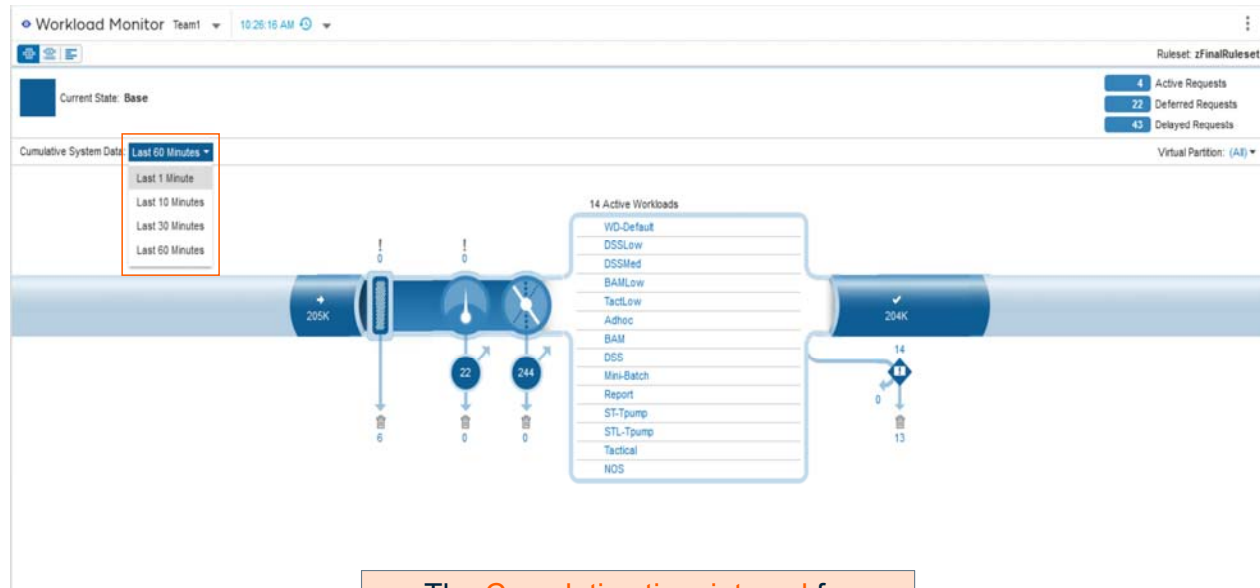
1. Arrivals
2. Filter Rejects
3. Filter Warnings
4. Meter Deferred
5. Meter Rejects
6. Meter Warnings
7. Throttle Delays
8. Throttle Rejects
9. Active Workloads
10. Completions
11. Exceptions
12. Change to WD
13. Aborts

The **WORKLOAD MONITOR** portlet allows you to drill down to detailed information along various points on the pipe display.

You can display information on:

1. Arrivals
2. Warnings
3. Throttle Delays
4. Filter Rejects
5. Throttle Rejects
6. Completions
7. Exceptions
8. Aborts
9. Change to WD

Workload monitor: Time interval

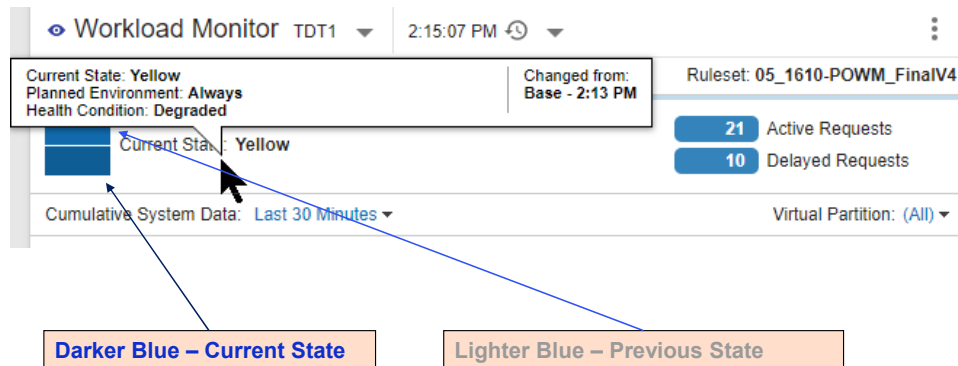


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The Cumulative time interval for reporting system data can be applied.

On the facing page, you can choose to change cumulative interval for system data.

Workload monitor: Current state



Moving the cursor over the State Matrix icon will display an information balloon about the **current state** and the last state change.

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In the pipe flow diagram, the current state of the system will also be displayed. Moving the cursor over the current state, displays details about the current state in an information balloon.

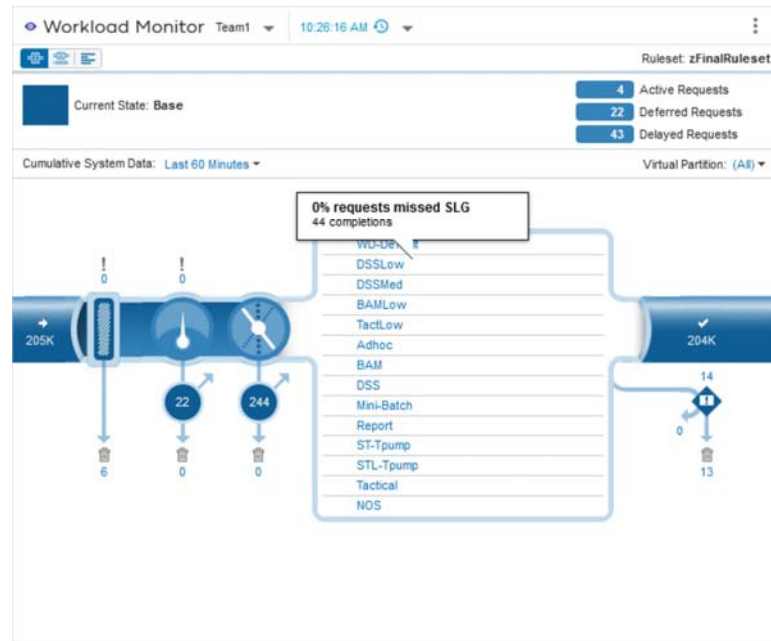
The state matrix icon in the toolbar shows changes in state, planned environment, or health condition during the cumulative sampling period. The state matrix icon uses color to show the following:

- **Dark blue** Active-state cell.
- **Medium blue** Previously active state cell
- **Light blue** Inactive-state cell

Note: During a state change, the cell representing the previous state changes from dark blue to medium blue. If there was a second state change during the sampling period, the previous state cell is shown in light blue. The number of cells in the state matrix icon depends on the state matrix of the monitored system. If a one-by-one state matrix is configured, the state matrix icon appears as one active cell.

Workload monitor: Workload status

Dragging the cursor over the Workload name will Display the **workload status** information balloon.

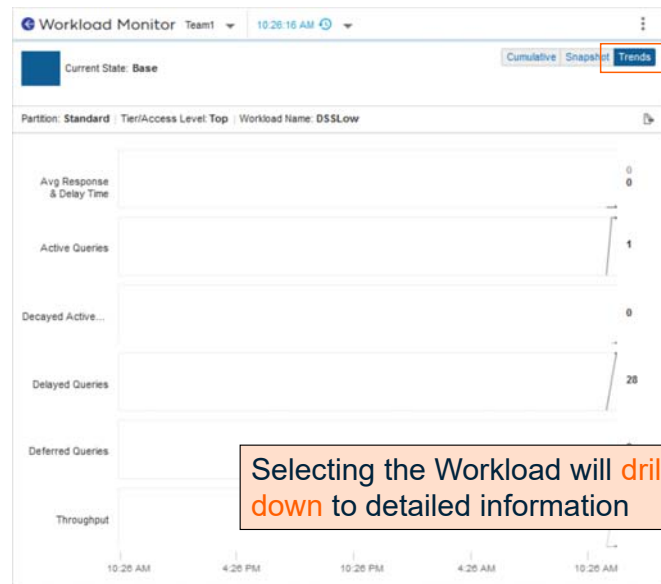
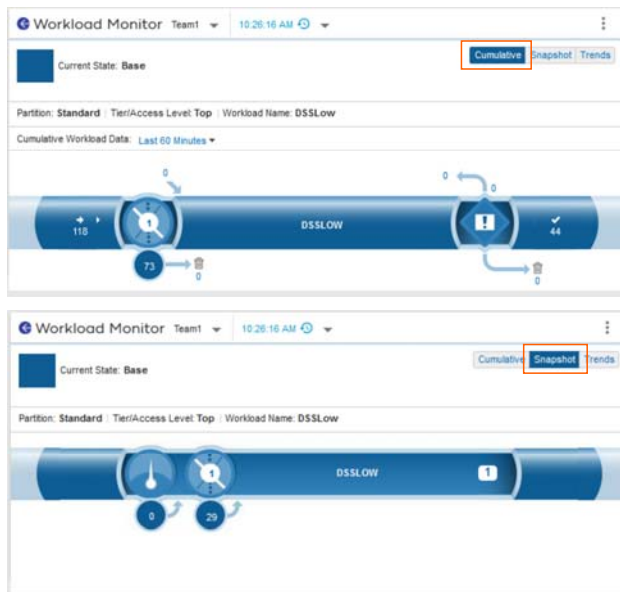


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In the pipe flow diagram, the current status of the workload can be display by moving the cursor to display an information balloon.

Workload monitor: Workload details

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > **Selected Workload**

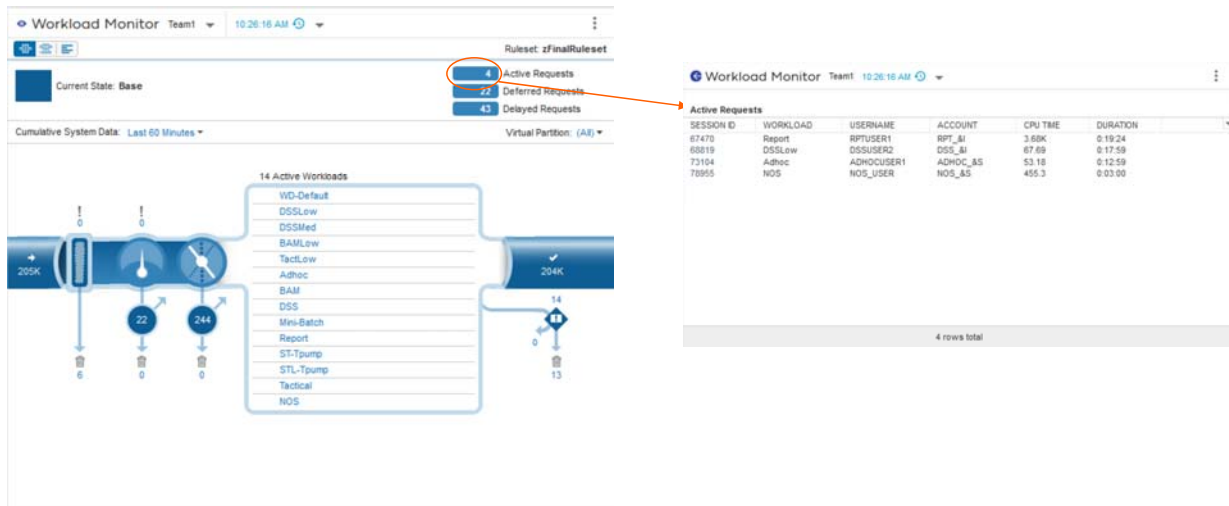


Selecting the Workload will **drill down** to detailed information

In the pipe flow diagram, selecting a workload will drill down on that specific workload and provide different detailed metrics.

Workload monitor: Active requests

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > **Active Requests**



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Click the **Active Requests** box to display a detailed list of the Active Requests

By selecting the active requests, you can drill down into a detailed view.

Workload monitor: Active requests details

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > **Active Requests** > **Session**

The screenshots show the following details for session 30107:

- Overview:** Status: Active, Time in state: 0:01:08, Total duration: 0:04:59, Spool space: 12,521,760, Spool AMP space: 691,264, Spool I/O: 2,772,662, Temp space: 0, Resused CPU: 71.5, Resused I/O: 452,434, Full: 0.18, Unnecessary I/O: 5.56.
- Dynamic Pipes:** Name: DDLJoin, Method: SQL Join, Classification mode: Auto, Virtual partition: Standard, User: DDLUSER1, Account: DDL_A, Partition: DDL_SQL, Requests: 2, Source: DDL_JOIN, DDL_JOIN: 172.32.0.80 TDAW01, 21498 R02-USER DDL_JOIN, 0.91 AMP.
- Explain:** Running step 6 of 12. We do an All-AMPS JOIN step from Spool 8867 (Last Used) to way of an all-row join, which is joined to table SALES_TRANSACTION_LINE_PPI. Spool 8867 and table SALES_TRANSACTION_LINE_PPI are joined using a dynamic hash join. The result goes into Spool 8868, which is built locally on the AMPs. Estimated Time: 0:05:54.733, Estimated Rows: 1372285, Actual Time: 0:00:53.616, Actual Rows: 1372285.
- Query Band:** PROFILE QUERY BAND, SESSION QUERY BAND, TRANSACTION QUERY BAND. QUERYNAME: DDLSQL.

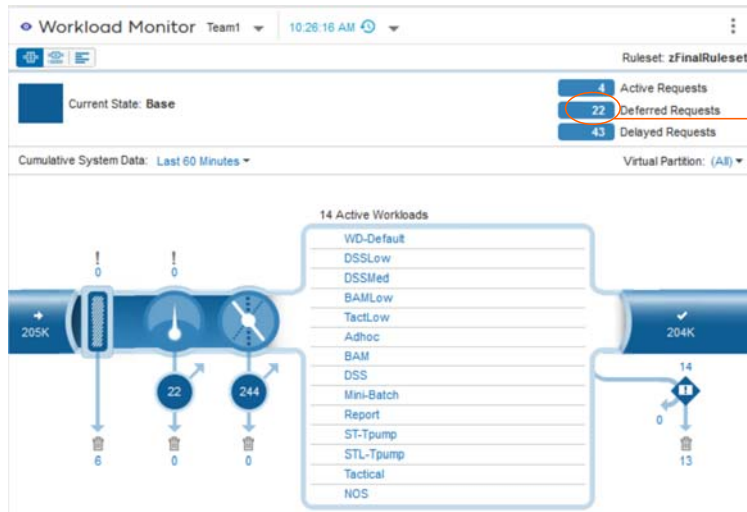
	CPU	AMP#	I/O	AMP#
Highest:	2.26	4	12716	4
2nd Highest:	2.18	3	12304	3
3rd Highest:	2.03	0	11953	0
Average:	0.92	-	5180	-
3rd Lowest:	0	8	0	8
2nd Lowest:	0	9	0	9
Lowest:	0	5	0	5
Session Skew:	59.2%	-	59.3%	-
Participating AMPs:	20			

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Selecting a specific session id will drill down to details regarding that specific session.

Workload monitor: Deferred requests

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > Deferred Requests



Click the **Deferred Requests** box to display a detailed list of the Deferred Requests by Workload and Meter, and Meter Counts

Workload Monitor
Team1
10:26:16 AM

Deferred Requests

BY WORKLOAD
BY METER
METER COUNTS

All currently deferred queries.

SESSION ID	VIRTUAL PART...	WORKLOAD	USERNAME	ACCOUNT	BLOCKING...	DELAY TR
66793	Standard	BAM	TACTUSER2	BAM_AS	0	0:24:51
66795	Standard	BAM	TACTUSER2	BAM_AS	0	0:24:51
66926	Standard	BAM	TACTUSER2	BAM_AS	0	0:10:26
66937	Standard	BAM	TACTUSER2	BAM_AS	0	0:10:26

Workload Monitor
Team1
10:26:16 AM

Deferred Requests

BY WORKLOAD
BY METER
METER COUNTS

All queries included in a Arrival Rate Meter counter. This is all queries, executing or deferred, that are managed by a Arrival Rate Meter.

SESSION ID	USERNAME	THROTTLE NAME	TMR HELD	METER DEF...
75072	TACTUSER2	BAMArm	0:44:50	Deferred
75073	TACTUSER2	BAMArm	0:44:50	Deferred
75074	TACTUSER2	BAMArm	0:44:50	Deferred
75075	TACTUSER2	BAMArm	0:44:50	Deferred
75076	TACTUSER2	BAMArm	0:44:50	Deferred
75077	TACTUSER2	BAMArm	0:44:50	Deferred
75078	TACTUSER2	BAMArm	0:44:50	Deferred
75079	TACTUSER2	BAMArm	0:44:50	Deferred
75080	TACTUSER2	BAMArm	0:44:50	Deferred
75081	TACTUSER2	BAMArm	0:44:50	Deferred
75082	TACTUSER2	BAMArm	0:44:50	Deferred
75083	TACTUSER2	BAMArm	0:44:50	Deferred
75084	TACTUSER2	BAMArm	0:44:50	Deferred
75085	TACTUSER2	BAMArm	0:44:50	Deferred
75086	TACTUSER2	BAMArm	0:44:50	Deferred
75087	TACTUSER2	BAMArm	0:44:50	Deferred
75088	TACTUSER2	BAMArm	0:44:50	Deferred
75089	TACTUSER2	BAMArm	0:44:50	Deferred
75090	TACTUSER2	BAMArm	0:44:50	Deferred
75091	TACTUSER2	BAMArm	0:44:50	Deferred
75092	TACTUSER2	BAMArm	0:44:50	Deferred
75093	TACTUSER2	BAMArm	0:44:50	Deferred
75094	TACTUSER2	BAMArm	0:44:50	Deferred

Workload Monitor
Team1
10:26:16 AM

Deferred Requests

BY WORKLOAD
BY METER
METER COUNTS

Active Arrival Rate Meter and their limits. Deferred query counts include queries that deferred by this Arrival Rate Meter

THROTTLE NAME	ACTIVE QU...	LIMIT	METER DEF...
66990	BAMArm	4	22
66991	BAMArm	4	22
66992	BAMArm	4	22
66993	BAMArm	4	22
66994	BAMArm	4	22
66995	BAMArm	4	22
66996	BAMArm	4	22
66997	BAMArm	4	22
66998	BAMArm	4	22
66999	BAMArm	4	22

By selecting the delayed requests, you can drill down into a detailed view.

Workload monitor: Deferred request details

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > **Deferred Requests** > **Session**

Workload Monitor Team1 10:26:03 AM

Session: 66793

OVERVIEW SQL EXPLAIN SKEW QUERY BAND DEFER

QUERY INFO

State: **Deferred**
Time in state: 0:19:24
Total duration: 0:19:24
Spool space: -
Hot AMP spool: -
Spool skew: -
Temp space: 0
Request CPU: -
Request IO: -
PJI: 0
Unnecessary IO: 0

WORKLOAD INFO

Name: BAM
Method: SLG Tier
Classification mode: Auto
Virtual partition: Standard

SESSION INFO

User: TACTUSER2
Account: BAM_SS
Partition: DBC/SQL
Requests: 2
Request Admission Time: -
Source: (TCP/IP) aa12 127.0.0.1 DBC 13879 EC2-USER BAM_BTQ 0 1 LSS

SNAPSHOT INFO

CPU use: 0%
Impact CPU: 0
Snapshot CPU ... 0%
Snapshot IO s... 0%

Workload Monitor Team1 10:26:03 AM

Session: 66793

OVERVIEW SQL EXPLAIN SKEW QUERY BAND DEFER

STATISTICS

Blocking Count: 0
Defer Time: 0:24:51
User Name: TACTUSER2

RULE INFO

Rule Name: BAMArm
Rule Type: ARM_DELAY_QUERY
Overridable: YES

Notice that for **Deferred Requests** the **SQL**, **EXPLAIN**, and **Query Band** tabs are disabled

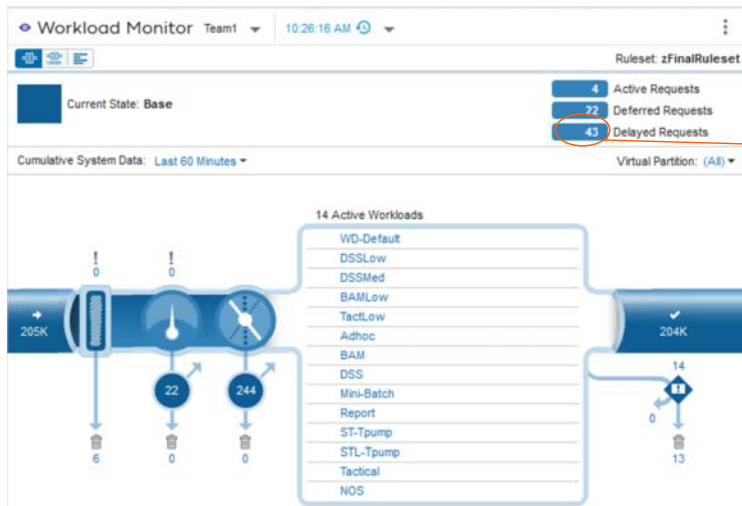
teradata.

Selecting a specific session id will drill down to details regarding that specific session.

From the request details, you can select Defer tab.

Workload monitor: Delayed requests

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > **Delayed Requests**



Click the **Delayed Requests** box to display a detailed list of the Delayed Requests by Workload and Throttle, and Throttle Counts

The screenshot shows the 'Delayed Requests' view for 'Team1' at 5:11:10 PM. It displays a list of delayed requests with columns: SESSION ID, VIRTUAL PART, WORKLOAD, USERNAME, ACCOUNT, BLOCKING, and DELAY TIME. The 'BY THROTTLE' tab is selected. Below the list, a table shows 'Throttle Counts' for various workloads and throttle types.

SESSION ID	VIRTUAL PART	WORKLOAD	USERNAME	ACCOUNT	BLOCKING	DELAY TIME
5539						
5540						
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5539						
5540						
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Workload monitor: Delayed request details

Portlet: **Workload Monitor** > Button: **Dynamic Pipes** > **Delayed Requests** > **Session**

Workload Monitor TDT1 5:05:38 PM 

Session: 31735

OVERVIEW SQL EXPLAIN QUERY BAND DELAY

QUERY INFO		WORKLOAD INFO	
State:	in Delayed	Name:	DSSLow
Time in state:	0:00:59	Method:	SLQ Tier
Total duration:	0:00:59	Classification mode:	Auto
		Virtual partition:	Standard
Spool space:		SESSION INFO	
Hot AMP spool:	-	User:	DSSUSER2
Spool spiler:	-	Account:	DSS_81
Temp space:	0	Partition:	DBCISQL
Request CPU:	-	Requests:	2
Request I/O:	-	Source:	
P/E:	0	(TCP/IP) 951 172.32 0.50 TDWAS	6425 EC2-USER DSS_BTO 0
Unnecessary I/O:	0	1 LSS	
SNAPSHOT INFO			
CPU use:	0%		
Impact CPU:	0		
Snapshot CPU...	0%		
Snapshot I/O s...	0%		


Workload Monitor
TDT1
5:05:36 PM



Session: **31735**

OVERVIEW
SQL
EXPLAIN
QUERY BAND
DELAY

STATISTICS

Blocking Count:	0
Delay Time:	0:01:40
Utility Throttle:	-
System Throttle:	-
Workload Throttle:	DELAY OCCURRED
Workload Group Throttle:	-

RULE INFO

Rule Name:	DSSLow
Rule Type:	WORKLOAD_DELAYED_QUERY
Overridable:	YES

teradata.

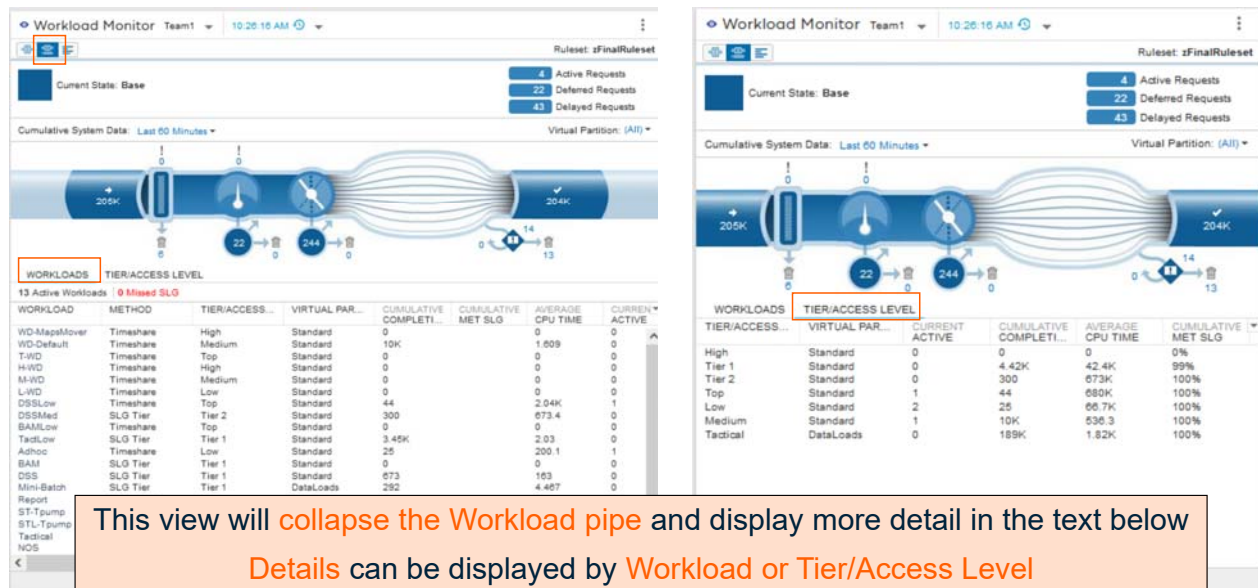
Selecting a specific session id will drill down to details regarding that specific session.

From the request details, you can select Delay tab.

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Workload monitor: Static pipe display

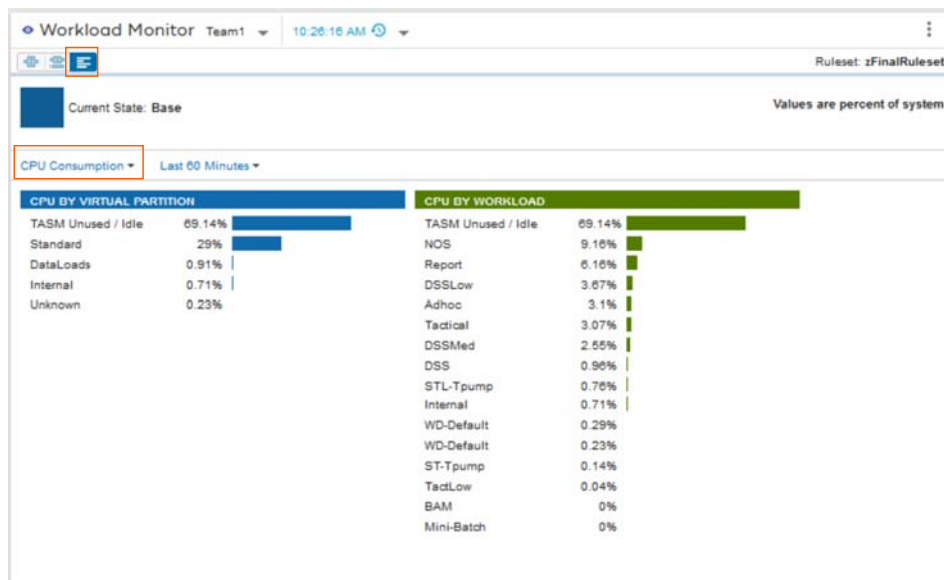
Portlet: **Workload Monitor** > Button: **Static Pipes**



Use the Static Pipes view to compare summary and detail workload metrics. Workloads can also be viewed within their Virtual Partition and data can be sorted by column.

Workload monitor: Distribution view: CPU consumption

Portlet: **Workload Monitor** > Button: **Distribution**

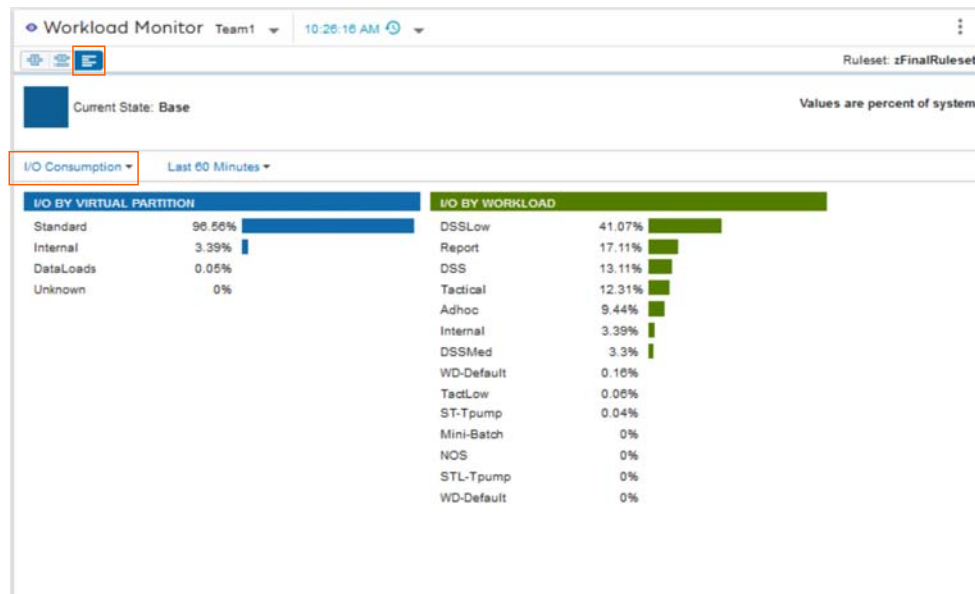


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Use the Distribution view to review Virtual Partition and Workload CPU consumption.

Workload monitor: Distribution view: I/O consumption

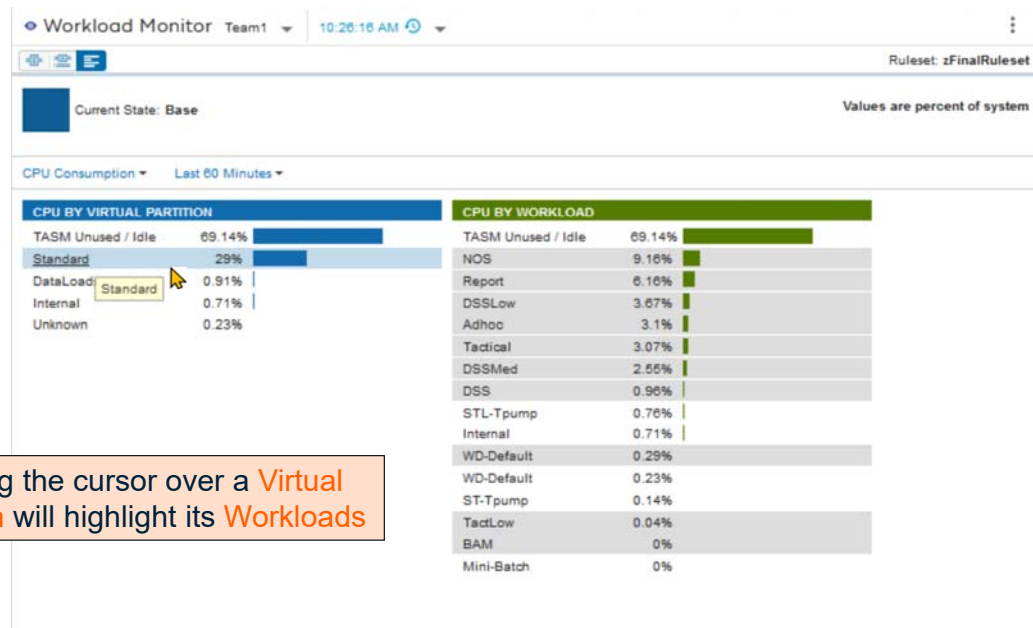
Portlet: **Workload Monitor** > Button: **Distribution**



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Use the Distribution view to review Virtual Partition and Workload I/O consumption.

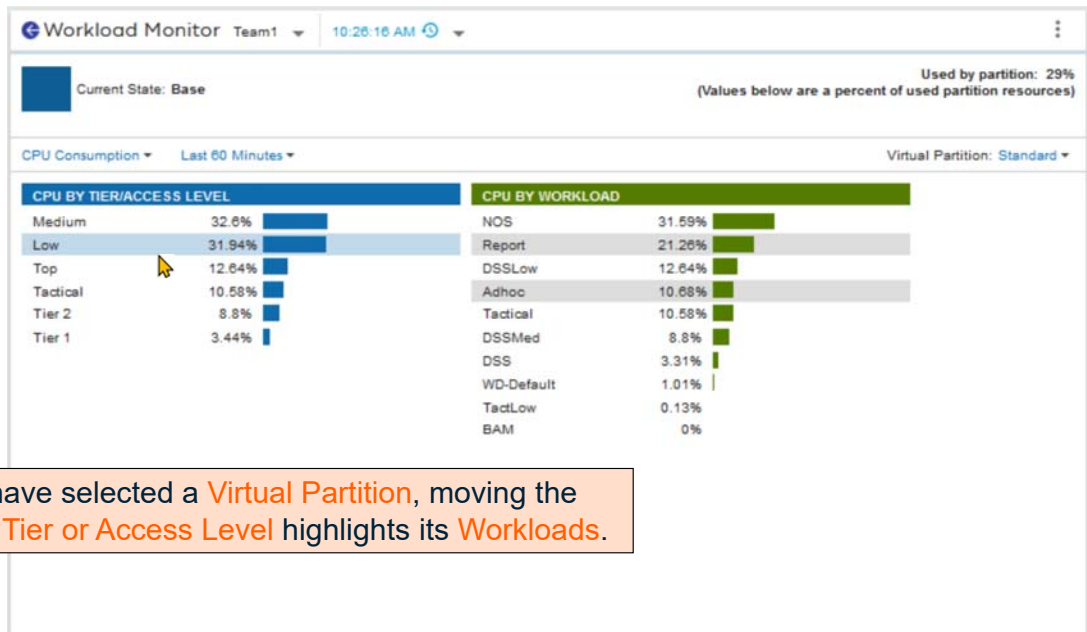
Workload monitor: Distribution highlights (1 of 2)



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The Distribution view displays workload CPU consumption percentages, allowing you to compare the CPU consumption for Virtual Partitions and Workloads assigned to that Virtual Partition.

Workload monitor: Distribution highlights (2 of 2)

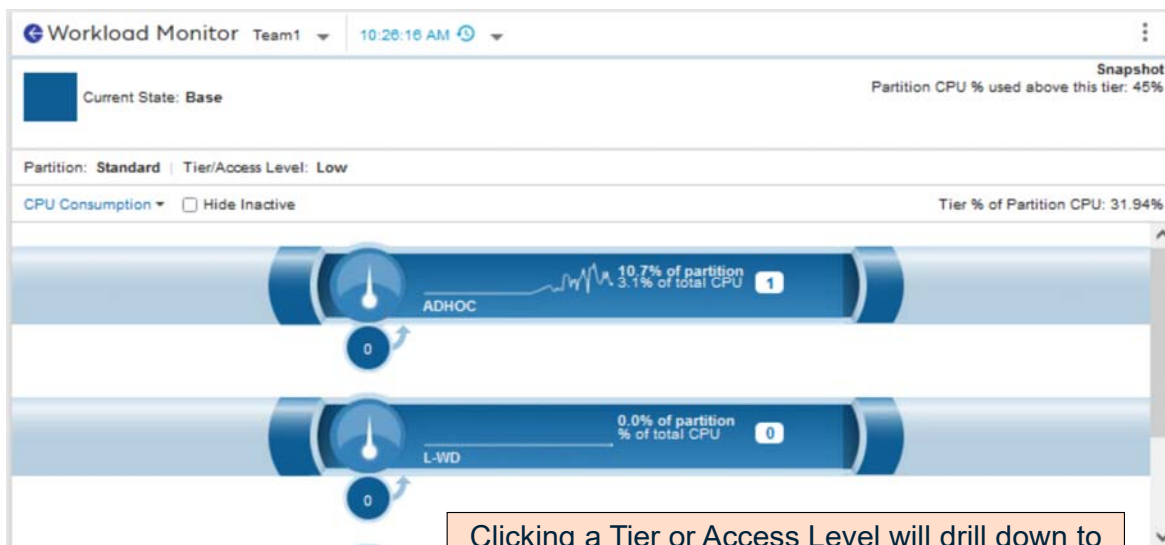


Once you have selected a **Virtual Partition**, moving the cursor over a **Tier or Access Level** highlights its **Workloads**.

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Workload monitor: Distribution details

Portlet: **Workload Monitor** > Button: **Distribution** > : **Selected Workload Method**



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Clicking a Tier or Access Level will drill down to a detail view for that specific Workload Method

Selecting a Virtual Partition with drill down to the Workload Methods and Workloads within that Virtual Partition.

Current topic: Dashboard

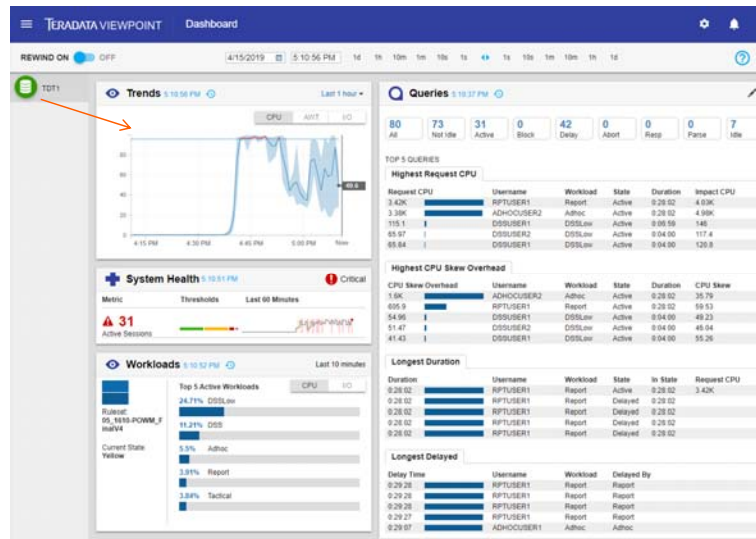
- Overview of workload monitoring portlets
- Workload health
- Workload monitor
- **Dashboard**
 - **The viewpoint dashboard**
 - **Dashboard: System health**
 - **Dashboard: Workloads**
 - **Dashboard: Queries**

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The viewpoint dashboard

The **DASHBOARD** provides access to the most commonly used information about a system including: System Health, Workloads, Queries, and Alerts.

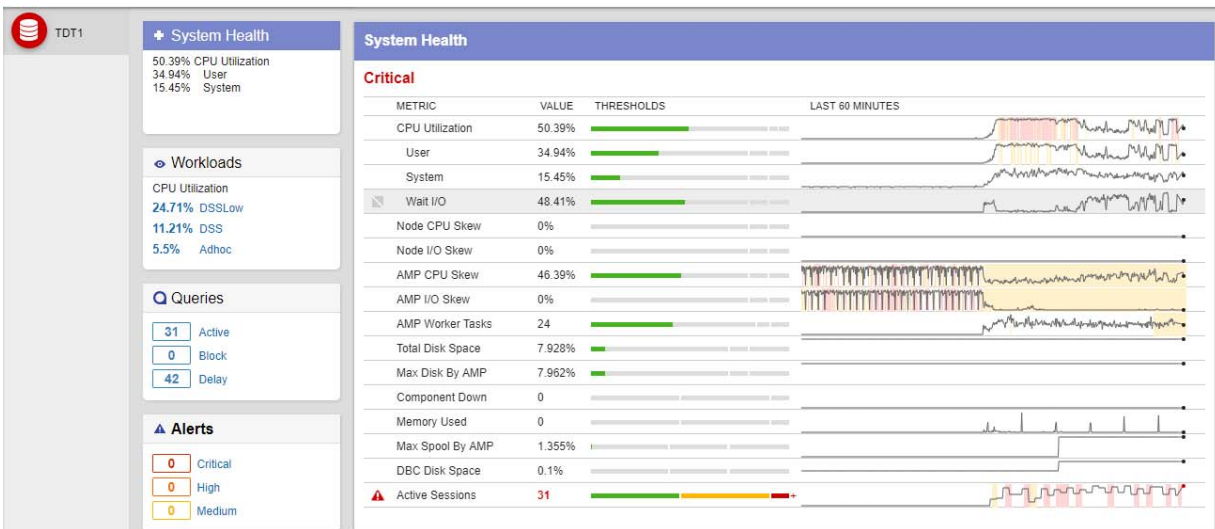


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The DASHBOARD provides access to the most commonly used information about a system including: System Health, Workloads, Queries, and Alerts.

Dashboard: System health

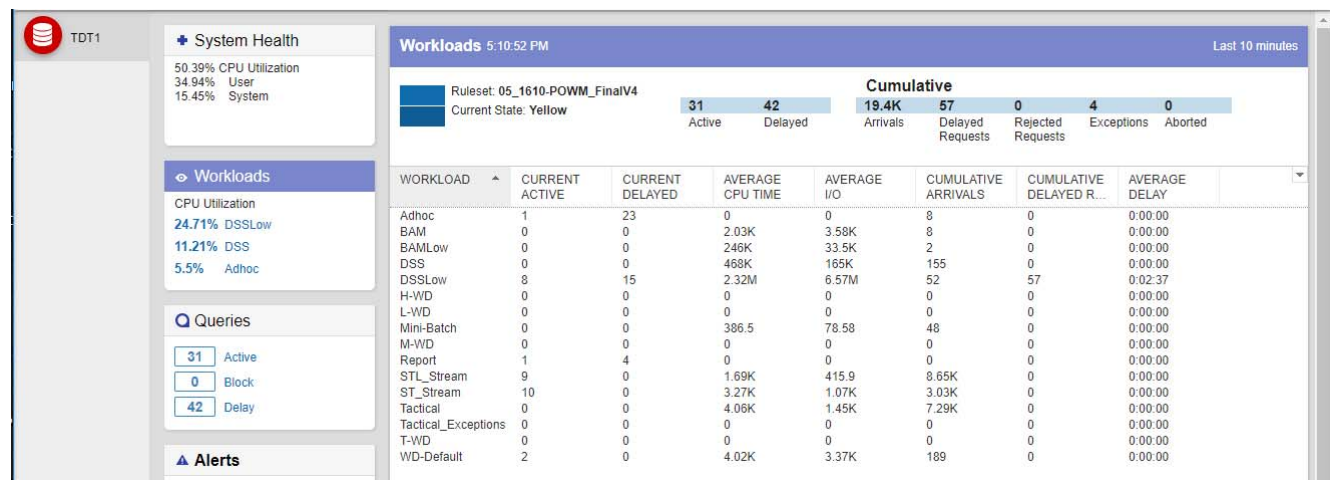
The **System Health** view displays icons to indicate the overall system health for the selected system.



The System Health view displays icons to indicate the overall system health for the selected system.

Dashboard: Workloads

The **Workloads** view allows you to monitor workload management activity in the Analytics Database.

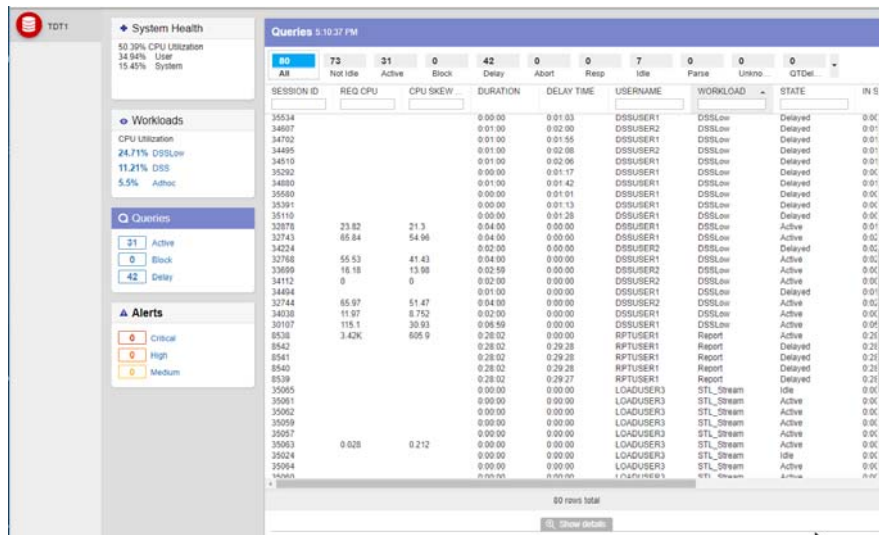


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The Workloads view allows you to monitor workload management activity in the Analytics Database.

Dashboard: Queries

The **Queries** view provides a detailed list of queries by session and/or state. Clicking a on specific session will display its details.



The screenshot displays the Teradata Query Monitor interface. On the left, there's a sidebar with 'System Health' (50.30% CPU Utilization, 34.94% User, 15.45% System), 'Workloads' (CPU Utilization: 24.71% DSSLow, 11.21% DSS, 5.5% Adhoc), and 'Alerts' (0 Critical, 0 High, 0 Medium). The main panel is titled 'Queries 5:10:32 PM' and shows a summary of query counts: 86 All, 73 Not Idle, 31 Active, 0 Block, 42 Delay, 0 Abort, 0 Resp, 7 Idle, 0 Parse, 0 Unkno., 0 GTDel. Below this is a table of queries with columns: SESSION ID, REQ CPU, CPU SKEW, DURATION, DELAY TIME, USERNAME, WORKLOAD, STATE, and IN S. The table lists various queries with their respective session IDs, resource usage, and states. At the bottom, it indicates '80 rows total' and a 'Show details' button.

SESSION ID	REQ CPU	CPU SKEW	DURATION	DELAY TIME	USERNAME	WORKLOAD	STATE	IN S
35534			0:00:00	0:01:03	DSSUSER1	DSSLow	Delayed	0:00
34607			0:01:00	0:02:00	DSSUSER2	DSSLow	Delayed	0:01
34762			0:01:00	0:01:55	DSSUSER1	DSSLow	Delayed	0:01
34495			0:01:00	0:02:08	DSSUSER2	DSSLow	Delayed	0:01
34810			0:01:00	0:02:06	DSSUSER1	DSSLow	Delayed	0:01
35292			0:00:00	0:01:17	DSSUSER1	DSSLow	Delayed	0:00
34880			0:01:00	0:01:42	DSSUSER1	DSSLow	Delayed	0:01
35560			0:00:00	0:01:01	DSSUSER1	DSSLow	Delayed	0:00
35391			0:00:00	0:01:13	DSSUSER1	DSSLow	Delayed	0:00
35110			0:00:00	0:01:28	DSSUSER1	DSSLow	Delayed	0:00
32876	23.82	21.3	0:04:00	0:00:00	DSSUSER1	DSSLow	Active	0:51
32743	65.84	54.06	0:04:00	0:00:00	DSSUSER1	DSSLow	Active	0:00
34224			0:02:00	0:00:00	DSSUSER2	DSSLow	Delayed	0:00
32768	55.53	41.43	0:04:00	0:00:00	DSSUSER1	DSSLow	Active	0:00
33609	16.18	13.08	0:02:59	0:00:00	DSSUSER2	DSSLow	Active	0:00
34112	0	0	0:02:00	0:00:00	DSSUSER2	DSSLow	Active	0:00
34464			0:01:00	0:00:00	DSSUSER1	DSSLow	Delayed	0:01
32744	65.87	51.47	0:04:00	0:00:00	DSSUSER2	DSSLow	Active	0:00
34038	11.97	8.752	0:02:00	0:00:00	DSSUSER1	DSSLow	Active	0:00
30107	115.1	30.93	0:08:59	0:00:00	DSSUSER1	DSSLow	Active	0:00
8538	3.42K	665.9	0:28:02	0:00:00	RPTUSER1	Report	Active	0:21
8542			0:28:02	0:29:28	RPTUSER1	Report	Delayed	0:21
8541			0:28:02	0:29:28	RPTUSER1	Report	Delayed	0:21
8540			0:28:02	0:29:28	RPTUSER1	Report	Delayed	0:21
8539			0:28:02	0:29:27	RPTUSER1	Report	Delayed	0:21
35065			0:00:00	0:00:00	LOADUSER3	STL_Stream	Idle	0:00
35061			0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00
35062			0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00
35059			0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00
35057			0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00
35063	0.028	0.212	0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00
35024			0:00:00	0:00:00	LOADUSER3	STL_Stream	Idle	0:00
35064			0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00
35065			0:00:00	0:00:00	LOADUSER3	STL_Stream	Active	0:00

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The Queries view provides a detailed list of queries by session and/or state. Clicking a on specific session will display its details.

Summary (1 of 2)

- The WORKLOAD HEALTH portlet displays **workload health information** and provides Filter and Sort menus allowing the customization of the displayed data
 - Data in the WORKLOAD HEALTH portlet is **refreshed every minute** to provide near-real-time reporting
 - The WORKLOAD HEALTH portlet displays workloads that:
 - Have **completed** processing according to their **Service Level Goals**
 - Have **missed** their Service Level Goals
 - Are **inactive** or **disabled**
 - Have **no** defined **Service Level Goals**
- The WORKLOAD MONITOR portlet allows you to monitor **workload activity**, management method and session data, and it provides:
 - Multiple **summary and details** views for presenting information
 - A state matrix icon that displays the **current state** of the Analytics Database
 - A choice of data **sampling periods**
 - The ability to **filter workloads** and sort columns

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The WORKLOAD HEALTH portlet displays workload health information and provides Filter and Sort menus allow you to customize the displayed data

The WORKLOAD HEALTH portlet displays workloads that:

- Have completed processing according to their Service Level Goals
- Have missed their Service Level Goals
- Are inactive or disabled
- Have no defined Service Level Goals

The WORKLOAD MONITOR portlet allows you to monitor workload activity, Management Method and session data

The WORKLOAD MONITOR provides:

- Multiple summary and details views for presenting information
- A state matrix icon that displays the status of the Analytics Database
- A choice of data sampling periods
- The ability to filter workloads and sort columns

Summary (2 of 2)

The **DASHBOARD** provides access to the most commonly used information about a system including: **System Health**, **Workloads**, **Queries**, and **Alerts**

When expanded, the Dashboard initially shows an overview for the selected system. For this at-a-glance system overview, there are **5 main content areas**:

1. **Trend graphs** for key metrics
2. **System Health** metrics that have exceeded thresholds
3. **Workload details** such as the current ruleset, state, and top active workloads
4. **Query details** showing counts of queries in each state and the **top 5 lists for queries** including
 - Highest **Request CPU**
 - Highest **CPU Skew** Overhead
 - Longest **Duration**
 - Longest **Delayed**
5. **Alert details** showing counts of alerts in each state

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The **DASHBOARD** provides access to the most commonly used information about a system including: System Health, Workloads, Queries, and Alerts.

When expanded, the Dashboard initially shows an overview for the selected system. For this at-a-glance system overview, there are 5 main content areas:

1. Trend graphs for key metrics
2. System Health metrics that have exceeded thresholds
3. Workload details such as the current ruleset, state, and top active workloads
4. Query details showing counts of queries in each state and the top 5 lists for queries including
 - Highest Request CPU
 - Highest CPU Skew Overhead
 - Longest Duration
 - Longest Delayed
5. Alert details showing counts of alerts in each state

Module 5: Workload Designer: General settings

Teradata Vantage optimizing the
Analytics Database through Workload
Management

teradata.

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Objectives

After completing this module, you will be able to:

- Describe how to establish Workload Designer general settings
- Describe the concept of a Bypass user
- Identify the system-wide parameters and options available in Workload Designer's General tab
- Describe the concept of Amp Work Tasks
- Explain what causes a system to go into a state of Flow Control

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Topics

- General settings: General, bypass, limits/reserves
- General settings: Other settings
- AMP worker tasks

teradata.



Current topic: General settings: General, bypass, limits/reserves

- **General settings: General, bypass, limits/reserves**
 - **General button: General tab**
 - **General button: Bypass tab**
 - **General button: Limits/reserves tab**
- General settings: Other settings
- AMP worker tasks

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General button: General tab

Portlet: **Workload Designer** > Button: **General** > Tab: **General**

The screenshot shows the 'Workload Designer' interface. At the top, there's a header with 'Team1' and 'FirstConfig'. Below this is a toolbar with icons for 'General', 'States', 'Sessions', 'Filters', 'Meters', 'Throttles', 'Workloads', and 'Exceptions'. The 'General' tab is selected. Underneath the toolbar, there are sub-tabs: 'GENERAL', 'BYPASS', 'LIMITS/RESERVES', and 'OTHER'. The 'GENERAL' sub-tab is active. It contains two input fields: 'Name: *' with the value 'FirstConfig' and a character limit of 'Up to 30 Characters'; and 'Description:' with a character limit of 'Optionally, up to 80 Characters'.

On the General Tab, enter the ruleset **Name** and **Description**

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A ruleset is a complete collection of related filters, throttles, events, states, and workload rules. You can create multiple rulesets, but only one ruleset can be active on the production server. After creating a ruleset, you can specify settings, such as states, sessions, and workloads, using the toolbar buttons. New rulesets are automatically locked so only the owner can edit the ruleset.

1. Specify a ruleset name, up to 30 characters.
2. [Optional] Enter a description up to 80 characters.
3. Click **Save**.

General button: Bypass tab

Portlet: **Workload Designer** > Button: **General** > Tab: **Bypass**

The screenshot shows the 'Workload Designer' application with the 'BYPASS' tab selected. The interface includes a top navigation bar with tabs: General, States, Sessions, Filters, Meters, Throttles, Workloads, and Exceptions. Below this, the 'BYPASS' tab is active, showing a list of items on the left and a list of items to bypass on the right. The 'Source Type' is set to 'Profile'. The 'Filter' dropdown is set to 'Username'. The 'Items' list on the left includes: ADHOC_Prof, BAM_Prof, DSS_Prof, Load_Prof, NOS_Prof, Report_Prof, Stream2_Prof, TDMAPS_USER, and Tactical_Prof. The 'These will bypass' list on the right includes: Usernames (with sub-items DBC and tdwm), Account Names, Account Strings, and Profiles. At the bottom, there are 'SAVE' and 'RESET' buttons.

On the Bypass tab, choose WHO will bypass **System Filter** and **Throttle** rules. (By Default, **DBC** and **tdwm** are always bypass users)

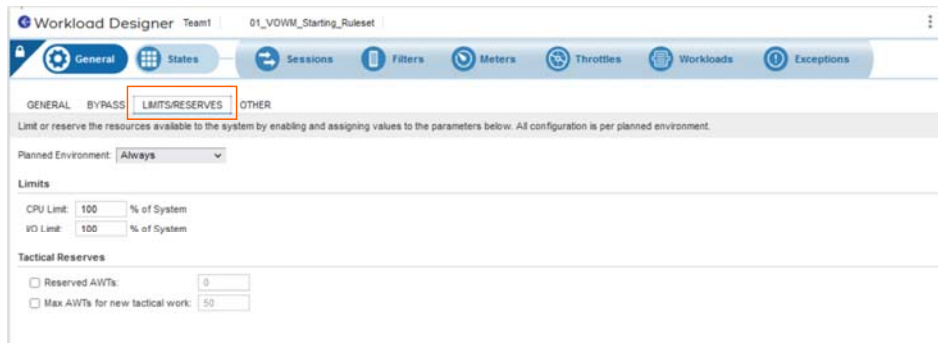
Note: Recommend to make the **Viewpoint** data collector user a bypass user

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Within the Bypass tab, you can designate particular users, accounts and profiles that should be exempted from Workload Management filtering and throttling at the system level. For example, you may designate a special administrative user bypass privileges so that the DBA/User can always access the system for immediate troubleshooting purposes. Note that Bypass does NOT exempt requests from being managed by the Workload it is classified into, including Workload Throttling.

General button: Limits/reserves tab

Portlet: **Workload Designer** > Button: **General** > Tab: **Limits/Reserves**



Workload Designer Team1 01_VOWM_Starting_Ruleset

General BYPASS **LIMITS/RESERVES** OTHER

Limit or reserve the resources available to the system by enabling and assigning values to the parameters below. All configuration is per planned environment.

Planned Environment: Always

Limits

CPU Limit: 100 % of System

I/O Limit: 100 % of System

Tactical Reserves

☐ Reserved AWTs: 0

☐ Max AWTs for new tactical work: 50

For each **Planned Environment**, choose to **enable CPU** and/or **I/O limits**

In addition, choose the number of **AWTs** that will be **reserved** for **expedited workloads**

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Capacity on Demand (COD) is implemented for both CPU and I/O, but in different ways. Because the SLES 11 operating system scheduler controls CPU, the CPU type of COD can rely on already-existing operating system structures and services. Since I/O is not managed by the SLES 11 operating system scheduler, I/O COD is by necessity handled differently

On SLES 11, strict limits on CPU consumption will only be offered at the Analytics Database level, for Capacity on Demand (COD) purposes. Capping the CPU at the Virtual Partition or the Workload level will not be available in the first release of the Linux SLES 11 operating system.

When a COD CPU limit has been defined, for example at 80%, will effectively take away resources from the Tdat Control Group. If a hard limit of 80% is applied to Tdat, all of the resources consumed under Tdat will be limited to 80% of the CPU that comes down from root.

Note that this 80% hard limit is only applied to the work that is running below Tdat, work being done on behalf of activity within the Analytics Database. Operating system utilities or other work running on the node external to the Analytics Database will get their resources at a higher level in the hierarchy, and this 80% limit will not be able to manage them.

I/O Capacity on Demand is applied at the disk level, using platform metering, a hardware option that has been available on Teradata hardware platforms since early 2010. The platform metering approach to COD is based on limiting I/O throughput to some specific number of MB/second, in the firmware itself. This limit does not vary based on whether the I/O is a read or a write, and it can be defined as low as 1% increments.

I/O COD is neither integrated with, nor is it a part of the new I/O prioritization infrastructure. While I/O prioritization adds a software level between the disk and the database, I/O COD is implemented completely within the disk hardware subsystem, with no interaction with the database. I/O COD only affects the drives where data in the Analytics Database is stored. The

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I/O limit does not affect the root drives of the system, or any devices that are not part of the Analytics Database. Because of that, the I/O COD limit is similar in scope to the CPU COD limit.

In addition, you can specify a number of AWTs that will be reserved for workloads assigned to the Tactical Workload Method.

Current topic: General settings: Other settings

- General settings: General, bypass, limits/reserves
- **General settings: Other settings**
 - **General button: Other tab**
 - **Intervals**
 - **Logging tables**
 - **Blocker**
 - **Workload priority**
 - **Utility limits**
 - **Additional settings**
- AMP worker tasks

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General settings: Other tab

The screenshot shows the 'OTHER' tab in the Teradata Workload Designer. The tab is highlighted with a red box. The settings are organized into several sections:

- Intervals:** A group of five settings, each with a dropdown menu set to '60' and the unit 'seconds':
 - Event Interval
 - Dashboard Interval
 - Logging Interval
 - Exception Interval
 - File Throttle Action Interval
- Blocker:** Two settings:
 - Block Cycles: A dropdown menu set to 'Off'.
 - Block Action: A dropdown menu set to 'Log'.
- Activation:** A section with the instruction 'Please select the features that will be enabled when this rule set is activated:' and two checked checkboxes:
 - Filters and Utility Sessions
 - System Throttles and Session Control
- Timeshare Decay:** A section with the instruction 'Timeshare decay automatically decreases the resource access rate of queries in a Timeshare workload over time.' and one unchecked checkbox:
 - Enable Timeshare Decay
- Prevent Mid-Transaction Throttle Delays:** A section with one unchecked checkbox:
 - Do not delay queries within transactions that are holding locks higher than 'acquire'
- Order the Throttle Delay Queue:** A section with two radio buttons, the first of which is selected:
 - By time delayed
 - By workload priority
- Utility Limits:** A section with one unchecked checkbox:
 - Support increased MLODCK job limits and increased AWT resource limits
- Define 'available AWTs' as:** A section with two radio buttons, the first of which is selected:
 - AWTs available for the Workload (WORKED) work type. The Workload (WORKED) work type is limited by default to 50 AWTs. If 50 AWTs are already in-use servicing Workload message types, there may still be AWTs in the unreserved pool that under this definition will not be considered available.
 - AWTs available in the unreserved pool for use by any work type. The number of AWTs available in the unreserved pool able to be used by all work types, not limited to Workload work types.

At the bottom right of the form, there are two buttons: 'SAVE' and 'RESET'.

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The other tab consolidates settings for Intervals, Blocker, Activate, Timeshare Decay and Prevent Mid-Transaction Throttle Delays.

Other tab: Intervals

Used to set intervals for workload management activities

- **Event Interval** specifies how often **event occurrences** are checked. Can be set at 5, 10, 30 or 60 second intervals
- **Dashboard Interval** specifies how often **workload statistics** are collected. Can be set from 60 to 600 seconds. Recommend to set at 60 seconds to **sync with the Workload Monitor** refresh interval
- **Logging Interval** specifies how often **workload and exception logs** are written from cache to disk. Note, if cache fills up sooner, it will be flushed to disk before logging interval is reached
- **Exception Interval** specifies how often **asynchronous exception** thresholds are checked, reasonable default interval being 60 seconds
- **Flex Throttle Action Interval** specifies how often the availability of system resources is checked. Must be a multiple of the Event Interval, only supported in Teradata Database 16.0 and later for SLES 11 EDW systems

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Intervals is used to define intervals for certain workload management activities.

Event Interval

The event interval is the interval of time between asynchronous checks for event occurrences. It can be set to 5, 10, 30 or 60 seconds.

Dashboard Data Interval

On an ongoing basis, Workload Management accumulates a variety of data about each workload that occurs within the interval of time specified by the dashboard data interval. The data is available for both short-term, real-time display via the Workload Monitor Portlet and for historical data mining from its long-term repository, TDWMSummaryLog. It is additionally used by the TDWMExceptions API to determine the amount of exception data to provide in its response to the end-user who calls it.

This workload data contains counts of arrivals, completions, delays, exceptions for each workload within each dashboard data interval. It collects average response time, CPU and IO usage consumption by workload and a running count of queries that meet their SLGs. It is recommended to set this interval to 60 seconds in sync with the default refresh interval of the Workload Monitor Portlet.

Logging Interval

A variety of historical log tables, including the TDWMSummaryLog data discussed above, are first stored in internal caches before physically writing the data permanently to the tables on disk. This technique assures low logging overhead. The logging interval is used to tell Workload Management how often to flush these accumulations from memory to disk.

The various historical log tables that are flushed on the logging interval include the TDWMSummaryLog, the DBQL detail log table (DBQLLogTbl), TDWMExceptionLog,

TDWMEventLog and TDWMEventHistoryLog information.

Note if any of these log caches fill up before the logging interval expires, it will flush to disk before the logging interval is reached. This simply specifies the 'maximum' time that will pass before this data is available on disk in the historical logs.

Exception Interval

The exception interval is the interval of time between asynchronous checking for exceptions. It can range from 1 to 3600 seconds, with 60 seconds being a reasonable default interval for identifying an exception within a long-running step.

Note: This does not impact the exception checks done at the end of each query step, but the exception checks done periodically during the course of the request when the request duration exceeds the exception checking interval.

Logging interval relationships

There is a relationship between the Event, Dashboard and Logging Intervals

- For event management to function correctly, Workload Designer will enforce that the **Event Interval** $\leq$ **Dashboard Interval** $\leq$ **Logging Interval**
- Assume the following interval settings:
 - Event Interval = 30
 - Dashboard Interval = 60
 - Logging Interval = 600
- Every 30 seconds, the collected **workload summary data** is moved to a “**completed cache**” and a new accumulation begins for the next interval
- Every 60 seconds, the 2 30-second event collections are rolled into a single dashboard “cache” for the **TDWMSummary API** usage, such as the **Workload Monitor portlet**, and rolled into the logging area
- Every 600 seconds, 10 60-second rolled up **dashboard collections** are written to a single row on disk for each active **workload**

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The dashboard interval determines the interval of time for the workload summary data accumulation. This accumulation is used by the Viewpoint Workload Health Portlet and other users of the TDWMSummary API. This same data is ultimately captured and written to the TDWMSummaryLog based on the logging interval. It is also needed for Event Detection at the frequency specified by the event interval. Therefore, workload summary data collection is managed by State/Event Management.

When the event interval expires, Event Management collects the information it needs from various sources, including workload summary data. It additionally saves the workload summary data for both the API and Logging function usage. For Event Management to function correctly, Workload Designer enforces the dashboard interval to be a multiple of the event interval and the logging interval to be a multiple of the dashboard interval.

An example is used to explain this relationship more clearly. Consider the following environment:

Event Interval = 30. Dashboard Interval = 60. Logging Interval = 600.

Every 30 seconds the collected workload summary data is moved into a “completed” cache and a new workload summary accumulation begins for the next event interval. When the dashboard interval expires at 60 seconds, there are 2 30-second collections rolled up into a single dashboard “cache” for the TDWMSummary API usage. The data is also moved into an area for eventual logging. Event management continues to collect data every 30 seconds and share with the dashboard area. Every 60 seconds the dashboard data is rolled up to the logging area. When the logging interval expires at 600 seconds, the 10 60-second dashboard collections that have been rolled up are written to a single row on disk for each active workload during the logging interval.

Logging tables

Workload Management writes rows to the following logs:

- **TDWMExceptionLog** – writes a row for each **exception** or **rejection** per request
- **TDWMEventLog** – writes a row for something of note that occurs not related to a request
- **TDWMSummaryLog** – writes a row for each **active workload** during the given logging period
- **TDWMEventHistory** – writes a row for each **activation or deactivation** of an event, event combination, health condition, planned environment, or state

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Workload Management writes rows to the following logs:

- TDWMExceptionLog – writes a row for each exception or rejection per request.
- TDWMEventLog – writes a row for something of note that occurs not related to a request.
- TDWMSummaryLog – writes a row for each active workload during the given logging period.
- TDWMEventHistory – writes a row for each activation or deactivation of an event, event combination, health condition, planned environment, or state.

Summary data is available to the Workload Monitor and Workload Health portlets. The types of data available include:

- Arrival Rate
- Response Time
- CPU Time
- Query Counts: Active (Concurrent), completed, failed due to error, rejected, delayed, encountered exceptions, and met SLGs.

Multi-Request transaction example

BT;

Request1; (This request obtains lock on TableA and completes)

Request2; (This request obtains locks on TableB and completes)

Request3; (TASM throttles this request and puts it in the Delay queue; locks on Tables are held)

ET;

Request4; (This request is blocked from accessing Tables A & B by requests that are no running; so the AMPS can't resolve the deadlock)

Note: TASM applies its rules to requests

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Before we discuss the Blocker option, here's an example of a Multi-Request Transaction.

Other tab: Blocker

Blocker is used to set Workload Management deadlock detection processing criteria for handling deadlock situations involving delayed queries.

Only applies to a **Multi-Request Transactions**, not a Multi-Statement Requests.

Blocker

Block Cycles: Off ▼

Block Action: Off ▼

Activation

Please select the features that will be

Blocker

Block Cycles: Off ▼

Block Action: Log ▼

Activation

Please select the features that will be enab

- **Block Cycles** specifies the number of deadlock detection cycles (exception interval) in which a query on the delay queue is identified as “blocker” of already executing queries before an action is taken on the delayed query. Valid values are Off or 1-3, with Off indicating no deadlock detection.
- **Block Action** specifies what kind of action to take on the delayed query. Choices are “Log”, “Abort” or “Release”. Log will always occur unless cycles is Off. Release is not an option for queries with throttle limits of 0.

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Blocker function allows Workload Management to take automatic action when a delayed query is identified as a “blocker” of running queries. Default is Off.

The Blocker function is used to specify the number of block detection cycles to execute before taking action on the delayed query causing the Workload Management deadlock situation. Valid values are zero through three. Zero indicates that no deadlock detection is used.

The Deadlock Action parameter indicates what kind of action to take on the delayed query after the required number of detection cycles. Options are "Log," "Abort," and "Release." Aborted or released queries are also logged. However, queries cannot be released if their throttle limit is zero.

It is recommended that you set this control to a value other than zero, and that you set **Deadlock Action** to *Release*. This gives the system a good chance to resolve the block in a normal manner first. After that time, if the blocking request is released, a lock needed by any other requests is freed.

The only downside is that the concurrency limits are a little softer (for example, if you set the concurrency limit to five, the system may occasionally run six or more queries). In addition, an exception action that moves a request to a workload with a concurrency limit defined could result in momentarily exceeding that new workload concurrency limit. Consider monitoring concurrency levels with dashboard or trend reporting for softness.

Note: You define the deadlock checking interval via the **Exception Interval** on **Intervals**.

Other tab: Other settings (1 of 2)

Activation

- Choose to **enable** Filters and Utility Sessions rules and System Throttles and Session Controls rules when the ruleset is activated

Activation

Please select the features that will be enabled when this rule set is activated:

- ☒ Filters and Utility Sessions
- ☒ System Throttles and Session Control

Timeshare Decay

- Enable **Timeshare Decay** option to decay queries automatically after predefined CPU or I/O thresholds are exceeded

Timeshare Decay

Timeshare decay automatically decreases the resource access rate of queries in a Timeshare workload over time.

- ☐ Enable Timeshare Decay

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Activation categories available when the ruleset is activated - **Filters and Utility Sessions** and **System Throttles and Session Control**.

Timeshare Decay option is available that will automatically apply a decay mechanism to Timeshare Workloads. This decay option is intended to give priority to shorter requests over longer requests. Only requests running in Timeshare will be impacted by this option. Decay is off by default.

If this option is turned on, the decay mechanism will automatically reduce the Access Rate of a running request, if the request uses a specified threshold of either CPU or I/O. Initially, the request is reduced down to an Access Level that is $\frac{1}{2}$ the original Access Level. If a second threshold is reached, the request will be further reduced to an Access Level that is $\frac{1}{4}$ the original Access Level. This process of Access Rate reduction includes the Low Access Level, and means that the Access Rate could be as low as 0.25 (Low typically has an Access Rate of 1) for some requests running in Low.

Characteristics of the decay process include:

- A single request will only ever undergo two decay actions, each resulting in a reduction of the request's Access Rate
- Decay decisions are made at the node level, not the system level
- There is no synchronization of the decay action between nodes, so it is possible that a Timeshare request on one node has decayed, but the same request on another node has not
- Decayed requests are not moved to a different workload, the way a workload exception might behave
- Once decay has taken place for a given request, both its access to CPU and to I/O will be reduced, not just the resource whose threshold was exceeded

Decay may be a consideration in cases where there are very short requests mixed into very long requests in a single Workload, and there is a desire to reduce the priority of the long-running queries. Keep in mind, however, that if decay is on, all queries in all Workloads across all Access Levels in Timeshare will be candidates for being decayed if the decay thresholds are met.

Workload classification based on estimated processing time may be effective without relying on the decay option for ensuring that queries expected to be short-running run at a higher Access Level, and queries that are expected to be long-running classify to a Workload in a lower Access Level.

Other tab: Other settings (2 of 2)

Prevent Mid-Transaction Throttle Delays

- Choose to (Throttle Bypass) for any queries within a Multi-Request Transaction to prevent blocking of active requests. This effectively makes the Blocker setting unnecessary

Prevent Mid-Transaction Throttle Delays

☐ Do not delay queries within transactions that are holding locks higher than 'access'

Order the Throttle Delay Queue

- Choose to order the Delay Queue by start time or by Workload priority

Order the Throttle Delay Queue

- ☒ By time delayed
☐ By workload priority

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Prevent Mid-Transaction Throttle Delays option prevents any queries within a Multi-Request Transaction from being delayed and to prevent blocking of active requests

Order the Throttle Delay Queue option gives the ability to order the delay queue from the default of time ordered to workload priority

By time delayed

The longer a query has been delayed, the sooner it will be executed.

By workload priority

The higher the workload priority of a query, the sooner it will be executed.

Workload priority order

Workloads can be ordered in the Delay Queue by Priority value using the following Workload Priority formulas.

Workload Method	Priority Value
Tactical	10000 + Virtual Partition allocation
SLG Tier 1	9000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 2	8000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 3	7000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 4	6000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 5	5000 + Virtual Partition allocation + SLG Tier allocation
Timeshare Top	4000 + Virtual Partition allocation
Timeshare High	3000 + Virtual Partition allocation
Timeshare Medium	2000 + Virtual Partition allocation
Timeshare Low	1000 + Virtual Partition allocation

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When workloads are ordered by priority, they are ordered based on the workload management method assigned to the workload. A priority value is calculated for each workload using the formulas in Table on the facing slide.

Workloads are ordered from high to low based on the priority value. Workload Management uses these formulas when assigning the session WD and when ordering the delay queue by priority.

Other tab: Utility limits

This option allows you:

- Support an increase of TPT Update jobs utilizing the **Extended Multiload** protocol from **30** to a maximum of **120**
 - With **15.10**, if using Extended Multiload protocol, this option **allows a higher concurrency** limit for MLOADX
 - **Prior to 15.10**, Workload Management **did not distinguish** between traditional Multiload and Extended Multiload
 - Extended Multiload protocol **uses SQL sessions** rather than Multiload sessions
- Support **user-defined AWT resource limits** for FastLoad, MultiLoad, MLOADX and FastExport utilities rather than the default of **60%** of the total AWTs

Utility Limits

- ☐ Support increased MLOADX job limits and increased AWT resource limits

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Utility Limits option allows TPT Update jobs utilizing the Extended Multiload protocol to be increased from 30 to a maximum of 120. The Extended MultiLoad Protocol (MLOADX) uses SQL sessions to load tables that traditional MultiLoad cannot process. MLOADX runs only when the standard MultiLoad protocol cannot be used.

Support user-defined AWT resource limits for FastLoad, MultiLoad, MLOADX and FastExport utilities rather than the default of 60% of the total AWTs.

Current topic: AMP work tasks

- General settings: General, bypass, limits/reserves
- General settings: Other settings
- **AMP worker tasks**
 - **What are AMP work tasks?**
 - **Reserved pools of AWTs**
 - **Work types**
 - **AMP message queues**
 - **BYNET message queue**
 - **Other tab: Define 'available AWTs' as**

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Before we discuss the last option on the other tab

Let's talk about AMP Work Tasks (AWTs) before we discuss the last option **Define 'Available AWTs' as.**

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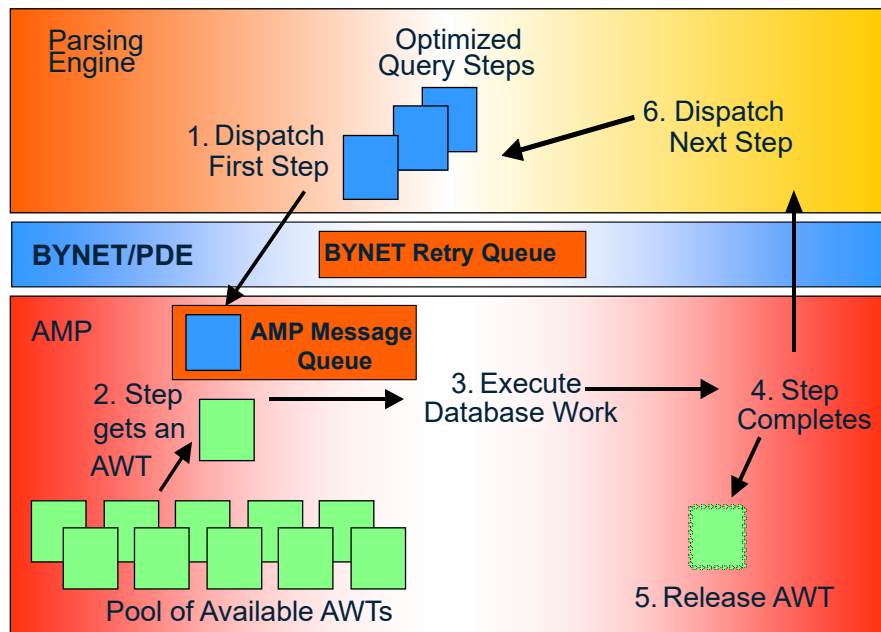


We need to examine AMP Work Tasks (AWTs) before we discuss the last option Define 'Available AWTs' as.

What are AMP worker tasks?

AWTs are assigned to each dispatched query step to perform database work. They are anonymous and not tied to a particular session or transaction.

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AMP worker tasks are execution threads that do the work of executing a query step once it is dispatched to the AMP. They are not tied to a particular session or transaction and are anonymous and immediately reusable. When a query step is sent to an AMP, that step acquires a worker task from the pool of available AWTs. All of the information and context needed to perform the database work is contained within the query step. Once the step is complete, the AWT is returned to the pool.

If all AMP worker tasks are busy at the time the message containing the new step arrives, then the message will wait in a queue until an AWT is free. Position in the queue is based first on work type and second on priority, which is carried within the message header. When the AMP message queue is full, messages will be blocked and put into the sender's BYNET retry queue.

Internally, separate queues are maintained for each message work type, and for MsgWorkNew. Separate queues are maintained for All-AMP steps and single AMP steps.

Message queue limit is number of nodes + 5 for configurations > 20 nodes; otherwise message queue limit is 20.

SQL Request sent from a host to Analytics Database is processed by a PE

- PE – parses request, does syntax check and generates join plan
- Dispatcher sends join plan steps to AMP via BYNET driver
- BYNET driver broadcasts all-AMP steps to all AMPs or sends a single point to point message to a single AMP

BYNET driver in receiving AMP puts request in Message Queue (mailbox)

When AWT is available, scheduler takes request out of Message Queue (by priority setting) and assigns it to an AWT step

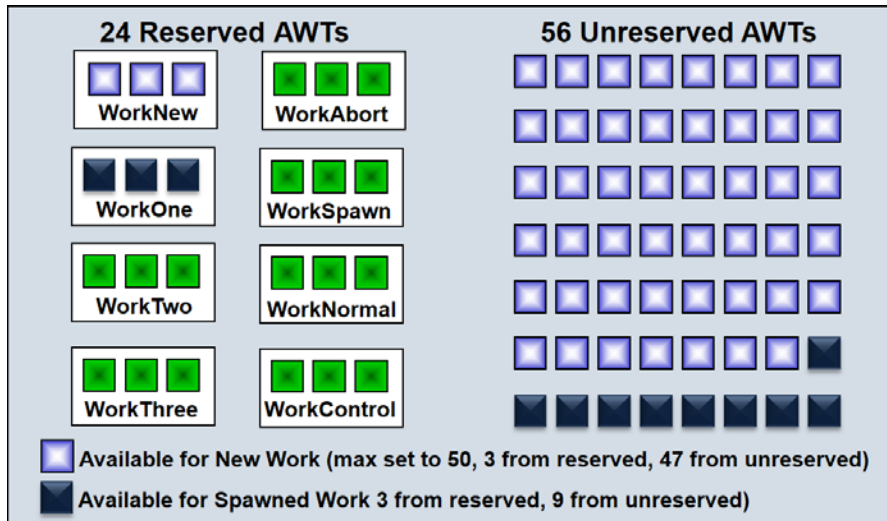
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- LIMIT of 50 AWTs for new dispatched steps
- Execution of the step can spawn a receiver task for row redistribution or unique secondary index handling

When AWTs are not available, requests remain queued up in Message Queue. When the Message Queue for a given message type reaches its limit of 20 for configurations with 16 or fewer nodes, or the number of nodes plus 5 for configurations larger than 16, additional messages of the same type sent to the AMP are rejected by the AMP and are queued into sending node's BYNET retry queue. For all-AMP messages, if one AMP's Message Queue is full, then the message is rejected by all AMPs. When messages go into the BYNET retry queue, the system is under "flow-control".

Messages in retry queue are retried at multiple of 40 ms intervals. First retry is at 40 ms, second is at 2×40 ms, third is at 3×40 ms, etc., up to 64×40 ms (2.56 seconds). Thereafter, all retries for the given message are done at 2.56 second intervals.

Reserved pools of AWTs



The reserve pools are logical, not physical.

No AWTs are set aside for a specific Work Type.

With a max of 50 AWTs that can be used for NewWork, up to 12 AWTs (3 reserved + 9 unreserved) will be available for first level of spawned work, WorkOne.

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Each SLG Tier can support multiple Workloads. When the DBA assigns a workload to a These reserve pools are logical, not physical. No AWTs are set aside specifically for MSGWORKONE, for example. Rather, internal counters keep track of the number of AWTs that are in use at any point in time. The AMP worker task resource manager makes sure that the number of unassigned AWTs never falls below the number that could support all reserves for all work types.

Suppose for a moment that in your workload AMP worker tasks are exclusively involved in supporting new query work and the first level of spawned work, such as row redistribution. Under those conditions, the maximum number of AWTs that could be used for MSGWORKNEW and MSGWORKONE combined could not exceed 62 (56 from the unreserved pool, plus 3 each from the reserve pools for those two work types). The remaining 18 AWTs would be held back as a reserve for each of the other 6 work types.

Different work types, each with their own reserve pool, exist to prevent resource deadlocks. If new user work, and its spawned work, were allowed to occupy all the AWTs in the system, then there would be no tasks available to service other important work. These reserve pools, combined with the hierarchy among work types, reinforces the ability of the Analytics Database to be self-managing and robust.

Having a limit of 50 on new work ensures that up to 12 AWTs will generally be available for the first level of spawned work. New work has its own reserve of 3. If it hits the limit of 50, it will have to draw on that reserve of 3, and only be allowed to use 47 AWTs from the unassigned pool. This allows the 9 remaining AWTs in the unassigned pool to be available for MSGWORKONE, the first level of spawned work, if needed. Because MSGWORKONE has its own reserve of 3 AWTs, it then has a total of 12 AWTs available at any point in time.

This limit of 50 on new work re-enforces the theory that it is more important to complete work

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already underway than to start something new. This limit restriction on new work is in place to encourage the completion of in-flight work, work which might depend on somebody else completing their assignment. If MSGWORKNEW were free to use all unreserved AWTs, it could make it difficult for work already started to complete.

Work types

Work types identify the importance of the work in descending priority

WorkNew	Step coming from the dispatcher (Work00)
WorkOne	1 ST level of spawned work from WorkNew (Work01)
WorkTwo	2 nd level of spawned work from WorkOne (Work02)
WorkThree	Special types of work (Work03)
WorkFour	Recovery management, control AMP (0) (Work04)
WorkAbort	Abort Processing (Work12)
WorkSpawn	End Transaction and spawned abort processing work (Work13)
WorkNormal	Urgent internal requests, response messages (Work14)
WorkControl	Most urgent internal requests (Work15)

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Work types identify the importance of the work in descending priority

- WorkNew – Steps coming from the dispatcher
- WorkOne – 1st level of spawned work
- WorkTwo – 2nd level of spawned work
- WorkThree – Special types of work
- WorkFour – Recovery management, control AMP (0)
- WorkAbort – Abort processing
- WorkSpawn – End transaction and spawned
- WorkNormal – Urgent internal requests, response messages
- WorkControl – Most urgent internal requests

AMP message queues

- Systems are typically configured with a total of **80** AMP Worker Tasks per AMP
 - **24** AWTs are reserved and available only for the 8 specific work types.
 - **56** AWTs are unreserved and available for any work type.
 - Maximum of **50** AWTs can be used for **Dispatched steps**.
 - Maximum of **62** AWTs are available for **Dispatched steps** and **1st level of spawned** work.
- When all 62 AWTs are in use, **new work is queued** in the AMPs message queue
- AMPs message queue is **prioritized by descending worktype** and within work type the queue is sequenced by the Priority Scheduler consumption-to-weight ratio (**virtual runtime**)
- For Broadcast and Multi-cast messages, if one AMP message queue is full, the message is rejected by all AMPs.
 - There is a separate message queue for each transmission type
 - Point to Point
 - Multi-cast
 - Broadcast
 - Maximum worktype messages than can be queued:
 - Maximum of **20** for systems of **16 nodes or less**
 - Maximum of **number of nodes + 5** for systems **greater than 16** nodes
 - HSN and AMP less nodes are excluded in determining the maximum number of nodes

Note: newer systems may be configured with more AWTs which will provide for a larger unreserved pool

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AMP Message Queues

SQL Request sent from a host to Analytics Database is processed by a PE.

- PE – parses request, does syntax check and generates join plan
- Dispatcher sends execution plan steps to AMP via BYNET driver
- BYNET driver broadcasts all-AMP steps to all AMPs or sends a single point to point message to a single AMP
- BYNET driver in receiving AMP puts request in Message Queue (mailbox).

When AWT is available, scheduler takes request out of Message Queue (by priority setting) and assigns it to an AWT step.

- LIMIT of 50 AWTs for new dispatched steps
- Execution of the step can spawn a receiver task for row redistribution or unique secondary index handling
- When AWTs are not available, requests remain queued up in Message Queue.

When the Message Queue for a given message type reaches its limit of 20 for configurations with 16 or fewer nodes, or the number of nodes plus 5 for configurations larger than 16 nodes, additional messages of the same type sent to the AMP are rejected by the AMP and are queued into sending node's BYNET retry queue. For all-AMP messages, if one AMP's Message Queue is full, then the message is rejected by all AMPs. When messages go into the BYNET retry queue, the system is under "flow-control".

BYNET retry queue

When an AMPs message queue is full for a worktype, **flow control** gates for that worktype close and messages will be put in the sending nodes BYNET retry queue

- System is in **Flow Control**
- BYNET retry queue is **not prioritized** (first in first out)
- BYNET **only delivers messages** if all AMPs can receive it
- Messages will be **retried** in multiple of **40ms** intervals
- First retry at **40ms**, 2nd at **80ms**, 3rd at **120ms**, up to **64 times (2.56 seconds)**
- Thereafter all retries are done every **2.56** seconds
- Dispatched messages could be accepted by the AMP if the flow control gates are open ahead of messages in the retry queue – **unfair algorithm**

teradata.

Rejected messages are put into the sending node's BYNET retry queue. When messages go into the BYNET retry queue, the system is under "flow-control".

Messages in retry queue are retried at multiple of 40 ms intervals. First retry is at 40 ms, second is at 2*40 ms, third is at 3*40 ms, etc., up to 64*40 ms (2.56 seconds). Thereafter, all retries for the given message are done at 2.56 second intervals.

Other tab: Define 'available AWTs' as

Starting with Teradata Database 16.0 you can designate the definition for **available AWTs**.

Define 'Available AWTs' as

- ☒ AWTs available for the WorkNew (Work00) work type
The WorkNew (Work00) work type is limited by default to 50 AWTs. If 50 AWTs are already in-use servicing WorkNew message types, there may still be AWTs in the unreserved pool that under this definition will not be considered available.
- ☐ AWTs available in the unreserved pool for use by any work type
The number of AWTs available in the unreserved pool able to be used by all work types, not limited to WorkNew work types.

AWTs available for the WorkNew (Work00) work type

- The WorkNew (Work00) work type is limited to 50 AWTs by default. If AWTs that can support WorkNew message are already in use servicing WorkNew message types, there may still be AWTs in the unreserved pool that will not be considered available.

AWTs available in the unreserved pool for use by any work type

- AWTs available in the unreserved pool for use by any work type. This is number of AWTs available in the unreserved pool able to be used by all work types, not limited to WorkNew work types

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Overview

The Viewpoint 16.00 Workload Designer portlet now includes capability to choose a different triggering algorithm for the system event, Available AWTs. Now you will be able to choose between:

- Current/Default: WorkNew (Algorithm #0)
- New: Entire unreserved pool (Algorithm #2)

Business Value

Prior to Teradata Database 16.00 Teradata Active System Management (TASM) would evaluate the Available AWT system event by looking at two different pools: the WorkNew pool and the entire unreserved pool. TASM would take the smaller of those two pools and compare it against the user-defined Available AWT threshold. If number of available AWTs from the smaller pools is equal or less than the threshold, TASM would trigger the event. Some customer sites found this method was too restrictive, so Teradata Database 16.00 systems and beyond provide an alternative way to interpret available AWTs in Workload Management.

Technical Overview

TASM has the ability to monitor various system resources and trigger an event defined by a user. The Available AWT system event type allows the DBA to trigger an action/state change when AWT shortage is detected. This system event currently checks and triggers if the WorkNew AWT pool or the overall unreserved pool falls below a user defined threshold on a specified number of AMPs for a period of time—qualification time. However, this model fails to take into account other work types that are available for new work such as new expedited user work (WorkEight AWTs).

In the existing mechanism, TASM triggers the Available AWT event when the user defined threshold is equal or less than the minimum of the WorkNew AWT pool or overall unreserved

pool:

MIN (AvailableForAll, WorkNewMax - WorkNewInuse)

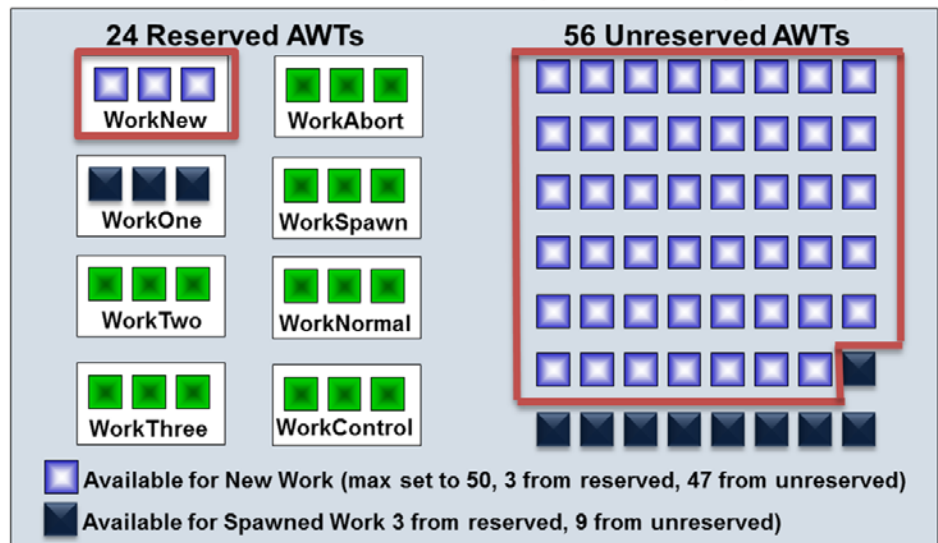
The issue with this model is that it limits the event triggering mechanism to a specific type of work. With workloads varying throughout the day for different business sectors, demands for AWTs for different work types are expected. Customers will benefit greatly if TASM allows users to define the triggering logic that fits their type of work and system configuration. With this added flexibility, TASM can trigger events that can meet customer's needs. So, with Teradata Database 16.0 and later there are two different triggering algorithms for the system event, Available AWT:

- Current/Default: WorkNew (Algorithm #0)
- New: Entire unreserved pool (Algorithm #2)

AWTs available for the WorkNew (work00) work type

The first option will define Available AWTs with this formula:

$$\text{AvailableAWTs} = \text{MIN} (\text{AvailableForAll}, \text{WorkNewMax} - \text{WorkNewInuse})$$



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Current/Default: WorkNew (Algorithm #0)

The chart on the opposite page describes the scenario where we are using the current/default: WorkNew (Algorithm #0). (Note: this scenario assumes that the system is configured with 80 AWTs and does not include expedited AWTs.) In this scenario the Available AWT system event is defined as:

$$\text{AvailableAWTs} = \text{MIN} (\text{AvailableForAll}, \text{WorkNewMax} - \text{WorkNewInuse})$$

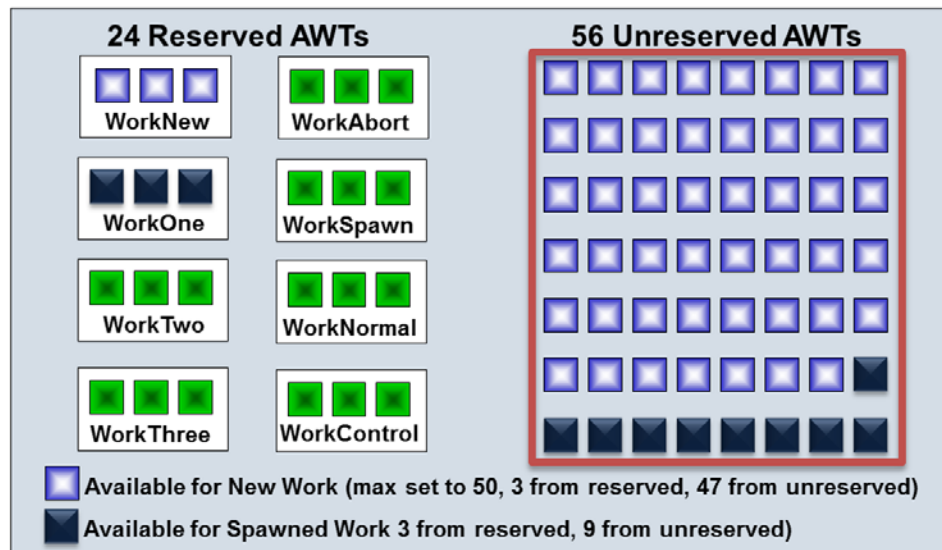
Thus, TASM looks at two pools and takes the smallest number of the two, and then it compares that number to the user-defined threshold.

In our scenario below the maximum number of AWTs available for the WorkNew is 50. Thus, the user-defined threshold for the Available AWT system event is evaluated against this pool of 50 AWTs. So, for example, if there are currently 48 AWTs servicing WorkNew message types, then there would be two Available AWTs.

AWTs available in the unreserved pool for use by any work type

$$\text{AvailableAWTs} = \text{Total AWTs} - (\text{Max}(\text{Min of each work type, InUse of each work type}))$$

The second option will define Available AWTs with this formula:



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New: Entire unreserved pool (Algorithm #2)

The chart on the opposite page describes the scenario where we are using the Entire unreserved pool (Algorithm #2). (Note: this scenario assumes that the system is configured with 80 AWTs and does not include expedited AWTs. It also assumes that we are not looking at AMP 0 which is configured one additional reserved AWT.) In this scenario the Available AWT system event is defined as:

$$\text{AvailableAWTs} = \text{Total AWTs} - (\text{Max}(\text{Min of each work type, InUse of each work type}))$$

In our scenario below the reserved number of AWTs is 24, leaving 56 AWTs for all work types. Thus, the user-defined threshold for the Available AWT system event is evaluated against this pool of 56 AWTs. So, for example, if there are currently 48 AWTs servicing any work type, then there would be 8 Available AWTs.

Summary

- Workload Designer General Icon contains the following tabs:
 - General
 - Bypass
 - Limits/Reserves
 - Other
- The preset default values often suffice; however, you may need to choose other values based on your customer's particular workloads

teradata.

Workload Management is a Goal-Oriented, Automatic Management and Advisement technology in support of performance tuning, workload management, capacity management and system health management.

Workloads provide the ability for improved control of resource allocation, improved reporting and automatic exception detection and handling.

Workload Designer General Icon contains the following tabs:

- General
- Bypass
- Limits/Reserves
- Other

The preset default values often suffice, however, you may need to choose other values based on your customer's particular workloads

Module 6: Workload Designer: Classifications

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Describe Workload Designer uses Classification criteria when creating rules
- Identify the different Classification criteria options

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Topics

- Classification overview
- Classification tab

teradata.



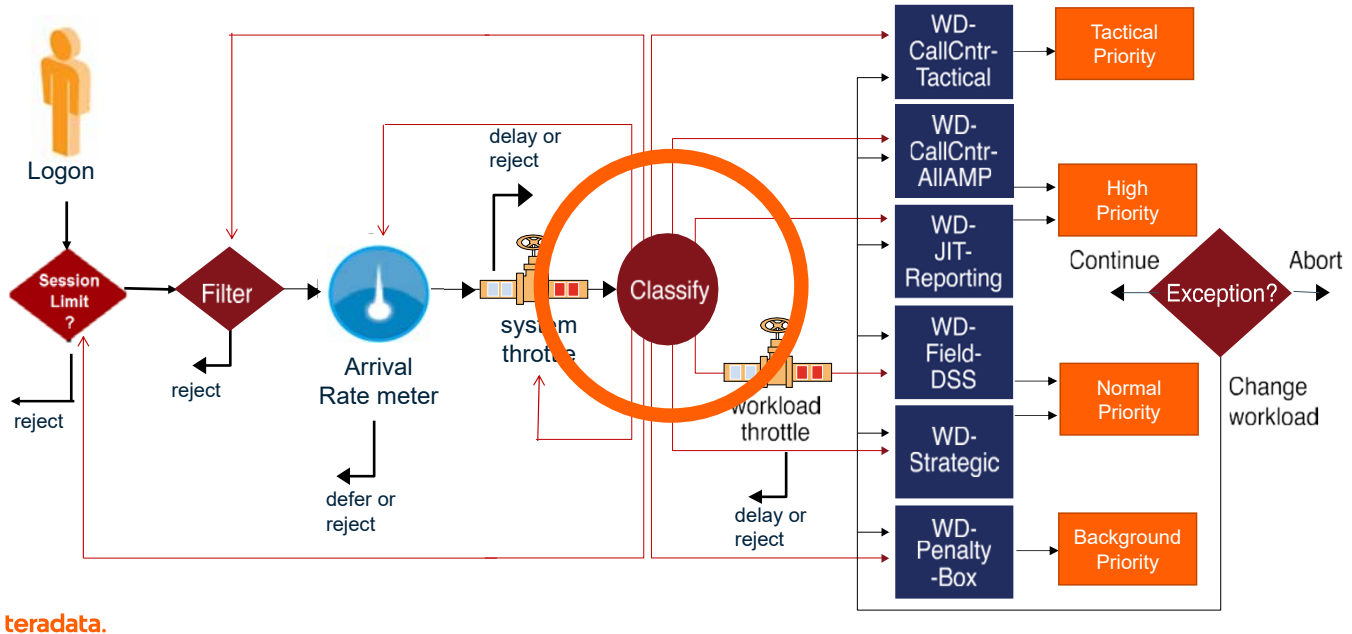
Current topic: Classification overview

- **Classification overview**
 - **Classification criteria**
 - **Criteria options**
 - **Criteria exactness**
 - **Classification criteria recommendations**
- Classification tab

teradata.



Workload management levels: Classify



There are seven different methods of management offered, as illustrated below:

- Methods regulated prior to the query beginning execution
- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level.
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level
- Methods regulated during query execution
- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

Classification criteria

Classification Criteria is used by Workload Management to determine which **Workload** a request should be assigned to or which **Session**, **Filter** and **Throttle** rules to apply

Classification Criteria options include:

- Request Source Criteria – **Who** submitted the request
 - QueryBand Criteria – Subset of **Who** to identify requests from common logons
 - Target Criteria – **Where** within the database the request will operate
 - Secondary Sub Criteria – Can be applied to Target Criteria to further define **What** type of processing can be performed on the object
 - Query Characteristics – **What** type of processing the request is expected to do
 - Utility Criteria – **Which** utility job is issuing the request
- If creating multiple Classification criteria for a workload, all criteria must be satisfied for a request to be classified to the workload
 - For classification purposes, a multi-statement request is a single request

teradata.

The basic classification criteria describes the *"who"*, *"where"*, and the *"what"* of a query. This information effectively determines which queries will run in a workload.

A query is classified into a Workload Definition (WD) if it satisfies all of the Classification criteria. Normally, you will only need to specify one or two criteria for a workload definition.

Avoid long include/exclude lists associated with "Who" and "Where" criteria. Consider the use of accounts (that combine many users into one logical group) or profiles to minimize long "Who" include/exclude lists.

A goal is to define your workload using as few classification criteria as possible. Fewer criteria mean less overhead for the dispatcher than with multiple classification criteria, although this is minimal.

A good approach is to start with a few broadly defined workloads that are defined only by who criteria. After the rule set has been in use for a while, use Teradata Workload Analyzer to determine if additional classification criteria are needed to correctly classify queries. For example, if you see long running queries in a tactical workload, you may want to add a criteria that excludes all-amp queries.

Classification criteria options

Request Source Criteria:

- Account String
- Account Name (Account String with PG information removed)
- Application
- Client Address
- Client ID (for Logon)
- Profile
- Username

QueryBand Criteria:

- Name/Value pair

Target Criteria:

- Database
- Table, View, Macro
- Stored Procedure
- User Defined Function and/or table operator
- User Defined Method
- QueryGrid Server Type
- Secondary Sub-criteria

teradata.

The classification criteria options include:

- Request Source Criteria (Who submitted the request?)
- Account String
- Account Name (Account String less the PG information.)
- Application
- Client Address
- Client ID (for logon)
- Profile
- Username
- QueryBand Criteria (QueryBand enables requests all coming from a single or common logon to be classified into different workloads based on the QueryBand set by the originating application.)
- Target Criteria (Where within the database will the request operate, and what type of step operation applies to that object)
- Data Objects (database, table, view, macro, stored procedure)
- User defined functions and/or table operator and user defined methods
- QueryGrid server object
- Secondary sub-criteria can be applied to Data Object accesses such as “Unconstrained Product Join against Table XYZ”
- Step (Intermediate or Spool) Row Count
- Estimated Step Processing Time
- Join Type (All or no Joins, All or no Product Joins, All or no Unconstrained Product Joins)
- Full Table Scan Access
- Data Block Selectivity
- Statement Type (DDL, DML, Select)

Classification criteria options (cont.)

Query Characteristics Criteria:

- Statement Type (DDL, DML, Select, Collect Statistics)
- Amp Limits (Single or Few AMPS)
- Step Row Count (Intermediate or Spool)
- Final Row Count
- Estimated Processing Time
- Join Type (Include only certain Join Types)
- Full Table Scan (Include or Exclude)
- Estimated Memory Usage
- Multi-Statement Requests
- Incremental Planning and Execution

Utility Criteria:

- Fastload (including subtypes – TPT Load Operator, JDBC Fastload, CSP Save Dump, and Non-Teradata Fastload)
- Multiload (including subtypes – TPT Update Operator, JDBC Multiload, and Non-Teradata Multiload)
- Fastexport (including subtypes – TPT Export Operator, JDBC Fastexport, and Non-Teradata Fastexport)
- Archive/Restore
- DSA Backup
- DSA Restore

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The classification criteria options include:

- Query Characteristics Criteria (What type of processing is it expected to do?)
- Statement Type (DDL, DML, Select, Collect Statistics)
- AMP Limits (Include single or few AMP queries only)
- Step (Intermediate or Spool) Row Count
- Final estimated Row Count
- Estimated Processing Time
- Join Type (All or no Joins, All or no Product Joins, All or no Unconstrained Product Joins; best practice: instead use as a sub criteria of a specific Target Criteria Object)
- Full Table Scan (best practice: use as a sub criteria of a specific Target Criteria Object)
- Estimated Memory Usage
- Multi-Statement Requests
- Incremental Planning and Execution
- Utility Criteria
- Which Utility (including subtypes such as JDBC vs. TPT) is issuing the request
- Archive/Restore
- For Teradata 15.10
- Replacing the BAR utility type with DSA Backup and DSA Restore to support managing DSA jobs differently
- MLOADX can be selected separately from Multiload

Classification criteria exactness

When defining Classification Criteria, consider the ability to **exactly characterize** a request into the appropriate workload

- **Request Source** classification criteria **exactly identifies WHO** the request came from.
 - However, sometimes it may not be granular and will need to be **supplemented** with other criteria, for example when a common logon is used
- **Queryband** classification criteria can be provided by the issuing application providing supplemental information about the request
- **Target** classification criteria **may or may not** provide exact identification of the request, depending on how the database environment was structured
- **Query Characteristics** criteria is **based on estimated** characteristics which may or may not accurately identify the request, depending on it's complexity
 - The **more complex** the request, the higher the probability of **misclassifications**
 - Combining Query Characteristics with other classification criteria can help **minimize misclassifications**
- **Utility Criteria** can exactly identify WHO issued the request

teradata.

The more exact the classification criteria, the higher the chances a query will be properly classified into the correct workload, reducing misclassifications and exception criteria necessary to identify misclassifications.

Classification criteria recommendations

Lead with Request Source and/or Queryband criteria since it is **most exact**, and add Target, Query Characteristics, and Utility criteria **as needed**

- Target criteria relies on assumptions
 - For example, access is to view X, therefore this must belong to the analysis workload
- Query Characteristics criteria relies on **estimates**
 - For example, **estimated processing time** is small, therefore this must belong in the tactical workload
 - Generally, very **short queries** have **more reliable** estimates
 - The **longer** and more complex the query, the **more unreliable** the estimates
 - Recommend to only use estimated processing time criteria to distinguish short queries from all others
- **Avoid** long Include/Exclude lists
 - Use criteria on a **higher level**, such as **database** vs. tables, **profile** vs. users or use **wildcards** (? – for single character or * – for multiple characters)

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Set the **Evaluation Order** to put Workloads with **more specific** criteria **ahead** of Workloads with **less specific** criteria

Lead with Request Source and/or Queryband criteria, and add Target, Query Characteristics , and Utility criteria only when necessary

General recommendations:

- Target criteria relies on assumptions
 - For example, access is to view X, therefore this must belong to the analysis workload
- Query Characteristics criteria relies on estimates
 - For example, estimated processing time is small, therefore this must belong in the tactical workload
 - Generally very short queries have more reliable estimates
 - The longer and more complex the query, the more unreliable the estimates
 - Recommend to only use estimated processing time criteria to distinguish short queries from all others
- Avoid long Include/Exclude lists
 - User criteria on a higher level, such as database vs. tables, profile vs. users or use wildcards
- Set the Evaluation Order to put the more specific criteria ahead of less specific criteria

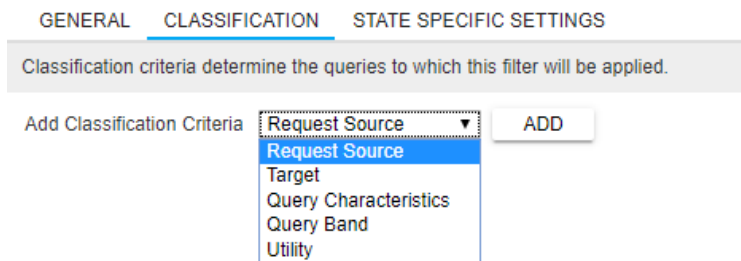
Current topic: Classification tab

- Classification overview
- **Classification tab**
 - **Where you'll find the classification tab**
 - **Request source criteria**
 - **Target criteria**
 - **Query characteristic criteria**
 - **Queryband criteria**
 - **Utility criteria**
 - **Multiple request source criteria**
 - **Some specific query characteristic criteria**

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Where you'll find the classification tab



The **Classification tab** will be **available when** creating any of the following rules:

- Session
- Filter
- Throttle
- Arrival Rate Meter
- Workload

The **Classification Criteria** drop down **menu selections may differ** based on the rule

teradata.

Workload Designer provides a common classification process for workloads, filters, throttles, query sessions, and utility sessions. Classification determines which queries use which rules. Analytics Database detects classification criteria before executing queries. The goal in creating a useful classification scheme is to meet business goals and fine-tune control of the Analytics Database.

Over time, modifications to the classification settings may need to be made in response to data monitoring, regular historical analysis, or changes. For example, classification groups may need to be created, or existing groups modified, if an application is added, two production systems are consolidated, or service-level goals are missed.

The classification tab will be available when creating any of the following rules:

- Session
- Filter
- Throttle
- Workload

Depending on the rule, the classification criteria drop down menu may differ.

Request source criteria

Portlet: **Workload Designer** > Button: **Sesn/Filtr/Thrlt/Wrkld** Button: **Create [+]** >
Tab: **Classification** > Button: **Add Criteria** > Source Type: **Username**

Source Type: **Username**

Username:

Wildcard: *

Usernames:

- AdhocUser1
- AdhocUser2
- ADW_DBA
- All
- console
- Crashdumps
- DBC
- Default
- DSSUser1
- DSSUser2
- External_AP

INCLUDE EXCLUDE

Selected:

Included

Excluded

REFRESH (This may take a while.)

OK CANCEL

The values displayed will depend on the Source Type chosen

Note: Wildcard capabilities are also available when defining Request Source Criteria

teradata. If a rule contains multiple values for the same request source type (e.g., multiple usernames) only one of the values needs to be matched

The request source classification type specifies who is making a request. You can classify filters, throttles, workloads, utility sessions, and query sessions by request sources such as account name or client IP address.

Multiple Request Source Types

- If a rule contains multiple values for the same request source type (e.g., multiple usernames) only one of the values needs to be matched
- If a rule contains multiple source types (e.g., username values and client IP addresses) then each of the multiple source types must be matched (e.g., a username value matches an expression in the username inclusion list and a client IP address value matches an expression in the client IP address inclusion list)

Target criteria

Portlet: **Workload Designer** > Button: **Filtr/Thrlt/Wrkld** Button: **Create [+]** >
Tab: **Classification** > Selector > **Target** > Button: **Add** > Target Type: **Database**

Edit Target Criteria

Choose a target type to display the list of available values for that type.

Target Type: **Database**

Database:

WILDCARD = *

Databases:

- AdhocUser1
- AdhocUser2
- ADW_DBA
- All
- console
- Crashdumps
- DBC
- dbcmngr
- Default
- DESIGN_DATA
- DESIGN_VM

INCLUDE EXCLUDE

Selected:

- Included

Excluded:

REFRESH (This may take a while.)

OK CANCEL

The values displayed will depend on the Target Type chosen

Target Type can only be used once per rule

Note: Wildcard capabilities are also available when defining Target Criteria

teradata.

The target classification type specifies the query data location. You can classify filters, throttles, or workloads by targets such as database, table, or stored procedure. You can add sub-criteria in V13.10 or later. If you add multiple sub-criteria to a single item, all sub-criteria conditions must be true in order for the query to be classified into the rule.

Available target types include database, table, macro, view, or stored procedure. A database selection list is displayed if table, macro, view, or stored procedure is selected. A target type can only be used once per rule. After a target type is used, it no longer appears in the menu.

Target Sub-Criteria

Portlet: **Workload Designer** > Button: **Filtr/Thrlt/Wrkld** Button: Create [+] > Tab: **Classification** > Selector: **Target** > Button: **Add** > Target Type: **View** > Selected Target Type: **Edit Subcriteria**

Edit Target Criteria

Choose a target type to display the list of available values for that type.

Target Type: **View**

Database: **OPTIMIZATION_VM**

View:

WILDCARD = *

Views:

- LOCATION
- LOCATION_TYPE
- MILEAGE
- PRICE_CHANGE_REASON
- REGION
- SALES_TRANSACTION
- SALES_TRANSACTION_LINE**
- STL_SUMMARY
- TRAIT
- TRAIT_GROUP
- TRAIT_VALUE

Selected:

INCLUDE	EXCLUDE
☒	☐
OPTIMIZATION_VM.SALES_TRANSACTION_LINE	

Edit Criteria

Query must meet all criteria to be classified

☒ Full Table Scan

☐ Include

☐ Exclude

☐ Join Type: **No Join**

☐ Estimated Step Row Count: x

☐ Estimated Step Row Count: x

☐ Estimated Step Processing Time: x seconds

☐ Estimated Step Processing Time: x seconds

☐ Estimated percent of table blocks accessed during the query: x

Only statements of the following type:

☐ DDL

☐ DML

☐ Select

OK CANCEL OK CANCEL

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Optionally, each selected target item can have sub-criteria. For example, if you select a database as the target, you could add sub-criteria so that it only applies if you are performing a full table scale. If you add more than one sub-criteria, they must all be present for the classification setting to be used. Target items containing sub-criteria display to the right of the item name.

Available sub-criteria include:

- **Full Table Scan.** Include or exclude full table (all row) scans.
- **Join Type.** Select a type, such as **No Join** or **Any Join**.
- **Estimated Step Row counts.** Set minimum or maximum rows at each step.
- **Estimated Step Processing Time.** Set minimum or maximum time at each step.

Query characteristics criteria (1 of 2)

Portlet: **Workload Designer** > Button: **Filtr/Thrlt/Wrkld** Button: **Create [+]** >
Tab: **Classification** > Selector: **Query Characteristics** > Button: **Add**

Edit Query Characteristics Criteria

Add one or more query characteristics by selecting a check box and entering associated criteria. A query must meet all of the selected criteria in order to be classified by this criteria.

☐ **Statement Type** Include Only: ☐ DDL ☐ SELECT
☐ DML ☐ COLLECT STATISTICS

AND

☐ **AMP Limits** ☒ Include only queries that use all AMPs
☐ Include queries that do not use all AMPs

AND

☐ **Estimated Step Row Count** ☐ Row Count ≥
☐ Row Count ≤

AND

☐ **Estimated Final Row Count** ☐ Row Count ≥
☐ Row Count ≤

OK CANCEL

teradata.

The query characteristic classification type describes query types and resource usage, such as statement type, row count, estimated processing time, or join type.

Query characteristics describe a query by answering such questions as what does the query do and how long will the query run.

Keep the following in mind when using query characteristics to classify information:

- Once a characteristic is selected, its value can be edited.
- Many characteristics have minimum and maximum values that can be set independently. You can set all values above the minimum, below the maximum, or between a minimum and a maximum.
- Query characteristic classification and utility classification are mutually exclusive. If you use one, the other option is not available.
- You can have one query characteristic classification per rule.

If you select **Join Type**, you can choose from **No Join**, **Any Join**, **Product Join**, **No Product Join**, **Unconstrained Product Join**, and **No Unconstrained Product Join**.

Query characteristics criteria (2 of 2)

Portlet: **Workload Designer** > Button: **Filtr/Thrlt/Wrkld** Button: **Create [+]** >
Tab: **Classification** > Selector: **Query Characteristics** > Button: **Add**

Edit Query Characteristics Criteria

Add one or more query characteristics by selecting a check box and entering associated criteria. A query must meet all of the selected criteria in order to be classified by this criteria.

☐ Estimated Step Processing Time

☐ Processing Time >
☐ Processing Time <

seconds
seconds

AND

☐ Estimated Total Processing Time

☐ Processing Time >
☐ Processing Time <

seconds
seconds

AND

☐ Join Type:

Include Only:

AND

☐ Full Table Scan

☒ Include queries that perform a Full Table Scan
☐ Include queries that do not perform a Full Table Scan

AND

☐ Memory Usage

Minimum Memory Use:

AND

☐ Incremental Planning

☒ Include queries that use incremental planning
☐ Exclude queries that use incremental planning

OK

CANCEL

teradata.

Queryband criteria

Portlet: **Workload Designer** > Button: **Filtr/Thrlt/Wrkld** Button: **Create [+]** >
Tab: **Classification** > Selector > **Query Band** > Button: **Add**

Edit Query Band Criteria

To add a Query Band, enter a query band name in the 'Name' field. Then choose one or more values from the 'Previously Used Values' list or type a value into the 'New Value' text box.

Name: -- Choose --

Previously Used Values

INCLUDE EXCLUDE

New Value:

Selected:

Included

Excluded

OK CANCEL

Multiple Queryband names will be “or”ed together for **Include** and “and”ed together for **Exclude**

teradata.

A query band contains name and value pairs that use pre-defined names (on Analytics Database) or custom names to specify metadata, such as user location or application version. The query band classification type describes the query band data attached to a query.

The query band classification type describes the query band data attached to a query.

Keep the following in mind when using query band to classify information:

- A name must be selected from the **Name** list or typed into the box.
- After picking a name, one or more values must be specified. The value can be selected from the **Previously Used Values** list or typed into the **New Value** box. Multiple values can be selected for the same name.
- The **Include** and **Exclude** buttons are available only after a name and value are specified.
- Multiple included query band names are connected with "and."
- Multiple excluded query band names are connected with "or."

Utility criteria

Portlet: **Workload Designer** > Button: **Filtr/Thrlt/Wrkld** Button: **Create [+]** >
Tab: **Classification** > Selector > **Utility** > Button: **Add**

Edit Utility Criteria

Select check boxes next to the utility names or subtypes to include in this filter.

- | | |
|---|--|
| ☐ FastLoad or MultiLoad | ☐ FastExport |
| ☐ FastLoad | ☐ Stand Alone FastExport |
| ☐ Stand Alone FastLoad | ☐ TPT Export Operator |
| ☐ TPT Load Operator | ☐ JDBC FastExport |
| ☐ JDBC FastLoad | ☐ Non-Teradata FastExport |
| ☐ CSP Save Dump | ☐ Backup Utilities |
| ☐ Non-Teradata FastLoad | ☐ Archive/Restore (ARC) |
| ☐ MultiLoad | ☐ Backup (DSA) |
| ☐ Stand Alone MultiLoad | ☐ Restore (DSA) |
| ☐ TPT Update Operator | |
| ☐ JDBC MultiLoad | |
| ☐ Non-Teradata MultiLoad | |
| ☐ MLOADX | |
| ☐ TPT MLOADX | |

Utility Criteria and Query
Characteristics Criteria are
mutually exclusive

OK CANCEL

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The utility classification type describes which utility submitted the query.

Keep the following in mind when using utility to classify information:

- Available utility types include **FastLoad**, **FastExport**, **MultiLoad**, **Archive/Restore**, **DSA Backup** and **DSA Restore**. Select a top level utility such as **FastExport** or a specific implementation of a utility such as **JDBC FastExport**.
- Utility classification and query characteristic classification are mutually exclusive. If you use one, the other option is not available.

You can have one utility classification per rule.

Multiple request source criteria

Workload Designer TDT1 | POWM Ruleset

Filter: Demo

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Classification criteria determine the queries to which this filter will be applied.

Add Classification Criteria Request Source ADD

Request Sources

Request Source
Include: (Usernames: DSSUser1)
Exclude:

Request Source
Include: (Profiles: DSS_Profile)
Exclude:

AND
AND
OR

With multiple Request Source criteria, you have the option to “AND or “OR” the criteria together. “OR” is only available for Username and Profile request criteria.

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SAVE RESET

There is an option when combining multiple Request Criteria to optionally choose to “OR” the criteria together. This could be used to setup Workload classification criteria to include these “Users” or these “Profiles”. Note: “OR” is only available for Username and Profile request criteria.

Data block selectivity

- Available as **Target sub-criteria selection** “Estimated percent of table blocks accessed during the query”
- Used to classify a query based on the **percent of the table** accessed
- The **ratio** between the optimizer’s estimated cost to access a **portion of the table** compared to the estimated cost to access the **entire table**
- Useful for **large partitioned tables** accessing a few partitions
- Can also be used for non-partitioned tables that are accessed used the **primary index or covering secondary index** access
- For **column partitioned tables**, the estimated cost is based on the **number of columns** that need to be read rather than rows
- For queries **counting all the rows** in a table the estimated cost reflects the **reading of cylinder indexes** rather than the data blocks
- Defining a Workload with **100% data block selectivity** may be helpful to identify queries doing unexpected **full table scans**

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The data block selectivity criteria can be used to classify a query based on the percent of the table accessed. Using current statistics, the optimizer estimates the cost to access the portion of the table needed by the query compared to the cost to process the entire table. The ratio between these costs is compared to the ratio that is defined in the data block selectivity criterion. These cost figures are those used within the optimizer and are not externalized. While this feature is particularly useful on large partitioned tables, it is also useful for non-partitioned tables as well.

Partitioned tables – For the typical single table access case using row-partitioned tables, access read costs are calculated as (number of partitions needed to be read for the query * cost-per-partition) and the total read cost as (total partitions in the table * cost-per-partition). The end ratio is equivalent to accessed partitions divided by total partitions.

There are a few situations that can affect accuracy, as follows:

- If statistics are not accurate (particularly statistics on PARTITION), the estimate of the number of partitions could be off, and can affect accuracy dramatically.
- Cost models for joins can add some additional sources of error. For example, the optimizer can assume that based on estimated number of row ‘hits’ in one table, the database will have a known distribution of partition ‘hits’ in the second table, which is not always the case.

Non-partitioned tables – Data block selectivity classification on non-partitioned tables can be useful when the access cost is predicted to be less than the cost of performing a full table scan. These cases include the following:

- Tables are accessed by way of a primary index.
- Covering secondary index access.

Projection of a fraction of all table columns in a column-partitioned table. The cost model for column-partitioned tables is based on the number of columns that need to read, rather than rows. Assume you have a column-partitioned table CP1 with columns (a1, b1, c1, d1, e1) and a query "SELECT a1 FROM CP1". Because only the a1 column needs to be read in order to answer the query, and assuming all the columns are the same size, the ratio of access cost to total cost could be expected to be 1/5.

Count (*) optimization. If there is a query that is counting all the rows in a table, the cost value reflects reading cylinder indexes rather than accessing the table data blocks themselves. The ratio returned in this case is the comparatively small cylinder index read cost estimate compared to the cost to read the entire table's data blocks. For example, "SELECT COUNT (*) FROM t1" will return cylinder index read cost in the data block selectivity ratio, while "SELECT COUNT(*) FROM t1 WHERE a1<5" uses the cost to read the entire table. In the latter case each row has to be read to apply the WHERE clause condition.

A probed table in a join where the probing table indicates few matches. The number of matches in one table of a join can be used to estimate the access cost of the other table. For example, assume there is a table t1 (a1, b1) where a1 is unique and a different table t2 (a2,b2) where a2 is unique, and a query specifies "t1 JOIN t2 ON a1=a2 WHERE a1 IN (1,2,3)". The optimizer knows there are 3 row hits in t2, and the access cost estimate of the retrieve step for t2 will indicate this. It could make sense to specify data block selectivity at 100% and use this feature to identify cases where non-table scans may be expected by the user, but full table scans are occurring. Setting up a workload to classify on 100% data block selectivity can help point to where the database design can be enhanced to improve performance.

Multi-Statement request

- The Multi-Statement Request Classification can be used to assign Multi-Statement Requests to a Workload or to exclude them from a Workload. The MSR Classification can also be used for Filter and Throttle rules
- Many modern BI tools group multiple SQL statements into one request. For example, at one customer site, the Tableau tool generates millions of heavy workload, Multi-Statement Requests each day
- If these larger, longer-running requests run concurrently, they can exhaust AWTs and cause shorter, and possibly higher priority, requests to be delayed
- These large requests now can be identified and classified into their own workload

Example of a Multi-Statement Request

```
SELECT * FROM Customer  
; DELETE FROM Customer WHERE Name = 'Inglis C' AND Custno = 10014  
; SELECT Name FROM Customer;
```

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Business intelligence tools, such as Tableau, group multiple SQL statements into one request.

Multi-Statement request classification settings

The Multi-Statement Request classification criterion lets you identify, isolate, and manage Multi-Statement Requests using the following options:

Option	Description
Exclude	Prevents Multi-Statement Requests from being included in a specific Workload.
Include	Classifies Multi-Statement Requests into a distinct workload so you can manage them separately from shorter requests (Default option).
Threshold statement count	Enter a value to define the minimum number of SQL statements that qualify as a Multi-Statement Request. The default is 2.

- This classification criterion applies to workload, filter, and throttle rules and can be combined with all existing classifications
- TASM ignores the following special statements when counting the statements in a request:
 - BEGIN TRANSACTION (BT)
 - COMMIT
 - END TRANSACTION (ET)
 - NULL
- The TDWMMSRCount field in DBC.QryLogV view shows the statement count in a multi statement request

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Estimated memory usage

- Used to classify queries based on the **estimated memory per AMP** that will be used by any **individual step**
- There are certain steps, such as **XML functions**, some types of aggregations and hash join steps that can cause a query to use **high levels of memory**
- Those queries can be identified and **classified** or **throttled accordingly**
- **Default thresholds** are determined automatically based on the number of nodes containing AMPs and the AMP buffer size
- **Estimates are for** peak memory usage for the **largest single step or sum of all steps executed in parallel**, not the sum of all individual steps for the query
- Recommended to be used primarily **for throttling purposes**

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The optimizer generates estimates on how much memory per AMP will be used by any individual step in a query. In particular, there are XML functions and some types of aggregations and hash join steps that can cause a query to use unusually high levels of memory.

The actual estimates made by the optimizer are not currently visible in the explain text. However, those estimates are passed to Workload Management using internal structures and are logged in DBQLogTbl.

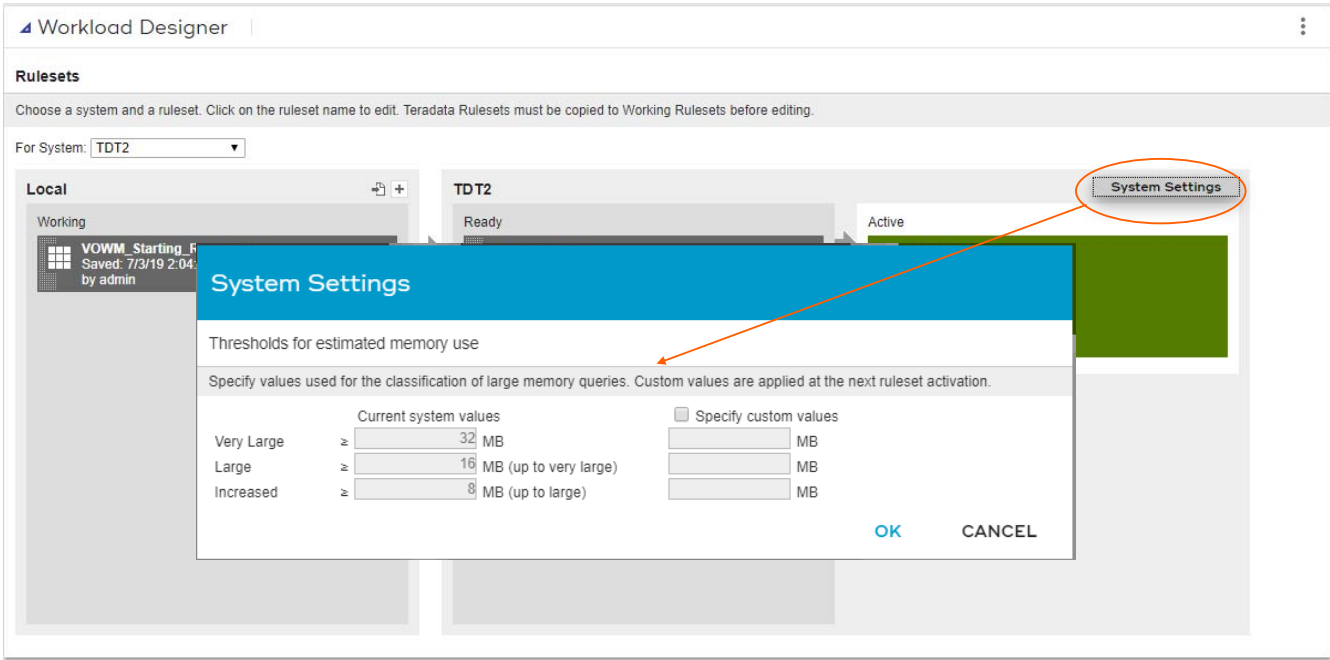
Queries that use excessive memory can be recognized so they can be classified or throttled accordingly. Default thresholds are determined for each platform automatically. These thresholds are based on the particular configuration and show up under “system settings” in Viewpoint, and are calculated based on the number of nodes (that contain AMPs) and the AMP buffer size. Note that the memory estimate is for the peak memory usage for the query and not the sum of all individual steps. Peak memory is either the largest single step or the sum of all steps executed in parallel, whichever is greater. DBQLogTbl can be used to view both estimated memory and actual memory used by a request.

The default thresholds are assigned labels: increased, large, and very large, which are defined in Workload Designer’s System Settings. (See next slide.)

Estimated memory classification for a throttle or a workload is then based on using one of those three labels.

It is recommended that this new classification option be used primarily for throttling purposes, rather than for workloads. Its purpose is to control the concurrency levels of queries that have memory-intensive steps, in order to contain the number of active queries that require unusually high levels of memory. However, it applies to all components that make use of common classification: workloads, throttles and filters.

Where to define values for estimated memory



Selecting System Settings

In Teradata Database 14.10 and later, you can specify values or use the default system values to throttle queries consuming excessive memory. In system settings, you can create memory groups that are available for query characteristic classification. The settings apply to the whole system, not individual rulesets. Changes take effect after the next ruleset is activated.

1. From the Workload Designer view, click **System Settings**.
2. Select the **Specify custom values** check box and enter a value for one or more of the following threshold options to set the estimated memory use:

Option	Description
-----	-----
Very Large	Must be greater than the Large threshold
Large	Must be greater than the Increased threshold
Increased	Must be less than the Large threshold

3. Click OK.

Incremental planning and execution

- Used to identify and classify queries that are being executed using dynamic plans
- The IPE framework provides a method to **reduce** the occurrence of **suboptimal plans** for **complex queries**
 - A complex request, once identified by the optimizer, is **broken into smaller pieces** referred to as **request fragments**
 - Request fragments undergo optimization one at a time, with the **first fragments** results used as **input into** the second fragment
 - Results returned from earlier request fragments are able to **provide more reliable information** for subsequent request fragments
 - Both optimizing and executing fragments take place **incrementally**
 - This contrasts with **non-IPE** queries, which are optimized as a single unit and produce **static plans**
- WorkLoad Management performs **classification based only on information in the first plan fragment**
- Specific **rule criteria that are ignored** when classifying an IPE query include:
 - Min and Max estimated row count
 - Min and Max estimated time
 - Full Table Scan
 - Join type

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IPE classification is an option that allows the identification of queries that are being executed as IPE queries when you want to manage them differently from non-IPE queries.

The IPE framework within the database provides a method to reduce the occurrence of suboptimal plans for complex queries. The basic approach is as follows:

- A complex request, once identified by the optimizer, is broken into smaller pieces referred to as *request fragments*.
- The request fragments undergo optimization one at a time, with the first fragment feeding its results as input into the second fragment. Both optimizing and executing fragments takes place incrementally. This is in contrast to traditional, non-IPE, queries, which are optimized as a single unit, and which produce a *static* plan.
- The plan generated by IPE is referred to as a *dynamic* plan. Results that are returned from earlier request fragments are able to provide more reliable information (such as hard values for input variables) for the planning of subsequent request fragments.

This can result in a more optimal overall plan and provide out-of-the-box performance benefit when processing the more complex queries on a platform. Starting with Teradata Database 15.0, the optimizer automatically looks for candidate queries and applies IPE, as appropriate, using dynamic plans that are built a fragment at a time. When using a dynamic plan, Workload Management performs classification based on the only information it has to work with – what is in the first plan fragment. Using an IPEspecific dynamic plan, Workload Management only has the information on the initial few steps of the query that is included in the first fragment, and until the optimizer builds the plan for the next fragment, information about the subsequent fragments of the query is unavailable.

Since Workload Management does not have the complete view of the query characteristics based solely on dynamic plans, Workload Management may not be able to apply all of its rule

criteria, particularly rules that include things like estimated step times, or types of joins that a given table might be involved in. All step level criteria for steps not within the first fragment are unknown until the subsequent fragments are optimized.

Taking this into account, and to minimize the need to change existing rule sets, Workload Management, by default, simply ignores step level criteria when faced with IPE query dynamic plans. Specific rule criteria that are ignored by Workload Management when attempting to classify an IPE query to a Workload Management object include the following:

- Min and Max Step estimated Row count.
- Min and Max Step estimated time.
- Full Table Scan.
- Join Type.
- IPE Request Criteria.

The intent of this new criterion is to allow sites that want to isolate any potential impact of IPE requests to do so. Once sites are comfortable with IPE behavior, it is expected that this criterion will be removed so that IPE requests can be treated as normal requests.

You can identify which requests are IPE queries, and which are not, by examining the DBC.DBQLogTbl table. The NumFragments field in that table is NULL for non-IPE requests and reports the number of fragments for IPE requests.

Summary

- Classification Criteria is used by Workload Management to determine which **Workload** a request should be assigned to or which **Session**, **Filter** and **Throttle** rules to apply
- Classification Criteria options include:
 - **Request Source Criteria** – **Who** submitted the request
 - **QueryBand Criteria** – Subset of **Who** to identify requests from common logons
 - **Target Criteria** – **Where** the request will operate
 - **Secondary Sub Criteria** – Can be applied to Target Criteria to further define **What** type of operation can be performed on an object
 - **Query Characteristics** – **What** type of operation will be performed by the request
 - **Utility Criteria** – **Which** utility job is submitting the request
- When defining Classification Criteria, consider the ability to **exactly characterize** a request into the appropriate workload
- **Lead with Request Source** and/or Queryband criteria, and **add** Target, Query Characteristics , and Utility criteria only **when necessary**

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Classification Criteria is used by Workload Management to determine which Workload a request should be assigned to or which Session, Filter and Throttle rules to apply

Classification Criteria options include:

- Request Source Criteria – Who submitted the request
- QueryBand Criteria – Identifies requests from common logons
- Target Criteria – Where the request will operate
- Secondary Sub Criteria – Can be applied to Target Criteria to further define What type of operation can be performed on an object
- Query Characteristics – What type of operation will be performed by the request
- Utility Criteria – Which utility job is submitting the request

When defining Classification Criteria, consider the ability to exactly characterize a request into the appropriate workload

Lead with Request Source and/or Queryband criteria, and add Target, Query Characteristics , and Utility criteria only when necessary

Module 7: Workload Designer: State matrix

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Describe the characteristics, components and purpose of the State Matrix
- Use the State Matrix Setup Wizard to create a State Matrix
- Modify the State Matrix

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Topics

- State matrix overview
- Setup wizard
- Lab activity

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Current topic: State matrix overview

- **State matrix overview**
 - **About the state matrix**
 - **Event actions and notifications**
 - **Alert setup**
 - **State transitions**
 - **Rulesets and work values**
 - **Default state matrix**
- Setup wizard: Planned environments
- Setup wizard: Health conditions
- Setup wizard: States
- Lab activity

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About the state matrix

- Workloads **do not always generate consistent demand** or maintain the same level of importance throughout the day/week/month/year
 - Tactical/DSS workloads need higher priority during the **day** and the Load/Batch workload needs higher importance at **night**
 - During **month-end processing**, when the month-end accounting workload is present, it must take precedence over all other workloads
- During periods of **degraded system health**, it may be more important to ensure Tactical workloads demands can be met at the expense of other workloads
 - For Example, when a **node fails**, lower priority workloads may need to be **throttled back** to make more resource available to higher priority work
- The **State Matrix** provides a way to automatically enforce **gross-level workload** management rules amidst these types of situations
 - It is a two-dimensional matrix, with Planned Environments and Health Conditions represented
 - The intersection of a **Planned Environment** and **Health Condition** is associated with a **State** and different ruleset working values

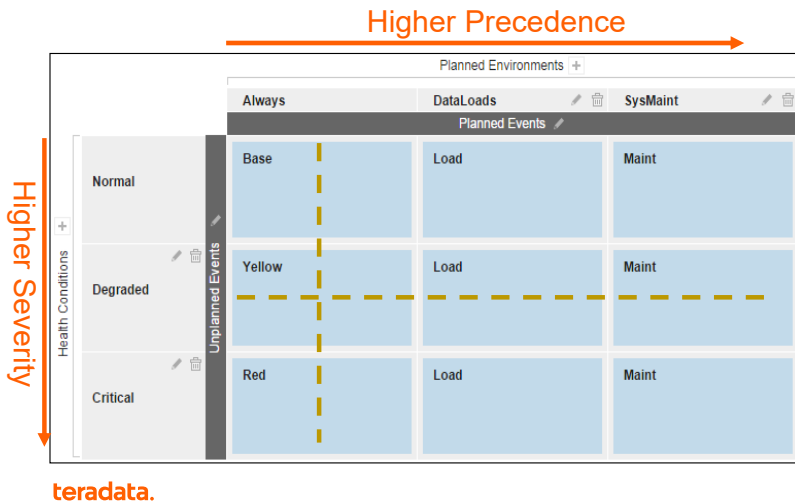
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Generally, workloads do not generate consistent demand, nor do they maintain the same level of importance throughout the day/week/month/year. For example, suppose there are two workloads: A query workload and a load workload. Perhaps the load workload is more important during the night and the query workload is more important during the day. Or perhaps there are tactical workloads and strategic workloads, and when the system is somehow degraded, it is more important to assure tactical workload demands are met, at the expense of the strategic work. Or finally, a year-end accounting workload may take precedence over all other workloads when present. The State Matrix allows a transition to a different working value set to support these changing needs.

The State Matrix allows a simple way to enforce gross-level management rules amidst these types of situations. It is a two-dimensional matrix, with Operating Environments and System Conditions represented, with the intersection of any Operating Environment and System Condition pair being associated with a State with different rule set working values. Multiple Operating Environment and System Condition pairs can be associated with a single State

- A state is a set of operating rules which control how TASM will allocate resources
- When rules are defined in the Workload Designer the rules can specify the “working values” for throttles, filters, limits, and controls that apply to all states (default) or that are state specific
- A state matrix defines the permutations of planned environments and health conditions
- Each state matrix cell specifies the state that TASM will use when the planned environment and health condition combination exists

State matrix



The State Matrix consists of two dimensions:

- **Health Condition** – (TASM ONLY) the condition or health of the system. Health Conditions are **unplanned events** that include system performance and availability considerations, such as number of AMPs in flow control or percent of nodes down at system startup
- **Planned Environment** – the kind of work the system is expected to perform. Usually indicative of **planned time periods or operating windows** when particular critical applications, such as load or month-end, are running
- **State** – identifies a set of **Working Values** and can be associated with one or more intersections of a Health Condition and Planned Environment
- **Current State** – the **intersection** of the current Health Condition and Planned Environment

The state matrix is a user-friendly way to manage states and events. The matrix is made up of two dimensions.

Health Condition: The condition or health of the system. For example, system conditions include system performance and availability considerations, such as number of AMPs in flow control or percent of nodes down at system startup.

Planned Environment: The kind of work the system is expected to perform. It is usually indicative of time periods or operating windows when particular critical applications, such as a load or month-end, are running.

Once you set up the state matrix, you can define the event combinations that activate each system condition and operating environment.

By default, the State Matrix is 1x1: i.e., a single planned environment defined for 24 hours x 365 days a year (“always”), and a single Health Condition, defined for the “normal” health of the system. This Planned Environment and Health Condition pair points to a single default State called “Base”. If there is more than one State associated with additional Planned Environment and/or Health Condition pairs, the system will adjust the rule set working values each time a state transition occurs.

State Matrix Precedence

- When TASM evaluates which planned environment should be active, it checks the events of the right most planned environment and moves left, stopping at the first planned environment with all events triggered
- When TASM evaluates which health condition should be active, it checks the events of the bottom health condition and moves up, stopping at the first health condition with all events triggered

- The intersection of the current planned environment and current health condition is the current state that TASM will use for resource management
- In the sample shown, TASM will use the Batch state when the planned environment is NightlyBatch and the health condition is Degraded or Normal

Event actions

- An Event is a set of conditions can be defined to TASM that, when the conditions are met, triggers an event action, event actions can cause either the **Health Condition** or **Planned Environment** to change allowing for a possible transition to **another State** when an event action is triggered
- The following are the different classes of events that can be detected individually or in combination:
 - **Period** – these are **intervals of time**. Workload Management monitors system time and activates the event when the period starts and deactivates when the period ends
 - **User Defined** – these are **planned events** that will be enabled in Workload Management via an OpenAPI call, they will last until disabled via an OpenAPI call or timed out
 - **System** – these are **unplanned events** related to various DBS components that degrade or fail, or resources that fall below a defined threshold for some period of time, they will last until the component is back up or until the resource is above the threshold for some minimum amount of time
 - **Workload** – these are **events** related to a specified workload that can be triggered by the number of active requests, arriving requests, AWT Wait Time, CPU utilization, and the number of queries in the delay queue, and time delayed
- If an event is detected, it will be logged in the **DBC.TDWMEventHistory** table

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An Event is a set of conditions can be defined to TASM that, when the conditions are met, triggers an event

Event Directives consist of the event definition and the associated actions that occur when triggered. The event definition can be based on the occurrence of a single event or a combination of events.

They consist of an event definition and the associated actions that occur when the event is triggered. Actions can be one or more of the following:

- Change Health Condition
- Change Planned Environment
- Take No Action

The following are the different classes of events that can be detected individually or in combination:

- **Period** – these are intervals of time. Workload Management monitors system time and activates the event when the period starts and deactivates when the period ends
- **User Defined** – these are external events that Workload Management will be informed by calling a stored procedure. They will last until rescinded or timed out.
- **System** – these are events related to various DBS components that degrade or fail, or resources that fall below a defined threshold for some period of time. They will last until the component is back up or until the resource is above the threshold for some minimum

amount of time.

Event notifications

Configure Notifications

ACTIVATED **DEACTIVATED**

	Start Action	End Action
Send Alert:	☒ NodeDown ▼	☐ NodeDown ▼
Run Program:	☐ ▼	☐ ▼
Post to QTable:	☐	☐

In addition to actions, one or more **automated notifications** can also be setup to occur when the event is activated and/or deactivated

Notification Types:

- Send Alert
- Run Program
- Post to QTable

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When you define an event, the automated action you would like to take if the event becomes active must be defined. Note that the detection is always logged to DBC.TDWMEventHistory.

You can choose from a number of different automated actions to occur when activated and/or (for notification actions only) when the event is no longer active:

- Notifications
- Post to a Queue
- Alert
- Run Program
- Automatically change the State by:
 - Changing Health Condition
 - Changing Planned Environment

Event notifications: Alert setup

Portlet: **Administration** > Button: **Alert Setup** > Setup Options: **Alert Presets** > Preset Options: **Action Sets** > Button : **Add new action set**

**Alert Setup** Configure the alert delivery settings and actions.

Setup Options

Delivery Settings

Alert Presets

Alert Exceptions

Database User Alerts

Preset Options

Core Hours

Action Sets

Groups

Action Sets

Adminis...

SM_LOG

+

□

□

□

□

Action Sets Details

Action Set Name:*

Times

☒ Core

☒ Evening

☒ Weekend

Actions

☒ Include in Alert Viewer

☐ Email recipients

☐ SNMP

☐ SQL

☐ BTEQ

☐ Run a program

Alert Setup is available in the Administrative portlets

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If selecting an Alert Notification, you may select from a list of Alert Actions Sets that have been previously defined with Viewpoint’s Alert Setup available under the Admin portlets.

Event notifications: Alert action set

The screenshot shows the 'Alert Setup' window with the 'Action Sets' tab selected. The 'Action Sets' list contains 'Adminis...' and 'SM_LOG'. A '+' icon is circled in the top right of the 'Action Sets' tab. A callout box points to the 'Times' section, which has checkboxes for 'Core', 'Evening', and 'Weekend', all of which are checked. The 'Actions' section has checkboxes for 'Include in Alert Viewer', 'Email recipients', 'SNMP', 'SQL', 'BTEQ', and 'Run a program', with 'Include in Alert Viewer' checked.

Setup Options	Preset Options	Action Sets	Action Sets Details
Delivery Settings	Core Hours	Adminis...	Action Set Name: * Node Down
Alert Presets	Action Sets	SM_LOG	Times
Alert Exceptions	Groups		☒ Core
Database User Alerts			☒ Evening
			☒ Weekend
			Actions
			☒ Include in Alert Viewer
			☐ Email recipients
			☐ SNMP
			☐ SQL
			☐ BTEQ
			☐ Run a program

Click the “+” button to configure the Delivery Settings and Actions for the Alert.

After defining an Alert, it will be available in the **Send Alert pull down menu**.

Note: Alerts are setup in the Viewpoint server and are **not specific to a system**. Specify an Action Set Name that contains your team name to distinguish it from other teams' alerts.

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Select the “+” icon to add an Alert Action Set.

Note, that an alert action set can be defined as active at different “times”. This could potentially conflict with the Planned Environment definitions established by Workload Management’s State Matrix, resulting in alerts not working at certain times of the day. Therefore it is recommend that the time assigned to an action set that will be used by Workload Management be set to active for all alert times so that the Workload Management setting will prevail.

Event notifications: Run program and post to Qtable

- Run Program:
 - Requires the “**Teradata Notification Service for Windows**” to be installed on a Windows server
 - Requires the “**Teradata Notification Service for Linux**” to be installed on a Linux server
 - The executable programs are then installed on these servers
 - Need to then define an Alert Action set to run
- Post to Qtable:
 - Queue Table actions will **write a message** containing the **name of the event** combination that triggered the queue table action
 - The message will be written to the **DBC.SystemQTbl**
 - **Optional textual entries** can be logged at the start action and end action
 - **Queue tables** can only be written to (**pushed**) once and read from and deleted (**consumed**) once
 - If you have multiple applications that would like to use the information pushed to the queue table, you will need to duplicate the information into multiple queue
 - Additional documentation about the SystemQTbl can be found in the Data Dictionary User Manual

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Running programs on Microsoft® Windows® requires the “Teradata Notification Service for Windows” to be installed on a Windows Server. Running programs on Linux® requires the “Teradata Notification Service for Linux” to be installed on a customer provided Linux server. The executable programs are then installed on these servers. For detailed information see the *Teradata Alerts Installation, Configuration, and Upgrade Guide*.

Queue Table actions write a message containing the name of the event combination that triggered the queue table action. If using the Queue Table action type, you need to consider and plan for the applications that would like to utilize this information. There may be several applications that you want to enhance to take advantage of the information being posted to this queue. However queue tables can only be written to (pushed) once and read from and deleted (consumed) once. If multiple applications would like to take advantage of the information pushed to the queue, you will need to duplicate the information into multiple queue tables.

State transitions

When the Health Condition and/or Planned Environment change, the **system can transition** to another State, adjusting the rule set working values to that of the new state.

Terminology:

- **Rule** – a single Filter, Throttle, or Workload Definition
- **Working Values** – the attributes of a rule that can change based on the active state
- **Rule Set** – full set of workload definitions, filters, throttles and priority scheduler settings
- **Working Value Set** – a complete set of Working Values for a Rule Set

Changing workload management behaviors via a State transition within a State Matrix is more efficient than changing behaviors by downloading and activating an entirely new rule set.

State Transitions will cause queries in the **delay queues to be re-evaluated** against the limits of the new working values.

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Whenever there is a state transition, the delay queues need to be re-evaluated against workload operating rule changes. Note: Internal performance measures were done to assess any processing overhead of state transitions. The measures confirmed that state transitions have a negligible impact on performance.

Note that changing workload management behaviors via state transitions within the state matrix is far more efficient than changing workload management behaviors by enabling an entirely new rule set. In the latter case, interaction with the Workload Management Administrator is required to download and activate the new rule set, and far more re-evaluations are required to existing requests, delay queues, priority scheduler mappings, etc. The latter delay on a very busy system has been measured in some situations to be several minutes vs the negligible overhead of state transitions.

Rule sets and working values (1 of 2)

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Rule Set			
Rule Name	Fixed Attributes	Base State Working Values	State Y Working Values
Filter A	No access to Trxn Table	Disabled	Enabled
Throttle X	Query Limit on Profile Adhoc	Enabled at 15	Enabled at 5
WD 1	Classify by Profile DSS, Exception if Skew > 25%, Action change to WD 2 Mapped to Timeshare Top	Query Limit at 10, Exception Enabled	Query Limit at 5, Exception Enabled
WD 2	...	...	...
Exception	Exception if CPU to Disk > 10, Action Abort on Select	Disabled	Enabled

Here is an example of a Rule Set and the fixed and variable attributes.

Rule sets and working values (2 of 2)

- **Fixed Attributes** do not change when the state changes
 - Classification criteria
 - Exception definitions and actions
 - Position in the Priority Scheduler hierarchy
 - Evaluation order of the workload
- **Working Values** can change to meet the needs of a particular Health Condition or Planned Environment
 - Working Values that are **State dependent** and can be changed are:
 - Enable or Disable a rule
 - ❖ Session Control
 - ❖ Filter
 - ❖ System Throttle
 - System or Workload Throttle Limits
 - Working Values that are **Planned Environment dependent** are:
 - Service Level Goals
 - Enable or Disable Exceptions
 - Workload SLG Tier share percent or Timeshare access level
 - Minimum response time

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Working Values can change to meet the needs of a particular State or Planned Environment

Working Values that are State dependent are:

- Enabled/Disabled
 - Session Control
 - Filters
 - System Throttles
- System or Workload Throttle Limits

Working Values that are Planned Environment dependent are:

- Service Level Goals
- Enabled/Disabled Exceptions
- Workload SLG Tier share percent or Timeshare access level
- Minimum response time

Displaying working values (2 of 2)

Portlet: **Workload Designer** > Button: **States** > Button: **Click to view by state**

The screenshot shows the 'Workload Designer' interface with the 'Filters' and 'Throttles' sections expanded. The 'Filters' section contains a table with columns: ENABLED, NAME, CLASSIFICATION, and a detailed configuration row. The 'Throttles' section contains a table with columns: ENABLED, NAME, CLASSIFICATION, and a detailed configuration row. Both sections also include a table with columns: NAME, DEFAULT SETTING, and YELLOW.

ENABLED	NAME	CLASSIFICATION
☒	No_Collect_Stats	PROFIL: ADHOC_Profile COLLECT_Stats: True
☒	No_PJ	PROFIL: ADHOC_Profile MIN_SR: 1000000 JOIN_TYPE: Unconstrained Product Join

NAME	DEFAULT SETTING	YELLOW
No_Collect_Stats	ENABLED	(Use Default)
No_PJ	ENABLED	(Use Default)

ENABLED	NAME	CLASSIFICATION
☐	Adhoc_Limit	PROFIL: ADHOC_Profile
☐	DSS_Limit	PROFIL: DSS_Profile
☐	Report_Limit	PROFIL: Report_Profile

NAME	DEFAULT SETTING	YELLOW
Adhoc_Limit	1, DELAY	(Use Default)
DSS_Limit	15, DELAY	(Use Default)
Report_Limit	1, DELAY	(Use Default)

Displays the rules and **working values** associated with the **state**

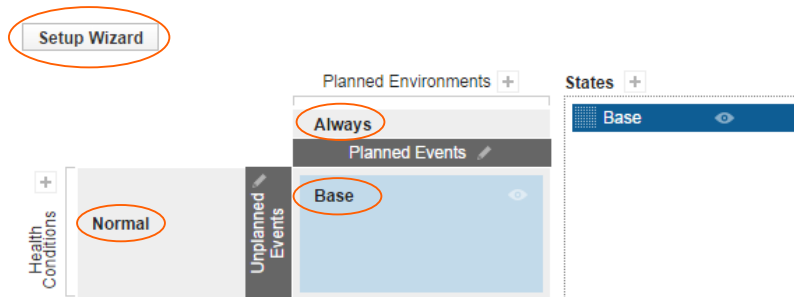
teradata.

After clicking the “eye” icon, a display of the rules and working values for that state will be displayed for:

- Sessions
- Filters
- Throttles

Default state matrix

Portlet: **Workload Designer** > Button: **States**



The default State Matrix is 1X1, consisting of:

- Health Condition of **Normal**
- Planned Environment of **Always**
- State of **Base**

Setup Wizard button will invoke the wizard to assist in creating the State Matrix

teradata.

By default, the State Matrix is 1x1: i.e., a single planned environment defined for 24 hours x 365 days a year ("always"), and a single Health Condition, defined for the "normal" health of the system. This Planned Environment and Health Condition pair points to a single default State called "Base".

Current topic: Setup wizard: Planned environments

- State matrix overview
- **Setup wizard: Planned environments**
 - **Getting started**
 - **Planned environments**
 - **Planned events**
 - **Period events**
 - **User defined events**
 - **Assigning planned events to planned environments**
- Setup wizard: Health conditions
- Setup wizard: States
- Lab activity

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Setup wizard: Getting started

Workload Designer TDT1 POWM Ruleset

State Wizard: Step 1 of 6

Getting Started

Your goal is to design a system regulation state matrix that represents your environment. A sample state matrix is shown below. Yours does not have to look like this, but if you are unsure of what to do, this would be a good starting point.

Health Conditions	Unplanned Events	Planned Environments	
		Always	BatchEnv
		Planned Events	
Normal		Base	OfflineState
DegradedC...		YellowState	OfflineState
CriticalC...		Red State	OfflineState

Planned Events
A planned event is anticipated, like a batch window or a scheduled time for month-end processing. One or more planned events can be combined into a planned environment.

Unplanned Events
An unplanned event is not scheduled and causes reactions in system regulation. A degraded system condition like a down node, or running out of AMP Worker Tasks are unplanned events that can cause a change in how a system allocates resources. One or more unplanned events can be combined into a health condition.

States
In the state matrix, planned environments and health conditions intersect to define states. A state is a set of operating rules and controls how your system will allocate resources. Each state matrix cell specifies the state the system will use when the planned environment and health condition combination exists. Reusing states is recommended. Defining too many states might become a maintenance burden.

CANCEL NEXT

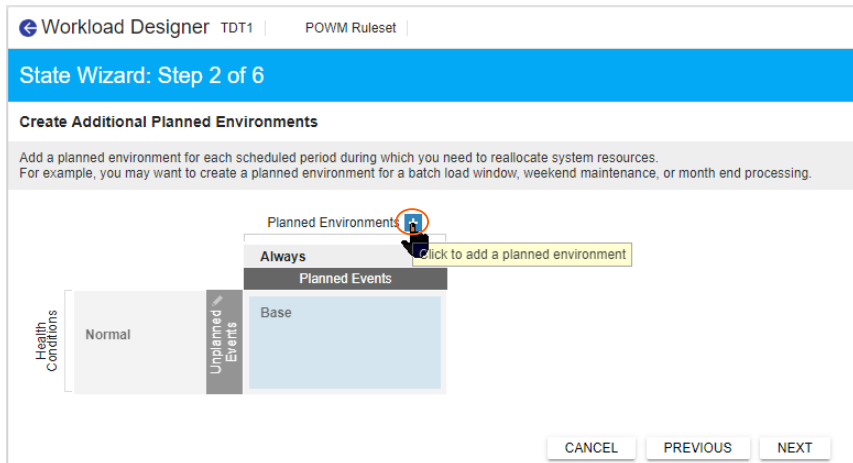
Describes the goal and components of the State Matrix

teradata.

From the initial State Matrix screen, clicking the Setup Wizard button will display the screen on the facing page. This will step 1 of 6.

Click the next button to go to Step 2.

Setup wizard: Planned environments



Click the “+” button to add one or more Planned Environments

teradata.

In Step 2 of the wizard, additional PLANNED ENVIRONMENTS can be added to the State Matrix.

Move the cursor over PLANNED ENVIRONMENTS to activate the + icon.

Click the + icon to add a PLANNED ENVIRONMENT.

Creating planned environments

Workload Designer TDT1 | POWM Ruleset |

State Wizard: Step 2 of 6

Create Additional Planned Environments

Add a planned environment for each scheduled period during which you need to reallocate system resources.
For example, you may want to create a planned environment for a batch load window, weekend maintenance, or month end processing.

Planned Environments	
Always	DataLoads
Base	

Planned Events	
Always	DataLoads
Base	

CANCEL PREVIOUS NEXT

After clicking the “+” button, a new Planned Environment will appear with the default name of “NewEnv”

- To change the default name, click the “pen” button
- To remove the Planned Environment, click the “trash can” button

teradata.

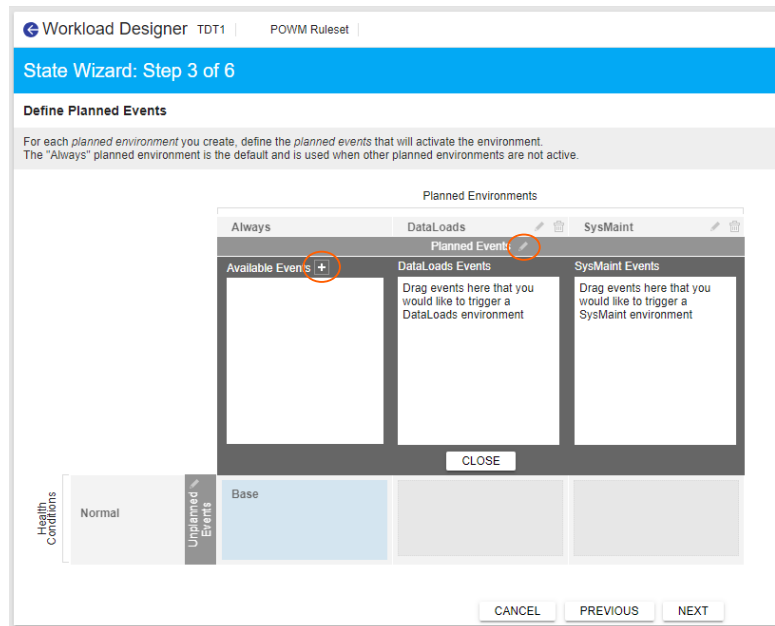
After clicking the “+” icon, a new Planned Environment will appear with the default name of “NewEnv”

To change the default name, click the “pen” icon. To remove the Planned Environment, click the “trash can” icon

Setup wizard: Planned events

Click the Planned Events “pen” button, and then click “+” button to create one or more Planned Events

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In Step 3 of the wizard, PLANNED EVENTS are going to be created.

Planned Events can be detected internal to Analytics Database such as specific time periods, or external to Analytics Database such as user defined event that load jobs are starting.

Click the + icon to create a PLANNED EVENT.

Creating period events

Period Events are **planned** and **scheduled**

To create a **notification only Event**, do not assign the Event to a Planned Environment

A **notification** can be sent when the Event **starts** and/or **ends**

Period Events should define contiguously

Daytime 8:00am to 5:00pm

Nighttime 5:00pm to 8:00am

Not

Daytime 8:00am to 4:59pm

Nighttime 5:00pm to 7:59am

Use the **Wrap Around Midnight** option to have a time range span midnight

teradata.

Create Event

Select Event Type

- ☒ Period Event
- ☐ User Defined Event
- ☐ Event Combination

Configure Event Details

Event Name:

Description:

☐ Create New Corresponding Planned Environment

Event Frequency and Duration

☒ Day of Week

☒ S	☒ S	☒ M	☒ T	☒ W	☒ T	☒ F	☒ S
---------------------------------------	---------------------------------------	---------------------------------------	---------------------------------------	---------------------------------------	---------------------------------------	---------------------------------------	---------------------------------------

☐ Day of Month

☒ 1	☒ 2	☒ 3	☒ 4	☒ 5	☒ 6	☒ 7
☒ 8	☒ 9	☒ 10	☒ 11	☒ 12	☒ 13	☒ 14
☒ 15	☒ 16	☒ 17	☒ 18	☒ 19	☒ 20	☒ 21
☒ 22	☒ 23	☒ 24	☒ 25	☒ 26	☒ 27	☒ 28
☒ 29	☒ 30	☒ 31				

Month of Year

☒ Jan	☒ Feb	☒ Mar	☒ Apr	☒ May	☒ Jun
☒ Jul	☒ Aug	☒ Sep	☒ Oct	☒ Nov	☒ Dec

☐ Start Time: AM End Time: AM

☒ Wrap around midnight

Configure Notifications

Start Action		End Action	
Send Alert:	☐ Administrator Default	☐ Administrator Default	
Run Program:	☐	☐	
Post to QTable:	☐	☐	

OK CANCEL

Period events are planned, scheduled events occurring on specific days and times, such as month-end financial processing. To create an event that only sends out a notification, create the event, but do not assign it to any planned environment. When the event occurs, the notification action you specified takes place.

You can define period events to indicate days and times when you would like a period event to be in effect. If the current time falls in the range of a period event, that event becomes active. When the current time falls outside of that time period, Workload Management deactivates the associated active planned environment, and other active events determine the current planned environment.

A period event can include:

- Time of day when the period event begins and ends
- Days and months when the period event is in effect

Creating user defined events

User Defined events are **triggered** based on **planned external conditions**.

To **Activate** and **Deactivate** the Event, execute the SQL statements given.

Events can **also** be **set with a duration time** the event will be active.

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Create Event

Select Event Type

- ☐ Period Event
- ☒ User Defined Event
- ☐ Event Combination

Configure Event Details

Event Name:

Description:

☐ Create New Corresponding Planned Environment

Activate/Deactivate Event

To activate this event, use this SQL request:

To deactivate this event, use this SQL request:

Configure Notifications

Start Action		End Action	
Send Alert:	☐ Administrator Default	☐ Administrator Default	
Run Program:	☐	☐	
Post to QTable:	☐	☐	

OK CANCEL

User-defined events let users trigger their own events. User-defined events can be planned or unplanned. To create an event that only sends out a notification, create the event, but do not assign it to any planned or unplanned environment. When the event occurs, the notification action you specified takes place.

There are three basic use cases for using user-defined event types:

To convey external system condition events:

As an example, consider that a single Analytics Database may be part of an enterprise of systems that may include multiple Analytics Databases cooperating in a dual-active role, various application servers and source systems. When one of these other systems in the enterprise is degraded or down, it may in turn affect anticipated demand on the Analytics Database. An external application can convey this information by means of a well-known user-defined event via open APIs to the Analytics Database. The Analytics Database can then act automatically, for example, by changing the system condition and therefore the state, and employ different workload management directives appropriate to the situation.

To convey business-oriented events:

Many businesses have events that impact the way an Analytics Database should manage its workloads. For example, there are business calendars, where daily, weekly, monthly, quarterly or annual information processing increases or changes the demand put on the Analytics Database. While period event types provide alignment of a fixed period of time to some of these business events, user-defined events provide the opportunity to de-couple the events from fixed windows of time that often do not align accurately to the actual business event timing.

For example, through the use of a period event defined as 6PM til 6AM daily, you could define an event combination that changes the Planned Environment to "LoadWindow" when the clock

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ticked 6PM. However the actual source data required to begin the load might be delayed, and therefore the actual load may not begin for several hours. Also, it is typical to define the period event to encompass far more hours than the actual business situation will require just to compensate for these frequently experienced delays. Even then, sometimes the delays are so severe that the period transpires while the load is still executing, leading to workload management issues.

However if instead of using a period event, you could define a user-defined event called "Loading". The load application could activate the event via an OpenAPI call prior to the load commencing, and de-activate it upon completion. The end result is that workload management is accurately adjusted for the complete duration of the actual load processing, and not shorter or longer than that duration.

Note that period events are not capable of operating on a business calendar, for example, that includes holidays, end-of quarter dates, etc. However they can be conveyed to the Analytics Database through user-defined events.

To enhance workload management capabilities through an external application:
An external application, through the use of PM/API and OpenAPI commands or other means, can monitor the Analytics Database for key situations that are useful to act on. Once detected through the use of the external application, the event can be conveyed to the Analytics Database in the form of a user-defined event, for example, to change the Health Condition and therefore the State of the system. (Generally utilizing an action type of notification has limited value-add here because the external application could have provided that notification directly without involving Workload Management. The real value is in automatically invoking a more appropriate state associated with the detected event.)

See the **Teradata Vantage™ Application Programming Reference** for more details on using the TDWMEventControl procedure.

Creating event combinations

More complex event definitions, consisting of logical expressions of **multiple single events** can be created

When considering the logical expression of an Event Combination, **consider simpler** rather than more complex expressions

Event Management is used to **facilitate Gross-Level** Workload Management. Complex Event Combinations are an indication you may be using Event Management in a way it was not intended

teradata.

The screenshot shows the 'Create Event' dialog box. Under 'Select Event Type', 'Event Combination' is selected. In 'Configure Event Details', there are fields for 'Event Name' and 'Description', and a checkbox for 'Create New Corresponding Planned Environment'. Below these are two panes: 'Available Events' and 'Event Combination Formula'. An 'INSERT >>' button is between them. The formula pane contains logical operators: AND, OR, NOT, and parentheses. At the bottom, the 'Configure Notifications' section has three rows: 'Send Alert', 'Run Program', and 'Post to QTable', each with a checkbox and a dropdown menu set to 'Administrator Default'. 'OK' and 'CANCEL' buttons are at the bottom right.

An event combination is a mix of two or more different events, such as period, system, and user defined events. Event combinations can be planned or unplanned. To create an event that only sends out a notification, create the event, but do not assign it to any planned environment. When the event occurs, the notification action you specified takes place.

When considering the logical expression of an event combination, consider simpler rather than more complex expressions. This simplification is further aided by the fact that you can have multiple event combinations cause the same change in Health Condition or Planned Environment.

Also consider that the added logical combination capabilities of OR-ing and parenthesis is really to facilitate future Event Types yet to become available in Workload Management. In practice, you will rarely need to even use the 'AND' capabilities unless combining a user-defined event with any of the Period, AMP Activity, Components Down or another user-defined type events. For example,

If daytime period AND LOADING (user defined event)

Remember that Event Management is to facilitate Gross-Level Workload Management, so if you find yourself using a lot of complex event combination logical expressions, you are probably trying to use Event Management in a way it was not intended, ie: very specific and granular workload management.

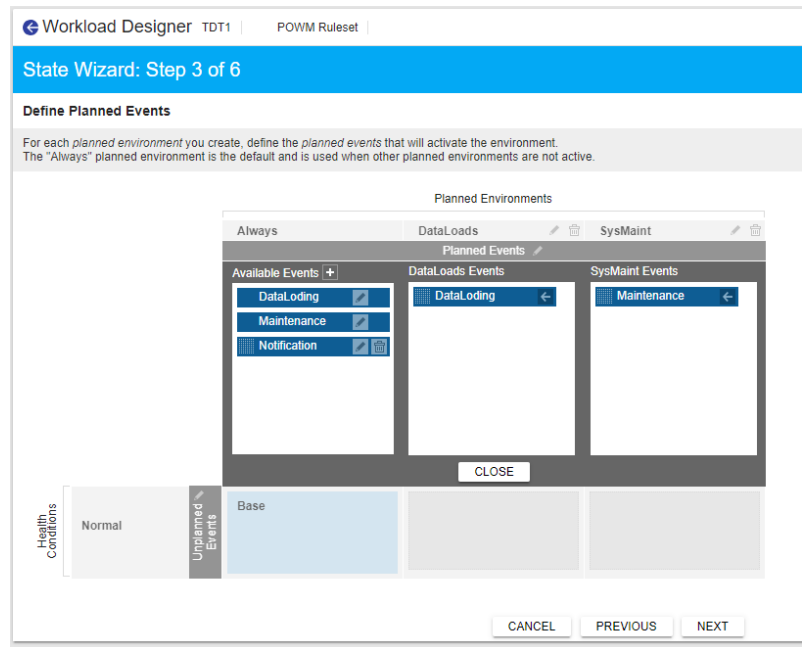
Assigning planned events

After creating your Planned Events, drag and drop them to the Planned Environment you want to be current when the event is detected.

If an Event is not assigned, it will be detected and the notification will be sent, but the no change in Planned Environment will occur.

Remember that the Planned Environment to the furthest right takes precedence.

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After creating your Planned Events, drag and drop them to the Planned Environment you want to be current when the event is detected.

By assigning an event to a Planned Environment, when the event is detected, the corresponding Planned Environment will be current.

If an Event is not assigned, it will be detected and the notification will be sent, but the no change in Planned Environment will occur.

If two events have date/times that overlap, and each one is assigned to a different Planned Environment, the Planned Environment to the furthest right takes precedence, making it the current Planned Environment.

Current topic: Setup wizard: Health conditions

- State matrix overview
- Setup wizard: Planned environments
- **Setup wizard: Health conditions**
 - **Health conditions**
 - **Unplanned events**
 - **Unplanned event types: System, I/O usage, workload**
 - **Assigning unplanned event to health conditions**
- Setup wizard: States
- Lab activity

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Setup wizard: Health conditions

Workload Designer TDT1 POWM Ruleset

State Wizard: Step 4 of 6

Create Health Conditions

Health Conditions define levels of system health. The default condition is *Normal*.
Add additional health conditions to change the allocation of resources when an event such as running out of AMP Worker Tasks, or flow control degrades system health.

Planned Environments	
	Always DataLoads SysMaint
Health Conditions	Normal
	Unplanned Events
	Base

CANCEL PREVIOUS NEXT

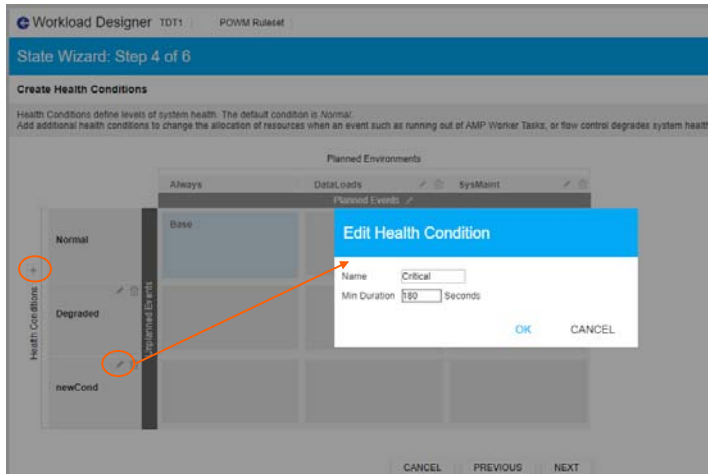
Click the “+” button to add one or more Health Conditions.

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In Step 4 of the wizard, additional HEALTH CONDITIONS can be added to the State Matrix.

Health conditions are levels of system health. The default system condition is “Normal

Creating health conditions



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After clicking the “+” button, a new Health Condition will appear with the default name of “NewCond”.

To change the default name, click the “pen” button.

Min Duration specifies the minimum amount of time the Health Condition **must remain Active** even if the event that triggered the Health Condition is no longer active.

Health Conditions activated by Unplanned Events, **recommend** to set the minimum duration to **10 minutes or greater**

When you create a Health Condition, you must give it a name and a minimum duration.

Minimum Duration: If some level of system resources hovers at the values that activate an event and that event causes a state change, a state change will occur each time the level goes above or below the threshold. To minimize this effect, a Minimum Duration must be entered.

This means that the health condition remain active for the Minimum Duration, even if the event that caused it is no longer active. If some other event combination comes true that activates a health condition with a higher severity, the higher severity system condition will become active immediately. The default Minimum Duration is 180 seconds.

Setup wizard: Unplanned events

Workload Designer TDT1 | POWM Ruleset

State Wizard: Step 5 of 6

Define and Map Unplanned Events

For each health condition that you create, define the unplanned events that will activate the condition. The Normal health condition is the default and is used when other health conditions are not active.

Planned Environments

Always	DataLoads	SysMaint
Base		

Health Conditions

- Normal
- Degraded
- Critical

Unplanned Events

Available Events +

Degraded Events

Drag events here that you would like to trigger a Degraded health condition

Critical Events

Drag events here that you would like to trigger a Critical health condition

CLOSE

CANCEL PREVIOUS NEXT

Click the Unplanned Events “pen” button, and then click “+” icon to create one or more Unplanned Events.

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In Step 5 of the wizard, UNPLANNED EVENTS are going to be created. Unplanned Events can be detected internal to the Analytics Database.

Creating system events

System Events are internally **triggered** based on **performance or availability**

Component Down Events are detected at system startup

System Wide Events are detected after a **qualified** amount of time has passed

AMP Activity Level Events are detected after a **qualified** amount of time has passed

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The screenshot shows the 'Create Event' dialog box. The 'System Event' radio button is selected. In the 'Configure Event Trigger' section, the 'System Event Type' dropdown is open, showing options like 'Node Down', 'AMP Fatal', 'Available AWTs', 'Gateway Fatal', 'PE Fatal', 'Flow Control', 'CPU Utilization', 'CPU Skew', and 'I/O Usage'. The 'End Action' dropdown is set to 'Administrator Default'.

The current release of TASM offers the following Event Types:

Component Down Event Types, detected at system startup:

- Node Down: Maximum percent of nodes down in a clique.
- AMP Fatal: Number of AMPs reported as fatal.
- PE Fatal: Number of PEs reported as fatal.
- Gateway Fatal: Number of gateways reported as fatal.

System Wide Event Types: To avoid unnecessary detections, these must also specify a qualification.

- CPU Utilization: Defines when the system CPU values are consistently outside defined CPU utilization threshold values. Must also set a qualification.
- CPU Skew: Maximum system wide skew. Must also set a qualification.

AMP Activity Level Event Types: To avoid unnecessary detections, these must also specify a qualification.

- Available AWTs: Minimum Number of AWTs available per AMP detected at any point within in the interval across all AMPs
- Flow Control: Number of AMPs in flow control.

Qualification: Qualification times can prevent very short incidents from triggering events. It is the time the condition must persist before the event is triggered.

- Simple: Specifies how long an event threshold must be met before an event is triggered.
- Immediate: Specifies that an event is triggered immediately after an event threshold is met.

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- Averaging: Specifies how long the rolling average of the metric value must meet the event threshold before an event is triggered.

System event types: Component down events

- **Node Down** – maximum percentage of nodes down within a clique
 - When a node fails, the VPROCs migrate to other nodes within the clique increasing the workload of those nodes still up
 - Very **large systems** are sized to maintain expected performance levels with nodes down, so a percentage of **>25%** may be a good threshold
 - For **smaller systems** a threshold of **24%** may be good

Configure Event Trigger

System Event Type:

Node Down ▼

% of clique:

24

- **AMP/PE/Gateway Fatal** – number of VPROCs that are fatal

Configure Event Trigger

System Event Type:

AMP Fatal ▼

Limit:

1

Configure Event Trigger

System Event Type:

PE Fatal ▼

Limit:

1

Configure Event Trigger

System Event Type:

Gateway Fatal ▼

Limit:

1

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The node down event type allows the definition of the maximum percentage of nodes down within a clique before the event triggers.

When a node goes down, its VPROCs migrate, increasing the amount of work required of the nodes that are still up. This translates to performance degradation. When the system is performing in a degraded mode, it is not unusual to want to throttle back lower priority requests or enable filters to assure that critical requests can still meet their SLGs.

Alternatively or in addition, you may want to send a notification so that follow-on actions can occur.

Specific Event Types are defined for AMP, PE (parsing engine) and Gateway VProcs. These events detect the specified VProc being fatal at system startup only. These event types are similar to node down except the user only defines the number of VProcs to trigger on.

Event qualification time (1 of 2)

- Some System Events require a **Qualification Time** before triggering an **Event Action**
- System Wide and AMP Activity System events are considered at **every Event Interval** setting (e.g., 60 seconds)
- Event **metrics** are **checked** using data that was accumulated from the **last event interval until the next** event interval
- **To avoid false detections**, these events must be qualified through a sustained metric reading
- There are three methods available to qualify events:
 - **Simple Qualification** – **requires the event to persist** beyond the threshold for the specified qualification time
 - **Averaging Qualification** – better choice **for events with highly fluctuating patterns**. It uses an un-weighted moving average to smooth out peaks and valleys and can better distinguish between a temporary utilization pattern and persistent utilization pattern
 - **Immediate Qualification** – **requires no persistence** and the event is detected once the threshold is met

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Event criteria metrics are checked for using data that has accumulated from the last event interval until the next event interval. To avoid false event detections, some events must be qualified through a sustained metric reading. There are up to three methods offered for qualifying events.

Simple Qualification: Simple qualification requires the event to persist for the specified qualification time based on consecutive event metric readings all beyond the specified threshold. The qualification time counter begins accumulating at the end of the interval where the event was first detected. Then, in order for the event to be qualified as active, the associated event must be continue to be detected repeatedly in any subsequent event checks until the qualification time counter is exceeded.

Averaging Qualification: The metrics associated with certain event types have highly-fluctuating patterns. These event types are more effectively detected using the averaging qualification method. Instead of requiring the metric to persist consistently beyond the specified threshold, it requires the moving average of the metric to measure beyond the specified threshold. In this way it smooth's out the peaks and valleys seen in these event metrics.

Immediate Qualification: Immediate essentially requires no persistence to activate the event. Once the associated metric measures beyond the specified threshold, the event is activated.

Event qualification time (2 of 2)

- Simple should be considered for the following event types:
 - Available AWTs
 - Flow Control
 - Workload AWT Wait Time
 - Workload Active Requests
- Averaging should be considered for the following event types:
 - System CPU Utilization
 - System CPU Skew
 - Workload CPU Utilization
 - Workload Arrivals
 - Workload SLG Response Time and SLG Throughput
- Simple or Averaging is not offered for the following event types:
 - Component Down events (Node, AMP, PE and Gateway)
 - User Defined events
 - Delay Queue Depth
 - Delay Queue Time
- Although it is available, **Immediate is not recommended** for use with any other events

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The slide lists recommendations on using Simple, Averaging and Immediate qualifications.

AMP activity level events: Available AWTs

Available AWTs – number of AWTs available on the specified number of AMPs.

Note: Remember that the total size of the Available AWT pool is defined in the Other tab of the General view.

- The number of AMPs that will be required to be at the specified threshold
- If **no AWTs are available** to support new work, messages will be queued
- Most systems reach **100% CPU** utilization with as little as **40 AWTs in use** servicing new or spawned work
- AWT usage can provide an **early indicator** of performance degradation

Configure Event Trigger

System Event Type: Available AWTs ▼

of AMPs: 1

Available AWTs: 0

Qualification Method: Simple ▼

Qualification Time: 1 Minute ▼

Trigger this event v Immediate old above is met for this time period.

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The AWT available event type allows the user to define the threshold for the number of AWTs that are available to support new work on the worst AMP in the system. The user can select one or more AMPs that will be required to be at that threshold for the event to be triggered.

Note that if a qualification time is given on this event, the threshold does not have to be maintained on the same AMP over the entire qualification interval. That is, if a threshold of two is defined, and this threshold is crossed by AMP A on one event sampling and AMP B on the next event sample, then the event condition is considered to be maintained across the two samples.

AMP activity level events: Flow control

Flow Control – the total number of AMPs that have been reported being in flow control during the event interval

- The number of AMPs that will be required to be at that threshold can be specified
- There is no difference if the AMP was in flow control for 1 millisecond or for the entire event interval
- It is not required that the same AMPs report being in flow control just the number of AMPs

Configure Event Trigger

System Event Type: Flow Control ▼

of AMPs: 1

Qualification Method: Simple ▼ Qualification Time: 1 Minute ▼

Simple

Immediate

i Trigger this event when the condition above is met for this time period.

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The Flow Control event type allows for the definition on the number of AMPs that have reported being in Flow Control during the sampling interval. This condition includes any time spent in flow control as well as currently being in flow control. There is no differentiation between being in flow control for 1 millisecond and being in flow control for the entire interval. As with the AWT available event, this event does not require that the same set of AMPs reports flow control between data samples.

System level events: CPU utilization

CPU Utilization – system-wide average of node CPU busy percentages

- Indicator of how busy the system is
- If CPU percentage is high, consider enabling throttles on low priority work
- If the CPU percentage is low, consider disabling throttles on low priority work

Configure Event Trigger

System Event Type: CPU Utilization ▼

CPU Percent: >= ▼

Qualification Method: Simple ▼ Qualification Time: 1 Minute ▼

Simple
Immediate
Averaging

Trigger this event when the condition above is met for this time period.

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The System CPU utilization event is based on the system-wide average of node CPU busy percentages. It is an indicator of how busy the system is: Does it have capacity to do more work, or is it effectively running at its peak capabilities?

System level events: CPU skew

CPU Skew – system-wide detection of **node skew**

- Used to detect skew due to **workload imbalance**
- For **coexistence systems**, adjust the threshold to accommodate for built-in system imbalance

Configure Event Trigger

System Event Type: CPU Skew ▼

Skew Percent: >=

Qualification Method: Simple ▼ Qualification Time: 1 Minute ▼

Simple
Immediate
Averaging

Trigger this event v... old above is met for this time period.

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Using exception processing, TASM can detect when an individual query is skewed so that a targeted action can be taken with regards to the skewed query. However, exception processing cannot detect a system-wide skew; such as, one associated with session balance issues or an application running on a single node of the configuration. The system skew event can detect when a skew occurs for any reason. When detected, a typical automated action is to send an alert to the DBA to investigate and act manually.

If you are using system skew events on a coexistence system, adjust the triggering threshold to a value appropriate for the built-in system imbalance. For example, suppose in a perfectly balanced workload environment, the typical utilization of 10 old nodes is 95% when 10 new nodes are maxed out at 100%. Here the built-in “system skew” level is $(100 - 95) / 100 = 5\%$. Set the system skew triggering threshold to a value that measurably exceeds 5%.

Creating workload events

Workload Events are **specific to a workload**

Note: SLG event types will only appear if the corresponding SLG has been specified for the workload

Create Event

Select Event Type

- ☐ System Event
- ☐ User Defined Event
- ☒ Workload Event
- ☐ Event Combination

Configure Event Details

Event Name:

Description:

Configure Event Trigger

Workload Name:

Workload Event Type:

Count:

Qualification:

Trigger this event when set for this time period.

Configure Notifications

	Start Action	End Action
Send Alert:	☐ NodeDown	☐ NodeDown
Run Program:	☐	☐
Post to QTable:	☐	☐

OK Cancel

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The current release of TASM offers the following Event Types:

Active Requests

Defines maximum or minimum number of queries that can be active at one time. Active Requests are not available for utility workloads.

Arrivals

Defines maximum or minimum per-second arrival rate for queries. Arrivals are not available with utility workloads.

AWT Wait Time

Defines minimum time a step in a request can wait to acquire an AWT.

CPU Utilization

Defines maximum or minimum CPU usage for a query.

Delay Queue Depth

Defines minimum number of queries in the delay queue.

Delay Queue Time

Defines a minimum for the time a request can be in the delay queue. The threshold can include or exclude system throttle delay time.

SLG Response Time

Workloads response time SLG was missed

SLG Throughput

Workloads throughput SLG was missed

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Workload event types

Workload Level Event Types are specific to a workload and should only be created on **key-indicator workloads** not all workloads

- **Active Requests** – monitor the **number of concurrent requests** that are **actively executing** within a workload
 - Does **not include** just logged on sessions or requests held in the delay queue sessions
 - Can be used to **detect** high or low **concurrency levels**
 - Useful to detect when the **penalty-box workload** concurrency level is too high
- **Arrivals** – the **total number** of SQL requests **classified into a workload**
 - Does **not include** change workload exceptions
 - Can be used to indicate arrival **surges** or **lulls**
- **AWT Wait Time** – monitors if a workload is encountering delays in obtaining an AWT
 - If a tactical workload is encountering a delay, then **possibly enable throttles** on low priority workloads
- **CPU Utilization** – monitors CPU utilization for specific workloads
 - Only monitor **key-indicator workloads** that fall above or below a defined threshold
 - Can **enable or disable throttles** on low priority workloads

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The Active Requests event type allows for monitoring the number of concurrent requests that are active within a specific WD. Concurrency is defined as active executing request, not just logged on sessions, and not including requests held in the delay queues.

A sustained concurrency level is an indicator of either unmanaged arrival rate surges and lulls or other situations that can lead to unusual concurrency. Higher active concurrency levels lead to exhaustion of resources. Primarily of concern is the exhaustion of critical shared resources such as AWTs, memory, and physical spool, as well as the undesirable effects associated with being at extremely high concurrency where you end up in flow control or congestion management.

The Arrivals event type allows for the definition of an event based on the number of SQL requests classified into a WD. The arrival rate is defined within the event and is expected to be consistent over the interval as it is tested on each event interval. The arrivals are the total number of SQL requests classified into a WD, without adjustment for change-WD exceptions, regardless of whether they get queued due to a throttle.

If the system is low on AWTs or even in flow control, it is difficult to determine if that is causing performance issues on your critical workloads. The AWT wait time event detection is more actionable because it can detect if a particular workload is encountering delays obtaining an AWT. Different automated actions are appropriate depending on the workload and the length of the delay. Longer delays are acceptable for lower priority work, but nearly any delay at all for tactical work can be unacceptable.

Compared to System CPU utilization event detection, WD CPU utilization detections are associated with a specific WD. This enables more specific and appropriate automated or manual actions to address the underlying metric.

CPU utilization detections enable the DBA to proactively solve issues before they worsen. For

example, when CPU utilization of a key-indicator workload exceeds or falls below the defined threshold, it can be due to a symptom, such as demand surge, system overload, etc. When detected, actions can be taken to address the underlying issue before it results in unacceptable performance.

Workload event types (cont.)

Workload Level Event Types are specific to a workload and should only be created on **key-indicator workloads** not all workloads

- **Delay Queue Depth** – monitor the **number of queries** currently delayed
 - Useful to **detect** when queries are encountering **long delays**
 - Possibly **check the throttle limits**
 - Could be useful to **inform users** to expect longer response times
- **Delay Queue Time** – Based on the **longest amount of time** a query has been waiting in the delay queue
 - Delay time can optionally include or exclude time incurred while delayed only on a system throttle
- **SLG Response Time** – monitors **service percent** of the queries within a workload
 - Can set a response time and percentage goal
 - Can adjust resource usage if goal is not being met
- **SLG Throughput** – monitors **number of queries per hour**
 - Is measured based on sufficient demand, arrival rate > throughput rate
 - If arrival rate is less than throughput rate this event will not be triggered

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The two Delay Queue event types allow the monitoring of TDWM delay queues. These events allow for monitoring of overall number of requests and the time held of the oldest request for queries classified into a specific WD.

It is useful to know when a workload is encountering long delays. Long delays are often indicative of longer response times and/or that work is backing up. There are two methods provided to detect when a workload is encountering long delays caused by throttle limits:

- Queue Time – Based on the amount of time the longest request in the delay queue has been waiting. Delay time can optionally include or exclude time incurred while delayed only on a system throttle.
- Queue Depth: Based on the number of entries currently delayed.

SLGs allow the DBA to gauge the success of a workload's performance, and to note trends with respect to meeting those SLGs. Most SLGs are based on response time with a service percentage, such as < 2 seconds 90% of the time. Occasionally, SLGs are based on throughput (completions), such as > 100 queries per hour. Many investigations are triggered based on knowing that SLGs were being missed, enabling the DBA to do what is necessary to bring the workload's performance back to SLG conformance. The SLG response time event type allows the DBA to trigger specified automated actions when a SLG has been missed.

Whether there is an issue with just one particular workload, an issue with a set of common workloads, or whether there is an issue with all workloads is a valuable distinction. Individual SLG misses or a set of common workload misses suggest a workload management problem that can be solved by adjusting resource usage through the various TASM control mechanisms. When generally all WDs are missing their SLGs, this suggests a capacity problem that cannot be solved through refinement of TASM controls.

If the event type is SLG throughput, a throughput miss is internally qualified based on sufficient

demand as measured by workload arrivals and inter-workload movement (due to change-WD exception actions). Generally speaking, missed throughput has two causes:

- System Overload – If Arrival Rate > Throughput SLG, then the cause of the missed SLG is system overload. Not only is the system falling behind and unable to keep up with arrivals to this workload, other competing workloads may be impacting the ability to at least deliver the throughput SLG.
- Under-Demand – If Arrival Rate $\leq$ Throughput SLG, then the cause of the missed SLG is under-demand. In other words, there is insufficient demand from the application servers to realize the throughput SLG. The system could be nearly idle and still miss the throughput SLG, so you should pre-qualify the missed SLG throughput event with arrivals > throughput SLG.

TASM only triggers the SLG throughput event if it is due to system overload. You will not see either of these events in Viewpoint Workload Manager unless the SLG has been defined for the workload.

Unplanned event guidelines

Events that **result in performance degradation**, such as Node Down, CPU Utilization or AWT Exhaustion, consider:

- **Throttling back** lower priority work
- **Reassigning Priority** Distribution
- **Enabling Filters** to reject non-priority requests
- **Send Notifications** for follow-on actions

For large systems designed to run with some amount of degradation (hundreds of nodes), set the threshold number high enough to only activate the Node Down event when the degradation exceeds what the system was sized for.

For AWT shortages, most systems will reach 100% CPU utilization prior to AWT Exhaustion. Consider creating an event to detect when the available AWTs fall **below 7-15 persistently** for at least **180 seconds** of qualification time.

For AMPs in Flow Control, consider a qualification time of **30-60** seconds if pure **DSS**. For Tactical work, consider a shorter qualification time of **15-30** seconds on at least **1-2% of the AMPs**.

If multiple Planned Environments or Health Conditions are made active by events, the active State is determined by the higher Planned Environment precedence or Health Condition severity.

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Node down guidelines, consider that when a node goes down, its VPROCs migrate, increasing the amount of work required of the nodes that are still up. That translates to performance degradation. When your system is performing in a degraded mode, it is not unusual to want to throttle back further lower priority requests or reassigned priority distributions or enabling filters to assure that critical requests can still meet their Service Level Goals. Alternatively or in addition, you may want to send a notification so that follow-on actions can occur.

If your system is designed to run with some amount of degradation (for example, many very large systems with hundreds of nodes may be sized expecting that there is always a single node down somewhere in the system) it is suggested to set the threshold such that the Node Down event will activate only when that degradation exceeds what was sized for. For example, if the example system above were sized to meet workload expectations as long as any 3 node or smaller clique did not experience a down node, you might set your Node_Down event to activate at a threshold > 25%. Assuming your system is NOT designed to expect nodes down (as is the case with many small to moderate sized systems), a good threshold to set Down_Nodes Event Type Threshold to is roughly 24%.

The Analytics Database has the reputation of being a throughput engine, of being able to perform well under stress, and to respond productively no matter what work is demanded of it. AWT Management is part of that success. Having a shortage of AWTs or being in flow control identifies that there is some degree of pent-up demand being experienced and that Analytics Database is managing it. Note that the Analytics Database does not shut down when you run out of AMP worker tasks or are in flow control. You can keep throwing work at Analytics Database, and messages that cannot be serviced at that time on that particular AMP will either wait or be unobtrusively re-sent.

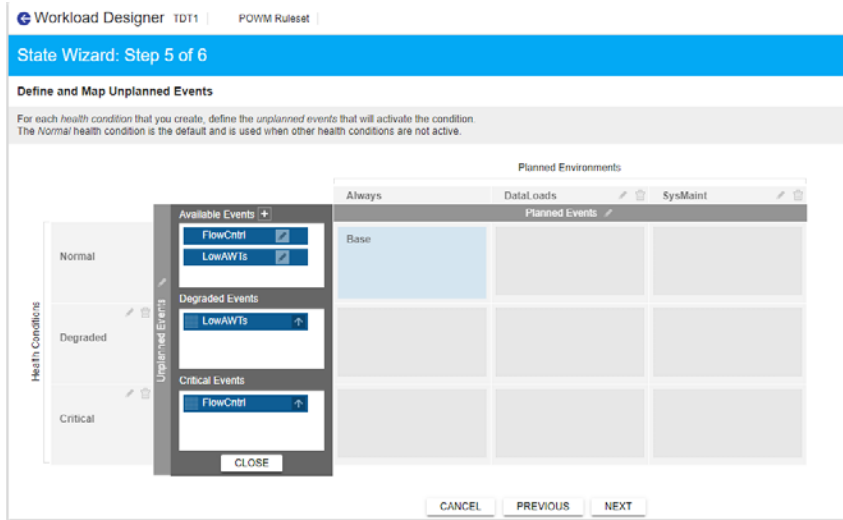
If AWT shortages are being detected, analyze your corresponding CPU and I/O utilization metrics. Most Vantage Analytics Database systems will reach 100% CPU utilization while there

are still plenty of AWTs available in the unreserved pool. Some sites experience their peak throughput when as little as 40 AWTs are in use servicing new or spawned work. By the time most systems are approaching a depletion of the pool, they are already at maximum levels of CPU usage. Most likely you are not bottlenecking on AWTs, rather you are already bottlenecked on CPU or I/O resources. Because AWT shortages are a rough gauge of concurrency level, it can be used effectively to provide notification of slow response time expectations.

Recommendation: Consider setting up an event combination for when Available_AWTs falls below a threshold of about 7-15 persistently for at least 180 seconds of qualification time.

Consider setting up an event for when AMPs are Flow Controlled for at least the 30-60 seconds of qualification time if the environment is pure decision support. If the environment is an ADW one containing a tactical work, consider a shorter qualification time of 5-10 seconds. Further qualify that to be on at least 1-2% of the AMPs to avoid insignificant detections.

Assigning unplanned events



After creating your Unplanned Events, drag and drop them to the Health Conditions you want to be current when the event is detected

If an **Event is not assigned**, it will be detected and the **notification** will be sent, but the no change in Health Condition will occur

Remember that the Health Condition in the lowest position takes precedence

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By assigning an event to a Health Condition, when the event is detected, the corresponding Health Condition will be current.

If an Event is not assigned, it will be detected and the notification will be sent, but the no change in Health Condition will occur.

If two events are detected at the same time, and each one is assigned to a different Health Condition, the Health Condition in the lowest position takes precedence, making it the current Health Condition.

Current topic: Setup wizard: States

- State matrix overview
- Setup wizard: Planned environments
- Setup wizard: Health conditions
- **Setup wizard: States**
 - **State guidelines**
 - **Creating and assigning states**
- Lab activity

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Setup wizard: States

Workload Designer TDT1 | POWM Ruleset

State Wizard: Step 6 of 6

Map States

Each cell in the state matrix represents the intersection of a health condition and a planned environment. Assign a state to each cell to determine how system resources are allocated. The same state can be used in multiple cells. For example, you might use the same state in your "BatchLoad" planned environment for all health conditions. For each cell below, drag a state name into the cell. If desired, you can create new states by clicking the plus icon at the top of the list of states.

		Planned Environments		
		Always	DataLoads	SysMaint
Health Conditions	Normal	Base	Drag states here that you would like triggered by Normal/DataLoads	Drag states here that you would like triggered by Normal/SysMaint
	Degraded	Drag states here that you would like triggered by Degraded/Always	Drag states here that you would like triggered by Degraded/DataLoads	Drag states here that you would like triggered by Degraded/SysMaint
	Critical	Drag states here that you would like triggered by Critical/Always	Drag states here that you would like triggered by Critical/DataLoads	Drag states here that you would like triggered by Critical/SysMaint

States 
Base

CANCEL PREVIOUS
FINISHED

Move the cursor over State to activate the "+" button

Click the "+" button to add additional States

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In Step 6 of the wizard, States are going to be created.

A *state* is the intersection of a health condition, which is composed of unplanned events, and a planned environment, which is composed of planned events. Creating states provides greater control over how the system allocates resources. When a health condition and a planned environment intersect, the state triggers system changes.

State guidelines

- Unless there is a **clear-cut need** for additional states, simply utilize the default state, “Base”
- **Consider** creating **additional Health Conditions or Planned Environments** first while still associated with the default state and monitor performance for each Health Condition/Planned Environment pair
- As the **need for additional states** becomes apparent, try to **keep the number** of Health Condition or Planned Environment related state transitions down to **1 to 3 per day**
- Reasons for additional states:
 - Consistent **peak workload** periods where priority management must be more strictly assigned to different workloads
 - The existence of **dependent processing periods** where the current processing period must complete before the next processing period can begin
 - Possibility that **system health degradation can impact** business **critical workloads** if they are not provided adequate resources by preventing and/or reducing lower priority workloads

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If there are not clear-cut needs for managing with additional states, as may be the case when you first dive into workload management, it is recommended to simply utilize the default state, “base”, referenced by the default <“Always”, “Normal”> planned environment and health condition pair. While the overhead to change states is minimal, workload management complexity is minimized when there are fewer states to consider. Monitoring for performance trends is also more clear-cut with fewer states.

As the need for additional states becomes apparent, add them, but keep the total number of states to a minimum. For example, do not have 24 planned environments with 24 states in a 24 hour day, but instead try to keep the number of planned environments related state transitions per day down to 1 - 3. Likewise, if you consider that there are eight unique ways for the system to be considered in degraded health, do not create eight unique health conditions and related states. Because the state matrix supports gross level, not granular level, system management, consider instead the degree of associated degradation, and create just 1 or two new health conditions to represent all eight system degradation scenarios.

So what are valid reasons for a distinct health condition, planned environment and state?

- There exists consistent peak workload hours (or days), where priority management must be more strictly assigned to the highest priority work, with background type work given little resources.
- The existence of Load or Query Windows where one workload must receive priority in order to complete within the critical window. For example, the accuracy and/or performance of subsequent query results in the next operating period depend on the completion of a load in the current operating period. If queries need to operate against data that is up to date, including the previous night's load, the previous night's load must be complete before the queries start. Or perhaps its simply a matter of performance, where queries need more

isolated access to the tables in order to perform to Service Level Goals, and a competing use of the table (as in a load or batch queries) result in longer response times.

- There exists a possibility that system or enterprise health degrading can impact the business if critical workloads are not provided adequate resources. When this occurs, priority management, filters and throttles can be employed to limit resources to lower importance work so that the critical workloads can be provided the resources they need.

Consider creating and monitoring new Health Condition(s) or Planned Environment(s) first while still associating it with an existing State. Monitor performance of each new <Health Condition, Planned Environment> pair first to see if associating it with an unique State is merited. For example, if you created a new Health Condition of “degraded” in a state matrix that already consisted of the default “always” Planned Environment to represent typical daytime activity, plus a load Planned Environment to represent typical nighttime activity, there now will exist two new <Health Condition, Planned Environment> pairs within the matrix. You may find that you only need to create a single new, unique state associated with one of those two new pairs, while the other can simply be associated with an appropriate, existing State. This helps keep the number of unique States to a minimum.

Creating states

Workload Designer TDT1 | POWM Ruleset

State Wizard: Step 6 of 6

Map States

Each cell in the state matrix represents the intersection of a health condition and a planned environment. Assign a state to each cell to determine how system resources are allocated. The same state can be used in multiple cells. For example, you might use the same state in your "BatchLoad" planned environment for all health conditions. For each cell below, drag a state name into the cell. If desired, you can create new states by clicking the plus icon at the top of the list of states.

The screenshot shows the 'Map States' step of the State Wizard. It features a state matrix on the left and a list of states on the right. The matrix has three rows for 'Health Conditions' (Normal, Degraded, Critical) and three columns for 'Planned Environments' (Always, DataLoads, SysMaint). Each cell in the matrix contains a state name or a description of the state. The 'Always' column has a 'Base' state. The 'DataLoads' and 'SysMaint' columns have states that are triggered by the 'Normal' health condition. The 'Degraded' and 'Critical' health conditions have states that are triggered by the 'Always', 'DataLoads', and 'SysMaint' planned environments. The list of states on the right includes 'Base', 'Load', 'Maint', 'Yellow', and 'Red'. A plus icon at the top of the list is circled in red, and a hand icon is shown clicking the 'Red' state.

Health Conditions	Planned Environments		
	Always	DataLoads	SysMaint
Normal	Base	Drag states here that you would like triggered by Normal/DataLoads	Drag states here that you would like triggered by Normal/SysMaint
Degraded	Drag states here that you would like triggered by Degraded/Always	Drag states here that you would like triggered by Degraded/DataLoads	Drag states here that you would like triggered by Degraded/SysMaint
Critical	Drag states here that you would like triggered by Critical/Always	Drag states here that you would like triggered by Critical/DataLoads	Drag states here that you would like triggered by Critical/SysMaint

States: +

- Base
- Load
- Maint
- Yellow
- Red

After clicking the "+" button, a State will appear with the default name of "NewState"

To change the default name, click the "pen" button

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Each combination of Health Condition and Planned Environment defines a corresponding state. A state is a unique workload environment that carries with it a working value set. In Workload Management, a rule, for example, can have a different throttle value for each possible state.

Using only a few states in the state matrix reduces maintenance time. However, consider adding states to the matrix to manage the following situations:

- Consistent, peak workload hours or days where priority management must be strictly assigned and enforced.
- Load or query times where priority tasks must finish within a specific time frame.
- Conditions where resources must be managed in a different way, such as giving higher priority to critical work when system health is degraded

Assigning states

Workload Designer TDT1 POWM Ruleset

State Wizard: Step 6 of 6

Map States

Each cell in the state matrix represents the intersection of a health condition and a planned environment. Assign a state to each cell to determine how system resources are allocated. The same state can be used in multiple cells. For example, you might use the same state in your "BatchLoad" planned environment for all health conditions. For each cell below, drag a state name into the cell. If desired, you can create new states by clicking the plus icon at the top of the list of states.

Health Conditions	Unplanned Events	Planned Environments		
		Always	DataLoads	SysMaint
		Planned Events		
Normal		Base	Load	Maint
Degraded		Yellow	Load	Maint
Critical		Red	Load	Maint

States +

- Base
- Load
- Maint
- Red
- Yellow

CANCEL FINISHED PREVIOUS

Drag and Drop to assign a State to a Health Condition/Planned Environment intersection

Click Finished to end the Setup Wizard

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After creating your States, drag and drop the state to the Health Condition/Planned Environment you want to transition to when either the Health Condition or Planned Environment changes.

Completed state matrix

SETUP WIZARD

		Planned Environments +			States +	
		Always	DataLoads	SysMaint		
		Planned Events				
Health Conditions +	Normal	Base	Load	Maint	Base	
	Degraded	Yellow	Load	Maint	Load	
	Critical	Red	Load	Maint	Maint	
Unplanned Events					Red	
					Yellow	

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Changes can be made by selecting any of the buttons (“+”, “pen” or “trash can”)

Current topic: Lab activity

- State matrix overview
- Setup wizard: Planned environments
- Setup wizard: Health conditions
- Setup wizard: States
- **Lab activity**
 - (refer to day 2: Lab guide)
 - **Edit the VOWM_starting_ruleset**
 - **Build the state matrix**
 - **Run the workload simulation**
 - **Capture the baseline performance metrics**

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The instructions for this lab activity are in the *Day 2 – Lab Guide*.

Summary

- State Matrix provides for dynamic and automatic workload management based on real-time business needs and system health
- State Matrix allows a simple way to enforce gross-level workload management.
- It is a two-dimensional matrix, with Planned Environment and Health Conditions represented, which the intersection being associated with a State
- The State Matrix allows a transition to a different working value set to support changing needs
- The State Setup Wizard can be used to step through the initial build of the State Matrix
 - Planned Environments
 - Planned Events
 - Health Conditions
 - Unplanned Events
 - States

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State Matrix provides for dynamic and automatic workload management based on real-time business needs and system health.

State Matrix allows a simple way to enforce gross-level workload management.

It is a two-dimensional matrix, with Planned Environment and Health Conditions represented, which the intersection being associated with a State.

The State Matrix allows a transition to a different working value set to support changing needs.

The State Setup Wizard can be used to step through the initial build of the State Matrix.

- Planned Environments
- Planned Events
- Health Conditions
- Unplanned Events
- States

Module 8: Workload Designer: Session control

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Discuss how to manage concurrent query sessions
- Discuss how to manage the number of concurrent utilities
- Discuss how to manage concurrent utility sessions

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Topics

- Sessions overview
- Query sessions
- Utility limits
- Utility sessions

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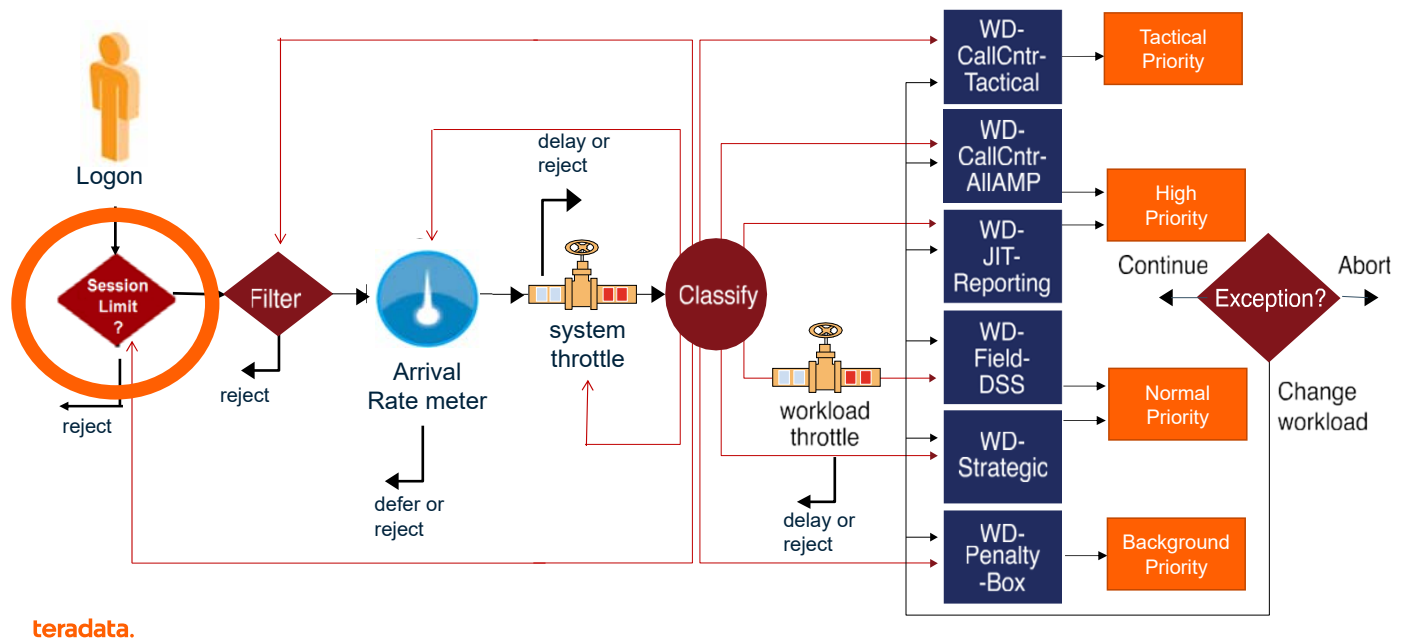
Current topic: Sessions overview

- **Sessions overview**
 - **Sessions view**
 - **Session control**
- Query sessions
- Utility limits
- Utility sessions

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Workload management levels: Session limits



There are seven different methods of management offered, as illustrated below:

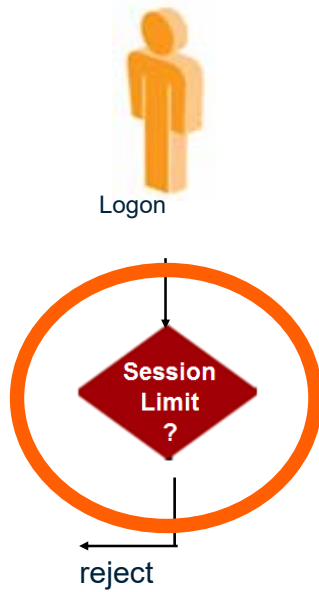
Methods regulated prior to the query beginning execution

- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level.
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level

Methods regulated during query execution

- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

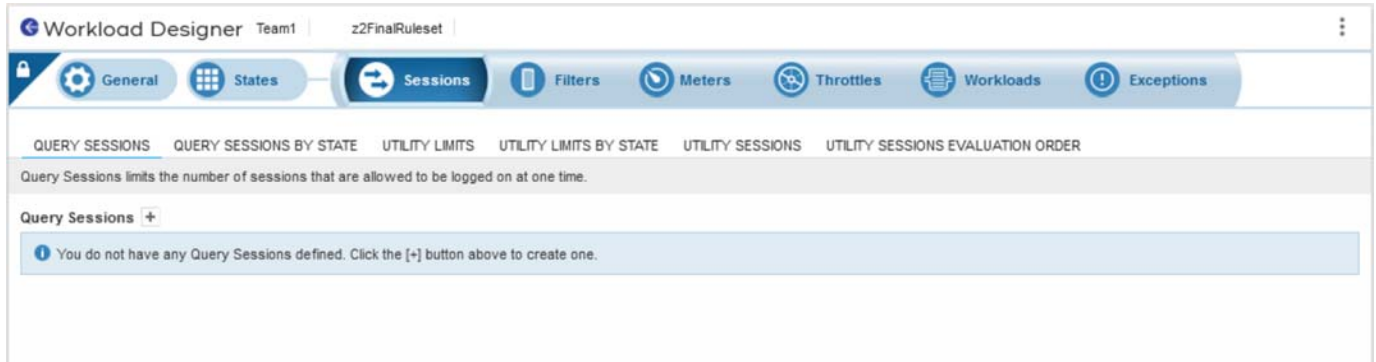
Query session throttles



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- The Query Sessions throttle limits the non-utility sessions that can log on at the same time
- The Query Sessions can only reject sessions during logon, it cannot delay logons
- If user's logon request is rejected the user must reattempt the logon
- Only request source classification criteria may be used in a Query Sessions throttle, since only information about the logon request is available

Sessions view



Sessions controls the number of Query Sessions, Utility Sessions and Utility Limits

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The **Sessions** view appears after you click the **Sessions** button on the ruleset toolbar

Session control

Session Control sets the default limits when creating and editing rulesets

- **Query Sessions** – Sets the **limits** on the number of **query sessions** a user can log on at one time
- **Query Sessions by State** – **Displays** the **limits** on the number of query sessions a user can log on **for each state**
- **Utility Limits** - Sets the **limits** on the number of bulk **utility jobs**
- **Utility Limits by State** – **Displays** the limits on the number of utilities for each **utility limit** rule in **each state**, and the System Default Utility Limits
- **Utility Sessions** – Overrides the system **limits** on the number of **sessions** a **specific utility** can use
- **Utility Sessions Evaluation Order** – **Precedence**, from highest to lowest, in which the utility session rules will be applied

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Session control limits information you can specify when creating and editing rulesets. The **Sessions** view appears after you click the **Sessions** button on the ruleset toolbar and has the following tabs:

- **Query Sessions** - Limits on the number of query sessions that can be logged on at one time. You can create, enable, clone, and delete query sessions on this tab.
- **Query Sessions by State** - Limits on the number of query sessions for each state. The default session limit for a state is listed, along with each state you have created and its assigned state specific session limit.
- **Utility Limits** - Limits on the number of utilities. You can create, enable, clone, and delete utility limits on this tab.
- **Utility Limits by State** - Limits on the number of utilities for each utility limit rule in each state. The default utility limit for a state is listed, along with each state you have created and its assigned state specific utility limit.
- **Utility Sessions** - Limits on the number of sessions a specific utility can use. You can create, enable, clone, and delete utility sessions on this tab.
- **Utility Sessions Evaluation Order** - Precedence, from highest to lowest, of utility session rules. Evaluation order determines the rule in which the utility job is placed if a utility job matches more than one utility session rule.

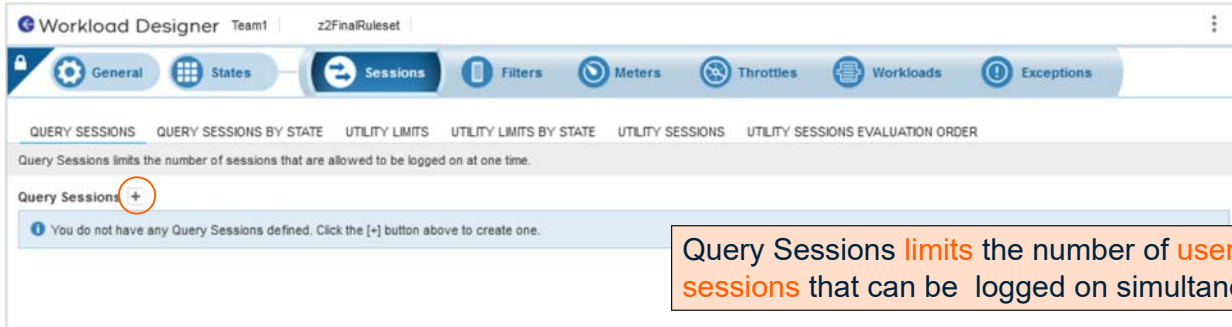
Current topic: Query sessions

- Sessions overview
- **Query sessions**
 - **Creating query sessions**
 - **Session limit rule types**
 - **Request source classification criteria**
 - **State specific settings**
 - **Query sessions by state**
- Utility limits
- Utility sessions

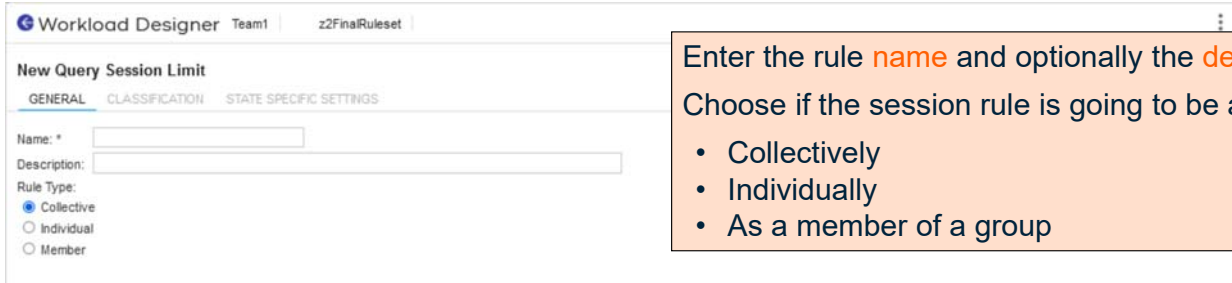
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Creating query sessions



Query Sessions **limits** the number of **user sessions** that can be logged on simultaneously



Enter the rule **name** and optionally the **description**

Choose if the session rule is going to be applied:

- Collectively
- Individually
- As a member of a group

Session limit rule types

- **Collective**

- Session limits will be applied to **all users**, as a collective group, that meet the classification criteria
- The **group** will get a maximum number of sessions

- **Individual**

- Session limits will be applied **individually** to each user that meet the classification criteria
- **Each user** will get a separate session limit

- **Member**

- Applies when **Account or Profile** is used as the **Classification Criteria** for the rule
- Session limits are **placed on individuals** within the group, no limit is placed on the account or profile
- Each member will get an **Individual** session limit

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Choose from the following three **Rule Type** options when you create a **Query Sessions** throttle:

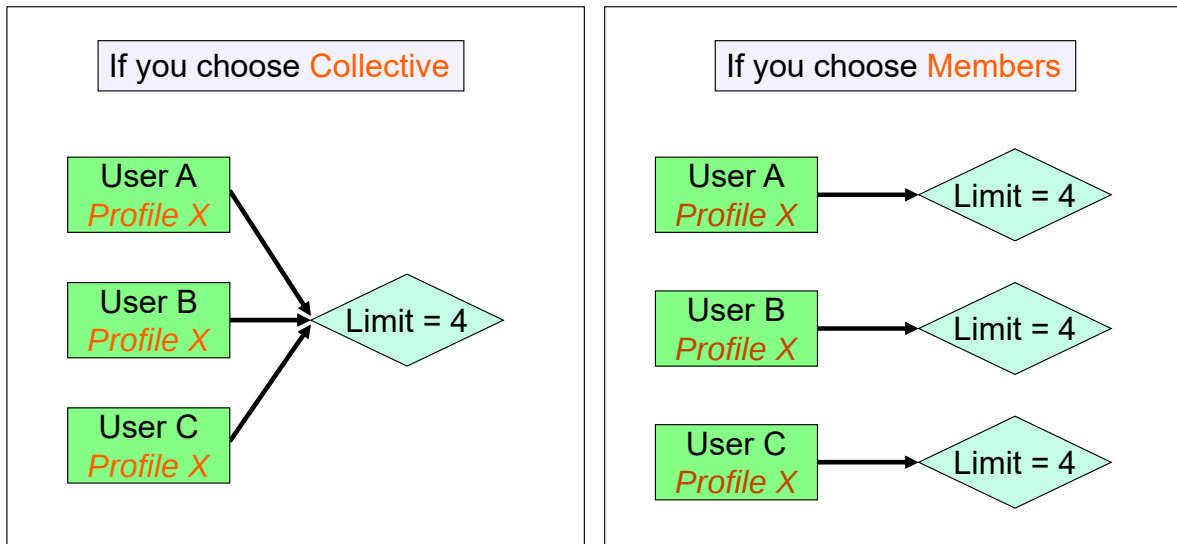
Collective: Provides the least granular control. TASM treats all requests that qualify for this throttle as one group. If the requests are delayed, the delay is based on the concurrency level of the entire group. For example, users Jay, Kay, and May are the classification criteria for a session throttle rule with a limit of 3 sessions. With the collective option, all the users together cannot have more than 3 sessions at the same time.

Individual: Each individual object that is associated with the throttle is treated independently of other, similar objects. Each object has a counter. For example, if users Jay, Kay, and May are the classification for a session throttle rule with a limit of 3 sessions, each of the users can have up to 3 sessions running at the same time.

Member: Provides the most fine control. TASM treats each user in a throttled object (such as account) independently of other users in that account. The intent of the Member option is to enable objects that represent groups of users, such as accounts and profiles, to be used as the classification criteria for the rule. The rule specifies the group, and any member of the group is covered by the rule. For example, users Jay, Kay, and May are associated with account ShoeDept. If there is a session throttle rule with a session limit of 3, classification criteria of account ShoeDept, and the Member option, then each of the users with the ShoeDept account can have up to 3 sessions running at the same time.

Collective and members example

Session Limit was applied to Classification Criteria of *Profile X*



teradata.

The facing page shows an example to contrast the difference between the collective and members options.

Collective:

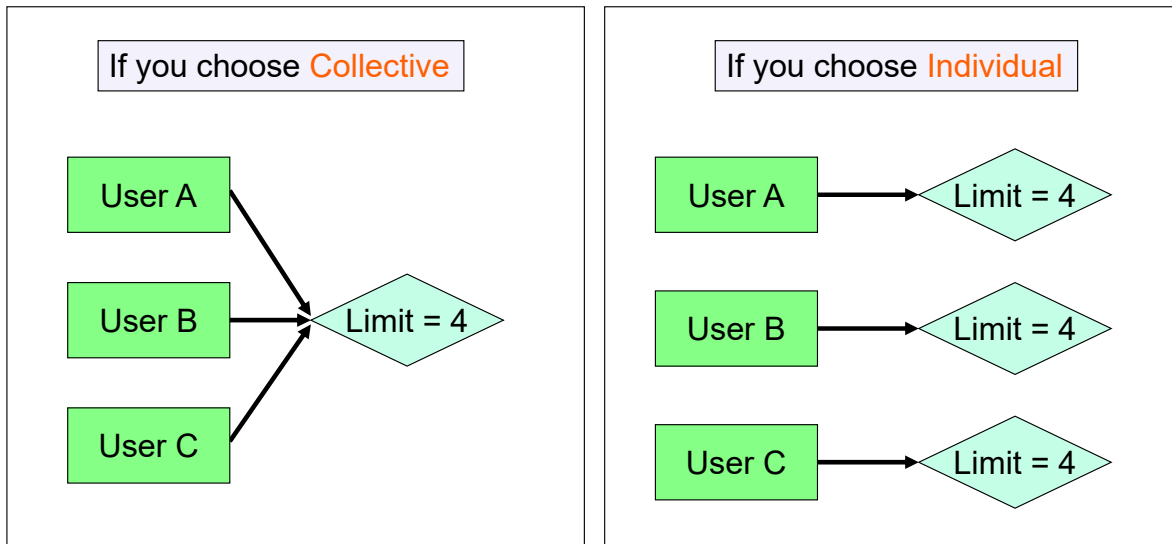
All requests that satisfy the classification criteria are throttled as a single group.

Member:

Each user that satisfies the classification criteria is throttled at the user level.

Collective and individual example

Session Limit was applied to Classification
Criteria of *Username = User A, User B, User C*



teradata.

The facing page shows an example to contrast the difference between the collective and individual options.

Collective:

All requests that satisfy the classification criteria are throttled as a single group.

Individual:

- Can only be defined with a single classification criteria request source type
- Each source type object that satisfies the classification criteria is throttled as an individual group
- If the classification criteria identifies only a single source type object, the throttle will behave the same as a collective rule type

Request source classification criteria

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Query Sessions** >
Button: **Create a Query Session [+]** > Tab: **Classification** > Button: **Add Criteria**

teradata.

Specify the **Request Source** classification criteria

Note: **Other** classification criteria are not applicable

Add the request source classification criteria that will be used to determine which requests the session limit rule applies

(Refer to slide text.)

State specific settings (1 of 2)

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Query Sessions** >
Button: **Create a Query Session [+]** > Tab: **State Specific Settings**

Workload Designer TDT1 POWM Ruleset

Query Session Limit: POWMQrySession

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Click on a state to set a session limit.

Planned Environments	
	Always
Normal	Base (use Default)
Degraded	Yellow (use Default)
Critical	Red (use Default)

Health Conditions

Default Settings

Session Limit:

☒ Unlimited

☐ 0

The Default Setting will apply to all states

Sessions exceeding the default limit will be rejected

teradata.

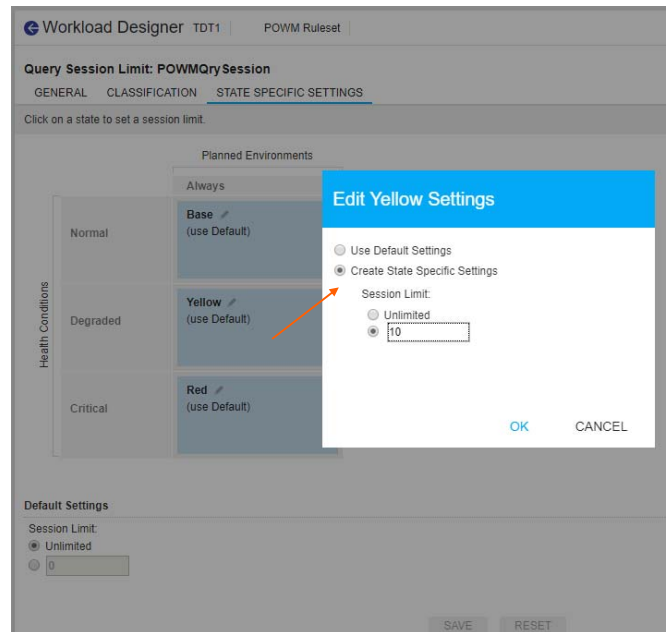
The default setting for sessions is unlimited. This can be changed to a specific limit. Sessions over the limit will be rejected.

State specific settings (2 of 2)

To **override the default setting** for a specific State, select the State's "pen" button

In the Edit dialog box, enter the **working value** settings for that state

teradata.



To override the default setting, move your cursor over the State to display the "pen" icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

Query sessions by state

Workload DesignerTeam1z2FinalRuleset

GeneralStatesSessionsFiltersMetersThrottlesWorkloadsExceptions

QUERY SESSIONSQUERY SESSIONS BY STATEUTILITY LIMITSUTILITY LIMITS BY STATEUTILITY SESSIONSUTILITY SESSIONS EVALUATION ORDER

Query Sessions limits the number of sessions that are allowed to be logged on at one time.

Query Sessions +

ENABLED	NAME	CLASSIFICATION
☒	VOWMQuerySessionRule	PROFIL: DSS_Profile

Query Sessions by State displays a summary of the session limits for each state

Workload DesignerTeam1z2FinalRuleset

GeneralStatesSessionsFiltersMetersThrottlesWorkloadsExceptions

QUERY SESSIONSQUERY SESSIONS BY STATEUTILITY LIMITSUTILITY LIMITS BY STATEUTILITY SESSIONSUTILITY SESSIONS EVALUATION ORDER

This view shows the maximum number of sessions that are allowed to be logged on at one time for each query session control rule in each state.

NAME	DEFAULT SETTING	BASE	RED	YELLOW
VOWMQuerySessionRule	Unlimited	(Use Default)	1	10

teradata.

View all created query sessions on the **Query Sessions By State** tab.

Current topic: Utility limits

- Sessions overview
- Query sessions
- **Utility limits**
 - **Creating utility limits**
 - **Utility limits classification**
 - **Supported utility protocols**
 - **State specific settings**
 - **Utility limits by state**
- Utility sessions

teradata.



Creating utility limits

Workload Designer Team1 z2FinalRuleset

General States Sessions Filters Meters Throttles Workloads Exceptions

QUERY SESSIONS QUERY SESSIONS BY STATE **UTILITY LIMITS** UTILITY LIMITS BY STATE UTILITY SESSIONS UTILITY SESSIONS EVALUATION ORDER

Utility Limits control the number of utilities that are allowed to be run at one time.

Utility Limits +

You do not have any Utility Limits defined. Click the [+] button above to create one.

Utility Limits control the **number of bulk utilities** that can execute simultaneously

Workload Designer Team1 z2FinalRuleset

New Utility Limit

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Name: * VOWMUtilityLimit

Description:

TASM will use the default Utility Limits if none are defined here.

To create Utility Limit, click the + button, and enter the rule **Name** and optionally the rule **Description** in the General tab

teradata.

A *utility limit* determines the number and type of utility jobs that can be run at one time.

- Considering using utility throttles to balance overall system resource use
- Utility throttles are useful for easily controlling the number of utilities that can run if different system states
- The classification criteria for utility throttles allows only the specification of utility names grouping and sub-types
- Utility throttles specify a job concurrency limit and delay/reject option

System-Wide Utility Limits (Throttles) allow the DBA to define the level of system-wide concurrency desired by utility type. The DBA can choose to either delay or reject a job when it exceeds the utility threshold. If delay is selected, utility jobs can be run in the order they are submitted (FIFO). If the reject option is selected, the application will stop without retrying.

A number of system utility limit rules are in place by default and replaced the old MAXLOADTASKS DBSControl Parameter of previous releases. It served to limit the number of concurrent load utilities in order to prevent AWT depletion by high load job concurrency levels. The Workload Management default utility limit rules serve this same purpose, but are able to do so in a more granular way than MAXLOADTASKS did, by applying the rules to individual load types rather than being bound by the load type and phase that requires the most AWTs to operate.

To create a query session rule:

- Enter a name.
- [Optional] Enter a description up to 80 characters.

Click **Save**.

Utility limits classification

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Utility Limits** >
Button: **Create a Utility Limit [+]** > Tab: **Classification**

Workload Designer TDT1 | POWM Ruleset

Utility Limit: POWMUtilityLimit

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Select check boxes next to the utility names or subtypes to which this limit will be applied.

☐ FastLoad or MultiLoad	
☒ FastLoad	☐ FastExport
☐ Stand Alone FastLoad	☐ Stand Alone FastExport
☐ TPT Load Operator	☐ TPT Export Operator
☐ JDBC FastLoad	☐ JDBC FastExport
☐ CSP Save Dump	☐ Non-Teradata FastExport
☐ Non-Teradata FastLoad	
☐ MultiLoad	☐ Backup Utilities
☐ Stand Alone MultiLoad	☐ Archive/Restore (ARC)
☐ TPT Update Operator	☐ Backup (DSA)
☐ JDBC MultiLoad	☐ Restore (DSA)
☐ Non-Teradata MultiLoad	
☐ MLOADX	
☐ TPT MLOADX	

Select the utilities to limit

teradata.

Add the classification criteria that will be used to determine which utilities the utility limit rule applies.

Supported utility protocols

- Some **non-Teradata utilities** implement **variations** of FastLoad, MultiLoad and FastExport protocols
- Workload Management's utility management features are also available to **non-Teradata utilities** that implement these protocols via the **TPT API**
- Workload Management's **recognizes** them as **TPT** Load/Update/Export operators
- Non-Teradata utilities that implement other variations of these protocols are called **non-conforming utilities**
- For **non-conforming utilities**, certain Workload Management's **features may be restricted**. (For example, Delay option is not supported for non-conforming utilities)
- **MLOADX** uses **SQL session protocol** not MLOAD session protocol

Protocols	Utility Names
FastLoad	FastLoad utility, TPT Load operator, JDBC FastLoad, CSP Save Dump
MultiLoad	MultiLoad utility, TPT Update operator, JDBC MultiLoad
MLOADX	TPT Update operator
FastExport	FastExport utility, TPT Export, JDBC FastExport
Backup/Restore	ARCMAIN, Data Stream Architecture (DSA/BAR)

teradata.

Some non-Teradata utilities implement variations of FastLoad, MultiLoad, and FastExport protocols. Workload Management's Utility Management features are also available to non-Teradata utilities that implement these protocols via the TPT API. Workload Management recognizes them as Teradata Parallel Transporter Load/Update/Export operator. The non-Teradata utilities that implement other variations of these protocols are called non-conforming utilities. For these, certain Workload Management's features may be restricted.

Note that the TPump utility and Teradata Parallel Transporter Stream operator implement the normal SQL protocol; therefore, they should be managed as SQL requests. Workload Management's Utility Management features described in this section do not apply to them.

State specific settings

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Utility Limits** >
Button: **Utility Create a Utility Limit [+]** > Tab: **State Specific Settings**

- You can set a **lower Default Job Limit value** that will apply to **all states**
- By default, additional jobs will be delayed, effectively disabling any utility **Tenacity** and **Sleep** parameters
- Delay option is not supported for non-conforming utilities. Non-conforming utilities will always be rejected even if Delay is selected
- **DBSControl parameters** MaxLoadTasks, MaxLoadAWT, MLOADXUtilityLimits, MaxLOADXTasks and MaxLOADXAWT are **disabled** when have an active ruleset is running in TASM

teradata.

Planned Environments	
Normal	Base (use Default)
Degraded	Yellow (use Default)
Critical	Red (use Default)

Default Settings

Job Limit: 30

☒ Delay ☐ Reject

Default maximum concurrency limits are:

- FastLoad: 30
- MultiLoad: 30
- FastLoad+MultiLoad: 30
- FastLoad+MultiLoad+FastExport: 60
- MLOADX: 30/120
- Backup Utilities: 350

In the Job Limit field, you can enter a value (lower than the default value) that specifies the maximum number of jobs for the specified type of utility that can run simultaneously.

If you specify Delay, any additional jobs will be put into the Delay queue. Once the number of running jobs falls below the specified threshold, then TASM will release the Delayed jobs.

If you specify Reject, then the utility will use its Tenacity and Sleep parameters. So, the utility will wait for the number of minutes in the Sleep parameter and will then attempt to start the job again. If unsuccessful, it try will again after the number of sleep minutes transpire, and it will continue trying until it reaches number of hours specified in the Tenacity parameter before it gives up.

(Refer to slide text.)

The limit you set here overrides the MaxLoadTasks value set using the DBS Control Utility.

Maximum concurrency limits are:

- FastLoad: 30
- MultiLoad: 30
- FastLoad+MultiLoad: 30
- FastLoad+MultiLoad+FastExport: 60
- MLOADX: 30
- Backup Utilities: 350

When you select **Delay**, Workload Management delays utilities that exceed the concurrency limit you specify until the limit is no longer exceeded. This effectively overrides the Tenacity and Sleep utility parameters.

When you select **Reject**, Workload Management immediately rejects utilities that exceed the concurrency limit you specified and the utilities Tenacity and Sleep parameter settings will be in effect.

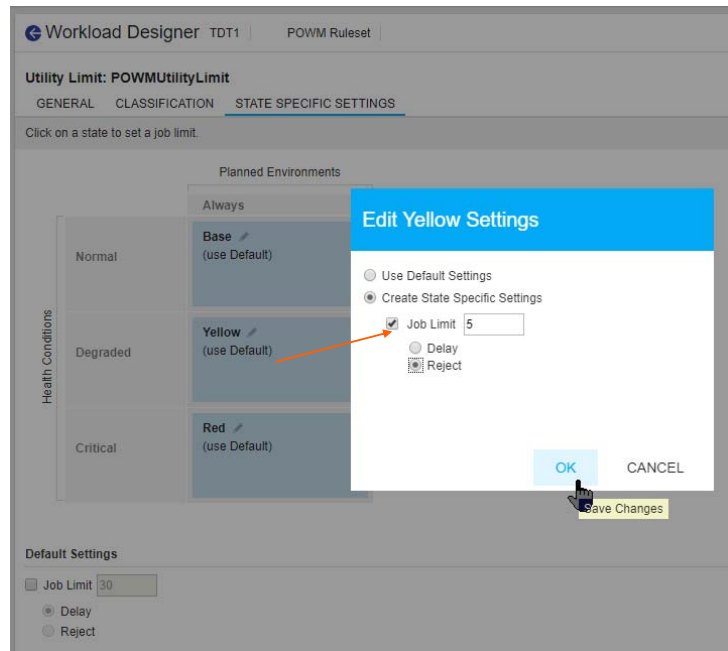
.

State specific settings (cont.)

To **override the default** setting for a specific State, select the State's "pen" button

In the Edit dialog box, enter the **working value** settings for that state

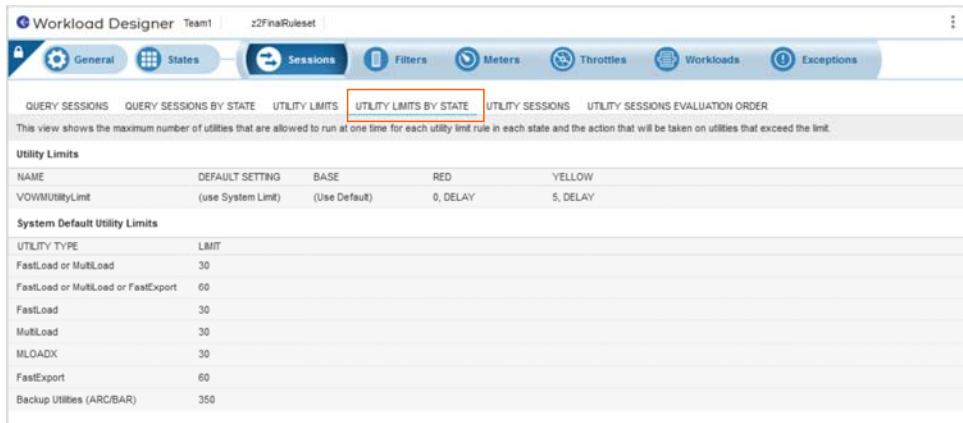
teradata.



To override the default setting, move your cursor over the State to display the "pen" icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

Utility limits by state

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Utility Limits by State**



NAME	DEFAULT SETTING	BASE	RED	YELLOW
VOVMUtilityLimit	(use System Limit)	(Use Default)	0, DELAY	5, DELAY

UTILITY TYPE	LIMIT
FastLoad or MultiLoad	30
FastLoad or MultiLoad or FastExport	60
FastLoad	30
MultiLoad	30
MLOADX	30
FastExport	60
Backup Utilities (ARC/BAR)	350

Utility Limits by State displays a summary of the utility limits for each state

Any Utility Limits created will override the System Default Utility Limits

teradata.

View all created utility limit rules on the **Utility Limits By State** tab. This view also displays the system maximum utility sessions.

(Refer to slide text.)

Utility Sessions provides session control for individual utility jobs.

A set of default session control rules are included in the Workload Management ruleset. Additionally, the DBA can create additional session control rules that override the default rules. While the default rules' classification criteria are limited to utility type and volume of data to load (which can be provided through the application script via QueryBand name of UtilityDataSize=Large, Medium or Small), the custom rules the DBA creates can be more granular by specifying the utility, its driver source, the request source (who issued the request) and the volume of data to load. If a utility job classifies into a custom DBA session control rule, it will override any default session control rule that it would otherwise have classified into.

Current topic: Utility sessions

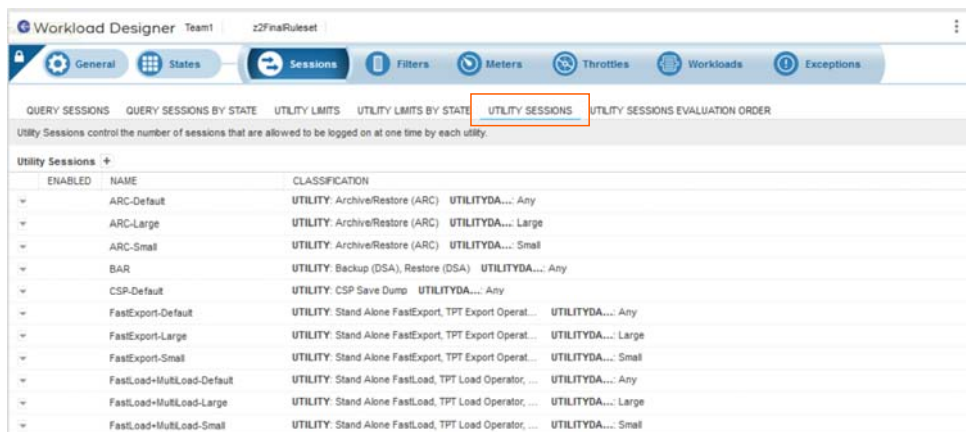
- Sessions overview
- Query sessions
- Utility limits
- **Utility sessions**
 - **The utility sessions tab**
 - **Utility protocols**
 - **Default utility session rules**
 - **Creating utility session**
 - **The UtilityDataSize Queryband**
 - **Utility session classification**
 - **Utility sessions evaluation order**

teradata.



Utility sessions

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Utility Sessions**



ENABLED	NAME	CLASSIFICATION
✓	ARC-Default	UTILITY: Archive/Restore (ARC) UTILITYDA... Any
✓	ARC-Large	UTILITY: Archive/Restore (ARC) UTILITYDA... Large
✓	ARC-Small	UTILITY: Archive/Restore (ARC) UTILITYDA... Small
✓	BAR	UTILITY: Backup (DSA), Restore (DSA) UTILITYDA... Any
✓	CSP-Default	UTILITY: CSP Save Dump UTILITYDA... Any
✓	FastExport-Default	UTILITY: Stand Alone FastExport, TPT Export Operat... UTILITYDA... Any
✓	FastExport-Large	UTILITY: Stand Alone FastExport, TPT Export Operat... UTILITYDA... Large
✓	FastExport-Small	UTILITY: Stand Alone FastExport, TPT Export Operat... UTILITYDA... Small
✓	FastLoad-MultLoad-Default	UTILITY: Stand Alone FastLoad, TPT Load Operator, ... UTILITYDA... Any
✓	FastLoad-MultLoad-Large	UTILITY: Stand Alone FastLoad, TPT Load Operator, ... UTILITYDA... Large
✓	FastLoad-MultLoad-Small	UTILITY: Stand Alone FastLoad, TPT Load Operator, ... UTILITYDA... Small

Utility Sessions are used to provide **session control** for individual utility jobs

Utility session limits will **disable** any utility **MINSESS** and **MAXSESS** parameters

teradata.

View all created utility limit rules on the **Utility Limits By State** tab. This view also displays the system maximum utility sessions.

Utility protocols

- Utilities may log on the following sessions:
 - Control SQL session that is used for executing SQL statements pertaining to utility work
 - Auxiliary SQL session which may be used for maintaining a restart log for recovery purposes
 - One or more utility sessions that are used to send/receive data to/from the Analytics Database
- All SQL sessions and requests issued through them are subject to Query Session limits, System Throttles and Workload Throttles
- Not all of the requests issued by a Utility will be associated with a Utility Classified Workload
- May need to have separate non-Throttled Workload for Auxiliary SQL

Utility Name	Control SQL Session	Auxiliary SQL Session	Utility Sessions
FastLoad, MultiLoad, FastExport	Yes	Yes	Yes
TPT Load, Update, Export operator	Yes	Yes	Yes
JDBC FastLoad, JDBC FastExport	Yes	No	Yes
CSP Save Dump	Yes	No	Yes
ARCMAN	Yes	No	Yes
DSA/BAR	Yes	No	No

teradata.

(Refer to slide text.)

Utility sessions are subject to different throttles than SQL sessions. The SQL sessions and requests issued through them are subject to these throttles:

- Query session
- Workload
- System

Utility sessions are subject only to utility session rules.

TASM and Teradata utilities treat the whole load/unload process as one Unit of Work. A Unit of Work starts with the BEGIN LOADING/MLOAD/EXPORT request on the control SQL session. It ends with the END LOADING/MLOAD/EXPORT request. This means that a utility Unit of Work spans multiple SQL requests. Only the first SQL request of a unit on the control SQL session satisfies the utility classification criterion and is assigned to a utility workload. Subsequent requests are assigned to the same utility workload. SQL requests outside the load/unload unit and SQL requests from the auxiliary SQL session can be assigned to different workloads. All applicable throttle counters are checked and incremented only at the beginning of a unit and decremented only at the end of the unit. For backup and restore, the entire script is treated as one unit.

Default utility sessions

- Default utility session rules are intended to select a **reasonable number of sessions** for every utility **based on** the system configuration
- Default utility sessions rules **can be modified** to fit specific requirements
- However, the **default** utility session rules cannot be deleted
- Each protocol has **up to three different default values** for different data sizes
 - Queryband name UtilityDataSize can be specified with a value of **Small**, **Medium** (default) or **Large**
- DSA architecture does not use utility sessions, the utility session parameter is used to specify the number of build processes
- The default system limit for DSA/BAR Max Build Processes is 5

teradata.

Utility Sessions provides session control for individual utility jobs.

A set of default session control rules are included in the Workload Management ruleset. Additionally, the DBA can create additional session control rules that override the default rules. While the default rules' classification criteria are limited to utility type and volume of data to load (which can be provided through the application script via QueryBand name of UtilityDataSize=Large or Small), the custom rules the DBA creates can be more granular by specifying the utility, its driver source, the request source (who issued the request) and the volume of data to load. If a utility job classifies into a custom DBA session control rule, it will override any default session control rule that it would otherwise have classified into.

Default utility session rules

Protocol	Default or Medium Data Size	Small Data Size	Large Data Size
FastLoad and MultiLoad	If NumAMPs <= 20 then NumAMPs else Min(20 + NumAMPs / 20), 100	Default * 0.5	Min(default * 1.5), NumAMPs)
FastLoad – CSP Save Dump	If NumNodes <= 10 then 4 per node elseif NumNodes <= 20 then 3 per node elseif NumNodes <= 30 then 2 per node else 1 per node	N/A	N/A
FastExport	If NumAMPs <= 4 then NumAMPs else 4	Default * 0.5	Default
ARC	If NumAMPs <= 20 then 4 else Min ((4 + NumAMPs / 50), 20	Default * 0.5	Min(default) * 1.5, 2 * NumAMPs)
DSA/BAR	1	N/A	N/A

teradata.

Vantage assigns a default number of sessions to each utility based on the following factors:

- The specific utility (for example, FastLoad or MLOADX)
- The number of AMPs in the system
- Data size (any, small, medium, large) The default data size is medium.

Tip: Enable the Viewpoint Monitored Systems portlet System Stats data collector so that the Workload Designer portlet can retrieve the correct number of AMPs in the system.

Creating utility sessions

Workload Designer TDT1 | POWM Ruleset

New Utility Sessions

GENERAL CLASSIFICATION

Select the Utilities to which this rule applies:

Name: *

Description:

☒ Session Based Utilities

☐ FastLoad or MultiLoad ☐ MLOADX

☐ FastLoad ☐ MultiLoad ☐ TPT MLOADX

☐ Stand Alone FastLoad ☐ Stand Alone MultiLoad

☐ TPT Load Operator ☐ TPT Update Operator

☐ JDBC FastLoad ☐ JDBC MultiLoad

☐ CSP Save Dump ☐ Non-Teradata MultiLoad

☐ Non-Teradata FastLoad

☐ FastExport ☐ Archive/Restore (ARC)

☐ Stand Alone FastExport

☐ TPT Export Operator

☐ JDBC FastExport

☐ Non-Teradata FastExport

Data Size: (Default system limit is 20 sessions)

Number of Sessions: (Default system limit is 20 sessions)

☐ Backup/Archive/Restore

Data Size: (Default system limit is 5 processes)

Build Processes: (Default system limit is 5 processes)

SAVE CANCEL

If a Data Size is specified, then the utility script needs to include a corresponding SET QUERY_BAND='UtilityDataSize=...;'

teradata.

The Utility Session System Default rules can be overridden by creating a Utility Session rule. To use the data size option, the utility must set the queryband name "UtilityDataSize" to a value of small, medium or large.

Create utility session: UtilityDataSize

```
.LOGTABLE logtable01;  
.LOGON tdpX/user,pwd ;
```

FastExport Script

```
SET QUERY_BAND = 'UtilityDataSize=SMALL;' for session;  
  
.BEGIN EXPORT;  
.EXPORT OUTFILE ExpData_fep MODE RECORD;  
  
SELECT * FROM table_1  
WHERE service_skill_target_id > 60000  
AND service_skill_target_id <= 80000;  
  
.END EXPORT ;  
  
.LOGOFF ;
```

Setting the Data Size to SMALL, MEDIUM or LARGE is subjective

teradata.

```
.LOGTABLE logtbale02;  
.LOGON tdpX/user,pwd ;
```

Multiload Script

```
SET QUERY_BAND = 'UtilityDataSize=LARGE;' for session;  
  
.BEGIN IMPORT MLOAD  
...;  
  
.LAYOUT DATAIN_LAYOUT;  
.FIELD start_datetime 1 CHAR(19);  
...  
  
.DML LABEL INSERT_DML;  
INSERT INTO  
  &DBASE_TARGETTABLE..&TARGETTABLE (  
    start_datetime = :start_datetime  
  ...  
  );  
  
.IMPORT INFILE ExpData  
  FORMAT FASTLOAD  
  LAYOUT DATAIN_LAYOUT  
  APPLY INSERT_DML;  
  
.END MLOAD;  
  
.LOGOFF &SYSRC;
```

To use the data size option, the utility must set the queryband name “UtilityDataSize” to a value of small, medium or large.

Create utility session: Classification

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Utility Sessions**
Button: **Create a Utility Session [+]** > Tab: **Classification**

The screenshot shows the 'Workload Designer' interface with the 'Utility Session: POWMUtilitySession' selected. The 'CLASSIFICATION' tab is active, displaying the text: 'Classification criteria determine the utility sessions to which this limit will be applied.' Below this, the 'Add Classification Criteria' section shows a dropdown menu with 'Request Source' selected, and an 'ADD' button. A mouse cursor is pointing at the 'Request Source' option. The 'Edit Request Source Criteria' dialog is open, showing a 'Source Type' dropdown set to 'Choose --'. The 'Match String' field contains 'wildcard *'. The 'Items' list is empty. The 'Selected' list contains 'Included' and 'Excluded'. The 'Include' and 'Exclude' buttons are visible. The 'OK' and 'CANCEL' buttons are at the bottom right.

Specify the **Request Source** or **Query Band** classification criteria

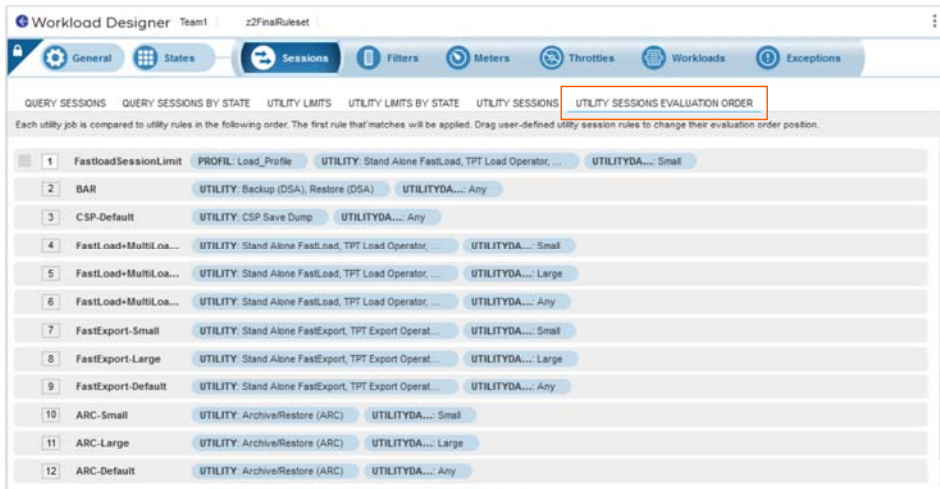
Note: **Other** classification criteria are not applicable

teradata.

Add the request source classification criteria that will be used to determine which request source the utility session limit rule applies.

Utility sessions evaluation order

Portlet: **Workload Designer** > Button: **Sessions** > Tab: **Utility Sessions Evaluation Order**



User Defined Utility Session rules can be ordered from more specific to less specific

System Defined Utility Session rules cannot be reordered

teradata.

You can set evaluation order for utility sessions.

If a utility job matches more than one utility session rule, evaluation order determines the rule in which the utility job is placed. The rule in the highest position on the **Utility Sessions Evaluation Order** tab is applied.

Summary

Session Control limits what you can specify when creating and editing rulesets

- **Query Sessions** – Sets the **limits** on the number of **query sessions** a user can log on at one time
- **Query Sessions by State** – **Displays** the **limits** on the number of query sessions a user can log on for **each state**
- **Utility Limits** - Sets the **limits** on the number of **bulk utility jobs**
- **Utility Limits by State** – **Displays** the **limits** on the number of utilities for each utility limit rule in **each state**, and the System Default Utility Limits
- **Utility Sessions** – **Overrides** the system **limits** on the **number of sessions** a specific **utility** can use
- **Utility Sessions Evaluation Order** – **Precedence**, from highest to lowest, in which the utility session rules will be applied

teradata.

Session Control limits what you can specify when creating and editing rulesets

- Query Sessions (V13.10 and later) – Sets the default limits on the number of query sessions a user can log on at one time
- Query Sessions by State (V13.10 and later) – Overrides the limits on the number of query sessions a user can log on for each state
- Utility Limits - Sets default limits on the number of utilities
- Utility Limits by State – Overrides limits on the number of utilities for each utility limit rule in each state
- Utility Sessions (V13.10 and later) – Overrides the system limits on the number of sessions a specific utility can use
- Utility Sessions Evaluation Order (V13.10 and later) - Precedence, from highest to lowest, in which the utility session rules will be applied

Module 9: Workload Designer: System filters

Teradata Vantage optimizing the
Analytics Database through Workload
Management

teradata.

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Objectives

After completing this module, you will be able to:

- Discuss how Workload Management can be used to improve response consistency and throughput in a mixed workload environment
- Describe the characteristics, components, and purpose of Filter rules
- Explain how to create Filter rules and when to use the various available options

teradata.



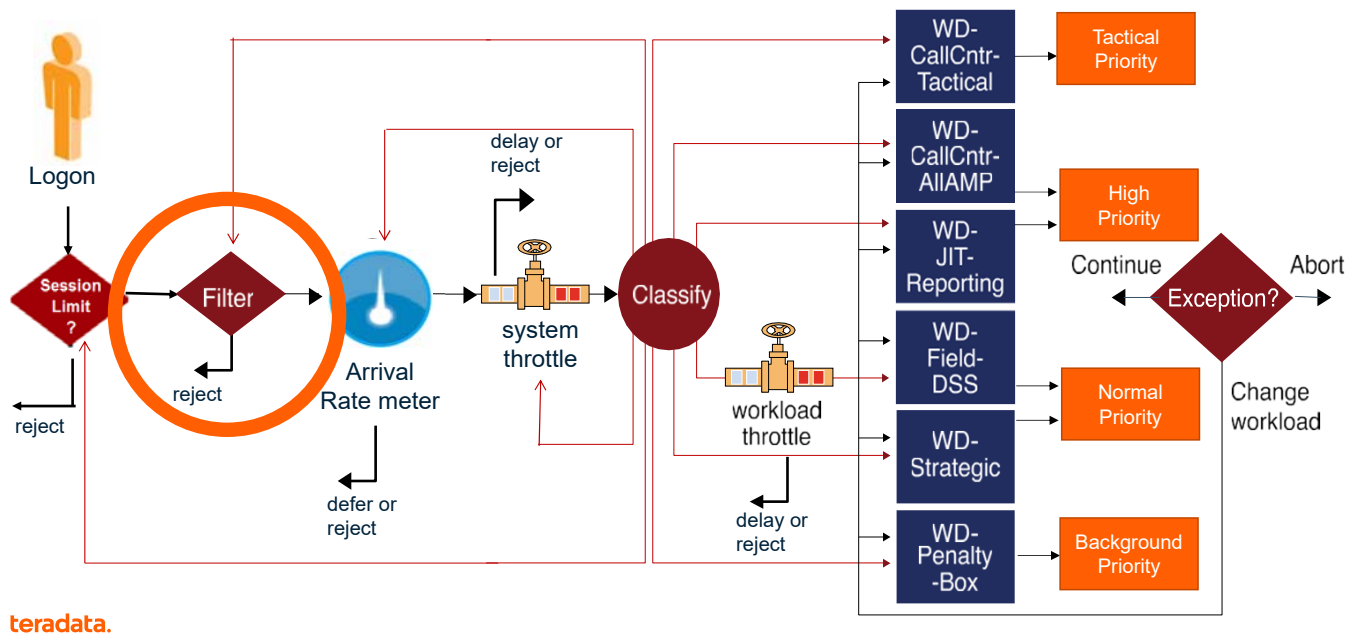
Topics

- Filters
- Bypass filters
- Creating filters
- Warning only
- Classification criteria
- State specific settings
- Enabled by state tab
- Using filters

teradata.



Workload management levels: Filter



There are seven different methods of management offered, as illustrated below:

Methods regulated prior to the query beginning execution

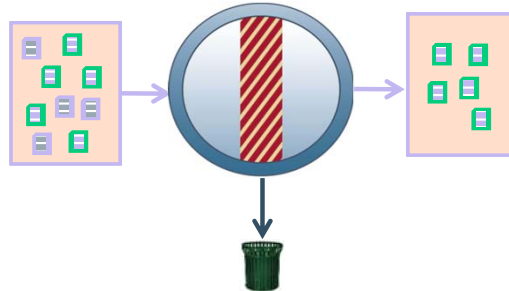
- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level.
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level

Methods regulated during query execution

- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

Pre-Execution controls: Filters

- Filters are applied system-wide and **reject a query** before the query starts running based on the classification criteria
- **Classification criteria** include:
 - Source, Target, Query Characteristics, and QueryBand
- The “**Warning Only**” option can be used for testing
- **System Bypass** privileges can be applied based on username, account name, account string, or profile, allowing queries from **Bypass** users to skip the check for Filter rules



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Filters are system-wide and allows the DBA to reject queries before they begin running based on the classification criteria. If it is determined that a certain type of request should never run during the day, for example, system filter rules are able to enforce that. Filters are very useful for rejecting problematic query requests (e.g., unconstrained product joins against large tables) before they consume resources and are eventually cancelled due to exception criteria

In order to restrict the impact and scope of a filter to a selective number of deserving queries, the administrator can apply a variety of qualifying criteria. These are the same criteria choices that can be used for workload classification purposes.

Filter rules need to be used with caution and forethought, and applied very selectively, as rejecting queries is a strong action to take and may be considered as inappropriate in an ad hoc environment. “Warning Only” option can be used for testing as queries are logged but not rejected (can be used for testing new filters).

System Bypass privileges can be applied based on username, account name, account string, or profile, **allowing queries from Bypass users to skip the check for Filter rules.**

Bypass filters

Workload Management allows selected Users submitting requests to **circumvent** all **Filter rules** by turning off filter rule checking for those users

- Purpose is to **give exceptions** to one or more Users from a group that has been associated with a filter
- There is **no partial bypass** for a subset of filters
- Grant Bypass is **checked at logon** time. If the User is determined to be bypassed, the user is flagged and no further checking is done
- Users submitting high priority requests, such as **tactical** queries, **Tpump** jobs, and the **Viewpoint data collector** user would be good candidates for being bypassed to avoid the overhead of filter rules checking



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You can identify users submitting requests as bypass to circumvent Workload Management Filter rules. This turns off the Workload Management rule checking for all of the requests issued within the context of the user's session. The set of users that are designated as Bypass are referred to as "unrestricted users".

User DBC and TDWM are automatically given bypass status.

The purpose of the Bypass User option is to give exceptions to a subset of users from a group, when the group as a whole has been associated with a rule. There is no partial bypass for a subset of rules. Bypass applies to all filter rules defined.

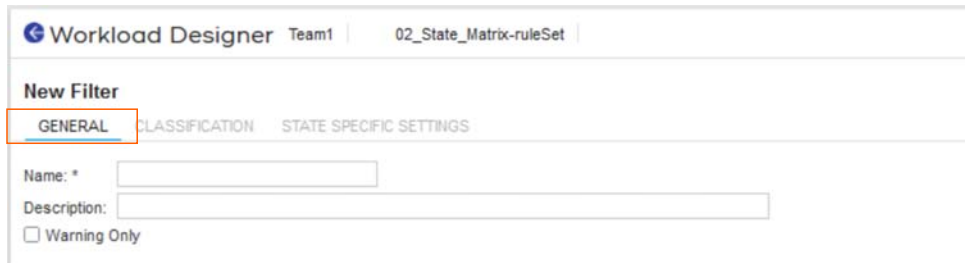
Whether or not a User is bypassed is checked at user logon time. If the User is determined to be bypassed when he/she first logs on, then no further check is done. Once a user's logon is flagged as bypassed, all queries will bypass filter rules each time they enter the system.

Users associated exclusively with active data warehouse workloads, such as tactical queries or TPump jobs, are solid candidates for being bypassed. While the overhead of rules checking is slight, a query that only performs a single-AMP operation, such a primary index read, could be impacted if 10 or 20 or 30 rules had to be checked before the query could execute.

Creating filters



Filters are used to reject queries that meet defined criteria



Warning mode applies the filter rule logs any violations but does not reject the query

teradata.

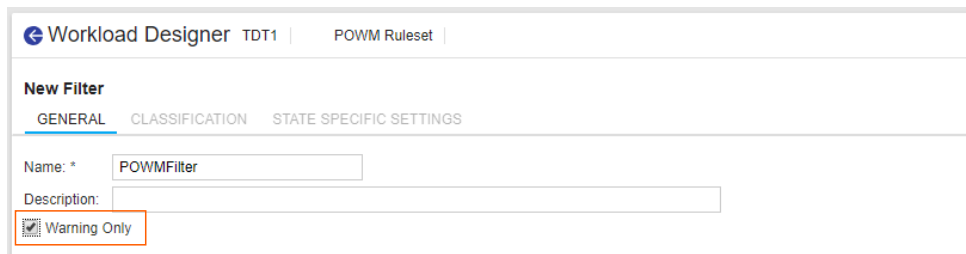
To create a new filter rule, click Filters on the toolbar and then click + icon to create a new filter rule.

Warning only

Warning Only allows you to **analyze the potential impact** of a filter without rejecting queries.

When a rule is in a warning mode, the following events occur:

- Queries are **evaluated** as if the rule is in normal mode
- Errors are **logged only** for queries that would potentially be rejected
- An error status code or **message is not returned** to the end-user
- Rule violations will be logged:
 - **DBC.DBQLogTbl** via the WarningOnly column and relevant information will be stored in the ErrorCode and ErrorText columns.
 - **DBC.TDWMExceptionLog** with relevant information in the ExceptionCode and ErrorText columns.



Workload Designer TDT1 | POWM Ruleset

New Filter

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Name: * POWMFilter

Description:

☒ Warning Only

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Database administrators can analyze the potential impact of filter rules by defining them as 'warning only.'

Once a filter rule is defined to be in warning mode, they will not actually be enforced. Instead, errors that would have been reported will be logged for impact analysis (WarningOnly flag).

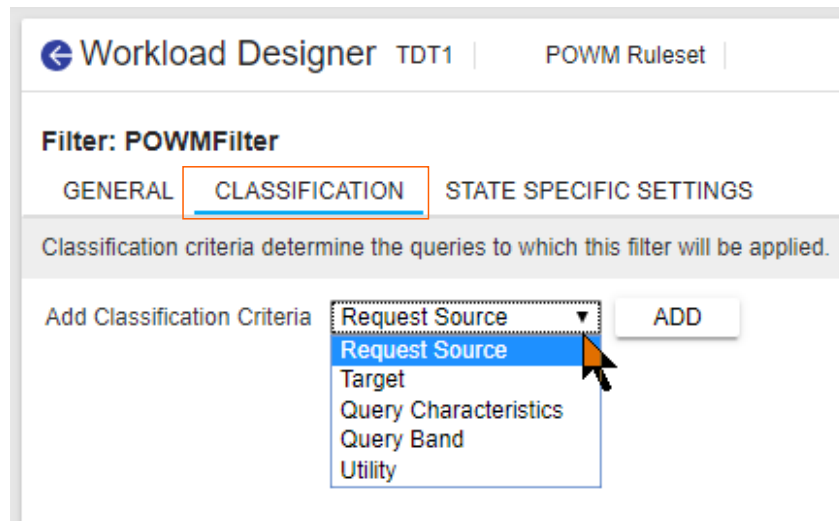
Currently the errors will be logged to the Database Query Log (DBQLogTbl table) with relevant information stored in ErrorCode and ErrorText columns.

Note: all exceptions, warning or not, will be logged in the DBC.TDWMExceptionLog.

Classification criteria

Portlet: **Workload Designer** > Button: **Filters** > Tab: **Filters** > Button: **Create a Filter [+]** >
Tab: **Classification**

Add the Classification
Criteria



Workload Designer TDT1 | POWM Ruleset |

Filter: POWMFilter

GENERAL **CLASSIFICATION** STATE SPECIFIC SETTINGS

Classification criteria determine the queries to which this filter will be applied.

Add Classification Criteria

- Request Source
- Request Source
- Target
- Query Characteristics
- Query Band
- Utility

ADD

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Add the classification criteria that will be used to determine which requests the filter rule applies.

State specific settings (1 of 2)

Portlet: **Workload Designer** > Button: **Sessions** > Button: **Create a Query Session [+]**
> Tab: **State Specific Settings**

Workload Designer TD11 POWM Ruleset

Filter: POWMFilter

GENERAL CLASSIFICATION **STATE SPECIFIC SETTINGS**

Click on a state to enable or disable this filter.

Planned Environments	
	Always
Normal	Base (use Default)
Degraded	Yellow (use Default)
Critical	Red (use Default)

Default Setting

☒ Filter is Enabled
☐ Filter is Disabled

By **Default**, the **Filter is Enabled** for every state

teradata.

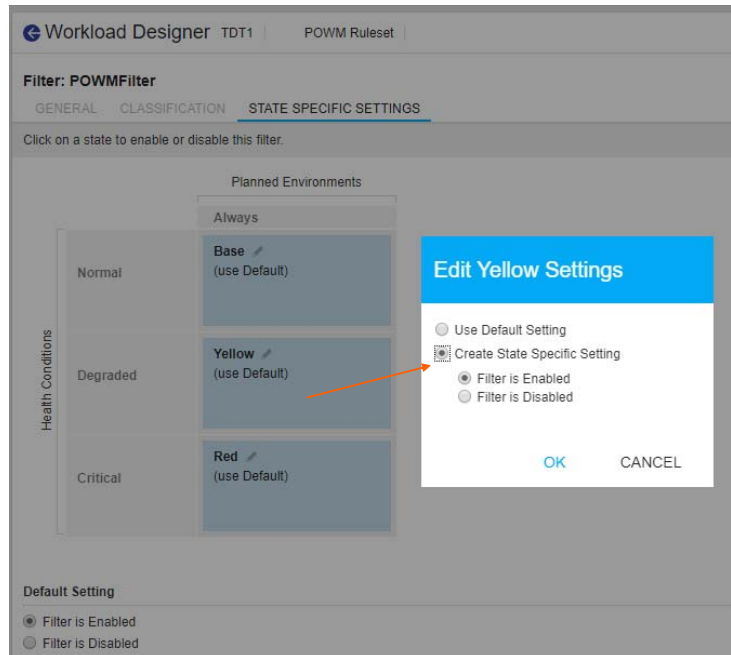
By default, the filter rule is enabled. The default setting can be set to disabled.

State specific settings (2 of 2)

To **override the default setting** for a specific State, select the State's "pen" button

In the Edit dialog box, enter the **working value** settings for that state

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To override the default setting, move your cursor over the State to display the "pen" icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

Enabled by state

Workload Designer Team1 02_State_Matrix-ruleSet					
General States Sessions Filters Meters Throttles Workloads Exceptions					
FILTERS ENABLED BY STATE					
NAME	DEFAULT SETTING	BASE	RED	YELLOW	
VOWMFilter	DISABLED	(Use Default)	ENABLED	ENABLED	

Displays the working values for each state

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The Enabled by State tab displays the working values for each state.

Using filters

- Filter rules can provide for more consistent response time and throughput by **rejecting** high resource consuming queries during peak activity periods
- Filter rules determine if query requests will be **accepted** or **rejected**
- Filter rules can consider “what” each request is doing
 - Only allow indexed access to specific tables during critical times by prohibiting **full table scans**
 - Prohibit **unconstrained product joins** estimated to exceed a large amount of time or return a large number of rows
 - Prohibit DDL statements, such as **Collect Statistics**, during high activity windows or during times when performance is degraded
 - Only allow access to “hot” data and prohibit access to “**cold**” data during specific operating windows
 - Prohibit poorly formulated queries that may require an **unreasonable share** of resources
- Filter rules can help in **preventing the exhaustion** of uncontrolled system resources, such as AMP Worker Tasks, CPU or memory

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Using Filters can have a benefit of improving response time consistency and throughput

- Only allow indexed access to specific tables during critical times by prohibiting full table scans from all users
- Prohibit full table scans against specified large tables, but allow for others
- Prohibit unconstrained product joins estimated to exceed a large amount of time or return a large number of rows
- Prohibit DDL statements, such as Collect Statistics, during high activity windows or during times when performance is degraded
- Only allow access to “hot” data and prohibit access to “cold” data during specific operating windows

Summary

In this module we covered how to:

- Discuss how Workload Management can be used to improve response consistency and throughput in a mixed workload environment.
- Describe the characteristics, components, and purpose of Filter rules.
- Explain how to create Filter rules and when to use the various available options.

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Module 10: Workload Designer: Arrival rate meter

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Describe how Workload Management can be used to improve response consistency and throughput in a mixed workload environment
- Describe the characteristics, components, and purpose of Arrival Rate Metering rules
- Explain how to create Arrival Rate Metering rules and when to use the various available options

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Topics

- Arrival rate meter overview
- Creating arrival rate meters

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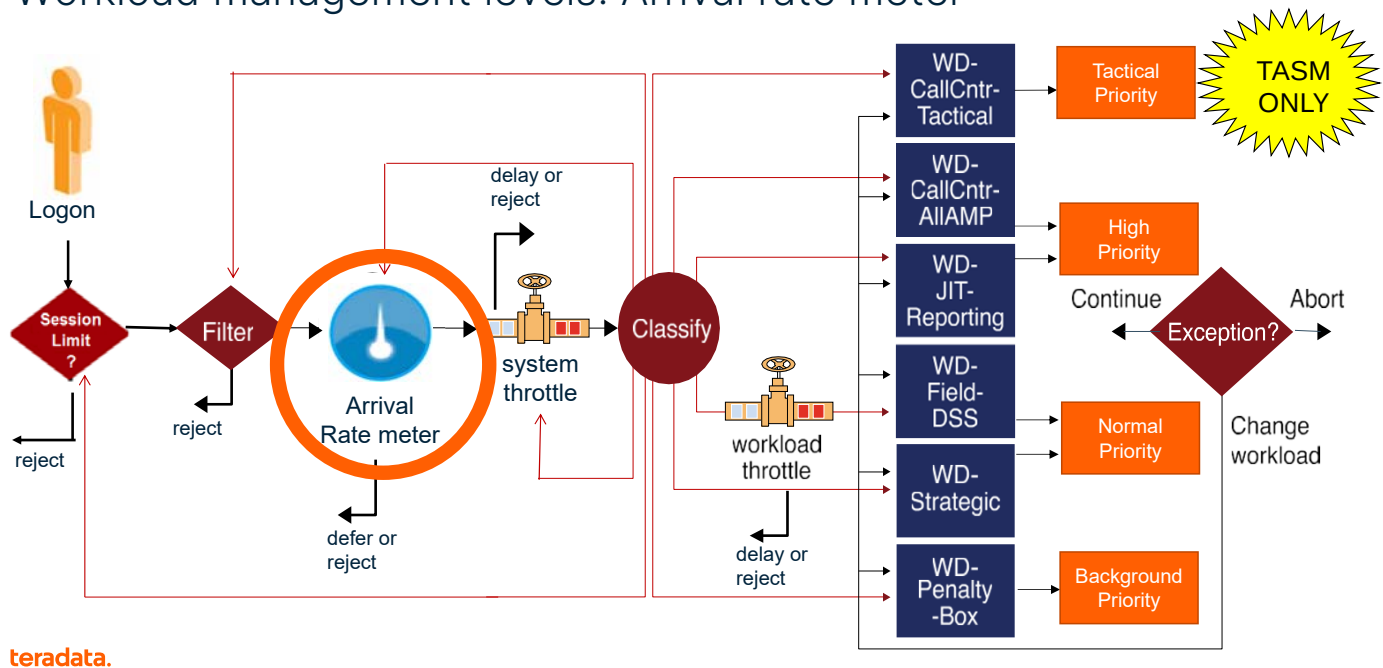
Current topic: Arrival rate meter overview

- **Arrival rate meter overview**
 - **What is arrival rate meter?**
 - **Deferring requests**
 - **Case study examples**
 - **ARM rule attributes**
 - **Current arrival rate example**
- Creating arrival rate meters

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Workload management levels: Arrival rate meter



Arrival Rate Meters (ARMs) are TASM rules that provide a way to regulate the flow of requests admitted by TASM into the system. An ARM rule specifies the maximum number of requests per time unit that TASM will admit into the system. Qualification criteria can be used to target specific types of requests.

TASM's filter and throttle processing sequence for a SQL request is enhanced to check for ARM. New Workload Management ruleset APIs allow you to automate the creation of a system throttle or arrival rate meter without using Viewpoint Workload Designer.

Benefits

- Prevents a predictable request surge from flooding the TASM delay queue and impacting service time of subsequent queries.
- Limits resource-intensive requests to a specific rate.
- Prevents you from submitting many extremely complex queries frequently.

Considerations

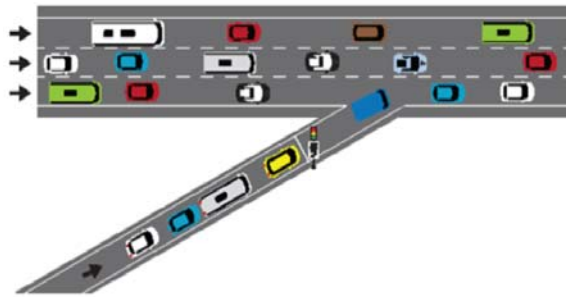
- Use arrival rate meters carefully after an evaluation period in warning mode.

Arrival rate meter

- An Arrival Rate Meter (ARM) is a new TASM rule type
- ARM rules provide a means to regulate the flow of SQL requests admitted by TASM into the system
- An ARM rule specifies the max number of SQL requests per time unit that TASM will admit into the system. Qualification criteria can be used to target specific types of requests (e.g. 3 queries per hour for queries with estimated processing time of more than 1 hour)

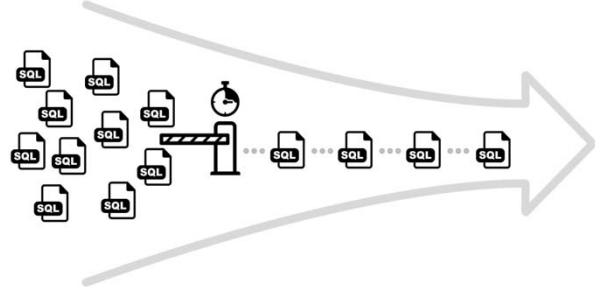


Freeway with Ramp Metering



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TASM Arrival Rate Meter



Arrival Rate Meters (ARMs) are TASM rules that provide a way to regulate the flow of requests admitted by TASM into the system. An ARM rule specifies the maximum number of requests per time unit that TASM will admit into the system. Qualification criteria can be used to target specific types of requests.

TASM's filter and throttle processing sequence for a SQL request is enhanced to check for ARM. New Workload Management ruleset APIs allow you to automate the creation of a system throttle or arrival rate meter without using Viewpoint Workload Designer.

Benefits

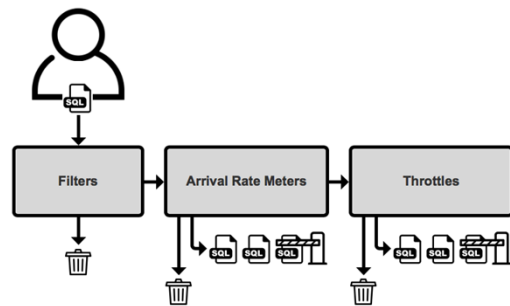
- Prevents a predictable request surge from flooding the TASM delay queue and impacting service time of subsequent queries.
- Limits resource-intensive requests to a specific rate.
- Prevents you from submitting many extremely complex queries frequently.

Considerations

- Use arrival rate meters carefully after an evaluation period in warning mode.

Deferring requests

- Order of processing: Filter → **ARM** → Throttle
- When an ARM limit is reached, queries will be held in the *Defer* queue (separate from the *Delay* queue)
- Once past the ARM check, a query is considered to be admitted into the system and is then subject to concurrency throttle rules
- A query can be held in the Defer queue and then in the Delay queue



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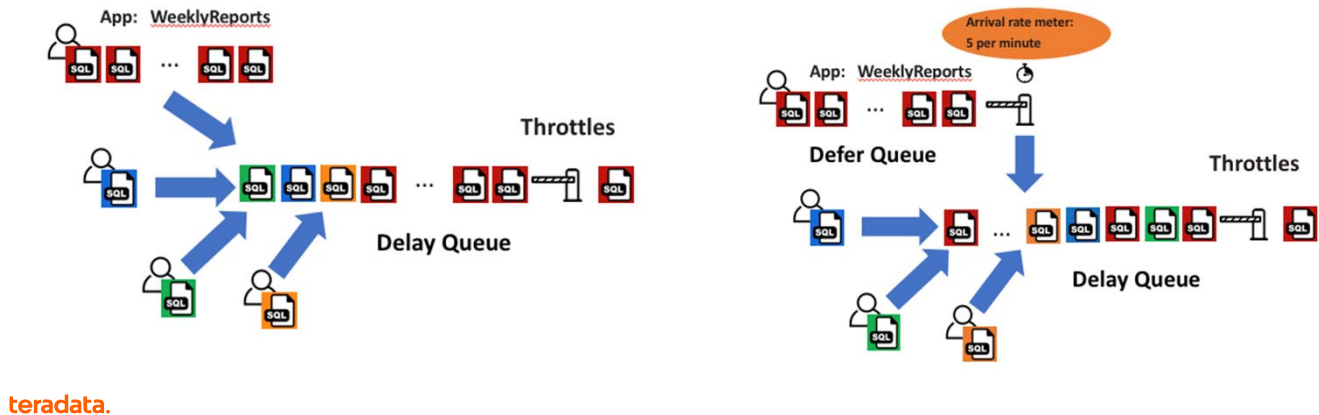
The TDWM order of processing is:

1. Filters
2. ARMs
3. Throttles

Arrival rate meter: Use case 1

Problem: Flood of requests from a specific application monopolizes the front of the delay queue. Subsequent requests added to end of the delay queue.

Solution: Create an ARM rule for this application so that other requests can get in between each time unit and can therefore be run sooner.



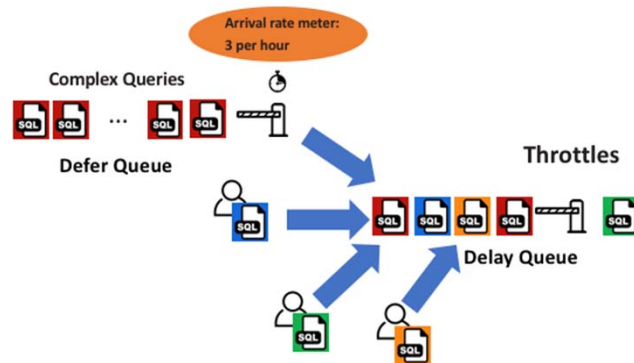
Problem: Flood of requests from a specific application monopolizes the front of the delay queue. Subsequent requests added to end of the delay queue.

Solution: Create an ARM rule for this application so that other requests can get in between each time unit and can therefore be run sooner.

Arrival rate meter: Use case 2

Problem: Control execution rate of complex queries to limit resource consumption.

Solution: Create an ARM rule for the complex queries to allow only a few each hour during weekdays and a higher limit on weekends.



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Problem: Control execution rate of complex queries to limit resource consumption.

Solution: Create an ARM rule for the complex queries to allow only a few each hour during weekdays and a higher limit on weekends.

Arrival rate meter: Rule attributes

Time Unit

Time Period for request rate (Options: second, minute, hour)

Limit

Max request rate per time unit (for example, 3 queries per minute)

Qualify Time

Number of seconds the current rate must continuously remain at the limit before the rule is enforced. (A zero value means the rule is enforced as soon as the limit is reached.)

Qualification Criteria

Same as Throttle: Request Source, Query Characteristics, Target and Query Band

Action

Action to perform when Limit and Qualify Time conditions are met. (Options: Delay, Reject, and Logging—warning mode)

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Arrival rate meter: Current arrival rate: Example

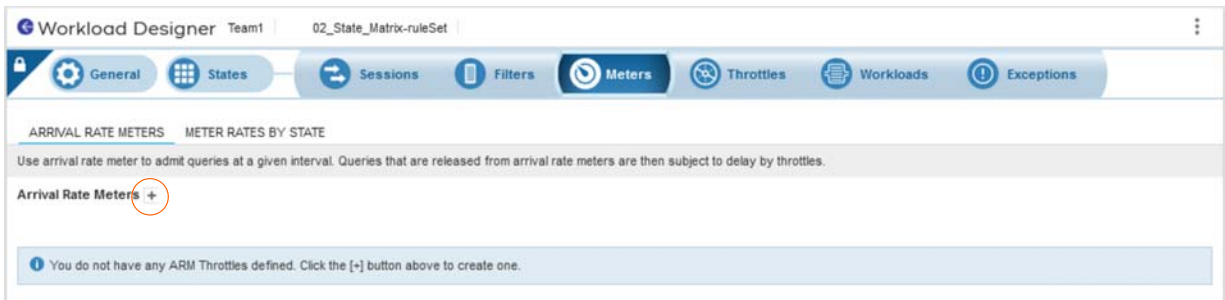
- The Current Arrival Rate is the number of admitted requests within the current time unit — updated every 1/6 of time unit
- Admitted request are no longer counted after one time unit regardless of its status.

Example: ARM limit of 10 requests per minute with 0 qualify time. (Current Arrival Rate is updated every 10 seconds.)

Interval Number	Interval Time	Expiration Time	Number of Arriving Requests	Current Arrival Rate	Comment
1	8:00:00 – 8:00:09	8:01:00	1	1	First request is admitted. Current rate is set to 1.
2	8:00:10 – 8:00:19	8:01:10	2	3	Second and third request are admitted. Current rate is set to 3.
3	8:00:20 – 8:00:29	8:01:20	2	5	Fourth and fifth request are admitted.
4	8:00:30 – 8:00:39	8:01:30	3	8	Sixth, seventh, and eighth request are admitted.
5	8:00:40 – 8:00:49	8:01:40	2	10	Ninth and tenth request are admitted.
6	8:00:50 – 8:00:59	8:01:50	1	10	Eleventh request is added to Defer Queue because current arrival rate has reached the limit.
7	8:01:00 – 8:01:09	8:02:00	3	10=>9=>10	The first request is no longer counted in the current rate. Current rate drops to 9; one deferred request (i.e., eleventh) is released; current rate goes back to 10. The 3 new requests arriving in this interval are added to Defer Queue (total of 3).
8	8:01:10 – 8:01:19	8:02:10	0	10=>8=>10	The second and third request are no longer counted in the current rate. Rate drops to 8; two deferred requests are released; current rate goes back to 10. No new request arrives in this interval. One request remains in Defer Queue.

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The arrival rate meters view



The following tabs on the Meters view display all meters information or a ruleset from different perspectives:

- **Arrival Rate Meters:** List of all the Meters, their rates, and classification criteria
- **Meter Rates by State:** List of all the states to which specific meters are applied

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Current topic: Creating arrival rate meters

- Arrival rate meter overview
- **Creating arrival rate meters**
 - **ARM general tab**
 - **Collective rule type**
 - **Disable manual release or abort**
 - **Classification criteria**
 - **State specific settings**
 - **Meter rates by state**

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ARM general tab

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Meters** >
Button: **Create an Arrival Rate Meter** [+] > Tab: **General**

Workload Designer Team1 02_State_Matrix-ruleSet

New Arrival Rate Meter

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Name: * VOWMArmRule

Description:

Rule Type: Collective: one queue for all matched queries

☒ Warning Only

☐ Disable manual Release or Abort

Enter the rule **name** and optionally the **description**

Rule Type is specifically set to **Collective** for all ARM rules

Warning mode applies the ARM rule, **logs any violations** but **does not reject or defer** the request

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Enter the rule **name** and optionally the **description**

Warning mode applies the ARM rule, **logs any violations** but **does not reject or defer** the request

You can also **disable** the ability to **manually release or abort requests** from the defer queue in the Query Monitor Portlet

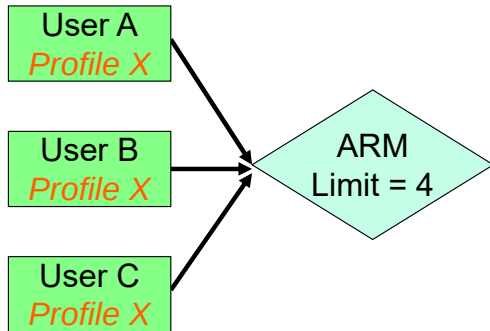
Rule Type is specifically set to **Collective** for all ARM rules

Collective

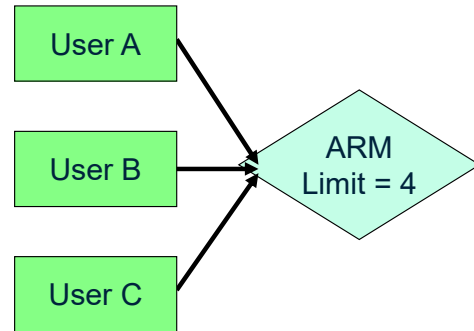
- Arrival Rate Meter rule will be applied to **all users**, as a collective group, that meet the classification criteria
- The **group** will get a maximum number of queries

Collective rule type with classification of profile

ARM rule was applied to Classification
Criteria of *Profile X*



ARM rule was applied to Classification
Criteria of *Username = User A, User B,
User C*



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The facing page shows an example to contrast the difference between the collective and members options.

Disable manual release or abort

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Meters** >
Button: **Create an Arrival Rate Meter** [+] > Tab: **General**



Workload Designer Team1 02_State_Matrix-ruleSet

New Arrival Rate Meter

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Name: * VOWMArmRule

Description:

Rule Type: Collective: one queue for all matched queries

☐ Warning Only

☐ Disable manual Release or Abort

Prevents manually releasing or aborting deferred requests in the Defer queue from the Query Monitor Portlet

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Select **Disable Manual Release or Abort** to prevent the manual aborting or releasing deferred queries in the delay queue via the Query Monitor portlet.

Classification criteria

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Meters** >
Button: **Create an Arrival Rate Meter [+]** > Tab: **Classification**

The screenshot shows the 'Workload Designer' interface. At the top, there's a breadcrumb trail: 'Workload Designer' > 'Team1' > '02_State_Matrix-ruleSet'. Below this, the title is 'Arrival Rate Meter: VOWMArmRule'. There are three tabs: 'GENERAL', 'CLASSIFICATION' (which is selected and highlighted with a red box), and 'STATE SPECIFIC SETTINGS'. A message states: 'Classification criteria determine the queries to which this Arrival Rate Meter will be applied.' Below this, there's a section 'Add Classification Criteria' with a dropdown menu currently showing 'Request Source'. The dropdown options are 'Request Source', 'Target', 'Query Characteristics', and 'Query Band'. To the right of the dropdown is an 'ADD' button.

Add the **Classification Criteria**

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Add the classification criteria that will be used to determine which requests the ARM rule applies.

State specific settings (1 of 2)

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Meters** >
Button: **Create an Arrival Rate Meter** [+] > Tab: **State Specific Settings**

Workload Designer Team1 02_State_MetricRuleSet

Arrival Rate Meter: VOWMRule

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Click on a state to set a meter limit.

Planned Environments	
	Always
Normal	Base (uses Default)
Degraded	Yellow (uses Default)
Critical	Red (uses Default)

Health Conditions

Default Settings

Default settings are applied to all states unless custom settings are applied.

Limit:

☐ Unlimited

☒ 5 Per minute

☒ Delay

☐ Reject

second
minute
hour

The **Default Setting** can be set to a **specific limit** and whether the requests exceeding the limit will be **Delayed** or **Rejected**

In the **For** pull-down menu select the Time Unit (Options: **second**, **minute**, **hour**)

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The **Default Setting** can be set to a **specific limit** and whether the requests exceeding the limit will be **Delayed** or **Rejected**

In the **For** pull-down menu select the Time Unit (Options: second, minute, hour)

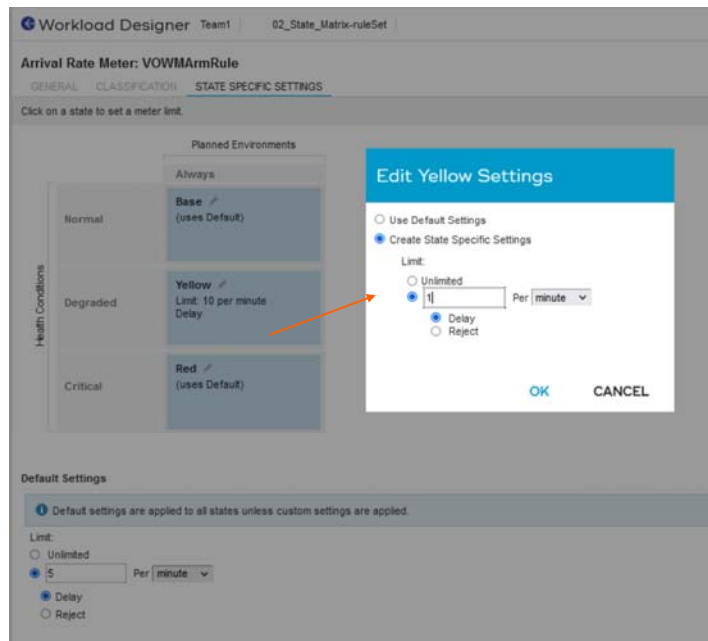
State specific settings (2 of 2)

To **override the default** setting for a specific State, select the State's "pen" button.

In the Edit dialog box, enter the **working value** settings for that state.

Notice that the Time Unit is not available in the settings for the individual states.

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To override the default setting, move your cursor over the State to display the "pen" icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State. Notice that the Time Unit is not available in the settings for the individual states. Each ARM rule is limited to a single time unit. Time units cannot be mixed across states. If a secondary state has a different time unit than the base state, the time unit for the secondary state is ignored and the time unit for the base state is used instead. A warning will be written to the DBC.TDWMEEventLog table.

Meter rates by state

Workload DesignerTeam102_State_Matrix-ruleSet

General

States

Sessions

Filters

Meters

Throttles

Workloads

Exceptions

ARRIVAL RATE METERS

METER RATES BY STATE

This view shows the maximum number of queries that are allowed to run concurrently for each meter in each state.

NAME	DEFAULT SETTING	BASE	RED	YELLOW
VOWMArmRule	5 per minute , DELAY	(Use Default)	0 per minute , REJECT	1 per minute , DELAY

Displays the working values for each state

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The Enabled by State tab displays the working values for each state.

Summary

In this module we covered how to:

- Describe how Workload Management can be used to improve response consistency and throughput in a mixed workload environment
- Describe the characteristics, components, and purpose of Arrival Rate Metering rules
- Explain how to create Arrival Rate Metering rules and when to use the various available options

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Module 11: Workload Designer: System throttles

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Describe how Workload Management can be used to improve response consistency and throughput in a mixed workload environment
- Describe the characteristics, components, and purpose of Throttle rules
- Explain how to create Throttle rules and when to use the various available options

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Topics

- Throttling overview
- The throttles tab
- Throttling examples and recommendations
- The AWT resource limits tab
- Lab activity

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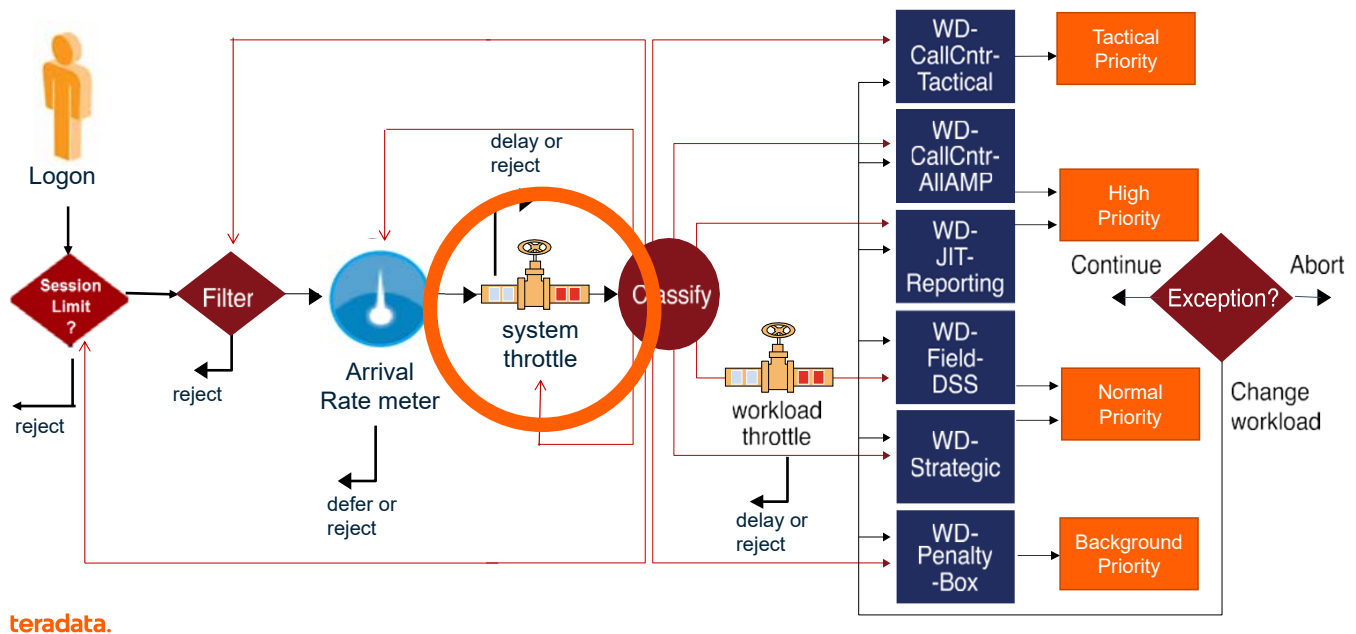
Current topic: Throttling overview

- **Throttling overview**
 - **Throttling levels**
 - **Throttling requests**
 - **Why use throttles?**
 - **Delay queue order**
 - **Bypass system throttles**
- The throttles tab
- Throttling examples and recommendations
- The AWT resource limits tab
- Lab activity

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Workload management levels: System throttle



There are seven different methods of management offered, as illustrated below:

Methods regulated prior to the query beginning execution

- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level.
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level

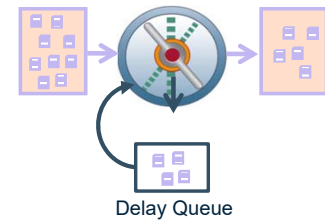
Methods regulated during query execution

- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

Throttling levels

The following are the different levels of throttling available:

- **Query Session Limits** – Used to limit the number of user sessions (covered in previous module)
- **Utility Limits** – Used to limit the number of bulk utilities (covered in previous module)
- **System Throttles** – Used to limit a subset of requests active on a system (covered in this module)
- **Virtual Partition Throttles** – Used to limit the number of requests active on a virtual partition with the exception of requests classified into non-throttled Tactical workloads (covered in this module)
- **Workload Throttles** – Used to limit the number of requests classified to the workload (covered in next module)
- **Workload Group Throttles** – Used to “collectively” limit the number of requests for a group of workloads. A Workload Throttle must first exist before it can be part of the group (covered in next module)



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The following lists different levels where these concurrency control rules that can be applied:

- Query Session throttles – Limit the number of sessions that are permitted to log on
- Workload throttles - An attribute of a workload and only control the requests that classify to the workload. Requests subject to these throttles can be may rejected or delayed
- System throttles – Control all or a subset of the requests active on the system. They may reject or delay requests. Standard “common” classification can be used to define which requests will qualify for a system throttle. All source criteria, target criteria, query band, or query characteristics criteria are available for use in defining a system throttle
- Group Throttles – A “collective throttle” limit for a group of WD rules. The WD rules that are a part of a group throttle must themselves be throttled in some state. If they are not throttled in the current state, their requests are still subject to the current Group Throttle limit. A WD rule can only belong to one Group Throttle

Virtual Partition Throttles – Defined to allow support for multi-tenancy. They limit the number of concurrent requests running in the Virtual Partition with the exception of requests classified into non-throttled tactical WDs.

Throttling requests

- Used to control the number of **concurrent requests**
- A **counter** is used to track the number of active requests
- When a new request is submitted for execution, the counter is **compared** against the limit
- If the counter is below the limit, the request is **allowed** to be executed immediately
- If the counter is equal to or above the limit, the request is either **delayed** or **rejected**
- Throttles can only control queries **prior to execution**
- Requests released from the delay queue **cannot be returned** to the delay queue
- The throttle delay queue can grow to be large as **16MB** which is large enough to accommodate about **40,000** delayed requests
- The requests in the delay queue are ordered by **time delayed** or by **workload priority**

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Throttles are used in controlling the number of concurrent requests. When a throttle rule is active, a counter is used to keep track of the number of requests (also referred to in this document as “queries”) that are active at any point in time among the queries under control of that rule. When a new request is ready to begin execution, the counter is compared against the limit specified within the rule. If the counter is below the limit, the request runs immediately; if the counter is equal to or above the limit, the request is either rejected or placed in a delay queue. Most often throttles are set up to delay requests, rather than reject them.

Once a request that has been delayed is released from the delay queue and begins running, it can never be returned to the delay queue. Throttles exhibit control before a request begins to execute, and there is no mechanism in place to pull back a request after it has been released from the delay queue. Requests are released from the delay queue if all applicable throttles are within limits.

The throttle delay queue can grow to be as large as 16MB, which is large enough to accommodate up to about 40,000 delayed queries.

Why use throttles?

- Using Throttles can have a **benefit** of improving **response consistency**, **throughput** and reducing a shortage of Amp Worker Tasks, CPU or memory
- **Limit** the number of **lower priority**, **long** running **queries** competing for limited system resources
- By limiting the number of concurrent queries, some queries may run longer, but overall **service percent and throughput** will **likely improve**
- To quiescent a system **for maintenance**, a throttle of limit 0 with delay, will allow all in-process work to complete and delay all new requests
- **After maintenance** work has been completed, the throttle can be disabled or limit set to another value to allow delayed requests to execute

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Using Throttles can have a benefit of improving response consistency, throughput and reducing a shortage of Amp Worker Tasks.

Limit the number of lower priority, long running queries competing for resources at one time.

By limiting the number of concurrent queries, some queries may run longer, but overall service percent and throughput will likely improve.

To quiescent a system for maintenance, a throttle of limit 0 with delay, will allow all in-process work to complete and delay all new requests.

After maintenance work has been completed, the throttle can be disabled or limit set to another value to allow delayed requests to execute.

Delay queue order

- Requests can be delayed by query **start time** or **workload priority** as specified on the **General** button > **Other** tab
- Workloads can be ordered in the Delay Queue by Priority value using the following **Workload Priority formulas**

Order the Throttle Delay Queue

☐ By time delayed

☒ By workload priority

Workload Method	Priority Value
Tactical	10000 + Virtual Partition allocation
SLG Tier 1	9000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 2	8000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 3	7000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 4	6000 + Virtual Partition allocation + SLG Tier allocation
SLG Tier 5	5000 + Virtual Partition allocation + SLG Tier allocation
Timeshare Top	4000 + Virtual Partition allocation
Timeshare High	3000 + Virtual Partition allocation
Timeshare Medium	2000 + Virtual Partition allocation
Timeshare Low	1000 + Virtual Partition allocation

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Starting with Teradata 15.10 there is a new option to order the delay queue by workload priority. A priority value is calculated for each workload based on the workload management method assigned to the workload. Requests in the delay queue are ordered from high to low based on the workload value. Ties are ordered by start time. If the option to order the delay queue by workload priority is not selected, the queue is ordered by query start time.

Bypass system throttles

Workload Management allows selected Users submitting requests to **circumvent** all **System Throttle rules** by turning off **System Throttle** rule checking for those users.

- Purpose is to **give exceptions** to one or more Users from a group that has been associated with a System Throttle
- There is **no partial bypass** for a subset of System Throttle
- Grant Bypass is **checked at logon** time. If the User is determined to be bypassed, the user is flagged, and no further checking is done
- Users submitting high priority requests, such as **tactical** queries, **Tpump** jobs, and the **Viewpoint data collector** user would be good candidates for being bypassed to avoid the overhead of System Throttle rules checking



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You can identify users submitting requests as bypass to circumvent Workload Management Throttle rules. This turns off the Workload Management rule checking for all the requests issued within the context of the user's session. The set of users that are designated as Bypass are referred to as "unrestricted users".

User DBC and TDWM are automatically given bypass status.

The purpose of the Bypass User option is to give exceptions to a subset of users from a group, when the group has been associated with a rule. There is no partial bypass for a subset of rules. Bypass applies to all System Throttle and Filter rules defined.

Whether or not a User is bypassed is checked at user logon time. If the User is determined to be bypassed when he/she first logs on, then no further check is done. Once a user's logon is flagged as bypassed, all queries will bypass System Throttle rules each time they enter the system.

Users associated exclusively with active data warehouse workloads, such as tactical queries or TPump jobs, are solid candidates for being bypassed. Throttles are best applied on low priority, heavy resource consuming requests.

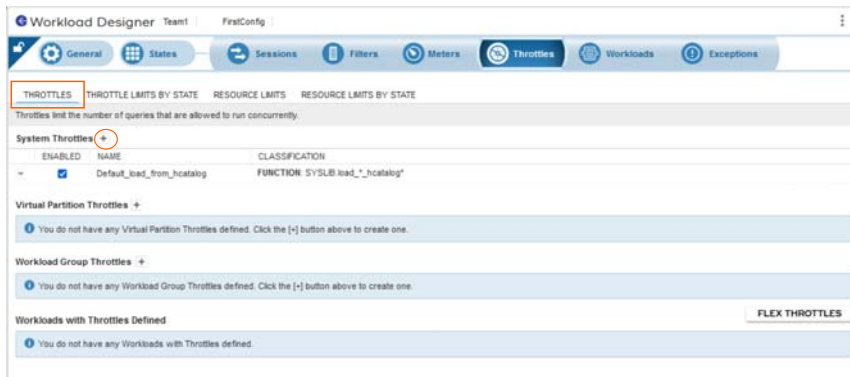
Current topic: The throttles tab

- Throttling overview
- **The throttles tab**
 - **Creating throttles**
 - **Creating system throttles**
 - **Rule types for system throttles**
 - **Disable manual release or abort**
 - **Classification criteria**
 - **State specific settings**
 - **Creating virtual partition throttles**
 - **State specific settings for virtual partition throttles**
 - **Throttle limits by state**
- Throttling examples and recommendations
- The AWT resource limits tab
- Lab activity

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Creating throttles



Throttles are used to **control** the **number** of queries that can **execute concurrently** and can be defined at the System, Virtual Partition, or Workload level.

Due to the potential impact of over-use of memory of certain SQL-H functions used for Hadoop, a pre-configured System Throttle to limit how many requests can use the SQL-H function is enabled.

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To create a new throttle rule, click Throttles on the toolbar and then click Create Throttle to create a new throttle rule.

The throttle delay queue starts at 4MB and can grow up to 16MB which is large enough to hold 40,000 queries.

Creating system throttles

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a System Throttle [+]** > Tab: **General**

Workload Designer Team1 FirstConfig

New Throttle

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Name: * VOMWSysThrottle

Description:

Rule Type:

- ☒ Collective: one queue for all matched queries
- ☐ Individual: one queue for each DB object
- ☐ Member: one queue for each user
- ☐ Disable manual Release or Abort

Enter the rule **Name** and optionally the **Description**.

Choose the **Rule Type** to apply:

- **Collective**
- **Individual**
- **Member**

and if the ability to **manually abort** or **release** requests from the delay queue will be disabled.

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On the General tab, enter the throttle rule name, up to 30 characters. Optionally, you can supply description of the rules, up to 80 characters.

System throttle rule types

- **Collective**
 - Throttle limits will be applied to **all users**, as a collective group, that meet the classification criteria
 - The **group** will get a maximum number of queries
- **Individual**
 - Throttle limits will be applied **individually** to each user that meet the classification criteria
 - **Each user** will get a separate query limit
- **Member**
 - Applies when **Account or Profile** is used as the **Classification Criteria** for the rule
 - Throttle limits are **placed on individuals** within the group, no limit is placed on the account or profile
 - Each member will get an **Individual** query limit

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Choose from the following three **Rule Type** options when you create a **System** throttle:

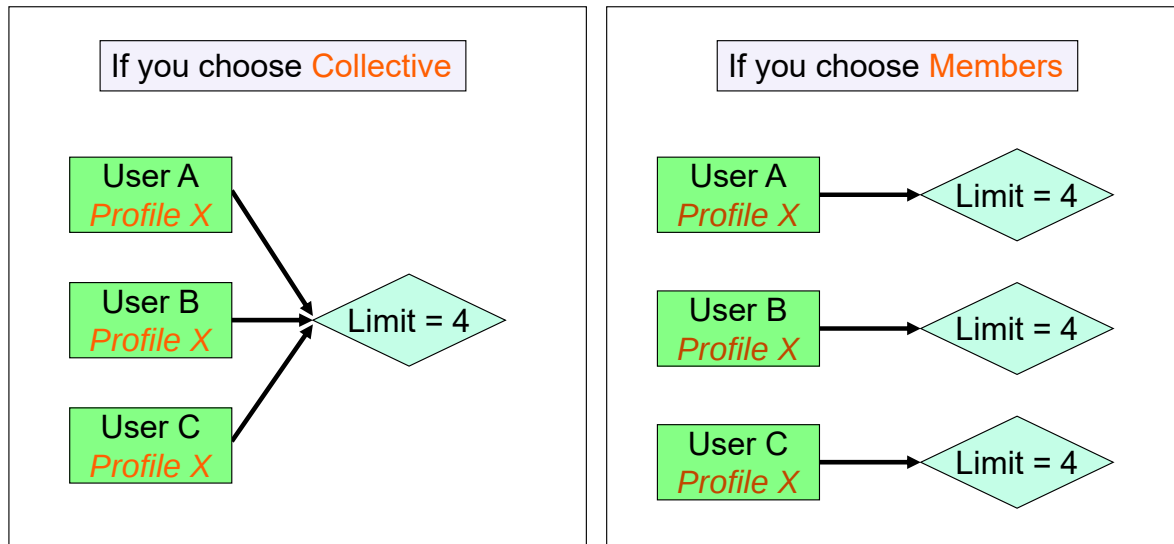
Collective: Provides the least granular control. TASM treats all requests that qualify for this throttle as one group. If the requests are delayed, the delay is based on the concurrency level of the entire group. For example, users Jay, Kay, and May are the classification criteria for a system throttle rule with a limit of 3 requests. With the collective option, all the users together cannot have more than 3 requests running at the same time.

Individual: Each individual object that is associated with the throttle is treated independently of other, similar objects. Each object has a counter. For example, if users Jay, Kay, and May are the classification for a throttle rule with a limit of 3 requests, each of the users can have up to 3 requests running at the same time.

Member: Provides the finest control. TASM treats each user in a throttled object (such as account) independently of other users in that account. The intent of the Member option is to enable objects that represent groups of users, such as accounts and profiles, to be used as the classification criteria for the rule. The rule specifies the group, and any member of the group is covered by the rule. For example, users Jay, Kay, and May are associated with account ShoeDept. If there is a system throttle rule with a session limit of 3, classification criteria of account ShoeDept, and the Member option, then each of the users with the ShoeDept account can have up to 3 requests running at the same time.

Collective and members example

Throttle Limit was applied to Classification Criteria of *Profile X*



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The facing page shows an example to contrast the difference between the collective and members options.

Collective:

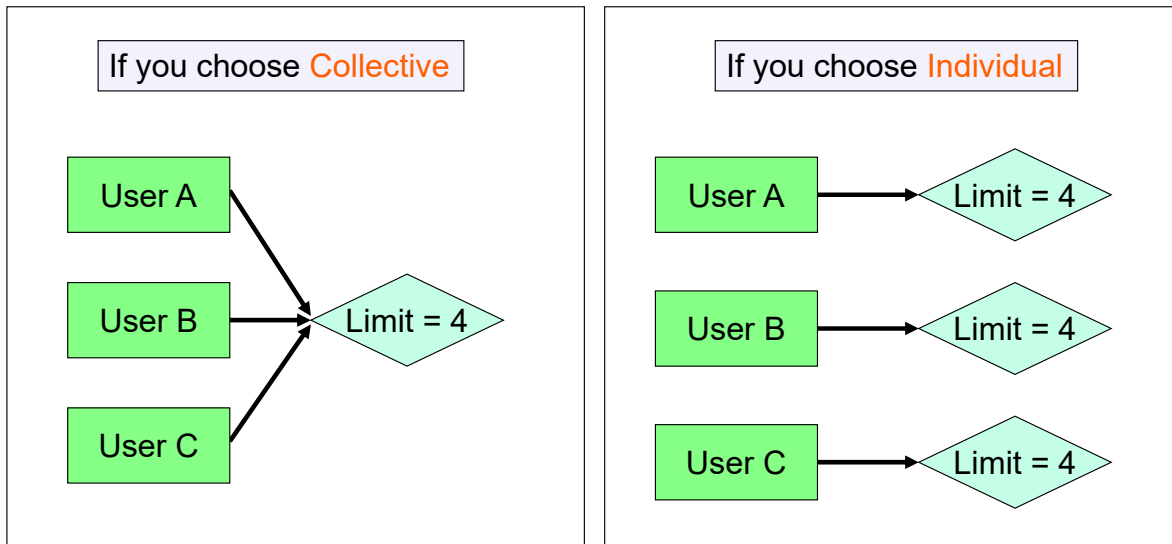
All requests that satisfy the classification criteria are throttled as a single group.

Member:

Each user that satisfies the classification criteria is throttled at the user level.

Collective and individual example

Throttle Limit was applied to Classification Criteria of *Username = User A, User B, User C*



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The facing page shows an example to contrast the difference between the collective and individual options.

Collective:

All requests that satisfy the classification criteria are throttled as a single group.

Individual:

- Can only be defined with a single classification criteria request source type
- Each source type object that satisfies the classification criteria is throttled as an individual group
- If the classification criteria identifies only a single source type object, the throttle will behave the same as a collective rule type

System throttle scenario: Collective and member

Scenario

An administrator wants to allow only five concurrent requests from users with the profile LawRpt to run concurrently system wide.

To be fair, the administrator wants to ensure no single LawRpt profile user can run more than two concurrent requests.

Solution

Create a system throttle with a classification of request source profile equal LawRpt, a rule type of **collective**, and a delay concurrency limit of **five**.

Create another system throttle with the same profile classification, but having a rule type of **member**, and a delay concurrency limit of **two**.

Request Time	User	Profile	Collective Limit 5	Member Limit 2	Delayed?
08:00:00	UL222	LawRpt	OK (requests active 0 < 5)	OK (requests active for UL222 0 < 2)	No
08:00:01	UL222	LawRpt	OK (requests active 1 < 5)	OK (requests active for UL222 1 < 2)	No
08:00:02	UL222	LawRpt	OK (requests active 2 < 5)	Delay, requests active for UL222 = limit	Yes
08:00:03	UL222	LawRpt	OK (requests active 2 < 5)	Delay, requests active for UL222 = limit	Yes
08:00:10	UL888	LawRpt	OK (requests active 2 < 5)	OK (requests active for UL888 1 < 2)	No
08:00:12	UL123	LawRpt	OK (requests active 3 < 5)	OK (requests active for UL123 1 < 2)	No
08:00:15	UL111	LawRpt	OK (requests active 4 < 5)	OK (requests active for UL111 1 < 2)	No
08:00:18	UL888	LawRpt	Delay, requests active = limit	OK (requests active for UL888 1 < 2)	Yes

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This example shows how you can create throttles to ensure that no single user of a group can dominate the limit placed on the group.

Disable manual release or abort

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a System Throttle [+]** > Tab: **General**

Workload Designer Team1 FirstConfig

New Throttle

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Name: * VOMWSysThrottle

Description:

Rule Type:

- ☒ Collective: one queue for all matched queries
- ☐ Individual: one queue for each DB object
- ☐ Member: one queue for each user
- ☒ Disable manual Release or Abort

Prevents manually releasing
or **aborting** throttled queries
in the delay queue.

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Select **Disable Manual Release or Abort** to prevent Analytics Database Administrators from aborting or releasing throttled queries in the delay queue via the Query Monitor portlet.

Classification criteria

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a System Throttle [+]** > Tab: **Classification**

The screenshot shows the 'Workload Designer' interface. At the top, there's a breadcrumb trail: 'Workload Designer' > 'Throttles' > 'Classification'. Below this, the title is 'Throttle: VOMWSysThrottle'. There are three tabs: 'GENERAL', 'CLASSIFICATION' (which is selected and highlighted with a red box), and 'STATE SPECIFIC SETTINGS'. A message states: 'Classification criteria determine the queries to which this throttle will be applied.' Below this, there's a section 'Add Classification Criteria' with a dropdown menu currently showing 'Request Source'. Other options in the dropdown are 'Request Source', 'Target', 'Query Characteristics', and 'Query Band'. An 'ADD' button is next to the dropdown.

Add the **Classification Criteria**.

Utility is **not** a **Classification choice**; they are controlled through Utility Limits.

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Add the classification criteria that will be used to determine which requests the throttle rule applies.

State specific settings

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles**>
Button: **Create a System Throttle [+]** > Tab: **State Specific Settings**

The **Default** Setting for every State is **Unlimited**.

The **Default Setting** can be set to a **specific limit** and whether the requests exceeding the limit will be **Delayed** or **Rejected**.

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The screenshot shows the 'Workload Designer' interface with the 'Throttle: VOWMSysThrottle' configuration. The 'STATE SPECIFIC SETTINGS' tab is selected. It displays a table for 'Planned Environments' with columns for 'Health Conditions' and 'Planned Environments'. The table has three rows: 'Normal', 'Degraded', and 'Critical'. Each row has a 'Base' setting with a checkmark and the text '(use Default)'. Below the table, the 'Default Settings' section is visible, showing 'Concurrency Limit' set to 'Unlimited' (selected with a radio button) and 'Delay' selected (selected with a radio button).

Health Conditions	Planned Environments
Normal	Base ✓ (use Default)
Degraded	Yellow ✓ (use Default)
Critical	Red ✓ (use Default)

Default Settings

Concurrency Limit:

☒ Unlimited

☐ []

☒ Delay

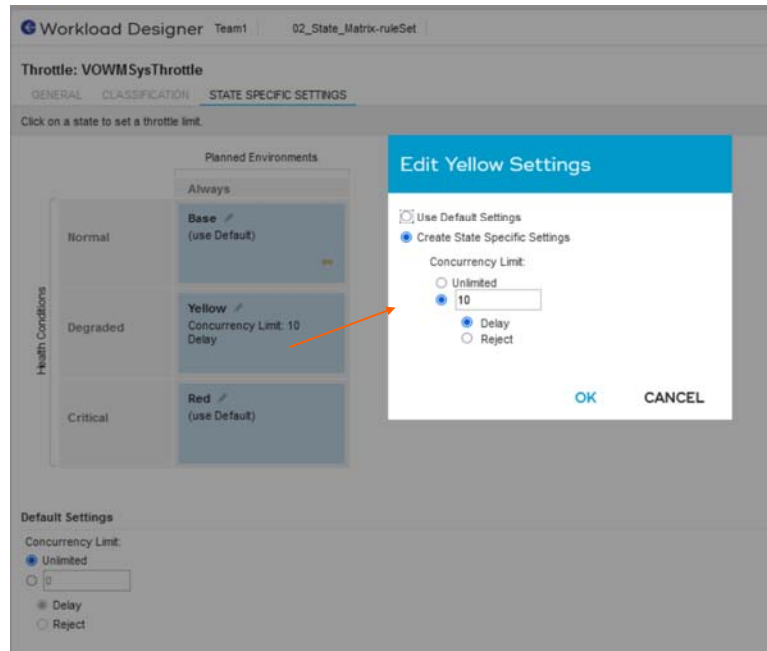
☐ Reject

The default setting for throttles is unlimited. This can be changed to a specific limit and whether requests exceeding that threshold will be delayed or rejected.

State specific settings (cont.)

To **override the default** setting for a specific State, select the State's "pen" button.

In the Edit dialog box, enter the **working value** settings for that state.



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To override the default setting, move your cursor over the State to display the "pen" icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

Creating virtual partition throttles

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a Virtual Partition Throttle** [+]

The screenshot shows the 'Workload Designer' interface with the 'Throttles' tab selected. The 'Throttles' section is highlighted with a red box. Below it, the 'System Throttles' section is visible. The 'Virtual Partition Throttles' section is also visible, with a red circle around the '+' button. The 'New Virtual Partition Throttle' dialog is open, showing the 'GENERAL' tab. The 'Name' field is filled with 'VOWMVpThrottle'. The 'Description' field is empty. The 'Virtual Partition' dropdown is set to 'Standard'. The 'Disable manual Release or Abort' checkbox is unchecked.

Enter the rule **name** and optionally the **description**.

Choose the **Virtual Partition**, and if the ability to manually **abort** or **release** requests from the delay queue will be disabled.

There will not be a Classification Tab since the limits apply to a Virtual Partition.

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On the General tab, enter the throttle rule name, up to 30 characters. Optionally, you can supply description of the rules, up to 80 characters.

- Virtual partition request throttles (i.e., virtual partition throttles) manage how many requests can run at the same time in a virtual partition
- Virtual partition throttles do not allow the specification of classification criteria, the setting simply controls the limit of requests for various states
- Virtual partition throttles can help influence the balance between multiple virtual partitions so one virtual partition does not capitalize the utilization of memory or AMP worker tasks (AWTs)
- Consider a relatively lower throttle limit for a virtual partition with complex requests with multiple parallel steps using more AWTs
- Consider a relatively higher throttle limit for a virtual partition with shorter, simpler requests using fewer AWTs

State specific settings

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles**>
Button: **Create a Virtual Partition Throttle [+]** > Tab: **State Specific Settings**

Workload Designer Team1 02_State_Matrix.ruleSet

Virtual Partition Throttles: VOWMvpThrottle

GENERAL STATE SPECIFIC SETTINGS

Click on a state to set a throttle limit.

Planned Environments	
Health Conditions	Normal
	Degraded
	Critical

Base (use Default)

Yellow (use Default)

Red (use Default)

Default Settings

Concurrency Limit:

☒ Unlimited

☐ 0

☐ Delay

☐ Reject

The **Default** Setting for every State is **Unlimited**.

The **Default Setting** can be set to a **specific limit** and whether the requests exceeding the limit will be **Delayed** or **Rejected**.

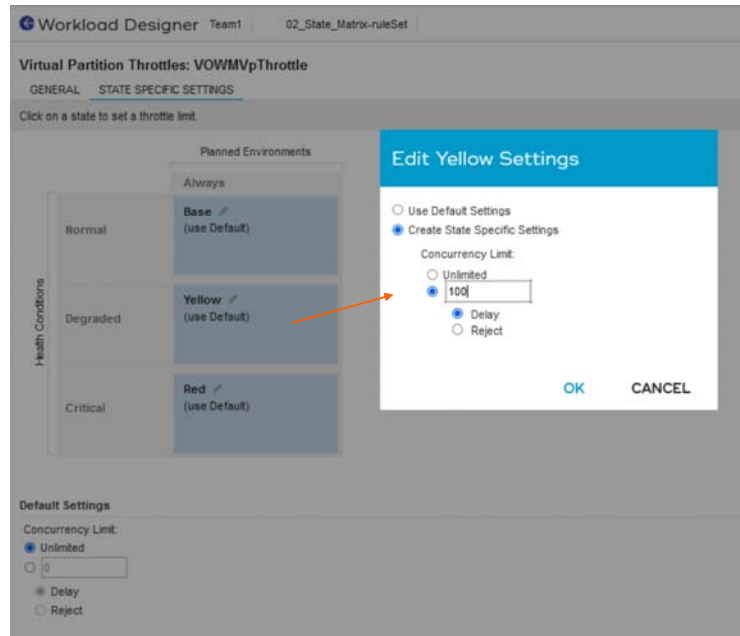
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The default setting for throttles is unlimited. This can be changed to a specific limit and whether requests exceeding that threshold will be delayed or rejected.

State specific settings (cont.)

To **override the default** setting for a specific State, select the State's **"pen"** button.

In the Edit dialog box, enter the **working value** settings for that state.



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To override the default setting, move your cursor over the State to display the "pen" icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

Throttle limits by state

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttle Limits by State**

Workload DesignerTeam102_State_Matrix-ruleSet

GeneralStatesSessionsFiltersMetersThrottlesWorkloadsExceptions

THROTTLES

THROTTLE LIMITS BY STATE

RESOURCE LIMITS

RESOURCE LIMITS BY STATE

This view shows the maximum number of queries that are allowed to run concurrently for each throttle in each state.

System Throttles

NAME	DEFAULT SETTING	BASE	RED	YELLOW
Default_load_from_hcatalog	2, DELAY	(Use Default)	(Use Default)	(Use Default)
VOWMSSysThrottle	Unlimited	(Use Default)	0, REJECT	10, DELAY

Virtual Partition Throttles

NAME	DEFAULT SETTING	BASE	RED	YELLOW
VOWMVpThrottle	Unlimited	(Use Default)	0, REJECT	100, DELAY

Workload Group Throttles

You do not have any Workload Throttles defined.

Workload Throttles

NAME	DEFAULT SETTING	BASE	RED	YELLOW
NOS	1, DELAY	(Use Default)	(Use Default)	(Use Default)

Displays the working values for each State.

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The Throttle Limits by State tab displays the working values for each state.

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Workload Designer: System throttles Slide 11-25

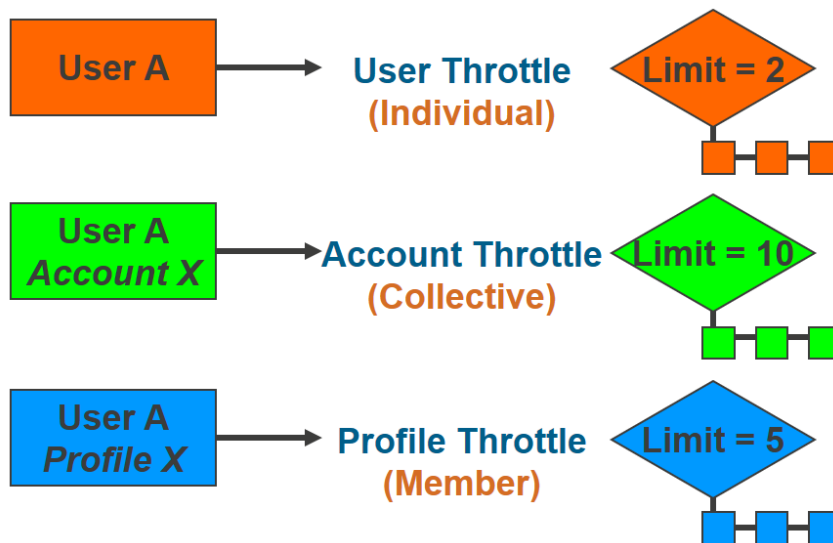
Current topic: Throttling examples and recommendations

- Throttling overview
- The throttles tab
- **Throttling examples and recommendations**
 - **Overlapping associations**
 - **Average response time examples**
 - **Throttle recommendations**
- The AWT resource limits tab
- Lab activity

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Overlapping associations



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User A logs on using Account X and Profile X and submits a query.

Which throttle limit will be applied?

When a request is subject to **more than one** throttle rule, the **most restrictive** thresholds will take precedence.

All throttle counters limits **must be below** their counters for a request to run.

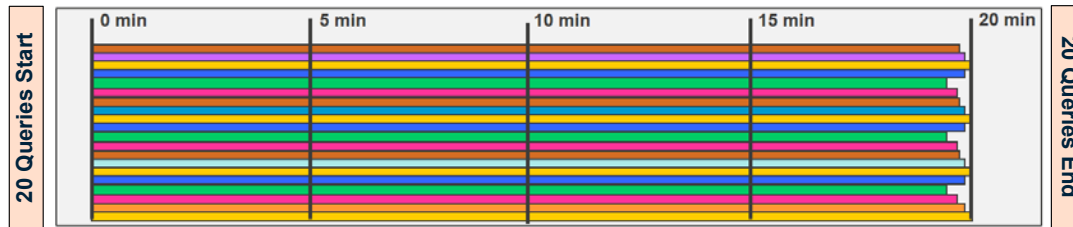
Any time you have a query which is under the control of more than one Throttle, and these different Throttles are each classified to the same User, all throttles must be satisfied for the query to be released for execution.

For example, if you created a Throttle rule associated with a specific User, a different Throttle rule associated with a specific Account, and a third one associated with a specific Profile. In other words, you set up three rules, each associated with a different classification. A query issued by *that* User within *that* Account and Profile will not be removed from the delay queue until it has satisfied all three limits. Only one Throttle limit per classification will be enforced, but because there are several different allowable classifications, a given query could be required to satisfy several Throttle rules before it runs.

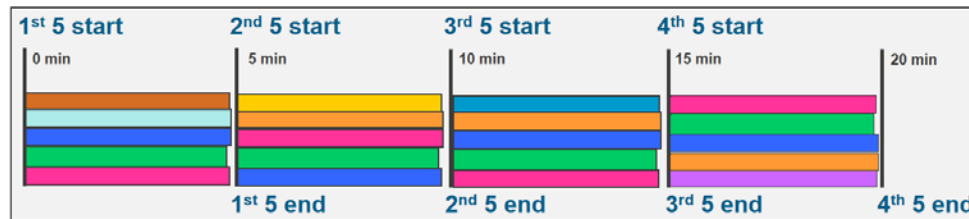
Which Throttle applies?

- What if... User A already has 2 queries running and submits a third? User Throttle applies.
- What if... User has no queries running, other Account Users have 10 queries currently running, and User A submits a query? Account Throttle applies. Would the Profile Throttle ever apply to User A. No, because the User Throttle will not allow User A to have more than 2 queries.

Average response time example



- Twenty 1-minute queries begin at the same time.
- Assume each query can use 100% of available system resources.
- Average response time will be **20 minutes**.



- Twenty queries with a query limit of 5 have an average response time of **12.5 minutes**
- Fewer active queries means faster response times for active queries

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The facing page shows a theoretical example of using object throttles to improve average response time.

Average response time example (cont.)

	<u>LT 20</u>	<u>20-40</u>	<u>40-60</u>	<u>60-80</u>	<u>GT 80</u>	(time in minutes)
No limit	0	6	25	35	54	# of queries that completed in that time range
60 limit	0	23	30	17	50	
40 limit	0	32	28	13	47	
20 limit	20	20	21	25	34	

- Test was executed with **120 complex queries** concurrently with and without query limits
- With a query limit of 20:
 - Some queries got less than 20-minutes response time
 - The percentage of queries in the worst-performing (GT 80) bucket is less
- With higher concurrency levels, queries get less resources and run longer

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A test was executed to better understand any advantages or disadvantages you might experience by delaying queries in a situation where all the work running on the platform was similar in nature.

In this test a total of 120 identical complex queries were executed, each using their own session, each session submitting its single query at the same time. This simulates a situation where 120 Users each log and issue a query at the same time. For all tests, a Throttle was defined in TDWM to limit concurrency at the Performance Group level. All the user sessions were in the same Performance Group. Four separate tests were run using these different thresholds:

- 20 sessions
- 40 sessions
- 60 sessions
- 120 sessions

The purpose of this test was to show that response time experiences for the average user can be improved by applying a limit to the concurrency, even though some users will experience longer run times than others due to being delayed in their start times.

The table shows a count of how many queries in each category completed within the ranges of time specified in the column headings.

This test shows that with limits of 20 concurrent queries in place, a greater number of end users will receive comparatively good query response times. For example, when the Workload Limit is set at 20, 1/6th of the users have very good turn-around, less than 20 minutes each, which is not achievable with either 40 or 60 concurrent queries. In addition, with that lower limit, fewer users must wait longer than 80 minutes for their queries to return an answer.

Throttle recommendations

- Throttles have their **highest impact** when applied against low priority, heavy resource consuming queries
- **Reducing the competition** for resource can improve overall service percent and throughput
- **Do not throttle high priority**, low resource consuming queries such as Tactical queries
- Throttles applied at a User and Group level can control an individual user from over dominating the group and apply **fairness** to the other users in the group

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Throttles have their highest impact when applied against low priority, heavy resource consuming queries.

Reducing the competition for resource can improve overall service percent and throughput.

Throttles can help avoid exhaustion of system resources such as AWTs.

Do not throttle high priority, low resource consuming queries such as Tactical queries.

Throttles applied at a User and Group level can control an individual user from over dominating the group and apply fairness to the other users in the group.

Current topic: The AWT resource limits tab

- Throttling overview
- The throttles tab
- Throttling examples and recommendations
- **The AWT resource limits tab**
 - **AWT resource limits**
 - **Creating AWT resource limits**
 - **Classification criteria**
 - **State specific settings**
 - **Resource limits by state**
- Lab activity

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AWT resource limits

- Prior to TD15.10, Workload Management enforces a default maximum AWT resource limit for FastLoad, MultiLoad, MLOADX, and FastExport of 60% of the total AWTs
- In TD15.10, Workload Management now supports a user-defined AWT resource limit and additional default AWT resource limits
- When a utility job is submitted, TASM checks the job's AWT requirement against the applicable AWT resource limits
- If the AWT resource limits are exceeded, the job is either delayed or rejected even if the applicable throttles have not been exceeded
- The following table shows the number of AWTs needed for a utility job to start:

Protocol	Required AWTs to Start
FastLoad	3
MultiLoad	2
FastExport No Spool	2
MLOADX	MIN (2, # Target Tables)
DSA Backup	2
DSA Restore	3

AWT resource limits are not checked for FastExport with Spool or ARCMAN.

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Prior to Teradata Database 15.10, Workload Management enforces one default AWT resource limit for FastLoad, MultiLoad, MLOADX, and FastExport utilities; that is, no more than 60% of the total AWTs can be used to support all of these utilities combined. Starting with Teradata Database 15.10, Workload Management supports user-defined AWT resource limits and enforces additional default AWT resource limits.

When a ruleset is activated, Workload Management dynamically creates default AWT resource limits if there are no user-defined AWT resource limits. The number of default AWT resource limits and their values depend on the setting of the "Support increased MLOADX job limits and increased AWT resource limits" option.

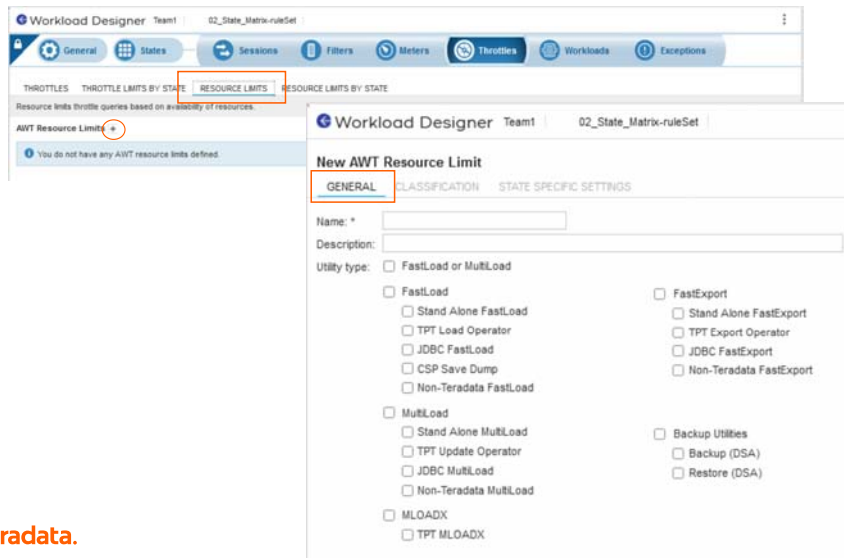
- If this option is not selected (default), the following AWT resource limits may be dynamically created:
 - Utility type: All FastLoad, FastExport, MultiLoad, MLOADX
 - AWT Limit: 60% of maximum AMP Worker Tasks
 - Utility type: DSA Backup, DSA Restore
 - AWT Limit: 70% of maximum AMP Worker Tasks
- If the "Support increased MLOADX job limits and increased AWT resource limits" option is selected, the following AWT resource limits may be dynamically created:
 - Utility type: All FastLoad, FastExport, MultiLoad, MLOADX
 - AWT Limit: 70% of maximum AMP Worker Tasks
 - Utility type: DSA Backup, DSA Restore
 - AWT Limit: 70% of maximum AMP Worker Tasks

When Workload Management is enabled, it overrides the dbscontrol general fields

MaxLoadTasks, MaxLoadAWT, MLOADXUtilityLimits, MaxMLOADXTasks, and MaxMLOADXAWT.

Creating AWT resource limits

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Resource Limits** >
Button: **AWT Resource Limits[+]** > Tab: **General**



AWT Limits can only be applied to utilities.

When Workload Management under SLES11, the DBSControl general fields **MaxLoadAWT** and **MaxMLOADXAWT** are ignored.

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To create a new AWT Resource Limit rule, click Throttles on the toolbar, select the **Resource Limits** tab and then click **+** icon to create a new AWT Resource Limit rule.

- AMP Worker Tasks (AWTs) are the threads inside each AMP that complete the work of the database software
- AWTs limit the number of active processes performing database work within each AMP
- The purpose of AWT resource limits is to prevent utility jobs, as a group, from using too many AWTs and possibly impacting non-utility work
- AWT resource limits can be defined in the Resource Limits tab of the Throttles view in the Workload Designer
- These TASM settings may override the DBS Control general fields MaxLoadTasks, MaxLoadAWT, MLOADXUtilityLimits, MaxMLOADXTasks, and MaxMLOADXAWT

Classification criteria

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Resource Limits** >
Button: **AWT Resource Limits[+]** > Tab: **Classification**

Workload Designer Team1 02_State_Matrix-ruleSet

AWT Resource Limit: VOWMResourceLimit

GENERAL **CLASSIFICATION** STATE SPECIFIC SETTINGS

Classification criteria determine the queries to which this limit will be applied.

Add Classification Criteria Request Source ADD

Request Source

Query Band

Add the **Request Source** and **Query Band** Classification Criteria.

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Add the Request Source and Queryband classification criteria that will be used to determine which requests the AWT Resource Limit rule applies.

State specific settings

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Resource Limits** >
Button: **AWT Resource Limits[+]** > Tab: **State Specific Settings**

Workload Designer Team1 02_State_Matrix-ruleSet

AWT Resource Limit: VOWMResourceLimit

GENERAL CLASSIFICATION STATE SPECIFIC SETTINGS

Set the max percent of AWTs that can be made available to selected utilities.

Planned Environments

	Always
Normal	Base (use Default)
Degraded	Yellow (use Default)
Critical	Red (use Default)

Health Conditions

Default Settings

Percent of AWTs: * %

☒ Delay ☐ Reject

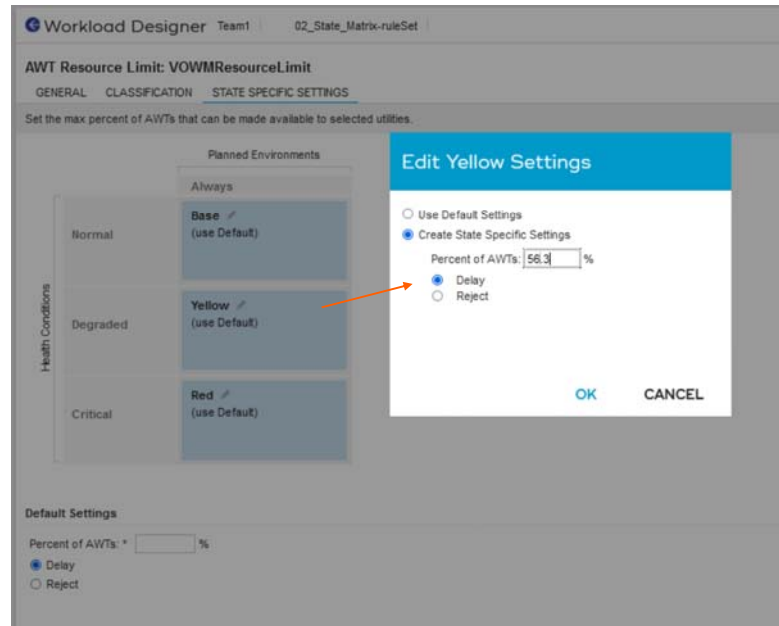
The limit value is entered as a **percentage** of the total AWTs.

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Specify the default percentage of total AWTs that will be available for utilities and if the utility will be delayed or rejected of percentage of AWTs are not available.

State specific settings (cont.)

If the total AWTs is 80, a limit of 45 AWTs is entered as 56.3% ($80 * 56.3\%$).



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To override the default setting, move your cursor over the State to display the “pen” icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

Resource limits by state

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Resource Limits** >
Button: **AWT Resource Limits[+]** > Tab: **Resource Limits by State**

Workload Designer Team1 02_State_Matrix-ruleSet

General States Sessions Filters Meters Throttles Workloads Exceptions

THROTTLES THROTTLE LIMITS BY STATE RESOURCE LIMITS **RESOURCE LIMITS BY STATE**

This view shows the maximum percentage of AWTs that may be available for each resource limit rule in each state. The lowest applicable limit shown in any section will be enforced.

AWT Resource Limits

NAME	DEFAULT SETTING	BASE	RED	YELLOW
VOWMResourceLimit	60%, DELAY	(Use Default)	(Use Default)	56.3%, DELAY

System Default AWT Resource Limits

UTILITY TYPE	LIMIT
All load utilities	60%
DSA	70%

Displays the working values for each state.

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The Resource Limits by State tab displays the working values for each state.

Current topic: Lab activity

- Throttling overview
- The throttles tab
- Throttling examples and recommendations
- The AWT resource limits tab
- **Lab activity**
 - (refer to day 3: Lab guide)
 - **Define filter rules**
 - **Define system throttle rules**
 - **Save and activate the ruleset**
 - **Run the Mixed-Workload simulation and capture performance metrics**

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Summary

- Set of Throttle Rules can be defined to determine if query requests will be accepted, **delayed**, **or rejected**
- Throttle rules can consider “**how many**” requests are going to be able to run **concurrently**
- Throttle rules can provide for **more consistent** service percent and throughput
- Throttles can help in **avoiding the exhaustion** of system resources, such as AMP Worker Tasks, CPU, or memory
- Throttles are best applied to **low priority**, heavy resource consuming requests
- The default number of AWTs available for utilities is **60%** of the total number of AWTs
- AWT **Resource Limits** can be used override the default number of AWTs available for utilities
- When Workload Management is enabled, the DBSControl general fields **MaxLoadAWT** and **MLOADXAWT** are **ignored**

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Set of Throttle Rules can be defined to determine if query requests will be accepted, delayed, or rejected.

Throttle rules can consider “how many” requests are going to be able to run concurrently.

Throttle rules can provide for more consistent service percent and throughput.

Throttles can help in avoiding the exhaustion of system resources, such as AMP Worker Tasks, CPU, or memory.

Throttles are best applied to low priority, heavy resource consuming requests.

The default number of AWTs available for utilities is 60% of the total number of AWTs.

AWT Resource Limits can be used override the default number of AWTs available for utilities.

When Workload Management is enabled, the DBSControl general fields MaxLoadAWT and MLOADXAWT are ignored.

Module 12: Workload Designer: Workloads and exceptions

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Use Workload Designer to create and modify workload definitions and exceptions
- Understand a workload definitions defining characteristics
- Understand the components of the defining characteristics

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Topics

- Workloads overview
- Creating a workload
- Flex throttles
- Workload group throttles
- Additional options for workloads
- Workload exceptions
- Additional workload tabs

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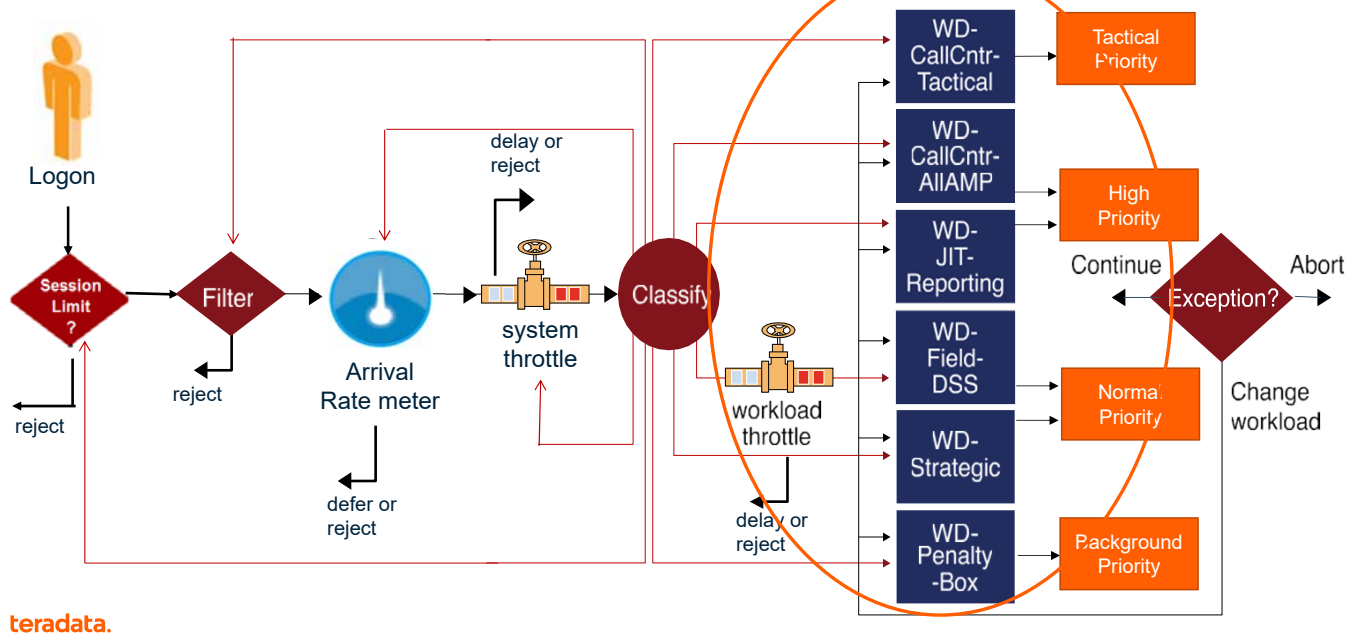
Current topic: Workloads overview

- **Workloads overview**
 - **What is a workload?**
 - **Advantages of workloads**
 - **Default workload**
- Creating a workload
- Flex throttles
- Workload group throttles
- Additional options for workloads
- Workload exceptions
- Additional workload tabs

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Workload management levels: Workloads



There are seven different methods of management offered, as illustrated below:

Methods regulated prior to the query beginning execution

- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level

Methods regulated during query execution

- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

What is a workload?

- A Workload is a **group of requests** with **common** characteristics
- Workloads are derived primarily from the **business** requirements (Users)
- Workloads can then be **supplemented** with **technical characteristics** (CPU, I/O, #AMPS, run time, etc.)
- Workloads consist of:
 - Fixed Characteristics
 - Classification Criteria – Characteristics that qualify a query to a workload
 - Exception Criteria – Operating rules a query is expected to adhere to during execution
 - Exception Actions – Automated action that will be taken when operating rules are violated
 - Workload management method
 - Working Values
 - Execution Rules – concurrency throttles, and exception enabling
 - Share percents
 - Service Level Goals – used to track workload performance
 - Minimum response time
- Workload Guidelines:
 - **5 System Workloads** – 1 default and 4 internal (T, H, M, L) workloads
 - Maximum **250** Defined Workloads
 - **Typical** initial number is between **10 to 30**

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A workload represents a portion of the queries that are running on a system. A Workload Definition (WD) is a workload grouping and its operating rules to assist in managing queries. The requests that belong to the same workload will share the same workload management controls. It consists of:

Classification Criteria: criteria to determine which queries belong to the workload. This criteria defines characteristics which are detectable prior to request execution. This is also known as the *"who"*, *"where"*, and *"what"* criteria of a request. For example, *"who"* may be an account name, *"where"* is the database tables being accessed, and *"what"* may be the type of statement (UPDATE), or estimated resource consumption being executed.

Exception Criteria: criteria to specify "abnormal" behavior for queries in this workload. This criteria is only detectable after a request has begun execution. If an exception criteria is met, the request is subject to the specified exception action which may be to lower the priority or abort the request.

Advantages of workloads

What are the **advantages** of Workload Definitions?

- Improved Control of **Resource Allocation**
 - Resource **priority** is given based on belonging to a particular workload
 - **Classification** rules permit queries to run at the **correct priority** from the start
 - Ability to **control** high resource **consumption** using **throttles**
- Improved **Reporting**
 - Workload definitions allow you to see **who is using the system** and how much of the various system resources
 - **Service level statistics** are reported for each workload
 - Real-time and long-term trends for workloads are available
- Automatic **Exception Detection** and Handling
 - After a query has started executing, a query that is running in an **inappropriate** manner can be automatically detected. Actions can be taken based on exception criteria that has been defined for the workload

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The reason to create workload definitions is to allow TASM to manage and monitor the work executing on a system.

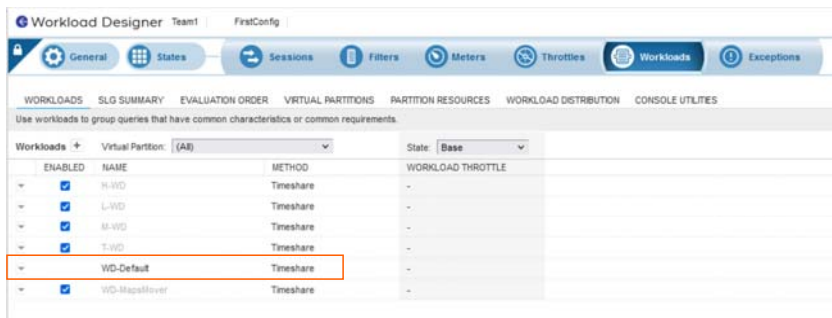
There are three basic reasons for grouping requests into a workload definition.

Improved Control – Some requests need to obtain higher priority to system resources than others. Resource priority is given based on belonging to a particular workload.

Accounting Granularity – Workload definitions allow you to see who is using the system and how much of the various system resources. This is useful information for performance tuning efforts, workload management and capacity planning.

Automatic Exception Handling – Queries can be checked for exceptions while they are executing, and if an exception occurs, a user-defined action can be triggered.

Default workload



The screenshot shows the 'Workloads' tab in the Teradata Workload Designer. The interface includes a top navigation bar with tabs for General, States, Sessions, Filters, Meters, Throttles, Workloads, and Exceptions. Below this is a sub-navigation bar with links for WORKLOADS, SLQ SUMMARY, EVALUATION ORDER, VIRTUAL PARTITIONS, PARTITION RESOURCES, WORKLOAD DISTRIBUTION, and CONSOLE UTILITIES. The main content area displays a table of workloads. The 'WD-Default' workload is highlighted with a red box.

ENABLED	NAME	METHOD	WORKLOAD THROTTLE
☒	H-WD	Timeshare	-
☒	L-WD	Timeshare	-
☒	M-WD	Timeshare	-
☒	T-WD	Timeshare	-
☒	WD-Default	Timeshare	-
☒	WD-Magallover	Timeshare	-

WD-Default is a system workload used for any queries that do not classify to any previously defined workloads.

It **cannot be disabled or deleted**.

The **classification criteria is none** and cannot be modified.

Recommended to use **WD_Default** for **unexpected requests**.

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The **WD-Default** workload definition is the default workload. It is automatically created and is used as a “No-Home WD”. Queries that do not match the characteristics of any other workload definition will run in this WD.

Note: The WD-Default definition cannot be disabled, deleted or edited.

The vast bulk of a workload mix fall into workloads as determined by accounting and priority needs. However, there is one mandatory workload that exists on all systems: WD-Default. If a request is submitted to the database but does not qualify to run in any of the other defined workloads, then it runs in WD-Default.

The recommended position is to reserve WD-Default for unexpected requests. Upon capture of the unexpected request to the Database Query Log (DBQL), the DBA can investigate its source and take appropriate action regarding such requests that may come in the future, such as, create a new workload, assign the request to an existing workload, or filter the request from future execution.

Current topic: The workloads view

- Workloads overview
- **The workloads view**
 - **Creating a new workload**
 - **Workload tabs**
 - **Classification criteria**
 - **Workload throttles**
 - **State specific settings**
- Flex throttles
- Workload group throttles
- Additional options for workloads
- Workload exceptions
- Additional workload tabs

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Creating a new workload (1 of 2)

The screenshot shows the Teradata Workload Designer interface. At the top, there's a header with 'Workload Designer', 'Team1', and 'FirstConfig'. Below this is a toolbar with icons for General, States, Sessions, Filters, Meters, Throttles, Workloads (highlighted), and Exceptions. Under the Workloads tab, there's a sub-header with 'WORKLOADS', 'SLG SUMMARY', 'EVALUATION ORDER', 'VIRTUAL PARTITIONS', 'PARTITION RESOURCES', 'WORKLOAD DISTRIBUTION', and 'CONSOLE UTILITIES'. A note says 'Use workloads to group queries that have common characteristics or common requirements.' Below this is a table with columns: 'ENABLED', 'NAME', 'METHOD', and 'WORKLOAD THROTTLE'. The table lists several workloads: H-WD, L-WD, M-WD, T-WD, WD-Default, and WD-MapsMover, all with 'Timeshare' method and '-' throttle. A red circle highlights the '+' button next to the 'Workloads' tab label.

ENABLED	NAME	METHOD	WORKLOAD THROTTLE
☒	H-WD	Timeshare	-
☒	L-WD	Timeshare	-
☒	M-WD	Timeshare	-
☒	T-WD	Timeshare	-
☒	WD-Default	Timeshare	-
☒	WD-MapsMover	Timeshare	-

Workloads are used to group requests with **similar characteristics**.

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To create a new workload rule, click **Workloads** on the toolbar and then click **Workloads +** button to create a new workload rule.

Creating a new workload (2 of 2)

The screenshot shows the 'Workload Designer' window with the 'GENERAL' tab selected. The 'Name' field is 'DSS'. The 'Description' field is empty. The 'Virtual Partitions' dropdown is set to 'Standard'. The 'Workload Management Method' section has three radio buttons: 'Tactical', 'SLG Tier', and 'Timeshare'. The 'Timeshare' radio button is selected and highlighted with a red box. Below the radio buttons is a note: 'Workload Management Method (Additional configuration recommended at Ruleset>Workloads>Workload Distribution)'.

Workload Management Method

determines the **priority** for CPU and I/O resources.

- **Tactical** – For short **high priority work** with response time requirements
- **SLG Tier** – For **important work** that should receive a **higher percentage of resources**
- **Timeshare** – For **average and lower priority work**

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To create a new workload rule, click Workloads on the toolbar and then click Create Workload to create a new workload rule.

On the General tab, enter the workload rule name, up to 30 characters. Optionally, you can supply description of the rules, up to 80 characters

Workload tabs

Workload: DSS

GENERAL CLASSIFICATION THROTTLES SERVICE LEVEL GOALS HOLD QUERY RESPONSES EXCEPTIONS

Name: * DSS

Description:

Virtual Partitions: Standard

Workload Management Method (Additional configuration recommended at Ruleset>Workloads>Workload Distribution)

☐ Tactical

☐ SLG Tier

☒ Timeshare

Workload Management Methods **SLG Tier** and **Timeshare** will have these tabs:

- General
- Classification
- Throttles
- Service Level Goals
- **Hold Query Responses**
- Exceptions

Workload Management Method **Tactical** will have these tabs:

- General
- Classification
- Throttles
- Service Level Goals
- Exceptions
- **Tactical Exceptions**

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Workload: Tactical

GENERAL CLASSIFICATION THROTTLES SERVICE LEVEL GOALS EXCEPTIONS TACTICAL EXCEPTION

Name: * Tactical

Description:

Virtual Partitions: Standard

Workload Management Method (Additional configuration recommended at Ruleset>Workloads>Workload Distribution)

☒ Tactical

☐ SLG Tier

☐ Timeshare

The tabs displayed on the Workload pane will depend on the Workload Management Method you select for the workload.

Workload Management Methods SLG Tier and Timeshare will have these tabs:

- General
- Classification
- Throttles
- Service Level Goals
- Hold Query Responses
- Exceptions

Workload Management Method Tactical will have these tabs:

- General
- Classification
- Throttles
- Service Level Goals
- Exceptions
- Tactical Exceptions

Classification criteria

Portlet: **Workload Designer** > Button: **Workloads** > Tab: **Workloads** >
Button: **Create a Workload [+]** > Tab: **Classification**

The screenshot shows the 'Workload Designer' interface for a workload named 'DSSLow'. The 'CLASSIFICATION' tab is selected and highlighted with a red box. Below the tab, a text box states: 'Classification criteria determine the workload into which each query will be classified.' Underneath, there is a section titled 'Add Classification Criteria' with a dropdown menu. The dropdown menu is open, showing options: 'Request Source' (selected), 'Request Source', 'Target', 'Query Characteristics', 'Query Band', and 'Utility'. An 'ADD' button is located to the right of the dropdown.

Specify the Classification Criteria.

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Add the classification criteria that will be used to determine which requests the workload rule applies.

Throttles state specific settings

Portlet: **Workload Designer** > Button: **Workloads** > Tab: **Workloads**>
Button: **Create a Workload** [+] > Tab: **Throttles**



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The **Default Setting** for every State is **Unlimited**.

The **Default Setting** can be set to a **specific limit** and whether the requests exceeding the limit will be **Delayed** or **Rejected**.

When you specify **Delay**, you have the option to **Enable Flex Throttles** for queries in this workload.

The default setting for workload throttles is unlimited. This can be changed to a specific limit and whether requests exceeding that threshold will be delayed or rejected.

- Workload request throttles (i.e., workload throttles) are the primary tool for limiting work on a system with a complex balance of workloads
- Workload throttles are easier to monitor and to assess request delay times
- If SQL requests in a stored procedure classify to a throttled workload, TASM throttles each request in the procedure individually as each request is submitted

State specific settings (cont.)

To **override the default** setting for a specific State, select the State's “**pen**” button.

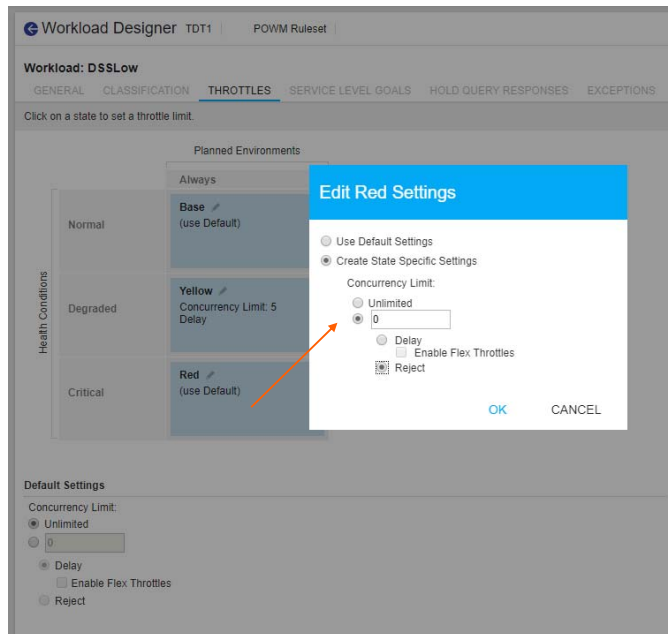
In the Edit dialog box, enter the **working value** settings for that state.

Using the **Reject** option effectively makes the Throttle function as a “**Filter by Workload**”.

Enable Flex Throttles means that queries that are delayed by this Workload Throttle can be automatically released from the Delay queue when system resource is available.

Note: In the following slides we'll discuss the Flex Throttle Feature.

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To override the default setting, move your cursor over the State to display the “pen” icon. Click the pen icon to display the Edit Value Settings dialog box. Enter the working value settings that will be applied for that specific State.

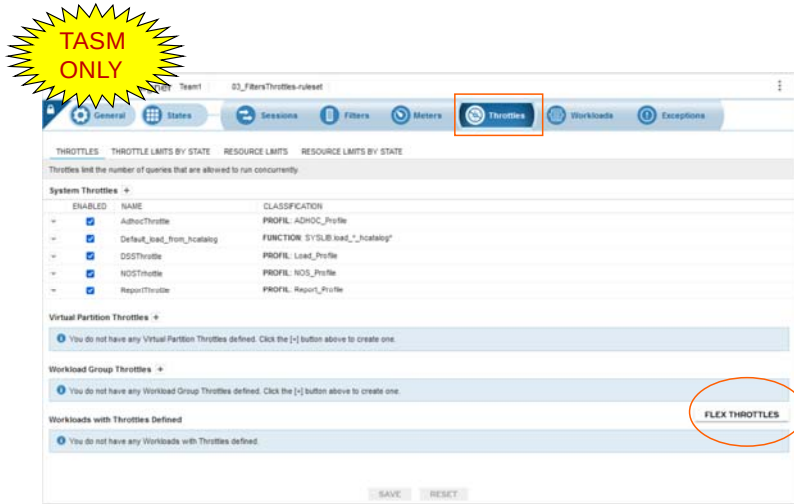
Current topic: Flex throttles

- Workloads overview
- The workloads view
- **Flex throttles**
 - **What are flex throttles**
 - **Characteristics of flex throttles**
 - **Enabling the flex throttle feature**
 - **Flex throttle examples**
- Workload group throttles
- Additional options for workloads
- Workload exceptions
- Additional workload tabs

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What are flex throttles?



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Starting with Viewpoint 16.00 **Workload Designer** portlet includes the capability to **automatically release queries** in the Delay queue when system resource is available.

Flex Throttles minimizes the manual management of the Delay queue. This feature provides the ability to:

- Automatically release queries from Delay queue based on triggering conditions a DBA defines
- Fully utilize previously unused resources
- Simplify ongoing manual monitoring and management of the Delay queue and concurrency limits
- Can be used to reduce the need for additional States in State Matrix

Starting with Viewpoint 16.00 Workload Designer portlet includes the capability to automatically release queries in the Delay queue when system resource is available.

Business Value

Flex Throttles will minimize the manual management of the Delay queue. This feature provides the ability to:

- Automatically release queries from Delay queue based on triggering conditions a DBA defines
- Fully utilize previously unused resources
- Simplify ongoing manual monitoring and management of the Delay queue and concurrency limits
- Can be used to reduce the need for additional States in State Matrix

Characteristics of flex throttles

The **Flex Throttle** feature attempts to utilize unused resources by automatically releasing work from the Delay queue based on triggering conditions a DBA defines. The flex action of releasing qualified queries from the delay queue is triggered from events that monitor AWT utilization and, optionally, CPU and/or I/O utilization.

The following is a list of characteristics related to Flex Throttle:

- Only applies to Workload Throttles
- Can be enabled or disabled in different states within a workload
- Workload Management, whether you are using TASM or TIWM, honors system throttles, system utility limits, workload group throttles and workload throttles with a limit of “0”
- Enabled at the ruleset level
 - Individual workload throttles must be selected to participate in Flex Throttle feature.
 - Flex Throttle is disabled by default
- Evaluation mode is available to assess the impact of the Flex Throttle feature before enabling it

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The Flex Throttle feature attempts to utilize unused resources by automatically releasing work from the Delay queue based on triggering conditions a DBA define. The flex action of releasing qualified queries from the delay queue is triggered from events that monitor AWT utilization and, optionally, CPU utilization.

The following is a list of characteristics related to Flex Throttle:

- Only applies to Workload Throttles.
 - Can be enabled or disabled in different states within a workload
- Workload Management, whether you are using TASM or TIWM, honors system throttles, system utility limits, workload group throttles and workload throttles with a limit of “0”
- Enabled at the ruleset level
 - Individual workload throttles must be selected to participate in Flex Throttle feature.
 - Flex Throttle is disabled by default.
- Evaluation mode is available to assess the impact of the Flex Throttle feature before enabling it

Enabling the flex throttles feature

Turns on the Flex Throttles feature.

Turns on Evaluation Mode.

Triggering Events:
Available AWTs (required)
CPU Utilization (optional)
I/O Usage (optional)

Actions:
Release qualified queries from the delay queue.

To turn on the Flex Throttles feature for the ruleset, click on the Enable Flex Throttles button. Clicking this button indicates that Flex Throttle Feature will release queries from the Delay queue for the Flex-enabled workloads.

Under the Triggering Events section there are three events that are available to trigger the Flex Throttle action: Available AWTs, CPU Utilization and I/O Usage. The Available AWTs is a required entry, and the CPU Utilization and I/O Usage are optional.

The parameters for the Available AWTs event include:

- **Number of AMPs with Available AWTs:** Specify the minimum number of AMPs with available AWTs. The event will be triggered if this number or more are available
- **Number of Available AWTs:** Specify the minimum number of available AWTs for the number of amps. The event will be triggered if this number or more are available
- **Qualification:** Specify the number of minutes for which both conditions must be met to trigger the action

The parameters for the CPU Utilization event include:

- **System CPU:** Specify the minimum system CPU utilization percentage. The event will be triggered if the CPU utilization is less than or equal to this value
- **Qualification:** Specify the number of minutes for which the minimum system CPU must be maintained to trigger the event. The average value of this metric must exceed the threshold for this time period

The parameters for the I/O Usage event include:

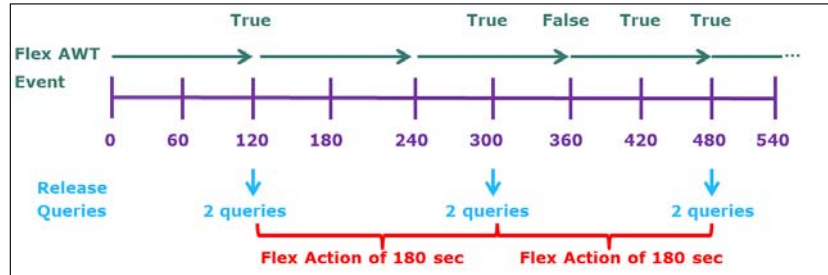
- **Bandwidth:** Bandwidth Threshold percentage that when exceeded will trigger the event

(default percentage is 80%, default operator is >=, valid range 1-1000%).

- **Monitored LUNs:** Percentage of targeted LUNs to monitor (default: 10% of the storage; 100% can be no more than 50 LUNs)
- **Triggered LUNs:** Percentage of the monitored LUNs that must meet the specified Bandwidth Threshold for the event to trigger (default: 1% of the monitored LUNs)
- **Qualification Method (Averaging Interval):** At the end of each Event Interval TASM will calculate the average of the bandwidth used for each monitored LUN. TASM will base the average calculation on the number of minutes specified in this field
- **Qualification Time:** When TASM first detects that the bandwidth threshold has been exceeded, the bandwidth must remain above the threshold for the number of minutes that are specified in this field. The value in this field specifies the number of minutes that must expire before for the event is triggered

Flex throttles example

This chart, describes the scenario where we have set up the Flex Throttle feature to release **two queries** from the delay queue. In this scenario we have the specified the Flex Throttle definition as follows:



- **Flex AWT Event:** When there are 3 AMPs with 3 or more AWTs available for **2 minutes**
 - (defined in Workload Designer > Throttles button > Throttles tab > **Flex Throttles screen**)
- **Flex Action:** Number of Queries to release: **2**
 - (defined in Workload Designer > Throttles button > Throttles tab > **Flex Throttles screen**)
- **Event Interval:** **60 seconds**
 - (defined in Workload Designer > General button > **Other tab**)
- **Flex Action Interval:** **180 seconds**
 - (defined in Workload Designer > General button > **Other tab**)

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The chart above describes the scenario where we have set up the Flex Throttle feature to release **2 queries** from the delay queue. In this scenario we have the specified the Flex Throttle definition as follows:

- **Flex AWT Event:** When there are 3 AMPs with 3 or more AWTs available for **2 minutes**
 - (defined in Workload Designer > Throttles button > Throttles tab > **Flex Throttles screen**)
- **Flex Action:** Number of Queries to release: **2**
 - (defined in Workload Designer > Throttles button > Throttles tab > **Flex Throttles screen**)
- **Event Interval:** **60 seconds**
 - (defined in Workload Designer > General button > **Other tab**)
- **Flex Action Interval:** **180 seconds**
 - (defined in Workload Designer > General button > **Other tab**)

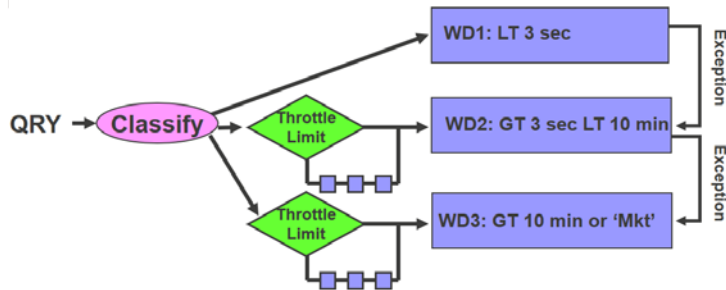
Current topic: Workload group throttles

- Workloads overview
- The workloads view
- Flex throttles
- **Workload group throttles**
 - **Workload throttles delay queue problem**
 - **Creating workload group throttles**
 - **State specific settings**
 - **Workload group throttles and demotions**
- Additional options for workloads
- Workload exceptions
- Additional workload tabs

teradata.



Workload throttles delay queue problem



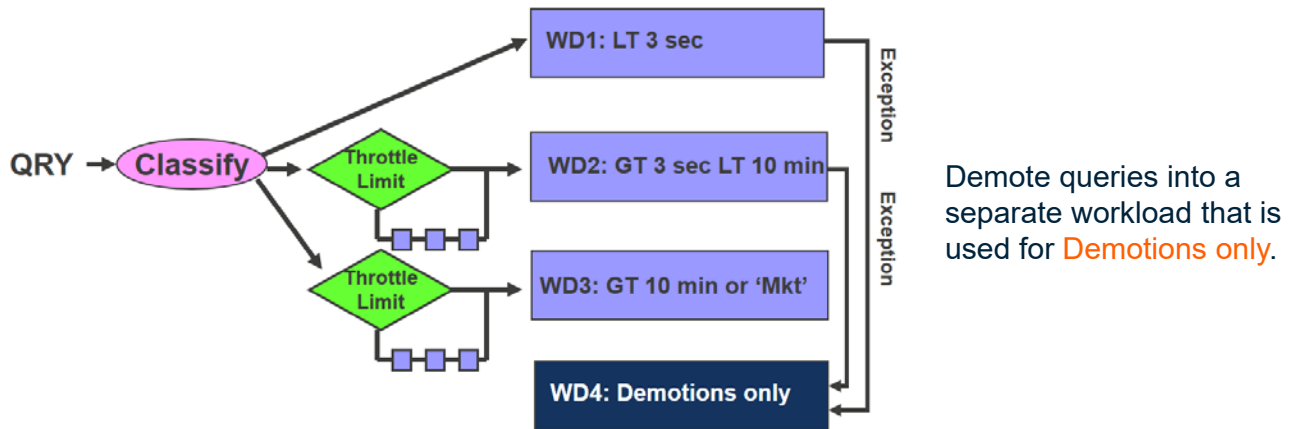
“Change to WD” Exception Action bypasses the WD Throttle Limit.

- Queries that have begun execution and are demoted into a WD due to an exception, cannot be interrupted and placed into a delay queue
- WD Throttle Limits adjust to demotions into the WD by raising the counters for each demotion
 - This could cause the throttle counter to exceed the throttle limit
 - This will further delay the normal release of queries from the delay queue
- Queries classified to WD2 or WD3 may never be released from the WD Delay Queue if there are lots of exceptions that push demotions into WD2 or WD3

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Another issue to be aware of is if you have Request Limits on a WD that is also in a Change Workload exception action. When queries are moved to the WD, the request limit counters will be incremented above the original Request Limit. Queries in the delay queue will not be released until the counter goes below the Request Limit. This may cause queries in the delay queue to be delayed longer.

Workload throttles delay queue solution



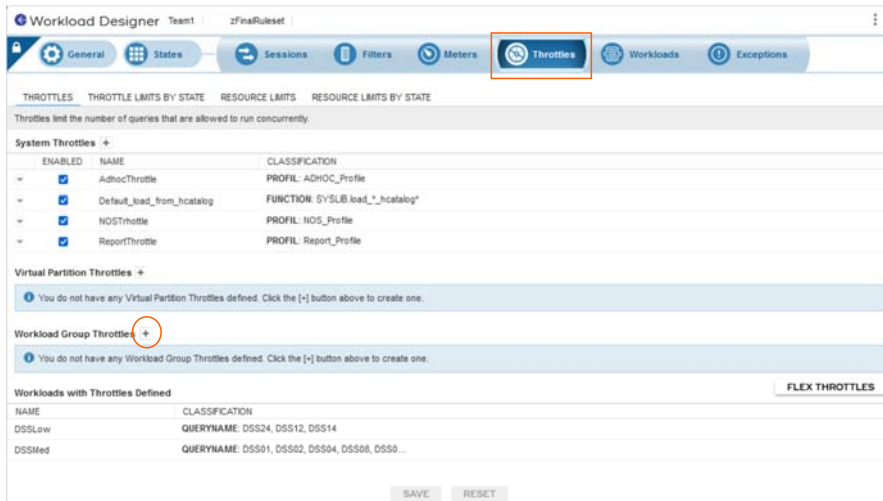
Note: A **Demotions only** workload should have **no classification** assignment, and thus would need to fall underneath the WD-DEFAULT workload in the Workload Evaluation Order.

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One solution to demoting queries into a workload with request limits is to demote queries into a separate workload that is used only for demotions.

Creating workload group throttles

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a Group Throttle** [+]



Workload Group Throttles control concurrency over a group of two or more workloads that already have throttles defined.

Concurrency **limits of both** the **workload throttle** and **group throttle** must be satisfied before a new query will be able to run.

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Group throttles were introduced in Teradata 14.10. They allow concurrency to be controlled over a group of two or more workloads.

In Viewpoint Workload Designer, all existing throttles can be seen by selecting the Throttles tab under the Throttles category. Throttles are broken down by system throttle, then group throttle, then workload throttle.

It is important to note that group throttles are *not* an alternative for using workload throttles. Group throttles can only be applied to workloads that already have workload throttles defined. Concurrency limits of both the workload throttle and the group throttle will have to be satisfied before a new query impacted by those throttles will be able to run.

In order to create a group throttle, use the Create Group Throttle button

Creating workload group throttles (cont.)

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a Group Throttle [+]** > Tab: **General**

Workload Designer TDT1 POWM_Ruleset

Group Throttle: POWMGroupThrottle

General State Specific Settings

Workload group throttles place aggregate limits on the number of queries that can be run by a group of workloads.

Name: POWMGroupThrottle

Description:

Available Workloads:
DSS

Workloads in this Group:
Adhoc
Report

teradata.

Workload Group Throttles can only consist of workloads that already have individual throttles defined with the delay option.

Workload Group Throttles do not contain any classification criteria, they rely on the classification criteria of workloads that are members of the group to control the number of queries allowed to execute.

A workload can only participate in one Group Throttle.

A workload is not given the option to participate in a group throttle until the group throttle has first been defined.

Workload Designer only offers an option to participate in a group throttle for workload throttles defined with the delay option. In addition, the workload throttle is only given the option to delay requests that would exceed its limit.

State specific settings

Portlet: **Workload Designer** > Button: **Throttles** > Tab: **Throttles** >
Button: **Create a Group Throttle [+]** > Tab: **State Specific Settings**

Workload Designer TOT1 POWM Ruleset v2

Workload Group Throttle: POWGroupThrottle

GENERAL STATE SPECIFIC SETTINGS

Click on a state to set a throttle limit.

Planned Environments:

Always

Base (Use Default)

Yellow (Concurrency Limit, Delay)

Red (Concurrency Limit: 0, Delay)

Health Conditions

Normal

Degraded

Critical

Default Settings

Concurrency Limit:

Unlimited

Delay

Workload Throttles

NAME	DEFAULT SETTING	BASE	RED	YELLOW
Adhoc	1, DELAY	(Use Default)	(Use Default)	(Use Default)
Report	1, DELAY	(Use Default)	(Use Default)	(Use Default)

SAVE RESET

Reject is not an option.

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Set the default limit and any state specific limits.

Workload group throttles and demotions

WORKLOAD	THROTTLE LIMIT	IF 2 WD1 QUERIES ARE DEMOTED TO WD2
WD1	6	6
WD2	3	5
Total Active	9	11

Without Group Throttles, concurrency limits of just the workload throttle must be satisfied before a new query will be able to run.

With Group Throttles, concurrency limits of both the workload throttle and group throttle must be satisfied before a new query will be able to run.

WORKLOAD	THROTTLE LIMIT	IF 2 WD1 QUERIES ARE DEMOTED TO WD2
WD1	6	4
WD2	3	5
Group Throttle	9	9
Total Active	9	9

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There is a desire to limit an application to a prescribed number of concurrent requests. However, the application's requests span more than one workload, with the higher priority workloads demoting into the lower priority workloads. All of the workloads have workload-level throttles with the delay option.

Prior to Teradata 14.10, when a query is demoted from a higher-level workload to a lower level one and both workloads have a workload throttle, the workload throttle counter from the higher priority workload is decremented and the counter on the lower-level throttle is incremented. The active query counts are accurate reflection of concurrency levels at any point in time. However, if the lower workload throttle is already at its concurrency limit, it may exceed its limit temporarily, as demoted queries are moved under its control. Demoted queries are never subject to delay, as they have already begun to execute. Under those conditions, the total number of requests active on behalf of the application can exceed what was intended as shown in the first table.

If there have been a lot of demotions in a short period of time and new queries have taken advantage of the freed-up query slots at the higher-level workload, the total number of queries active across the application will continue to increase. Group throttles can help to keep the number of active queries for the entire application in line with expectations, regardless of demotions.

In the second table, a group throttle that combines WD1 and WD2 will be limited to nine queries at a time. Although the counter for WD1 has been reduced by two due to the demotions, WD1 is not able to release two queries from its delay queue because of the presence of the group throttle. The group throttle is already at its limit of nine, so until two queries among WD1 and WD2 complete, WD1 will stay below its limit.

If a query in both WD1 and WD2 complete at the same point in time, the workload which has had

a query in its workload throttle delay queue for the longest time will be able to release a query to run.

Current topic: Additional options for workloads

- Workloads overview
- The workloads view
- Flex throttles
- Workload group throttles
- **Additional options for workloads**
 - **Workload service level goals**
 - **Hold query responses**
 - **Minimum response time**
- Workload exceptions
- Additional workload tabs

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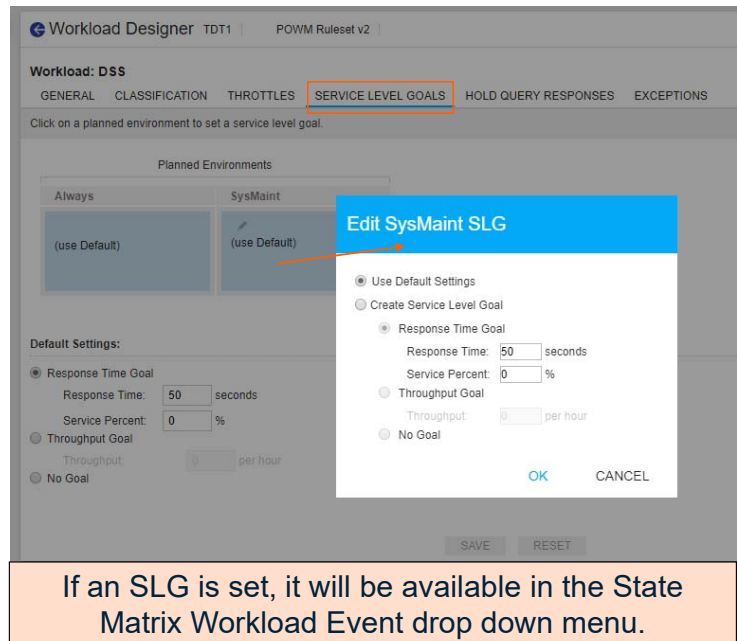
Workload service levels goals

For each **Planned Environment**, you can specify the desired Service Level Goal targets.

The SLG settings include **ONE** of these:

- **Response Time Goal**
 - **Response Time** – the desired average response time for queries in this workload
 - **Service Percent** – the percentage of queries expected to meet the response time
- **Throughput Goal**
 - The expected **number of queries** that will be executed in the workload **per hour**

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Workloads at all tiers (tactical, SLG, and timeshare) may have optional Service Level Goals specified in each workload's definition within the Workload Designer.

Service Level Goals can indicate the expected response times or throughput rates for a workload and can be helpful to document the original performance expectations of a workload and can simplify reporting of how workloads are performing against those original goals.

A goal is something you plan to achieve. If you have no goal, it suggests you have nothing you plan to achieve. As applied to workload management, your workloads will likely require a goal to reach critical performance objectives, whereas other workloads may require no goal because their performance levels are mostly irrelevant.

SLGs identified and proactively reported against improve the insight into the system and enable better management of workloads. When SLGs are not being met, there are several avenues to try to bring them into conformance:

- Performance Tuning
- Workload Management
- Capacity Planning
- Unrealistic Goal

Establishing service level goals

- It is recommended to establish SLGs for **important workloads** such as those with Tactical priority
- SLGs should be **realistic and attainable** as well as support the business and technical needs
- SLGs may **evolve** over time as needs change or knowledge increases
- Established SLGs can be used to **proactively report against actual** performance to enable better management of workloads
- SLGs being missed can prompt you to **analyze workload** requests to make them more efficient using less CPU and I/O
- SLGs can be used to identify if **priorities** need to be **reduced** for those workloads meeting their SLGs by a large margin and **increased** for those workloads not meeting their SLGs
- SLGs can be used to for **Capacity Planning** purposes to predict when additional system capacity will be required
- SLGs can be used to determine if the goals are **technically unrealistic** with the existing system capacity

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In general, it is good practice to establish Service Level Goals (SLGs) for the important workloads, but especially the tactical workloads. TASM helps to establish a goal-based-orientation not only by encouraging you to set goals, but also by helping you establish and evolve those goals so that they reflect the needs of the business. SLGs and how actual performance compares to those goals are communicated clearly in the workload dashboard and can be a subject of or a column in many of the workload trend reports provided by Teradata Manager.

SLGs are measurable. They can be set on either:

- response time at a particular service percent, (e.g., 2 seconds or less 80% of the time), or
- throughput, (e.g., 1000 queries per hour)

To maximize the effectiveness of SLGs, they should be realistic and attainable, as well as support the business and technical needs of the system. But when SLGs have never been set for a workload, it is difficult to know what value will best represent the business and technical needs of the system. So how do you determine what the right value is for the SLG?

There are several approaches that can be taken, but keep in mind that the SLG may evolve over time as needs change or knowledge increases:

- Known business need – For example, a web application is used by many demanding but inexperienced users. Experience has shown that users will kill and restart a request if it does not respond within 5 seconds, further aggravating a peak load situation that is causing their slow response times in the first place. This customer established a SLG of 4 seconds to avoid the aggravated demand
- Unknown need – For example, an important application currently has no established response time goal, and therefore user satisfaction has been difficult to measure. They

know when things are bad based on an increase in user complaints to IT, but they do not necessarily know what response time point triggers the dissatisfaction. Consider drawing an initial “line in the sand” based on typical actual response times obtained (either equal to or, for example, up to twice the typical actual). Once that initial goal is set, measure and monitor SLGs, adjusting as necessary and as determined by cross-comparing the SLG vs. complaints or business targets missed

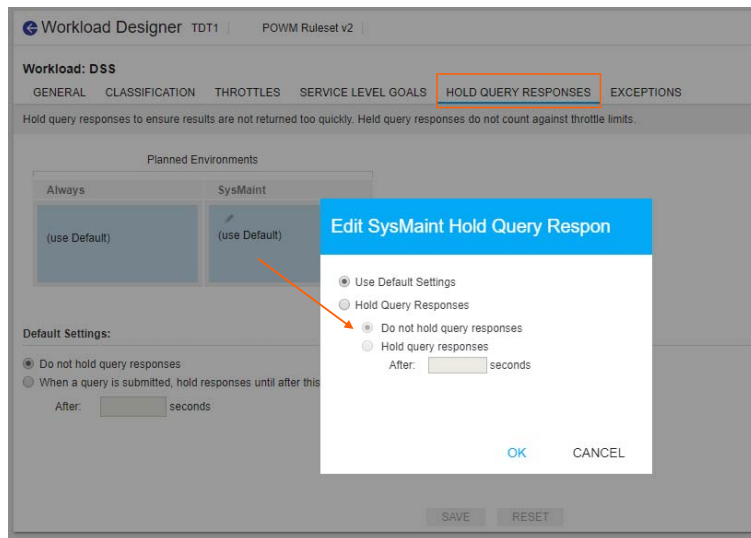
Hold query responses

Portlet: **Workload Designer** > Button: **Workloads** > Tab: **Workloads** >
Button: **Create a Workload [+]** > Tab: **Hold Query Responses**

Specify the **default Minimum Response Time** for **non-tactical** workloads.

For each **Planned Environment**, you can specify the MRT.

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From the Hold Query Responses tab specify the minimum response time for a non-tactical workload. The MRT can vary by Planned Environment.

To manage user expectations about request response times, a minimum response time may be specified for each request in a workload. TASM prevents requests from responding before the specified time by delaying the return of a request's answer set.

Various scenarios for holding query response:

- The Teradata Vantage system is being upgraded to double the current number of nodes to provide capacity for a new web application being added to SLG tier 1 in the coming months along with organic growth of other workloads
- The administrator is concerned that the queries for the Dashboard workload will now complete twice as fast and the Dashboard users may submit twice the workload, consuming some of the extra capacity before the new web application is implemented
- The administrator is also concerned that the Dashboard users may get an unrealistic expectation that this faster response time is the new normal and will be complain when the new web application workload is added
- The Dashboard workload runs in SLG tier 2 and currently has a service level goal of a response time in five seconds

Solution:

- Set a hold query responses time of four seconds for the Dashboard workload
- This will provide the Dashboard workload users with a consistent experience, as they had prior to the upgrade, while still meeting the workload's service level goal of five seconds
- After the new web application is added the hold query response time can be removed

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Minimum response time

- Starting with **TD15.10**, you can specify a **Minimum Response Time (MRT)** for a workload
- Queries in the workload will be **prevented** from **returning** their response **before** the specified **MRT**
- This can be used to achieve more consistent response times
- Can be used to **prevent unrealistic expectations** after a system upgrade and before the system becomes fully loaded
- MRT Characteristics are as follows:
 - Can be set for **SLG Tier** and **Timeshare** workloads
 - Value can be from **1 to 3600** seconds and can vary based on **Planned Environment**
 - Set commands, Transaction statements and EXEC commands are **not held**
 - **Stored Procedure calls** are **not held**, but each statement within the procedure is treated as a separate request and can be subject to an MRT
 - If a request triggers an **exception** that changes the workload, the **MRT** value of the **final workload** will be used
 - MRT is **calculated** by subtracting the query start time from the current time and includes any time spent in the delay queue
 - Query will be displayed with a state of **RESPONSE-HELD**

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Starting with Teradata 15.10, you can specify a minimum response time (MRT) for a workload. Queries in the workload will be prevented from returning their response before the MRT specified. This feature can be used to achieve more consistent response times and to prevent users from getting unrealistic expectations after such cases as an upgrade of hardware before the system becomes fully loaded.

One use case is when a system has capacity added but the DBA doesn't want certain classes of users to get a big benefit in response time and start submitting more work. The extra work would consume capacity ahead of plan. MRT could hold response times to historical expectations and keep the users from loading up the system with what is typically less important work.

The use cases for MRT are primarily for queries that have service level expectations, where consistency is a clear goal and where end users notice and care about elapsed times. This would mainly be for workloads where queries or short reports with similar profiles are being executed.

The characteristics of the MRT feature are as follows:

- A minimum response time can be set for SLG tier and timeshare workloads with a value from 1 to 3600 seconds, and the value can vary by operating environment
- SET commands, stored procedure calls, transaction statements, and EXEC commands are not subject to the minimum response time. For stored procedures, the CALL is not held, but each statement within the procedure is treated by TASM as a separate request (classified individually so each can be delayed and have a different workload). So, each request within the procedure can be subject to a MRT if it classifies to a workload with an MRT. This is also for true for statements within a macro
- The MRT value of the final workload is used if the request encountered an exception that changed the workload

- To determine if a query should be held to meet the MRT, the database calculates the elapsed time by subtracting the query start time (*DBQLogTbl.StartTime*) from the current time. The elapsed time includes any time the query was on the delay queue. If the elapsed time is less than the minimum response time value, the response is held
- A query is placed on hold at the point where the database would normally send the response to the client: AMP steps have completed, locks are released, and TASM throttle counters have been decremented
- The PE state, RESPONSE-HELD, as shown in the Viewpoint Session Monitor portlet indicates the response is being held until the minimum response time is met. A request can be aborted in the RESPONSE-HELD state; however, there is no mechanism to release a held request

Once a query is on hold, the minimum response time value for the request is fixed and will not change due to TASM operating environment or rule set changes.

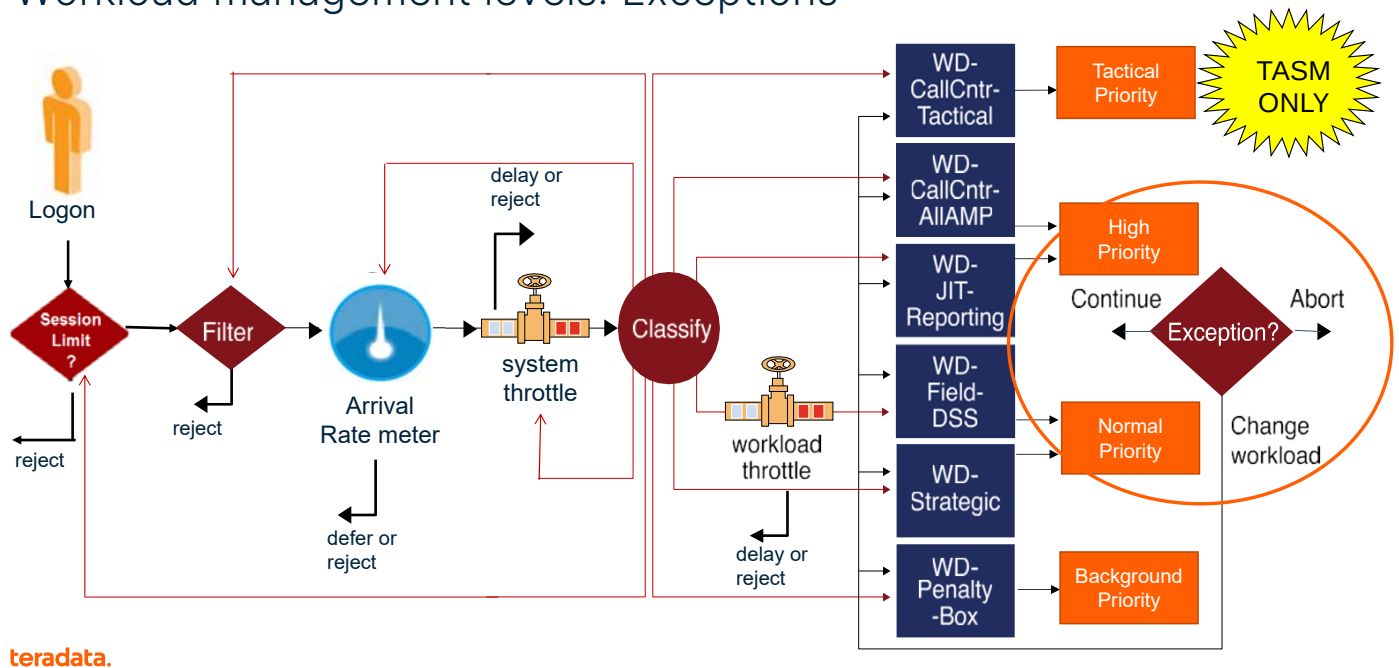
Current topic: Workload exceptions

- Workloads overview
- The workloads view
- Flex throttles
- Workload group throttles
- Additional options for workloads
- **Workload exceptions**
 - **Creating exceptions**
 - **Unqualified and qualified exceptions**
 - **Asynchronous and synchronous exception monitoring**
 - **Exception actions**
 - **Assigning exceptions**
- Additional workload tabs

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Workload management levels: Exceptions



There are seven different methods of management offered, as illustrated below:

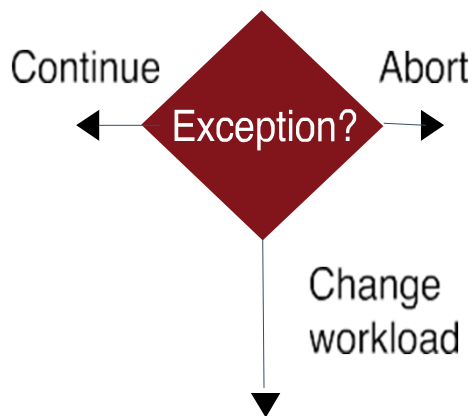
Methods regulated prior to the query beginning execution

- Session Limits can reject Logons
- Filters can reject requests from ever executing
- Arrival Rate Meter can defer requests by managing the number of requests admitted into the system per specified time period
- System Throttles can pace requests by managing concurrency levels at the system level
- Classification determines which workload's regulation rules a request is subject to
- Workload-level Throttles can pace the requests within a particular workload by managing that workload's concurrency level

Methods regulated during query execution

- Priority Management regulates the amount of CPU and I/O resources of individual requests as defined by its workload rules
- Exception Management can detect unexpected situations and automatically act such as to change the workload the request is subject to or to send a notification

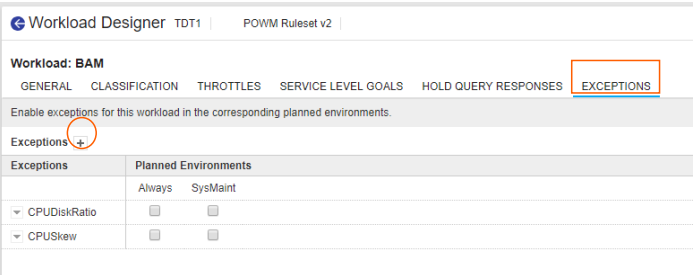
Exceptions



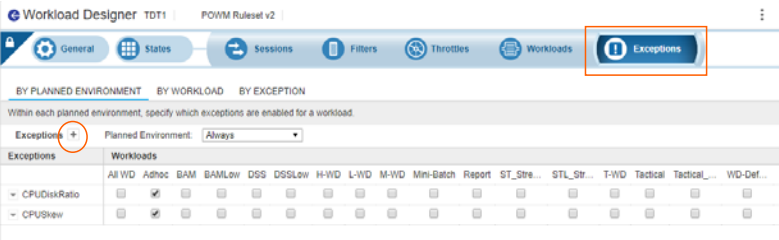
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- If a request exceeds expected limits administrators can use exceptions to define those thresholds and the actions TASM should take when the limit is exceeded
- Within Workload Manager exceptions can be maintained in the Exceptions view or within the definition of the workload they pertain to
- The Exceptions view provides perspectives by planned environment, by workload, and by exception to easily view and modify the planned environments and workloads an exception is enabled for
- If an exception definition includes multiple exception criteria, each criteria must be met for the exception to be triggered (i.e., the criteria are ANDed together)
- When an exception is triggered, the request may be aborted, changed to a different work, or a notification may be initiated

Creating exceptions



Exceptions can be created from the Exception tab for a specific workload.



Exceptions can also be created from the Exception Button on the Ruleset Toolbar.

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Exceptions can either be created by selecting the Exceptions tab for the workload and then selecting the Create Exception button, or by selecting the Exceptions button on the ruleset toolbar and then selecting the Create Exception button.

Creating exceptions (cont.)

Multiple criteria can be specified for an exception.

When multiple criteria are specified, they **all must be exceeded** to trigger the exception.

Exception Criteria **thresholds** should be set **outside** the boundary of the “**normal**” range.

Each Exception can invoke an **Action** and/or a **Notification**.

teradata.

New Exception

Use this form to define the criteria that will trigger this exception and to define the action(s) that will be taken.

General

Name: * CPUUsage

Description:

Criteria

Exception Criteria: -- Choose --

- Maximum Spool Rows
- Spool Usage (bytes)
- Number of AMPs
- Blocked Time
- Elapsed Time
- I/O Count
- CPU Time
- CPU Skew Percent
- I/O Skew Percent
- CPU Skew Difference
- I/O Skew Difference
- CPU Disk Ratio
- I/O Physical Bytes

Unqualified

Qualified

Unqualified

Only

☐ Abort

☐ Abort Selects Only

☐ Change Workload to: Adhoc

Notifications:

☐ Send Alert:

☐ Run Program:

☐ Post to QTable:

Exceptions consist of criteria and actions to trigger automatically when the criteria occur. The exception criteria options include:

Threshold-based criteria, that trigger as soon as the threshold is exceeded.

- Maximum Spool Rows
- IO Count
- Spool Size
- Blocked Time
- Response Time
- Number of AMPs
- CPU Time
- I/O Physical Bytes

Qualified criteria, that trigger after the situation is sustained for a qualification time.

- CPU milliseconds per I/O
- Skew: IO or CPU Skew or Skew Percentage

Each exception consists of either a single exception criterion or multiple criteria. When there are multiple criteria in an exception, they must all be exceeded to trigger this exception's actions. The values selected for exception criteria primarily depend on what is typical for the requests within the workload, and identifying boundary values for when the variation from that typical value has grown too large.

Unqualified exception thresholds

- **Maximum Spool Rows:** The maximum number of rows in a spool file
- **IO Count:** The maximum number of logical disk I/O's performed by the request
- **Spool Usage (bytes):** The maximum size of a spool file (in bytes)
- **Blocked Time:** The length of time the request is blocked by another request
- **Elapsed Time:** (including or excluding delay and blocked times) The length of time the request took to complete
- **Number of Amps:** The number of AMPs that participate in the request
- **CPU Time:** The total amount of CPU time (in hundredths of seconds) consumed by the request
- **I/O Physical Bytes:** The total amount of physical bytes transferred by the request

Exception action is taken as soon as the criteria threshold is met.

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You can define the Exception Criteria that are used to set thresholds for resource utilization. You can define one or more of the following exception criteria:

Unqualified Criteria: The following criteria are effectively “ANDed” together. If all the specified criteria are met, then the exception action is triggered.

- **Maximum Spool Rows** – the maximum number of rows in a spool file or result
- **IO Count** – the maximum number of disk I/O's performed on behalf of the request
- **Spool Size** – the maximum size of a spool file
- **Blocked Time** – the length of time the request is blocked by another request
- **Elapsed Time** – the length of time the request has been running
- **Number of AMPs** – the number of AMPs that participate in the request
- **CPU Time** – the maximum number of CPU seconds of processing time consumed by the request
- **I/O Physical Bytes** – the maximum I/O physical bytes transferred by the request

Qualified exception conditions

The following are **Qualified Conditions** that are based on the **Qualification Time**:

- **Qualification Time**: The number of **CPU seconds** the exception condition **must persist** before an action is triggered
- **IO Skew Difference**: The maximum **difference in logical I/O Counts** between the average and most skewed AMP
- **CPU Skew Difference**: The maximum **difference in CPU seconds** consumption between average and most skewed AMP
- **IO Skew Percent**: The **percentage difference in logical I/O Counts** between the average and most skewed AMP
- **CPU Skew Percent**: The **percentage difference in CPU seconds** consumption between the average and most skewed AMP.
- **CPU Disk Ratio**: The ratio of CPU milliseconds to logical disk I/O's (aka, **Product Join Indicator – PJI**)

Exception is **action is taken** only if the **condition persists** for duration of the **Qualification Time**.

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These conditions must exist for a period before the action is triggered. The following criteria are also “ANDed” together. If all the criteria are met, then the exception action is triggered. Also note that the Unqualified and Qualified Criteria are “ANDed” together as well.

- **Qualification Time** – the length of time (in CPU seconds) the condition must continue before an action is triggered
- **IO Skew** – raw number that represents the maximum difference in disk I/O counts between the average and the busiest AMP
- **IO Skew Percent** – the percentage difference in disk I/O counts between the average and the busiest AMP
- **CPU Skew** – raw number that represents the maximum difference in CPU consumption (in seconds) between the average and the busiest AMP
- **CPU Skew Percent** – the percentage difference in CPU consumption (in seconds) between the average and the busiest AMP
- **CPU millisec per IO** – the number of CPU milliseconds per disk I/O

Because skew and high CPU milliseconds per IO are situations that could occur momentarily in any legitimate request, the accumulated CPU qualification time must be specified to avoid false detections.

The exception criteria metrics are checked for using the resource usage data that has accumulated from the last exception interval until the next exception interval. The exception criterion must persist until the specified CPU qualification seconds have accumulated. The qualification time counter begins accumulating at the end of the first exception check time that detected the high CPU Milliseconds per IO.

For example, the CPU Skew must be more than 25% for more than 600 CPU seconds before the action is triggered.

Recommendation: Specify Skew as a percentage rather than a specific value.
A good value to consider is where skew percent is larger than 25%.

Recommendation: What is a good value for CPU Milliseconds per IO?
An anticipated range of appropriate CPU Milliseconds per IO values to set typically varies between 3 and 10. A typical request tends to fall between 1 and 2. A legitimate small-table product join request tends to fall between 2 and 3. High CPU queries are generally > 3.

For automated exception purposes and because there are some legitimate queries that can exceed a value of 3, it is recommend to start with a value of about 5, and fine-tuning that value as guided by performing workload analysis.

Qualification time

- Qualification Time **must be specified** for all Qualified Exceptions
- Qualification Time is **expressed in CPU seconds** rather than Clock seconds because CPU seconds are not subject to concurrency load conditions
- Qualification Time counter **begins** accumulating at the end of the first exception interval check that detected the exception condition
- The exception condition **must persist** for the duration of the Qualification Time
- If the exception is **not detected in subsequent** exception interval checks before the Qualification Time is exceeded all previous **accumulations are cleared**
- **Skew** and **CPU Disk Ratio** are exceptions that **could occur momentarily** in a legitimate query
- Qualification Time is used to avoid **false detections**

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Because skew and high CPU milliseconds per IO are situations that could occur momentarily in any legitimate query, the accumulated CPU qualification time must be specified to avoid false detections.

The exception criteria metrics are checked for using the resource usage data that has accumulated from the last exception interval until the next exception interval. The exception criterion must persist until the specified CPU qualification seconds have accumulated. The qualification time counter begins accumulating at the end of the first exception check time that detected the high CPU Milliseconds per IO. If the associated exception is not detected in any subsequent exception checks before the qualification time counter is exceeded, all previous detections are cleared. The qualification time counter is restarted from zero after the next detection.

Why not use wall-clock time to qualify these exceptions? Consider if a qualification wall-clock time of 30 minutes could be set. Now, consider a query that is badly skewed and runs on a lightly loaded system. Regardless of the skew, the query can complete the skewed portion of the query in 25 minutes, insufficient to qualify as a legitimate exception. However, the next day the same query runs on a heavily loaded system where the CPU cycles must be shared with more concurrent queries. The skewed portion of the query might perhaps run for an hour or longer if there were no exception specified but with the exception specified, would be detected and acted upon after the 30-minute wall-clock qualification time. This inconsistency in managing the skew is considered unacceptable by most. By instead assigning qualification time to CPU time, the concurrency load is irrelevant. The same skewed query will be detected in a heavily loaded system and a lightly loaded system.

Exceptions example

New Exception

Unqualified

Use this form to define the criteria that will trigger this exception and to define the action(s) that will be taken.

General

Name: * CPUUsage

Description:

Criteria

Exception Criteria: CPU Time +

CPU Time (sum over all nodes) 60 CPU-Seconds

(*) Qualification Time: CPU-Seconds

Actions

☒ Notification Only

☐ Abort

☐ Abort Selects Only

☐ Change Workload to: Adhoc

Notifications:

☐ Send Alert:

☐ Run Program:

☐ Post to QTable:

OK CANCEL

Edit Exception Criteria

Qualified

Use this form to define the criteria that will trigger this exception and to define the action(s) that will be taken.

General

Name: * CPUSkew

Description:

Criteria

Exception Criteria: -- Choose -- +

CPU Skew Percent 25 % (*)

AND

CPU Skew Difference 50 (*)

(*) Qualification Time: 60 CPU-Seconds

Actions

☐ Notification Only

☒ Abort

☐ Abort Selects Only

☐ Change Workload to: BAM

Notifications:

☐ Send Alert:

☐ Run Program:

☐ Post to QTable:

OK CANCEL

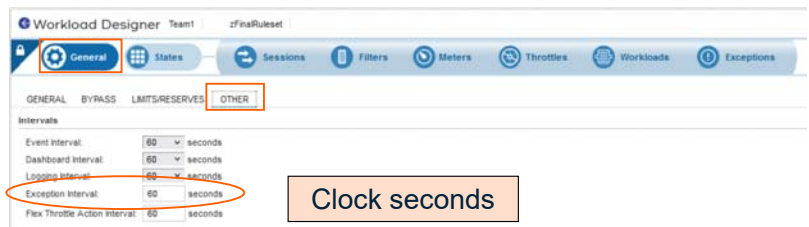
The facing page shows an exception created for CPU Time and CPU Time per Node. If a request assigned to the Tactical workload is detected, it will be moved to the BAM workload.

Exception monitoring

TASM performs **two types** of exception monitoring:

- Synchronous (monitors **Unqualified Criteria**)
 - Each query is monitored at the **end of each step** in its execution plan
- Asynchronous (monitors **Unqualified and Qualified Criteria**)
 - Each query is monitored at the **Exception Interval** in the Intervals section of General's Other tab
 - Primarily done to **monitor** queries that have **long running steps**, which otherwise would not catch the exception condition
 - At each Exception Interval, TASM will issue a Monitor Session command and collect snapshot data

Qualified Conditions are ONLY detected asynchronously.



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TASM checks for exception conditions at the following times.

- Synchronously – at the end of each AMP step
- Asynchronously – at the configurable time interval (1-3600 seconds); this value is set within TASM using the General Settings → Other Tab → Exception Interval

Asynchronous exception monitoring example

Assume a three AMP system with a CPU Skew Percent exception criteria of 30% and Qualification Time of 500 CPU seconds.

Exception Interval	CPU percent at end of Exception Interval			Accumulated CPU Seconds	Skew Percent	Action Taken
	AMP1	AMP2	AMP3			
1	100	99	98	0	1%	None
2	98	99	100	0	1%	None
3	86	100	88	0	9%	None
4	45	100	44	0	37%	Qualification Timer initiated
5	43	100	44	187	38%	None
6	44	100	45	376	37%	None
7	44	100	43	563	38%	Exception Detected

- At the end of each Exception Interval, a snapshot is taken
- Skew is beginning at exception interval 3
- Skew percent is met at exception interval 4 and the qualification time is started
- Skew persists through exception interval 7 where qualification time is exceeded, and exception is detected

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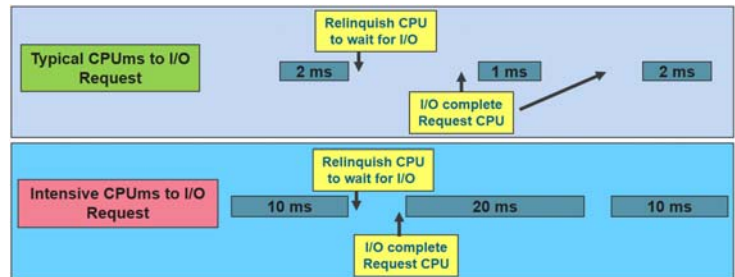
Here on this artificial 3 AMP system, the user specified he wants to see skew percentage exceed 30% consistently for 500 accumulated CPU seconds before the exception will be detected:

In this example, there is no skew in intervals 1 and 2. A skew is beginning in interval 3; however, the skew percentage criterion is not met until interval 4, where the qualification timer is initiated. The skew persists through interval 7 when the required accumulated CPU qualification time has transpired, and an exception is taken.

CPU disk ratio

- High CPU Disk ratio can be the result of **unconstrained product joins** or high number of **Duplicate Row checks**
- However, it can also be the result of other **legitimate** more CPU intensive operations such as **large aggregation** on highly distinct columns
- Typical query tends to fall between **1 and 2 milliseconds** per I/O
- Legitimate **small-table product join** query tends to fall between **2 and 3 milliseconds** per I/O
- High CPU queries are generally greater than **3 milliseconds** per I/O

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CPU Disk Ratio is calculated as $(\text{TotalCPUTime (seconds)} * 1000 / \text{Total Logical I/Os})$

Recommend to start with a Ratio value of **5** and tune with further workload analysis.

The exception, CPU milliseconds per IO, is a useful way of detecting queries that have an unusually high ratio of CPU processing relative to logical I/Os incurred. A good example of this is an accidental unconstrained product join being performed on a very large table. (This metric is sometimes called the product join indicator or PJI for this reason; however other legitimate queries such as some processing-intense full table scans can also be very CPU intensive.) Because of their very high CPU usage, these types of queries can more readily steal CPU resources from other higher priority workloads, impacting the effectiveness of the Priority Scheduler to favor higher priority requests.

To elaborate on the problem, consider a situation where an extremely CPU intensive request running at lower priority competes with a higher priority request that has the typical CPU vs. IO demand. The typical CPU vs. IO demand request does not always use the full time slice of CPU assigned to it by the Priority Scheduler before having to relinquish the time slice to perform an IO. It then must wait until the IO is complete before it can even get back onto the queue for its next time slice. However, an extremely CPU intensive request will use its full time slice of CPU before relinquishing the CPU to the next request in the queue, and immediately get back onto the queue to await its next turn for CPU.

So even though the very CPU intensive requests may have lower priority, it may at times over-consume CPU compared to less CPU intensive but higher priority requests. Many customers have found that by detecting and aborting such requests helps keep overall Priority Scheduler more effective. (Alternatively, changing WD to a very low priority has only sometimes been successful for reasons seen in this illustration. Even if Change WD is chosen, the details will now be captured within the DBQL for later follow-up.)

Skew detection

- Skew exception monitoring is detected on a **per-request** basis
- Skew is calculated based on the number of **Active AMPs** in the request step, not necessarily all the AMPs in the system
- **Group AMP** request steps that uses the same resources per AMP involved will not cause a skew to be detected
- Viewpoint's **System Health portlet** skew detection is based on the **session** as a whole and is calculated based on **all AMPs** in the system
 - System Health's skew detection is best used for detecting skew based on the session not individual requests
- Skew is only detected **asynchronously** at the end of each exception interval

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Exception monitoring skew is detected on a per-request basis. It is calculated based the number of AMPs active in the request step, which is not necessarily all the AMPs in the system. For example, a group-AMP request that uses the same processing time per AMP involved would not cause a skew to be detected even though the AMPs not involved show zero processing.

Workload Exception Monitoring should be distinguished from Viewpoint's System Health skew detection as they are different. Viewpoint's System Health is looking for skew for the session. It is calculated based on all the AMPs in the system, regardless of how many AMPs are involved within individual requests associated with the session. While Viewpoint's System Health skew detection can be very effective for session-level detection, it is not appropriate for detecting request-level skew. Viewpoint's System Health skew detection is best used to detect an imbalance of processing due to the current mix of requests associated with a session in each time interval. It is thought that if a session issues many few or group-AMP operation requests, statistical probability should result in an even balancing of that processing across all AMPs given sufficient time passage. When this is not the case, Viewpoint can detect and alert on the issue. Exception processing cannot detect this 'system-wide' issue due to its per-request orientation.

Skew is calculated at the end of each exception interval and considers the CPU or I/O consumed only during that interval in computing skew percent of skew value. If skew is detected, it must persist for at least the qualification CPU time specified

Skew is NOT calculated synchronously at the end of request steps. Asynchronous exception checking is the sole method used to detect skew. Due to potential conflicts in the way asynchronous and synchronous calculations are derived (end of intervals vs. end of steps), synchronous skew detection is disabled in favor of the more critical and valuable asynchronous monitoring. To explain this, consider that skews are often isolated to individual steps. If a step is short and skewed, it would likely not pass the required CPU Qualification time. Generally, short

steps that are skewed are not critical to eliminate in order to keep the system healthy. Alternatively, if the step is long, with synchronous checking Teradata DWM would not be able to detect the exception until the end of the step, which defeats the purpose of detecting the skew. Asynchronous checking is therefore the chosen method for detecting skew.

If operating in a coexistence environment, skew detection is based on normalized CPU skew calculations. Consider detecting skew based on I/O, since the CPU normalization done for coexistence is based on generalized node-to-node CPU differences, whose differences may vary a bit from customer workload to customer workload.

Skew detection (cont.)

- Skew can be detected as either a **percentage** and/or **difference** metric.
- Skew as a **percentage** is calculated as:
 - $((\text{HighAMP} - \text{AvgAMP}) / \text{HighAMP}) * 100$
 - 0% means no skew, >0% indicates skew
- Skew as a **difference** is calculated as:
 - $\text{HighAMP} - \text{AvgAMP}$
 - 0 means no skew, >0 indicates skew
- To detect skew, TASM issues a **Monitor Session** command to collect snapshot data at each **exception interval**
- With the newer **multi core** hyper threaded systems (e.g., **32 logical CPUs**) the **impact** of skew on parallel efficiency of the system is minimal
- Skewing will mainly affect just the skewed query and have **minimal impact on other** executing queries
- Long term **solution** is to **tune** the query or through **physical design** modifications
- All detected requests will be **logged** and can be addressed as necessary after-the-fact

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Skew can be detected either using a percentage metric and/or a “difference in amount processed” metric.

- Skew as a difference value:
 - CPU: $\text{HighAMPCPU} - \text{AvgAMPCPU}$
 - IO: $\text{HighAMPIO} - \text{AvgAMPIO}$

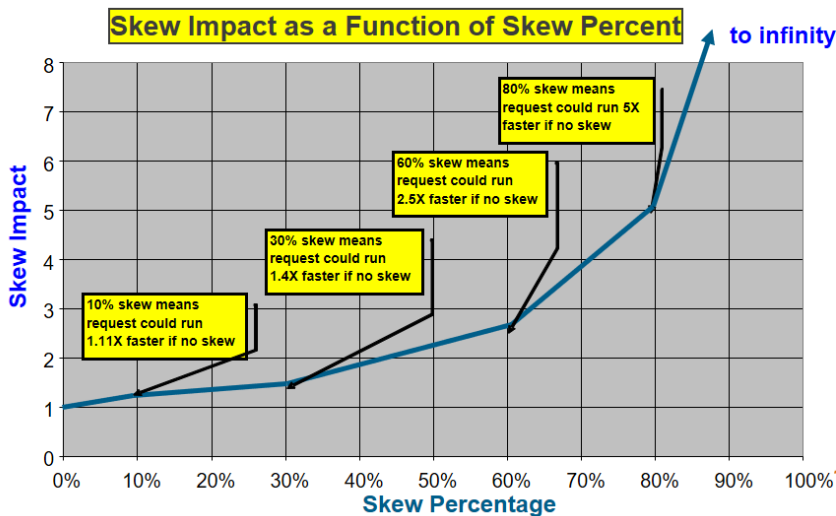
A value of 0 means there is no skew. A value > 0 indicates skew that has accumulated to a larger and larger value as long as the skew continues, up until the accumulated CPU qualification time has transpired.

If using skew difference, beware of an issue in detection accuracy: Whenever multiple applications issue monitor session commands, the internal collection cache is flushed and accumulations restart at the shortest interval being used. Say another application is set to refresh (submit Monitor Session) every 30 seconds and the exception interval is set to 60 seconds. The application will receive new data and reset the cache every 30 seconds. When TASM issues the monitor session command, it will contain not 60 seconds, but just 30 seconds of accumulated data. In order to keep the monitor session data used by TASM as complete as possible, it is recommended that other PM/PC applications that use monitor session do so at an interval greater than the exception interval. That way the other PMPC applications will always get the PMPC stats collected by TASM, maintaining TASM’s accuracy in computing skew difference values. In practice, for this and other reasons such as relativity, using skew difference is a less desirable way to detect skew than using skew as a percentage.

- Skew as a percentage:
 - CPU: $((\text{HighAMPCPU} - \text{AvgAMPCPU}) / \text{HighAMPCPU}) * 100$
 - IO: $((\text{HighAMPIO} - \text{AvgAMPIO}) / \text{HighAMPIO}) * 100$

A value of 0% means there is no skew. A value $> 0\%$ indicates skew, where the larger the number, the worse the skew is.

Skew impact



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- Skew impact is **calculated as** $((\text{HighAMP} - \text{AvgAMP}) / \text{AvgAMP}) + 1$
- The impact of skew **grows exponentially** although it's difficult to visualize using skew percent
- For example, If HighAMP = 1000 and AvgAMP = 700, skew impact would be $(1000 - 700) / 700 + 1 = 1.43$ and skew percent would be 30%
- Skew impact of **1.43x** means a query will take **43% longer** to complete vs. no skew
- A good **value to consider for skew percent is 25%** which would have an impact of running **1.33x longer**

The impact of that skew grows exponentially although that's difficult to visualize using the percentage format. To visualize, consider the impact of a skew using the formula:

- Skew impact = $((\text{HighAMP} - \text{AvgAMP}) / \text{AvgAMP}) + 1$.

i.e.: If HighAMP = 1000, AvgAMP = 700, skew impact would be $((1000 - 700) / 700) + 1 = 1.43$

A skew impact of 1.43X means a request will take 43% longer to complete vs. if there were no skew at all. The skew percentage here would have shown 30%.

A good value to consider is where skew percent is larger than 25%, i.e.: has an impact of running more than 1.33X longer as compared to an environment without skew

False skew

- Generally, skews can be **detected successfully** just using skew percentage
- However, in some situations, just using **skew percent** can lead to a **false detection** of skew
- For example, consider a situation where **HighAMPCPU = 3** and **AvgAMPCPU = 2**
 - Skew percent is calculated as **33%** and skew impact is **1.50x**
 - This **indicates** a skew that should be acted upon
 - However, the **skew difference** value is **only 1 CPU second**
 - This could be **due to a very short step** where any skew is insignificant
 - This could also be **due to an extremely heavy concurrency load**
 - **Either** situation is **not significant**
- To **avoid** these types of false skew it is recommended:
 - Use a combination **CPU qualification seconds** plus **skew percentage**
- If skew percentage plus CPU qualification seconds is resulting in the False skew:
 - Use a combination of **skew percentage** AND'd together with **skew difference**
 - For example, if skew percentage exceeds **25%** AND skew **difference exceeds 50 seconds of CPU processing time**

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In general skews can be detected successfully using just the skew percentage metrics. However, there exist some situations which could lead to a false detection of skew.

For example, consider a situation where HighAMPCPU = 3, AvgAMPCPU = 2. Skew percent is 33% and skew impact is 1.50X, suggesting a skew worth acting on. However, the skew difference value is only 1 CPU second. The low CPU difference metrics could be a result of a very short step where any skew that occurs is insignificant, or an extremely heavy concurrency load that limits the ability of the metrics to accumulate very high, at the same time as demonstrating that any skew that is occurring is not significantly hindering the other workloads, who are still getting a very large share of the CPU cycles.

To avoid these types of false skew, it is recommended that skew detection is set up as follows:

- The required CPU qualification seconds will help assure that the skew detected is real rather than a momentary situation
- When skew percentage (plus CPU qualification seconds) alone is resulting in false skew detections, use a combination of skew percentage AND'd together with skew difference value. For example:
 - If skew percentage exceeds 25% AND the skew difference exceeds 50 seconds of CPU processing time

Exception actions

If all the exception criteria are met, one of the following automated exception actions must be performed:

- **Notification Only:** Sends Notification only, no other action is performed
- **Abort:** Abort the query
- **Abort On Select and Log:** Abort the query if it contains only Select statements within the current transaction
- **Change Workload To:** Moves the request to the workload specified

Note that the detection is **always logged to the `DBC.TDWMExceptionLog`** and the SQL is captured in the **`DBC.DBQLSqlTbl`**.

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After defining the exception criteria, you will define the exception action you want TASM perform when the exception is detected.

- Notification Only
 - Sends Notification only, no other action is performed
- Abort
 - Abort the query
- Abort On Select and Log
 - Abort the query if it contains only Select statements within the current transaction

Note that the detection is always logged to the *DBC. TDWMExceptionLog* and the SQL is captured in the *DBC.DBQLSqlTbl*.

Change workload exception action

- Exceptions can be applied to one or more workloads and enabled or disabled for each Planned Environment
 - For example, during batch processing a query causing an exception might be aborted, but during online processing, allowed to complete
- If the Exception Criteria is Elapsed Time or Blocked Time, the Change to Workload action will not be available
 - This is indicative of a system-wide condition that is impeding the request
 - Better to send automated alert to DBA for investigation
 - Starting in TD15.10, there is an option to exclude Blocked Time and Delay Time for the Elapsed Time criteria

Exception Criteria: Elapsed Time

Elapsed Time (hh:mm:ss)

☐ Exclude blocked time

☐ Exclude throttle delay time

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Exceptions can be applied to one or more workloads and enabled or disabled for each Operating Environment. For example, at night, a request causing an exception might be aborted, but during the day, that same request would be permitted to run.

However, if the exception action is 'change workload', the exception must be enabled or disabled consistently for all defined Operating Environments. This is because we must assure that a request that runs to completion consistently resolves to the same workload regardless of when, or what state the system was in when they were run. This is to maintain consistency in accounting and management of the request. Consider that exceptions are an extension of classification, in that classification can properly classify only to the extent that the information it knows of the request before it begins execution is sufficient to properly place the request in the appropriate workload. But sometimes that information must be supplemented by exception conditions, which key off of additional information it obtains after the request is under execution. When an exception occurs, it provides an opportunity for automated re-classification of the request to its correct workload, and therefore automatically adjusts the workload operating rules to those of the now-correct workload assignment for the request.

Sometimes it is desirable for the request that encountered an exception to be managed differently in one operating environment vs. another automatically, without alerts or aborts. Changing to different workload just because the operating environment is different is not the correct solution. Instead, the workload operating rules of the 'changed-to' workload should simply be different for those resulting states.

Abort exception action

- Consider the **implications** of an Abort exception action
 - If a full table update request is aborted, the **rollback** will take several times longer to undo the updates
 - Consider **Abort on Select** to avoid lengthy rollback times
- Aborting a request in a **multi request job**, where the results of one request feed into the next request using temporary tables, could result in inaccurate results
- Aborting a request based on **false skew** detection
- **Caution is advised in aborting requests**, as an aborted request may have substantial business value

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Before deciding to use an automated abort for requests that encounter an exception, consider the possible implications of that abort.

For example, say the request about to be aborted happened to be a full table update that is nearing completion with 30 minutes down, 5 minutes to go. A rollback would take several times longer to undo than simply completing the request in the first place. For this example, consider the option “Abort on Select” to avoid aborting any full table update requests.

Another example would be a request that is part of a multi-request report, where the results of one request feed into the next request using temporary tables. Aborting one of the requests could result in inaccurate results for the report.

Yet another example involves the case with skew detection and the chance of aborting the wrong query due to a false skew detection.

Even if it is known that the workload will not receive these types of requests, caution is still advised regarding the use of the abort option. A query may be aborted that has great value to the business.

Exception action conflicts

- Since workloads can have **multiple exceptions** applied, it is possible for multiple exceptions to be **detected simultaneously** for a single request
- TASM will perform the **all** corresponding **exception notifications**
 - Alert
 - Run Program
 - Post to Qtable
- Conflict occurs between the exception actions **Abort** and **Change to Workload**
- TASM uses the following **criteria to resolve** conflicting actions:
 - **Aborts** always **take precedence** over any Change to Workload actions
 - If the conflict is **between two different** Change to Workload actions
 - Moved to workload with the **lowest priority**
 - If both are both are same priority, it is **alphabetical**

TASM will log all “**Change to Workload**” actions **not taken as overridden** in the **exception log**.

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With the ability to have multiple exceptions apply to a workload, it is possible for multiple exceptions to be detected simultaneously against a single executing query. TASM will perform all the corresponding exception actions as IonRg as they do not conflict. I.e., TASM always executes all Raise Alert, Run Program and Post to QTable exception actions since they cannot conflict. Associated logging always occurs as well.

A conflict occurs when two exception actions to be performed are either:

- Abort and change WD or
- Change to different WDs (e.g., change to WD-A and change to WD-B).

TASM follows these rules to resolve conflicting exception actions automatically when necessary:

- Aborts take precedence over any Change Workload exception actions.
- Within any conflicting exception list, precedence is given to the change workload with the timeshare then SLG tier then finally tactical workload management method. If two change workloads are both timeshare, the lower access level has higher precedence. If both have same access level, then they are sorted alphabetically by workload name. If the conflicting change workloads are both using the SLG tier workload management method, the precedence is first determined by the lower SLG tier and if the SLG tier is the same, the workload with the lower workload share percent value. If both have same workload share percent value, then they are sorted alphabetically by workload name.

In all cases, TASM logs all other Change Workload exception actions as overridden.

Exception notifications

If all of the exception criteria are met, in addition to the exception action, one or more of the following Notifications can optionally be performed:

- **Alert:** Sends the selected alert
- **Run Program:** Executes the selected program
- **Post to Qtable:** Posts the string entered in the box to the *DBC.SystemQTbl*

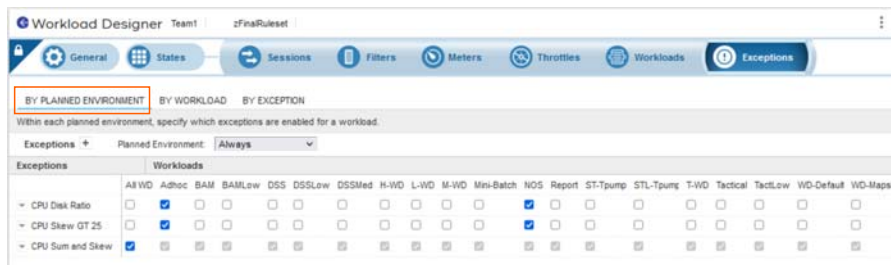
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After defining the exception criteria, you can optionally define one or more notifications that you want TASM perform when the exception is detected.

- Alert
 - Sends the selected alert
- Run Program
 - Executes the selected program
- Post to QTable
 - Posts the string entered in the box to the DBC.SystemQTbl

Enabling exceptions by planned environment

Portlet: **Workload Designer** > Button: **Exceptions** > Tab: **By Planned Environment**



Exceptions can be enabled or disabled **By Planned Environment**.

If the Exception Action is “Change to Workload”, the Exception must be applied to all Planned Environments.

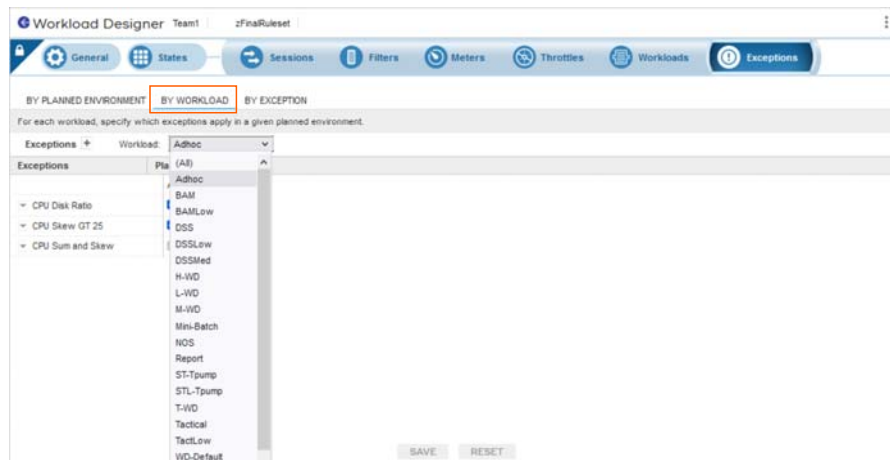
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After defining the exception you can then decide which Planned Environments the exception will be enabled. You also have the option of editing the exception rule or deleting the exception rule completely.

Note: the “change to workload” exception will always be enabled for all Planned Environments

Enabling exceptions by workload

Portlet: **Workload Designer** > Button: **Exceptions** > Tab: **By Workload**



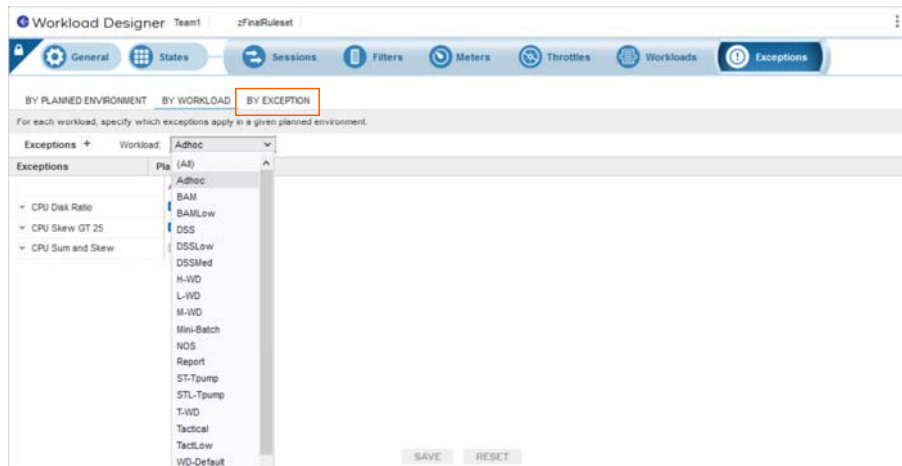
Exceptions can be enabled or disabled **By Workload** for one or more workloads.

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After defining the global exception you can then decide which Workloads the exception will be enabled. You also have the option of editing the exception rule or deleting the exception rule completely.

Enabling exceptions by exceptions

Portlet: **Workload Designer** > Button: **Exceptions** > Tab: **By Exception**



Exceptions can be enabled or disabled **By Workload** for one or more workloads.

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After defining the exception you can then decide workloads the exception will be enabled. You also have the option of editing the exception rule or deleting the exception rule completely.

Tactical workload exception

- Workloads assigned to the Tactical Workload Management Method receive the highest priority and are intended for highly-tuned tactical queries
- Typically, those queries require very small amounts of CPU and I/O resources
- If more resource-intensive less critical queries are mis-classified into a Tactical workload it is important to move those queries to another non-tactical workload
- Resource consumption is measured in two ways:
 - CPU seconds (default value is 2 seconds per node)
 - I/O physical bytes transferred (default is 200 MB per node)
- A query will be moved to a different non-tactical workload on a node when either CPU per node or I/O per node threshold is met
- It is possible for a request to be running in a different workload on one node while other nodes are running the query in the original workload
- The request is moved on all nodes when the CPU or I/O “sum over all nodes” threshold is met
- If there are multiple Tactical workloads, each can have different exception thresholds

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Workloads assigned the tactical workload management method receive the highest priority and are intended for highly-tuned tactical queries which are differentiated as those requiring less than some sub-second amount of CPU processing. If more resource-intensive, less critical queries begin executing in a tactical workload (because TASM can't distinguish them through classification criteria) it can be important to use exception processing to move the requests to another workload as quickly as possible.

Tactical exceptions are accomplished in two ways: 1.) CPU consumed by the query, and 2.) I/O physical bytes transferred. The Priority Scheduler detects when a request reaches the CPU or the I/O threshold limit and moves the request to the Change Workload on that node.

The tactical exceptions consist of the following:

- Tactical CPU time threshold values: CPU (sum over all nodes) and CPU per node. The default value for CPU per node is 12 seconds and the default value for the CPU (sum over all nodes) is the number of nodes times the CPU per node value.
- Tactical I/O Physical Byte threshold values: I/O (sum over all nodes) and I/O per Node. The default value for I/O per Node is 1 GB and the default value for the I/O (sum over all nodes) is the number of nodes times the I/O per node value.
- A change workload action
- Optional notification actions

A request will be moved to a different workload on a node when either the CPU per node or the I/O per node threshold is reached. This means that it is possible for a request to be running on a different workload on one node (having been demoted), while all other nodes are running the request in the original workload. This could be the case if there is heavy skew on one node, so that one node exceeds the per node threshold of either CPU or I/O. The request will be moved on all nodes and the notification actions performed when the CPU (sum over all nodes) or the I/O (sum over all nodes) threshold is reached.

If DBQLogTbl logging is enabled, the number of nodes that reach the CPU per node threshold value is logged in the TacticalCPUException field and the number of nodes that reach the I/O per node threshold value is logged in the TacticalIOException field.

The sum over all nodes CPU and I/O thresholds are automatically set at the CPU per node setting times the number of nodes in the configuration by default.

Tactical exception

Workload Designer TDT1 | POWM Ruleset v2 |

Workload: **Tactical2**

GENERAL CLASSIFICATION THROTTLES SERVICE LEVEL GOALS EXCEPTIONS **TACTICAL EXCEPTION**

All tactical workloads have an associated tactical exception. Tactical exceptions apply across all planned environments.

Criteria

Actions are triggered when both thresholds in one of the following two criteria are met.

Tactical CPU Time

CPU (sum over all nodes): CPU-Seconds

CPU per Node: CPU-Seconds

Tactical I/O Physical Byte

I/O (sum over all nodes): MB

I/O per Node: MB

Action

Change Workload:

Notifications

☐ Send Alert:

☐ Run Program:

☐ Post to QTable:

Workloads using the Tactical Workload Management Method will automatically have a **required** Tactical Exception for CPU and I/O.

Tactical Management Method is intended for **highly tuned** queries with **minimal** resource usage.

The only action to is **move** the query **to** another **non-Tactical** workload.

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For Tactical workloads the Tactical Exception will be available to modify the tactical exception CPU and I/O thresholds.

Current topic: Additional workload tabs

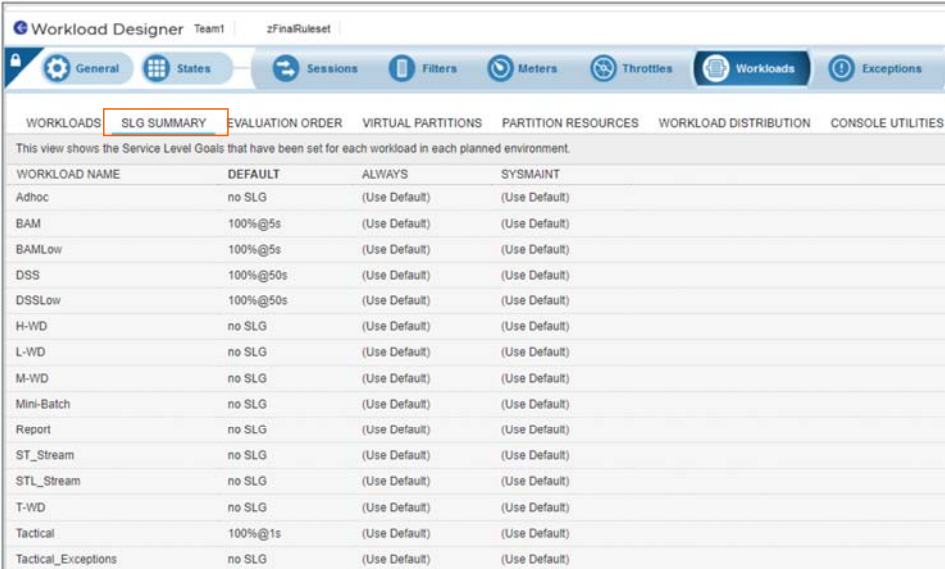
- Workloads overview
- The workloads view
- Flex throttles
- Workload group throttles
- Additional options for workloads
- Workload exceptions
- **Additional workload tabs**
 - **SLG summary**
 - **Workload evaluation order**
 - **Console Utilities**

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SLG summary

Portlet: **Workload Designer** > Button: **Workloads** > Tab: **SLG Summary**



The screenshot shows the 'Workload Designer' interface with the 'SLG SUMMARY' tab selected. The table below lists the Service Level Goals for various workloads across three categories: DEFAULT, ALWAYS, and SYSMANT.

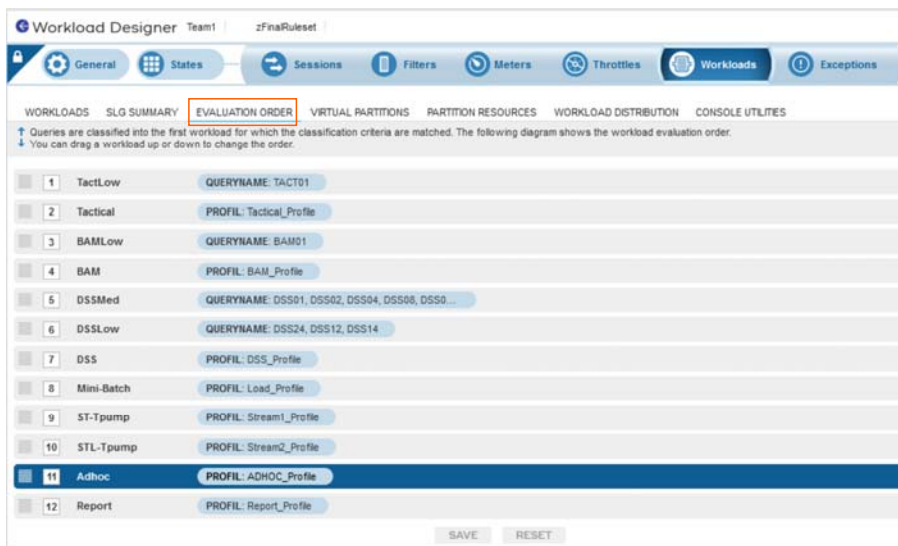
WORKLOAD NAME	DEFAULT	ALWAYS	SYSMANT
Adhoc	no SLG	(Use Default)	(Use Default)
BAM	100%@5s	(Use Default)	(Use Default)
BAMLow	100%@5s	(Use Default)	(Use Default)
DSS	100%@50s	(Use Default)	(Use Default)
DSSLow	100%@50s	(Use Default)	(Use Default)
H-WD	no SLG	(Use Default)	(Use Default)
L-WD	no SLG	(Use Default)	(Use Default)
M-WD	no SLG	(Use Default)	(Use Default)
Mini-Batch	no SLG	(Use Default)	(Use Default)
Report	no SLG	(Use Default)	(Use Default)
ST_Stream	no SLG	(Use Default)	(Use Default)
STL_Stream	no SLG	(Use Default)	(Use Default)
T-WD	no SLG	(Use Default)	(Use Default)
Tactical	100%@1s	(Use Default)	(Use Default)
Tactical_Exceptions	no SLG	(Use Default)	(Use Default)

Displays the **Service Levels Goals** of each workload for each Planned Environment.

Service Levels Goals that have been set for each workload for each Planned Environment are summarized. The edit or delete any of them edit the workload and select the Service Level Goals tab.

Workload evaluation order

Portlet: **Workload Designer** > Button: **Workloads** > Tab: **Evaluation Order**



Workload Management **evaluates** a request against a **Workloads classification criteria in sequence**.

The request is **assigned to the first** workload that **meets the classification criteria**.

Set the Evaluation order of the Workloads with **higher priority before lower priority** and more **specific before less specific** criteria.

Drag and Drop the Workload **to change** the order.

Within the Workloads definitions of the Workload Designer the Evaluation Order tab specifies the order in which workload classification criteria are matched. The classification criteria shown summarize the specification for each workload.

To change the order, drag and drop the workload. Workloads with more specific classification criteria need to be specified higher in the list.

Order of evaluation can help to manage logic of many criteria and/or workloads. When a match is found, the workloads later in the list are not considered.

This can be both an advantage and disadvantage: If a request could be classified into two or more workloads, the order of evaluation will dictate that the one first in the list is the one that “wins”. The disadvantage is if the order of evaluation has not been set up properly. Requests may be classified incorrectly into workloads if the more-specific classification is put after the less-specific classification.

Console utilities

Portlet: **Workload Designer** > Button: **Workloads** > Tab: **Console Utilities**

CONSOLE UTILITY	WORKLOAD NAME
CheckTable	WD-Default
Ferret	WD-Default
Filer	WD-Default
Query Configuration	WD-Default
Query Session	WD-Default
Reconfiguration	WD-Default
Recovery Manager	WD-Default
Replication	WD-Default
Table Rebuild	WD-Default

PERFORMANCE GROUP	WORKLOAD NAME
H	H-WD
L	L-WD
M	M-WD
R	T-WD

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Console Utilities and Performance Groups by default are mapped to **WD-Default**.

Console utilities do not require a logon.

Consider creating generic SYSADMIN workloads to map Console Utilities and Performance Groups (e.g., WD-SysAdminH, WD-SysAdminM, etc.).

Console Utilities such as *scandisk*, *checktable*, etc. do not logon and run through normal paths through the Parser/Dispatcher like request requests do. (For example, there is no user associated with a *checktable* job because there is no logon.) Therefore, they bypass the classification step that assigns requests to workloads and their AGs. That means a special way is needed to assign those console utilities to the appropriate workload and AGs. That is the intent of the screens seen when CONSOLE UTILITIES is clicked.

The Performance Group to Workload mapping table maps a Performance Group name to a workload. If a workload has not been determined for the utility, the Performance Group to Workload Mapping table is used to get the workload mapped to the default priority for the utility. All utilities default to 'M,' except the Recovery Manager utility, which defaults to 'L.'

Utilities that default to 'M' run in the workload mapped to 'M' in the Performance Group to Workload Mapping table. The Recovery Manager utility runs in the workload mapped to 'L.'

Summary

- Workload rules are used to classify requests with similar characteristics
- Workloads are derived primarily from business requirements are supplemented with technical requirements
- Workload rules consist of:
 - Classification criteria
 - Concurrency throttles
 - Operating rules
 - Exception Actions
 - Service Level Goals
- Maximum 250 workloads including the 1 default and 4 internal workloads, typical initial number is between 10 and 30
- Advantages of workloads:
 - Improved resource control
 - Improved reporting
 - Automated exception detection and handling

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Workload rules are used to classify requests with similar characteristics

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Maximum 250 workloads including the 1 default and 4 internal workloads, typical is between 10 and 30

Advantages of workloads:

- Improved resource control
- Improved reporting
- Automated exception detection and handling

Module 13: Refining workload definitions

Teradata Vantage optimizing the
Analytics Database through Workload
Management

teradata.

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Objectives

After completing this module, you will be able to:

- Analyze Workload Performance Metrics using DBQLogTbl
- Export DBQL data to a CSV file using Teradata Studio
- Import CSV file into MS Excel
- Create a Pivot Table
- Use the Pivot Table to identify “misbehaving” queries within a workload
- Discuss the reasons for splitting a Workload

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Topics

- Capturing workload performance metrics
- Creating a pivot table
- Using the pivot table to analyze workload performance metrics
- Refining the workloads
- Lab activity

teradata.



Current topic: Capturing the workload performance metrics

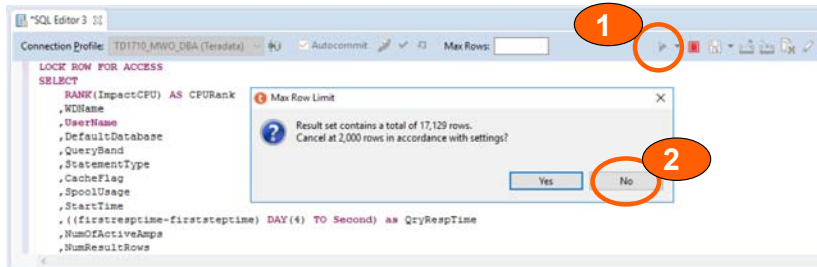
- **Capturing workload performance metrics**
 - **Run the AllQueries.sql**
 - **Export the result set to a CSV file**
- Creating a pivot table
- Using the pivot table to analyze workload performance metrics
- Refining the workloads
- Lab activity

teradata.



Run SQL in Teradata Studio

Run the **AllQueries.sql** script to capture the DBQLog table rows for the BAM, Tactical, and DSS workloads.



1. Click the **Execute** button
2. Select **No** in the Max Row Limit dialog box

teradata.

Run the AllQueries.sql script to capture the DBQLog table rows for the BAM, Tactical, and DSS workloads

LOCK ROW FOR ACCESS

SELECT

```

    RANK(ImpactCPU) AS CPURank
,WDName
,UserName
,DefaultDatabase
,QueryBand
,StatementType
,CacheFlag
,SpoolUsage
,StartTime
,((firstresptime-firststeptime) DAY(4) TO Second) as QryRespTime
,NumOfActiveAmps
,NumResultRows
,ParserCPUTime AS ParserCPU
,AmpCpuTime AS SumCPU
,TotalIOCount AS SumIO
,CASE WHEN (AmpCpuTimeNorm/(HASHAMP()+1)) =0 THEN 0 ELSE
MaxAmpCpuTimeNorm/(AmpCpuTimeNorm /(HASHAMP()+1))
    END (DEC(8,2)) AS CPUSkw
,CASE WHEN (TotalIOCount/(HASHAMP()+1)) =0 THEN 0 ELSE
Maxampio/(TotalIOCount/(HASHAMP()+1))
    END (DEC(8,2))AS IOSkw
,CASE WHEN TotalIOCount = 0 THEN 0 ELSE (AmpCpuTime *1000)/TotalIOCount
    END (DEC(8,2)) AS PJI
,CASE WHEN AmpCpuTime = 0 THEN 0 ELSE TotalIOCount /(AmpCpuTime *1000)
    END (DEC(8,2)) AS UII
    
```

teradata.

```

,MaxAmpCpuTime * (HASHAMP()+1) /*no. of Amps*/ AS ImpactCPU
,((FirstRespTime - StartTime) DAY(4) TO SECOND) AS QryRunTm
,(EXTRACT(DAY FROM FirstRespTime) * 86400 +
EXTRACT(HOUR FROM FirstRespTime) * 3600 +
EXTRACT(MINUTE FROM FirstRespTime) *60 +
EXTRACT(SECOND FROM FirstRespTime) -
EXTRACT(DAY FROM starttime) * 86400 -
EXTRACT(HOUR FROM starttime) * 3600 -
EXTRACT(MINUTE FROM starttime) *60 -
EXTRACT(SECOND FROM starttime) ) AS QryClockRunTm
,(CASE
WHEN QueryText LIKE '%BAM01%' THEN 'BAM01'
WHEN QueryText LIKE '%BAM02%' THEN 'BAM02'
WHEN QueryText LIKE '%BAM03%' THEN 'BAM03'
WHEN QueryText LIKE '%BAM04%' THEN 'BAM04'
WHEN QueryText LIKE '%BAM05%' THEN 'BAM05'
WHEN QueryText LIKE '%Tact01%' THEN 'Tact01'
WHEN QueryText LIKE '%Tact02%' THEN 'Tact02'
WHEN QueryText LIKE '%Tact03%' THEN 'Tact03'
WHEN QueryText LIKE '%Tact04%' THEN 'Tact04'
WHEN QueryText LIKE '%Tact05%' THEN 'Tact05'
WHEN QueryText LIKE '%Tact06%' THEN 'Tact06'
WHEN QueryText LIKE '%Tact07%' THEN 'Tact07'
WHEN QueryText LIKE '%Tact08%' THEN 'Tact08'
WHEN QueryText LIKE '%Tact09%' THEN 'Tact09'
WHEN QueryText LIKE '%Tact10%' THEN 'Tact10'
WHEN QueryText LIKE '%Tact11%' THEN 'Tact11'
WHEN QueryText LIKE '%Tact12%' THEN 'Tact12'
WHEN QueryText LIKE '%DSS10%' THEN 'DSS10'
WHEN QueryText LIKE '%DSS11%' THEN 'DSS11'
WHEN QueryText LIKE '%DSS12%' THEN 'DSS12'
WHEN QueryText LIKE '%DSS13%' THEN 'DSS13'
WHEN QueryText LIKE '%DSS14%' THEN 'DSS14'
WHEN QueryText LIKE '%DSS15%' THEN 'DSS15'
WHEN QueryText LIKE '%DSS16%' THEN 'DSS16'
WHEN QueryText LIKE '%DSS17%' THEN 'DSS17'
WHEN QueryText LIKE '%DSS18%' THEN 'DSS18'
WHEN QueryText LIKE '%DSS19%' THEN 'DSS19'
WHEN QueryText LIKE '%DSS1%' THEN 'DSS01'
WHEN QueryText LIKE '%DSS20%' THEN 'DSS20'
WHEN QueryText LIKE '%DSS21%' THEN 'DSS21'
WHEN QueryText LIKE '%DSS22%' THEN 'DSS22'
WHEN QueryText LIKE '%DSS23%' THEN 'DSS23'
WHEN QueryText LIKE '%DSS24%' THEN 'DSS24'
WHEN QueryText LIKE '%DSS25%' THEN 'DSS25'
WHEN QueryText LIKE '%DSS2%' THEN 'DSS02'
WHEN QueryText LIKE '%DSS3%' THEN 'DSS03'
WHEN QueryText LIKE '%DSS4%' THEN 'DSS04'
WHEN QueryText LIKE '%DSS5%' THEN 'DSS05'
WHEN QueryText LIKE '%DSS6%' THEN 'DSS06'
WHEN QueryText LIKE '%DSS7%' THEN 'DSS07'

```

```

        WHEN QueryText LIKE '%DSS8%' THEN 'DSS08'
        WHEN QueryText LIKE '%DSS9%' THEN 'DSS09'
        ELSE 'Other'
    END) AS Query_Name
    ,querytext
    ,queryband
    ,cacheflag
    ,s.SqlTextInfo
    ,s.SqlRowNo

FROM DBC.DBQLogTbl lg

LEFT OUTER JOIN DBC.DBQLSQLTbl s
    ON lg.procid = s.procid
    AND lg.queryid = s.queryid

WHERE lg.UserName = 'tactuser1'
OR lg.UserName = 'tactuser2'
OR lg.UserName = 'dssuser1'
OR lg.UserName = 'dssuser2'

ORDER BY WDName;

```

Export SQL result set

The screenshot shows the SQL Editor window with a query and the Teradata Result Set Viewer window displaying the results. The 'View' menu is open, and the 'Export' option is highlighted. Red circles and numbers 1 and 2 indicate the steps: 1. Click the View Menu button, 2. Select Export.

SQL Editor 3

Connection Profile: TD1710_MWQ_DBA (Teradata) | Autocommit | Max Rows:

LOCK ROW FOR ACCESS

```
SELECT
  RANK(IMPACTCPU) AS CPURank
  , WdName
  , UserName
  , DefaultDatabase
  , QueryBand
  , StatementType
  , CacheFlag
  , SpoolUsage
  , StartTime
  , ((firstresptime-firststateptime) DAY(4) TO Second) as QryRespTime
  , NumOfActiveAmps
  , NumResultRows
```

Teradata Result Set Viewer

	CPURank	WdName	UserName	DefaultDatabase	QueryBand	StatementType	CacheFlag	SpoolUsage
1	10016	null	TACTUSER1	OPTIMIZATION_VM	aT> QUERYNAME=TACT07	Exec		null
2	10016	null	TACTUSER1	OPTIMIZATION_VM	aT> QUERYNAME=TACT07	Exec		null
3	10016	null	TACTUSER1	OPTIMIZATION_VM	aT> QUERYNAME=TACT07	Exec		null
4	10016	null	TACTUSER1	OPTIMIZATION_VM	aT> QUERYNAME=TACT07	Exec		null
5	10016	null	TACTUSER1	OPTIMIZATION_VM	aT> QUERYNAME=TACT07	Exec		null

Teradata SQL History

	Timestamp	Source	User ID	Destination	Result	Row Count	Note	SQL Statement	SQL Type	Length	Statements	Statement Type	Elapsed Time	DBM
11	2021-11-11 14:48	T_BA	m...	Tera...	Exec...	52517		SELECT ...	TER-TA	4969	1	Select ...	00:02:27.812	00:00:00

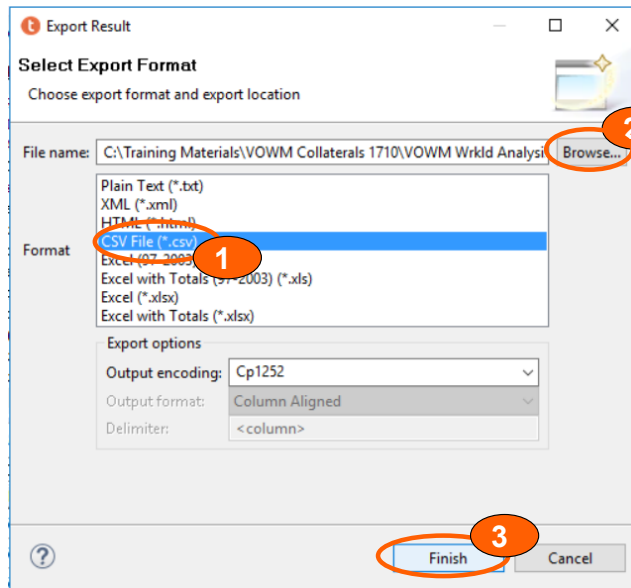
Export

1. Click the **View Menu** button

2. Select **Export**

teradata.

Select the CSV file format



1. Select **CSV File**
2. Click **Browse** to select the folder where you want to save the file
3. Click **Finish**

teradata.

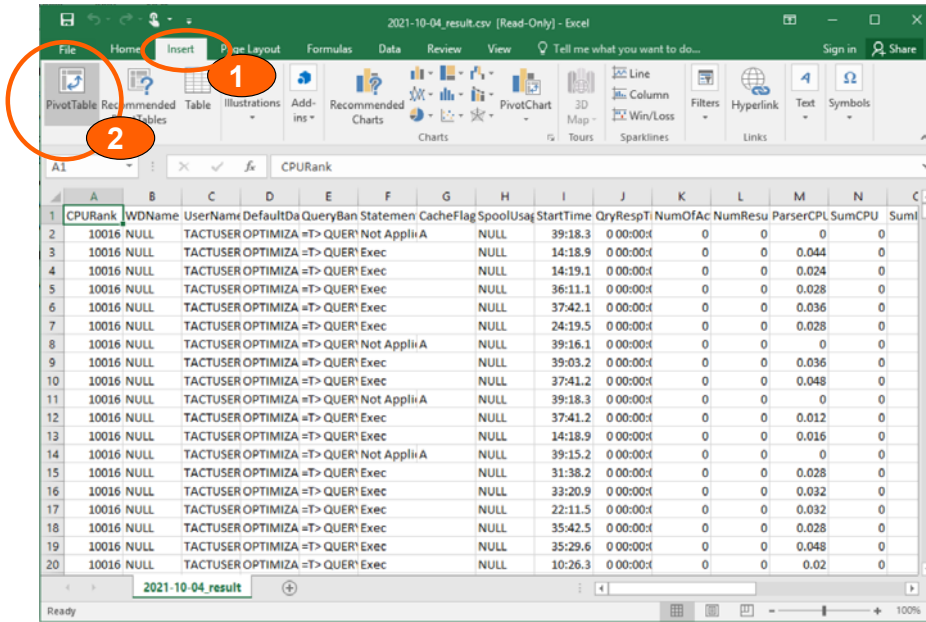
Current topic: Creating a pivot table

- Capturing workload performance metrics
- **Creating a pivot table**
 - **Open the CSV file in excel**
 - **Create a pivot table**
- Using the pivot table to analyze workload performance metrics
- Refining the workloads
- Lab activity

teradata.



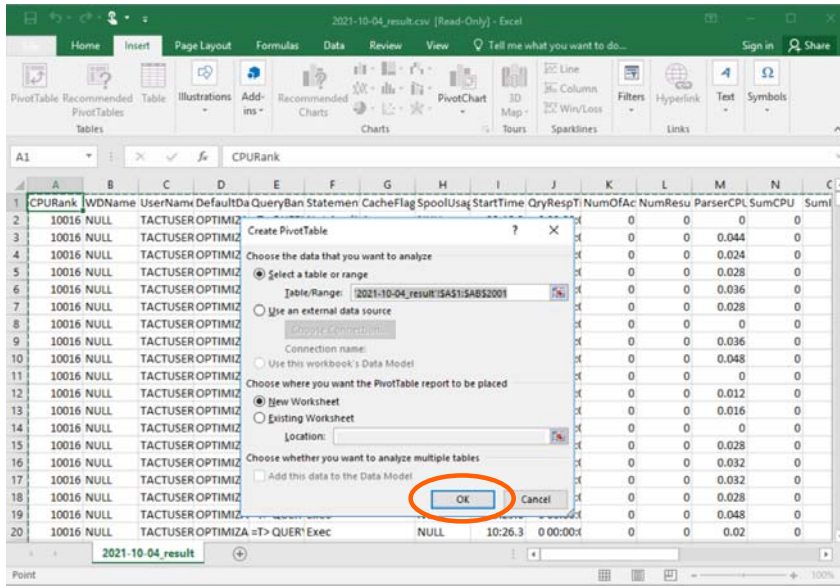
Select pivot table



1. After opening the CSV file in Excel, select the Insert menu
2. Select Pivot Table

We have modified the Analytics Database macro for this report to get the column Query Name. This is a key column for our pivot table.

Create pivot table

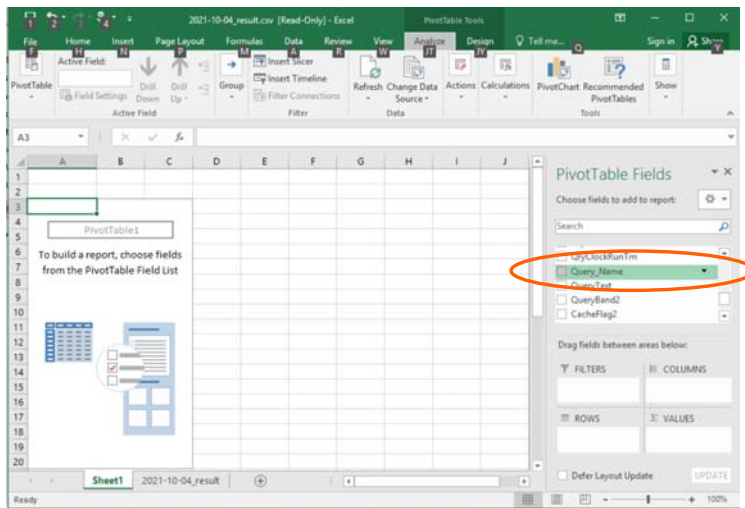


Accept the Table/Range default (all rows and columns) and place the Pivot Table in a New Worksheet, click **OK**.

teradata.

Accept the default table/range and create a new worksheet.

Populating the pivot table

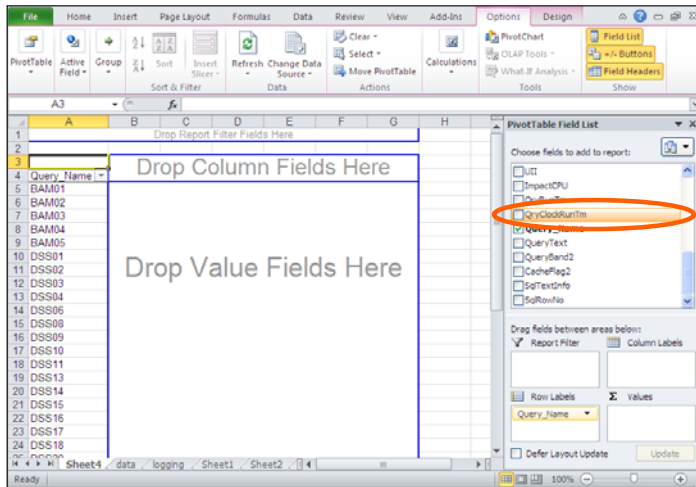


Select Query_Name in the PivotTable Field List.

teradata.

The Pivot Table initially is empty. From the Pivot Table Field List, drag and drop fields onto the template. First drag and drop the query name to the rows fields.

Populating the pivot table (cont.)



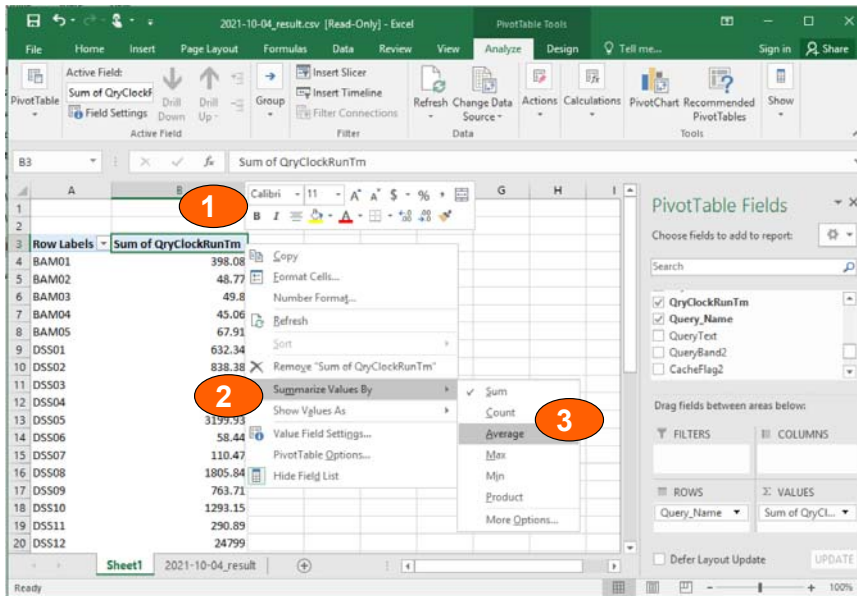
Select a field that you want to aggregate from the PivotTable Field List. In this field we are selecting QryClockRunTm.

NOTE: When you perform your analysis you can choose other fields, such as SumCPU, SumIO, NumberOfActiveAMPS, etc.

teradata.

The Query Name needs to be in the Rows Fields and the Query Run Time to be in the Columns Fields.

Changing value field settings

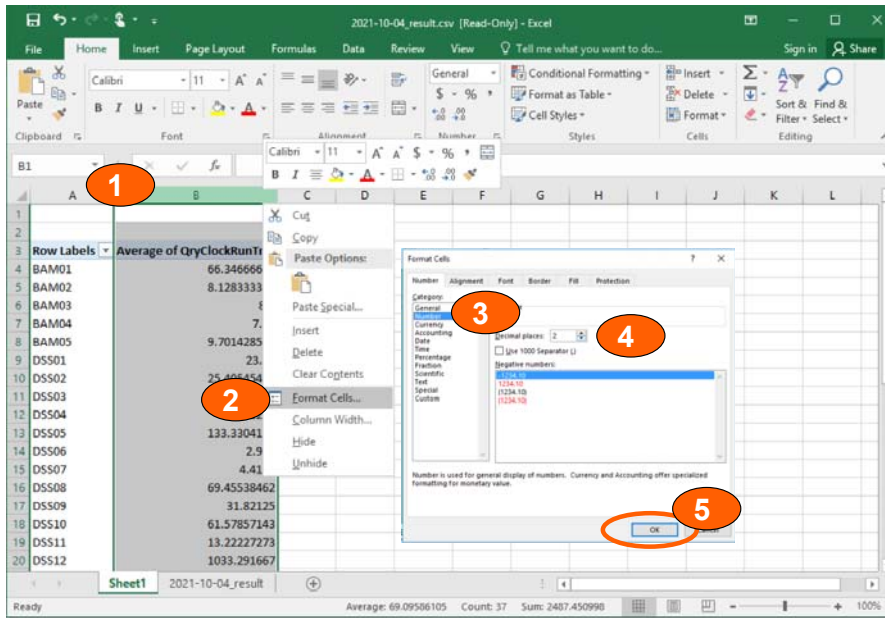


1. Right Click on **Sum of QryClockRunTm** cell
2. Select Summarize Values By
3. Select Average for the column, and notice that the name of the cell changes automatically to Average of QryClockRunTM

teradata.

The default display is total response time. Use the Value Field Settings to change the display to average response time.

Formatting data cells



1. Right Click on Cell B
2. Select Format Cells...
3. Select Number in the Category field
4. Select "2" in the Decimal places selector
5. Click OK

Use the Format Cells to change the format display of the average response time.

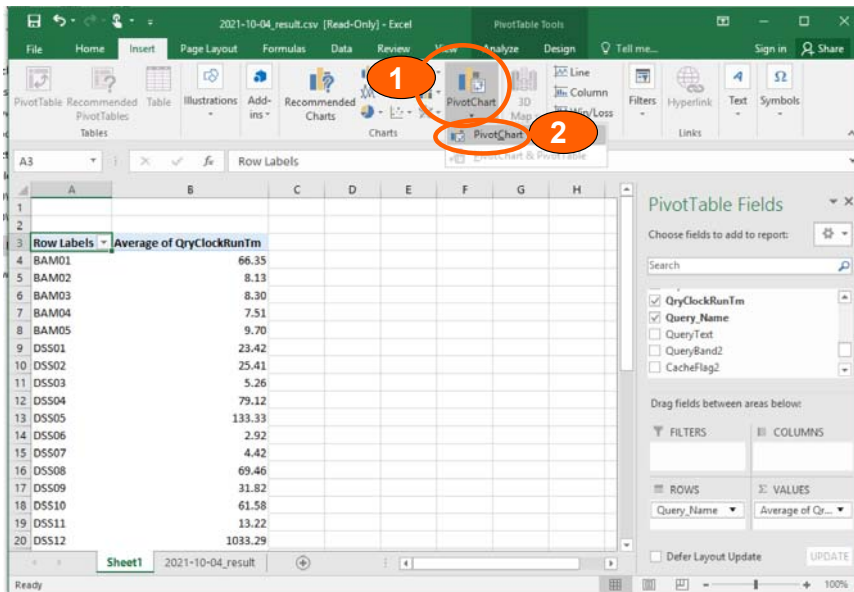
Current topic: Using the pivot table to analyze workload performance metrics

- **Capturing workload performance metrics**
 - Creating a pivot table
- **Using the pivot table to analyze workload performance metrics**
 - **Creating a pivot chart**
 - **Filtering the queries displayed in the chart**
 - **Identifying outlier queries**
 - **Creating a new workload**
 - **Adjusting the workload evaluation order**
- Refining the workloads
- Lab activity

teradata.



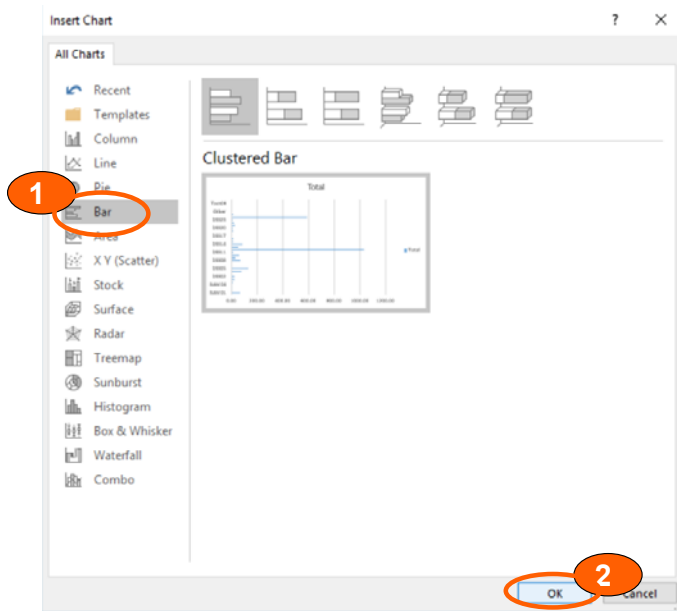
Create a pivot chart



1. Click the PivotChart button
2. Select PivotChart

teradata.

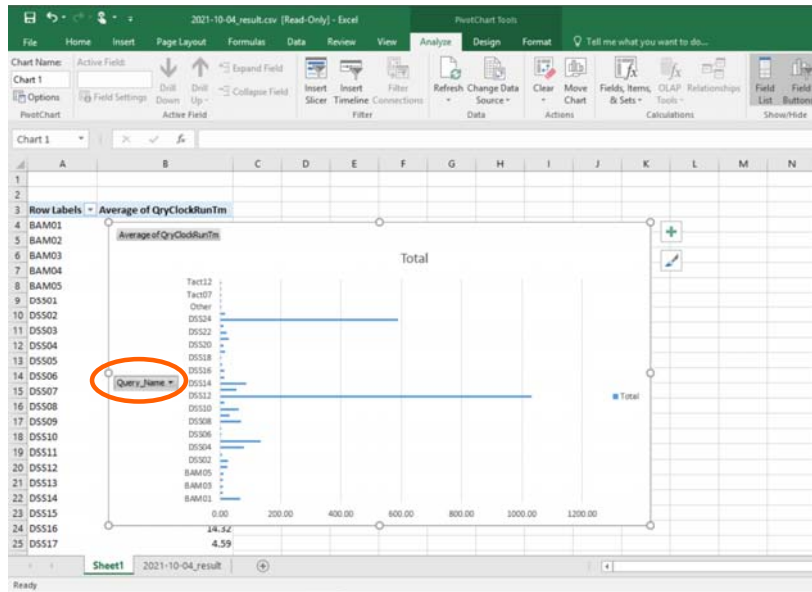
Select a style for your pivot chart



1. Click on one of the Chart options. In this example, we selected Bar
2. Click OK

teradata.

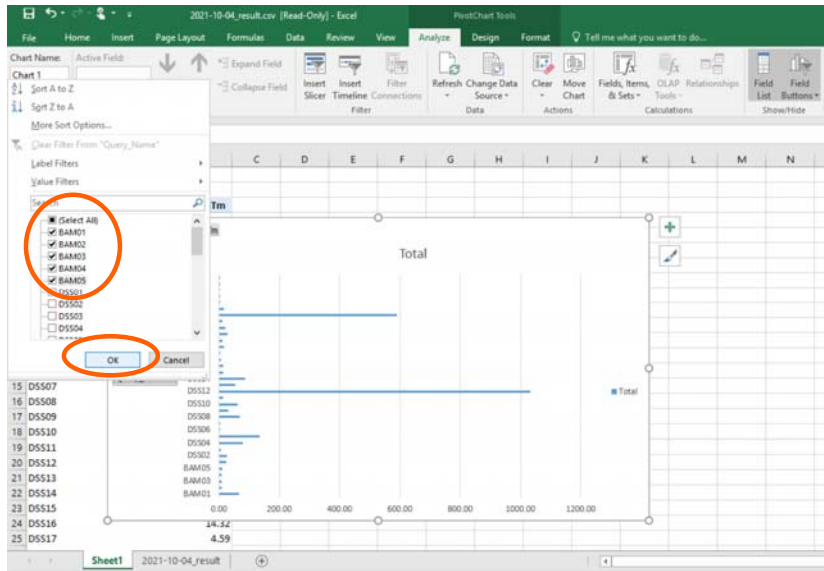
Filter the queries displayed in the pivot chart



Click Query_Name button to filter the queries displayed in the chart

teradata.

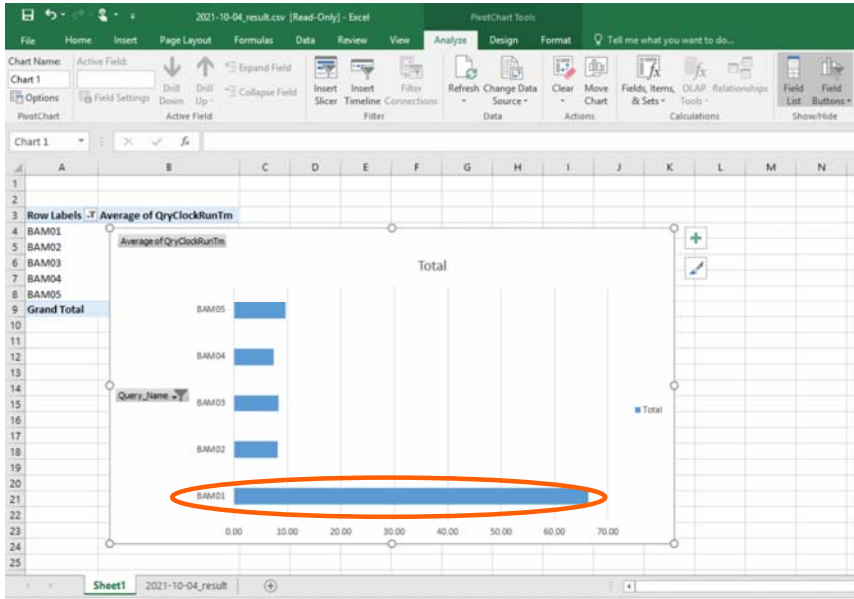
Filter the queries displayed in the pivot chart (cont.)



Click Query_Name button to filter the queries displayed in the chart

teradata.

Identify the outlier query (or queries)



Make note of the query (or queries) that are outliers and are candidates for splitting into another Workload. In this example we have noted that BAM01 is an outlier which we can split into its own Workload.

teradata.

Current topic: Refining the workloads

- Capturing workload performance metrics
- Creating a pivot table
- Using the pivot table to analyze workload performance metrics
- **Refining the workloads**
 - **Workload refinement**
 - **Creating a new workload**
 - **Adjusting the workload evaluation order**
- Lab activity

teradata.



Workload refinement

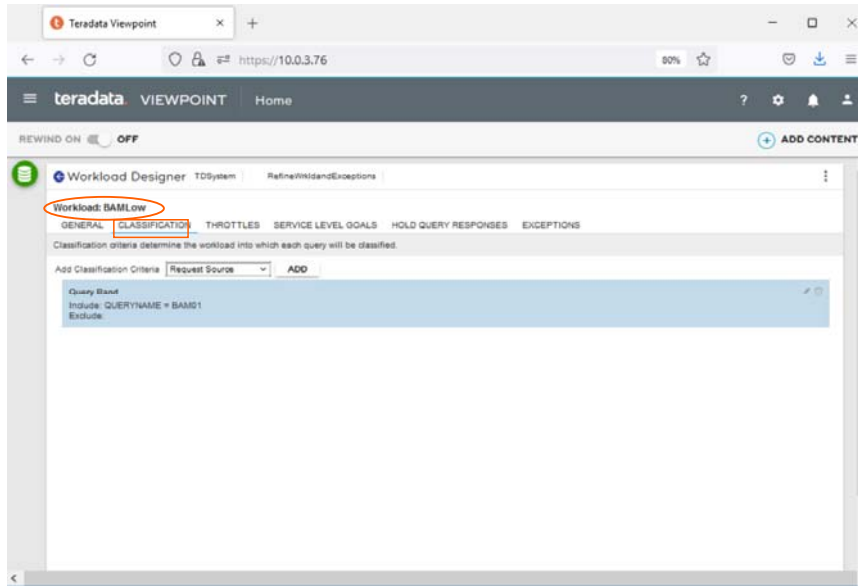
If workloads are missing the defined Service Level Goals, refinements to the Workload Definitions may be necessary.

- Refine the workloads defining characteristics:
 - Adjust the Classification Criteria or Exception Criteria
 - Add additional Classification Criteria or Exception Criteria and Actions
 - Split the Workload into multiple Workloads
- Add Concurrency Limits:
 - Reduce the resource consumption and impact on other Workloads
- Adjust Workload Mapping and Priority Distribution
 - To be discussed in Module 16, “Workload Designer – Mapping and Priority”

teradata.

The facing page lists the refinements you can make if SLGs are not being met.

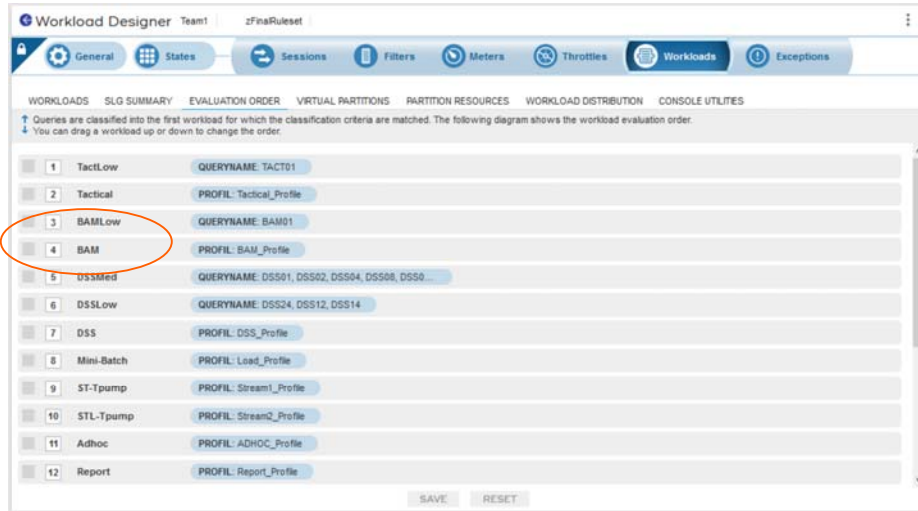
Create a new workload



Create a new Workload for the outlier query (or queries). In this example we have created a Workload called BAMLow. And we are using Query Band as the request source, QUERYNAME=BAM01.

teradata.

Adjust the workload evaluation order



Place the **BAMLow** Workload above the **BAM** Workload in the Evaluation Order. Remember that the more-specific workload definition needs to be above the less-specific workload definition.

teradata.

Workloads: Refinement

Missing Service Level Goals may require refinement of Workload rules

- Refine Workload defining characteristics
 - Adjust classification and/or exception criteria working values
 - Add additional exception conditions and actions
 - CPU Disk Ratio
 - Skew
 - Sum CPU, I/O and Spool thresholds
 - CPU per node
 - Split a problem workload into multiple workloads
- Add Throttles to control concurrency
 - Lessen impact on other workloads

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Missing Service Level Goals may require refinement of Workload rules Refine Workload defining characteristics

- Adjust classification and/or exception criteria working values
- Add additional exception conditions and actions
 - CPU Disk Ratio
 - Skew
 - Sum CPU, I/O and Spool thresholds
 - CPU per node
- Split a problem workload into multiple workloads

Add Throttles to control concurrency

- Lessen impact on other workloads

Current topic: Lab activity

- **Capturing workload performance metrics**
- Creating a pivot table
- Using the pivot table to analyze workload performance metrics
- Refining the workloads
- **Lab activity**
 - (refer to day 4: Lab guide)
 - **Refine workload definitions**
 - **Define exception rules**
 - **Create workload throttle rules**
 - **Save and activate the ruleset**
 - **Run the Mixed-Workload simulation and capture performance metrics**

teradata.



Summary

In this module we covered how to:

- Analyze Workload Performance Metrics using DBQLogTbl
- Export DBQL data to a CSV file using Teradata Studio
- Import CSV file into MS Excel
- Create a Pivot Table
- Use the Pivot Table to identify “misbehaving” queries within a workload
- Discuss the reasons for splitting a Workload

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Module 14: Workload Designer: Mapping workloads to priority levels

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Discuss how the Linux SLES 11 Completely Fair Scheduler works
- Describe Teradata's Priority Scheduler for SLES 11 interacts and leverages the capabilities of the Completely Fair Scheduler
- Identify the different Workload Management Methods
- Use Workload Designer to assign Workloads to a Workload Management Method

teradata.



Topics

- The completely fair scheduler
- Priority scheduler overview
- Tactical workload management method
- SLG tier workload management method
- Timeshare workload management method
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- Lab activity

teradata.



Current topic: The completely fair scheduler

- **The completely fair scheduler**
 - **Linux scheduler**
 - **Control groups**
 - **Resource shares**
 - **Virtual runtime**
- Priority scheduler overview
- Tactical workload management method
- SLG tier workload management method
- Timeshare workload management method
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- Lab activity

teradata.



Linux SLES 11 scheduler

- All operating systems have a built-in “scheduler” that is used to determine which tasks to run on which CPU
- Teradata’s Priority Scheduler is built on top of the Linux SLES 11 scheduler to manage tasks and other activities supporting database work
- SLES 11 offers an entirely new OS scheduler from SLES 10
- Linux refers to this new scheduler as “The Completely Fair Scheduler”
 - The concept of “time slices”, CPU run queues and Relative Weights are gone
 - The new scheduler operates on a higher degree of fairness, accuracy and balance when multiple requests are running on the system
 - It implements priorities using a hierarchy where the position in the hierarchy will influence the share of resources a task receives
 - It can group similar tasks and provide CPU at the group level instead of exclusively at the task level
 - Grouping allows resources to be shared first at the group level and then within the group at the task level
 - The Control Group in Linux equates to a Workload in the Analytics Database

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All operating systems come with a built-in “scheduler” that is responsible for deciding which tasks run on which CPU and when. The Teradata Database has always built on top of the operating system scheduler its own “Priority Scheduler” to manage tasks and other activities that run in or support the database work. Having a database-specific Priority Scheduler has been a powerful advantage for Teradata users, because it has allowed different types of work with varying business value and urgency to be managed differently.

With SLES 11, Linux offers an entirely new operating system scheduler. The Teradata’s Priority Scheduler is built on top of this new Linux scheduler

The SLES 11 scheduler has no concept of “time slices” in the way the SLES 10 scheduler has. The CPU run queues are gone, as are the relative weights. This new operating system is focused on delivering fairness with a high degree of accuracy and providing balance when multiple requests are running on the platform. Linux refers to its new scheduler as the Completely Fair Scheduler. Just as with the SLES 10 Linux scheduler, this new scheduler operates first and foremost on individual tasks. Also, like the earlier facility, it runs independently on each node in an MPP configuration.

One key characteristic of the Linux Completely Fair Scheduler is that it implements priorities using a hierarchy. Think of it as a tree structure. The level that a task is positioned in this tree will influence the share of resources that that the task receives at runtime.

There is another key characteristic of the Completely Fair Scheduler that is particularly important to the Analytics Database: The new scheduler can group tasks at the operating system level that have something in common.

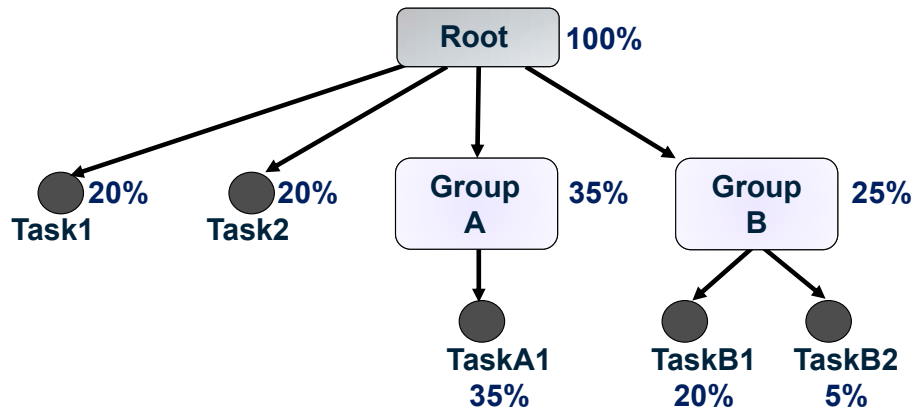
(Refer to last two bullets in slide)

Linux has recognized that there may be advantages to grouping similar tasks first, and then providing the CPU at the group level instead of exclusively at the task level, so it provides a new capability to do this bundling. In the Analytics Database, this grouping capability is able to be readily used to represent all the tasks within one request on a node, or all the tasks executing within one Workload on the node.

When task grouping is used, two levels of resource sharing will take place: First at the group level, and then within the group at the task level. Both groups and tasks can co-exist in a priority hierarchy within SLES 11.

Control groups

- **Control Groups** are used to **group tasks** that share **common characteristics**, such as belonging to the same Workload or Request
- Control Groups are placed into a **hierarchy** and can have control groups below them
- Resources flow from **top to bottom** of the hierarchy



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The way this task grouping is implemented in the SLES 11 scheduler is by means of the “Control Group” mechanism. Control Groups allow partitioning and aggregating of tasks that share common characteristics (such as belonging to the same Workload or the same request) into hierarchically-placed groups. Think of the Control Group as a family unit, with the several members of the family living in the same house, talking the same language, running up bills and expenses in common.

Control Groups can give rise to additional Control Groups below them, which may contain their own hierarchies. Each request running on the Analytics Database node will have multiple tasks (for example, one per AMP) that need to be recognized and prioritized as a unit.

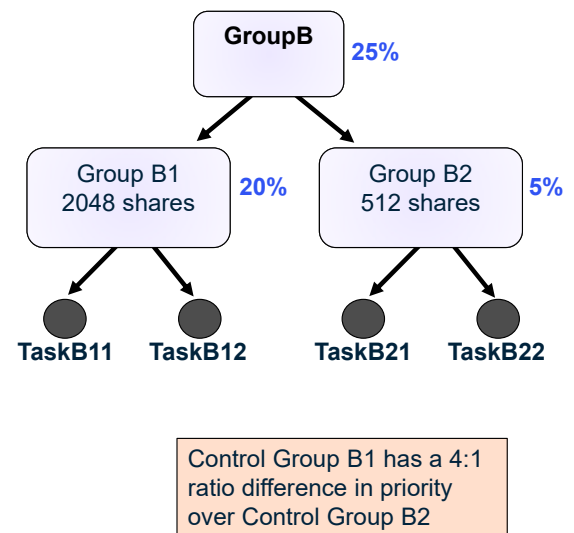
In previous versions of Linux, it was necessary to simulate this desired grouping characteristic on top of the operating system scheduler. That artificial overlay is no longer necessary with SLES 11.

Conceptually, resources flow within this Linux tree structure from the top of the Control Group hierarchy to the bottom, with resources allocated equally among the groups of tasks that fall under a Control Group. This is similar to all members of a family sharing a monthly paycheck equally among themselves. Such a Control Group tree provides a blueprint for how resources will flow through the system. The tree represents the plan for how the differently prioritized entities will share resources among them.

(Refer to slide diagram)

Resource shares

- Shares are used to determine the **portion of resources** that will be made available to a Control Group
- Shares **override the default state** of giving all children under a parent equal portions of resources
- The Linux “Completely Fair Scheduler” recognizes and supports differences in priority based on the **level in the hierarchy and number of assigned shares**
- At **runtime**, shares are **used as weight** or importance given to a group of tasks or individual task to **determine** which task will **receive CPU next**
- Teradata’s Priority Scheduler is used to manage **share assignments** for Control Groups which are represented as **Workload priorities**



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In Linux SLES 11, numbers called “shares” determine the portion of resources that will be made available to a Control Group compared to everything else under that parent at that level. If there is only one Control Group or task at a given level, it will get 100% of what flows down to that level from the level above.

Shares are simply numbers that when compared to other similar numbers reflect differences in priority of access to CPU to the operating system. When an administrator, or external program such as Teradata Priority Scheduler, applies different number of shares to different Control Groups at the same level in the tree structure, as shown in the diagram, then that will influence priority differences for the tasks within those Control Groups.

(Refer to slide text.)

Shares can be assigned to Control Groups using basic operating system commands. However, the new Priority Scheduler manages the share assignments for the Control Groups representing Analytics Database work, based on choices made by the administrator at setup time. High priority Workloads will be represented by Control Groups with a greater number of shares compared to low priority Workloads.

For example, 2048 shares were assigned to the B1 Control Group and 512 shares to the B2 group, setting up a 4:1 ratio difference in priority for the tasks running in those groups. That leads to the tasks within those two groups receiving a 4:1 ratio in runtime access to resources.

The Completely Fair Scheduler recognizes and supports differences in priority based on:

1. Level in the hierarchy
2. The number of assigned shares and their relative value compared to other groups or tasks under the same parent

At runtime shares are used to determine the weight (or importance) given to a group of tasks, or to an individual task. This weight is used in conjunction with other details to determine which task is the most deserving to receive CPU next.

Virtual runtime

- A Virtual Runtime is **calculated** for each **Control Group** and **Task** that is waiting for CPU
- The Virtual Runtime is calculated by **dividing** the number of **CPU seconds** the task has spent on the CPU already **by** the **number of resource shares** assigned
- The contrast in different tasks virtual runtimes will **influence** not only which **task will run next**, but also how long a given task will be allowed to run
- The **lower the virtual runtime** compared to the virtual runtimes of other tasks, the **higher** the proportion of **time on the CPU** will be given to the task
- A fundamental **goal** of the Linux SLES 11 “Completely Fair Scheduler” is to get **all virtual runtimes to be equal** where no single task is out of balance with what it deserves
- Teradata’s Priority Scheduler is used to determine a task’s weight based on the Workload the task is assigned, its’ level in the hierarchy and the share percent assigned to the Workload

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A virtual runtime is calculated for each task and for each Control Group that is waiting for CPU, based on the weight of a task alongside of the amount of CPU it has already been given.

(Refer to slide text.)

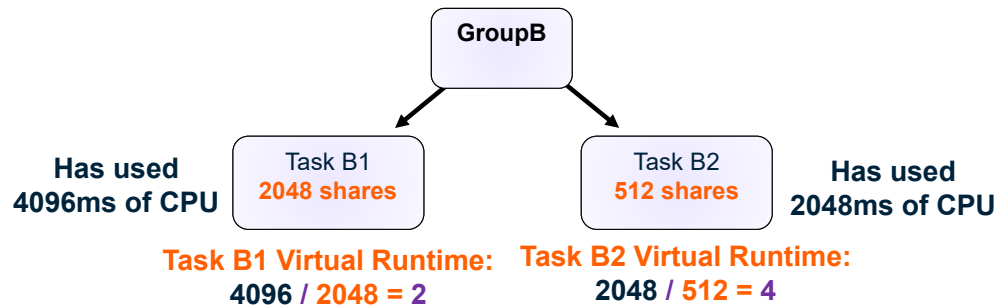
The virtual runtime is determined based on dividing the number of CPU seconds that the task has spent on the CPU already by the number of shares assigned. If this were a task originating from a Teradata Database request, the number of shares assigned to the task would depend on how its Workload priority was established by the administrator.

The contrast in different tasks’ virtual runtimes in the red-black tree will influence not only which task will run next, but how long a given task will be allowed to run, once it is given access to CPU. If its virtual runtime is very low compared to the virtual runtimes of other tasks waiting to run, it will be given proportionally more time on the CPU, in an effort to get all virtual runtimes to be equal. This is a fundamental goal of the Linux Completely Fair Scheduler.

The operating system scheduler tries to reach an ideal plateau where no single task is out of balance with what it deserves. Teradata Priority Scheduler provides input based on DBA settings that will be used to determine a task or a Control Group’s weight, based on such things as the Workload where the task is running and its level in the Control Group hierarchy, as well as the share percent the administrator has assigned the Workload.

Virtual runtime (cont.)

- Virtual Runtime accounting is at the **nano-second** (billionth) level which allows CPU time to be split up between candidate tasks as close to “ideal multi-tasking hardware” as possible
- This provides for supporting **finer priority contrasts** between tasks and **better predictability** of the Teradata Priority Scheduler
- The task with the **smallest Virtual Runtime** will **receive CPU next**, all other conditions being equal



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Each CPU tries to service the neediest task first, allowing the tasks with the lowest virtual runtime to execute before others. Virtual runtime accounting is at the nano-second level.

Determining what runs next is where the Linux Completely Fair Scheduler name most applies: The Completely Fair Scheduler always tries to split up CPU time between candidate tasks as close to “ideal multi-tasking hardware” as possible.

Current topic: Priority scheduler overview

- The completely fair scheduler
- **Priority scheduler overview**
 - **Teradata priority scheduler**
 - **Hierarchy of control groups**
 - **TDAT control group**
 - **Virtual partitions**
 - **Preemption**
 - **Remaining control group**
- Tactical workload management method
- SLG tier workload management method
- Timeshare workload management method
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- Lab activity

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Teradata SLES 11 priority scheduler

- The new Teradata Priority Scheduler **utilizes** the **Control Group** structure inherent in the **Linux SLES 11** “Completely Fair Scheduler”
- Because it is so **closely aligned** with the core features of the OS, it provides more flexibility and performance with **less overhead** than the previous SLES 10 Priority Scheduler
- **Workload** as defined in Workload Designer **becomes** the priority object visible to Linux SLES 11 “Completely Fair Scheduler” as a **Control Group**
- The new PSF is Workload based, the internal mapping of a Workload to a Performance Group is no longer done
- A smaller set of “controls” will be more direct and intuitive, and will allow TASM to **more accurately** and automatically **manage workloads** with less iteration than what was needed previously

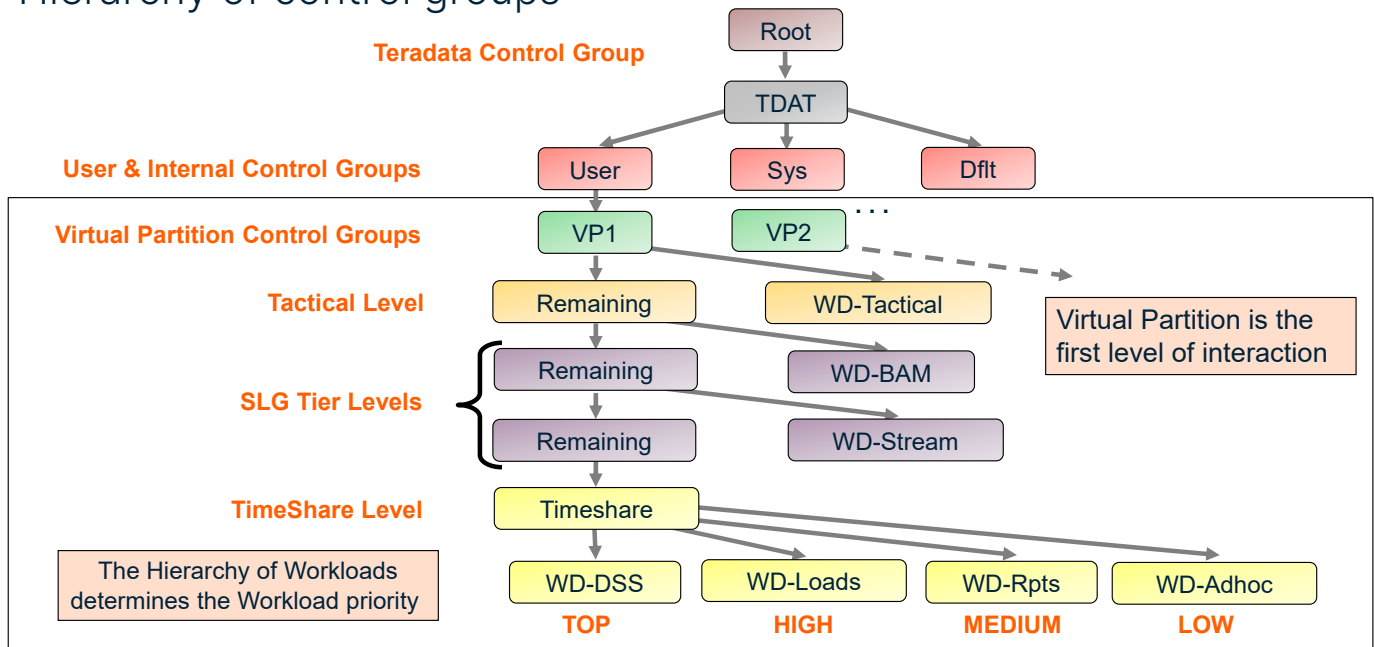
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Teradata’s SLES 11 Priority Scheduler offers a simpler and a more effective approach to managing resources compared to the previous facility. It utilizes the Control Group structure inherent in the Linux SLES 11 Completely Fair Scheduler to organize the various priority constructs. Because it is so closely aligned with the core features of the underlying operating system, the new SLG Driven Priority Scheduler provides greater flexibility and power with less overhead than what came before.

In order to understand how the operating system Control Groups have been used to advantage in the new Priority Scheduler architecture, let’s examine a few of the basic priority components and how they fit together.

First, the SLG Driven Priority Scheduler is Workload-based. While the previous Priority Scheduler linked Teradata Active System Management Workloads to Priority Scheduler Performance Groups under the covers, here the “Workload” as it is defined in Viewpoint Workload Designer becomes the priority object visible to the operating system. The translation layer is eliminated. Once the Workload is properly defined within Workload Designer, the operating system will treat the Workload as something it is intimately familiar with--just another Control Group.

Hierarchy of control groups



This page shows an example of how Priority Scheduler builds on the Control Group concept to define different priority levels.

- Workloads are defined within a consumption hierarchy that determines the relative percentage of resources that can be utilized
- The top hierarchy level that an administrator can define is a virtual partition
- Within a virtual partition workloads are created in one of three workload management methods which provide more details of the resource consumption hierarchy
 - Tactical tier
 - Service Level Goal (SLG) tiers
 - Timeshare tier

Hierarchy of control groups (cont.)

- All the Analytics Database generated tasks will be managed by the **TDAT Control Group**
- Critical **internal tasks** will execute in the **Sys (System)** and **Dflt (Default)** control groups immediately under TDAT
- **Control Group User** is the starting point for **Virtual Partitions and Workloads** that will support user database work.
- **Resources** will **flow** from the **top** of the hierarchy to the bottom
- Control Groups and their tasks at the **higher levels** will have their resource needs satisfied **before** Control Groups at **lower levels**
- **Workload Designer** will be used to **identify** what **level**, in the established hierarchical tree, a Workload will be located
- A Workload will be instantiated as a Control Group in the hierarchy

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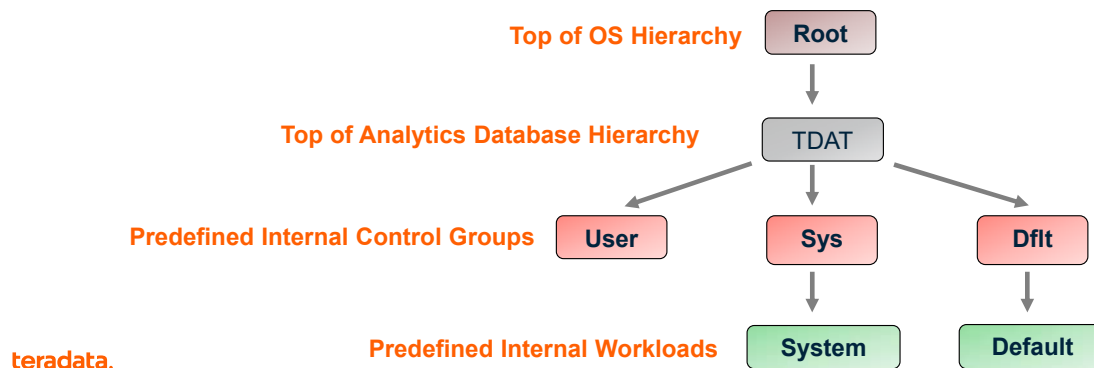
All the Analytics Database-generated tasks will be managed by Control Groups that exist under the high-level Tdat Control Group. Critical internal tasks and activities will execute in the Sys and Dflt Control Groups immediately under Tdat. They are expected to use very little resource, allowing all the remaining resources to flow to everything that is underneath the Control Group named User. The User Control Group is the starting point for the hierarchy of Virtual Partitions and Workloads that will support Analytics Database work.

Conceptually, resources flow from the top of this tree down through the lower levels. Control groups and their tasks at the higher levels will have their resource needs satisfied before Control Groups and their tasks at lower levels.

Using Workload Designer, the DBA will indicate where in this already-established tree structure each Workload will be located. More important Workloads will be assigned higher, in the Tactical and SLG Tiers, and less important Workloads will be assigned lower, in the Timeshare level. Each defined Workload will be instantiated in the hierarchy as a Control Group.

TDAT control group

- All **Analytics Database activity** will occur under the **TDAT** Control Group
- There are three predefined internal Control Groups under **TDAT**
 - User – Tasks supporting **user-initiated Analytics Database work**
 - Sys – Highly-critical **internal Analytics Database work**, similar to what used to run in the system Performance Group (Allocation Group 200) in SLES 10
 - Dflt – Critical Analytics Database tasks **not associated** to a given **user request**



All Analytics Database activity will occur under the Control Group named Tdat. Three Control Groups are defined below Tdat to differentiate user-submitted work from the internal Analytics Database work and other default work.

The three predefined Control Groups under Tdat, along with the different activities they own are:

- **User:** Tasks supporting user-initiated Analytics Database work
- **Sys:** Highly-critical internal Analytics Database work, similar to what runs in the System Performance Group (Allocation Group 200) in the SLES 10 Priority Scheduler
- **Dflt:** Critical Analytics Database tasks not associated to a given user request

Virtual partitions

- By **Default**, a single **Virtual Partition** exists named **Standard**
- **Up to 10** Virtual Partitions (VP) can be defined, but a single VP is expected to be adequate to support most priority setups
- Each VP can **contain** their own **Control Group hierarchy (Tactical, SLG Tier and Timeshare)**
- The **share percent assigned** to a VP, will **determine** how the **CPU** is **initially allocated** across multiple VPs
- If there are spare resources not able to be consumed within one VP, another VP will be able to consume more than it's assigned share percent unless **hard limits** are specified



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Virtual partitions are somewhat similar to Resource Partitions in the previous Priority Scheduler. In the Control Group hierarchy, Virtual Partitions are nodes that sit above and act as a collection point and aggregator for all or a subset of the Workloads.

A single Virtual Partition exists for user work by default, but up to 10 may be defined, if needed. Due to improvements in Priority Scheduler capabilities, a single Virtual Partition is expected to be adequate to support most priority setups. Multiple virtual partitions are intended for platforms supporting several distinct business units or geographic entities that require strict separation.

Virtual Partitions provide the ability to manage resources for groups of Workloads dedicated to specific divisions of the business. When a new Virtual Partition is defined, the administrator will be prompted to define a percentage of the Analytics Database resources that will be targeted for each, from the Viewpoint Workload Designer screens. This percent will be taken out of the percent of resources that flow down through the User Control Group.

Once defined, each of these Virtual Partitions can contain their own Control Group hierarchies. Each Virtual Partition hierarchy can include all allowable priority levels from Tactical to Timeshare.

In the initial version of the SLG Driven Priority Scheduler, there is no capability to set a hard limit on how much resource can be consumed by a Virtual Partition. That functionality is expected to be available in a future release. However, the share percent given to a Virtual Partition in this current release will determine how the CPU is initially allocated across multiple Virtual Partitions. If there are spare cycles not able to be used within one Virtual Partition, another Virtual Partition will be able to consume more than its defined percent specifies.

Users of the new Priority Scheduler only need to be concerned about the user-defined Virtual

Partitions, of which one will be provided by default. Below the surface, however, is an already-established internal Virtual Partition which is used to support internal work that is somewhat less critical than the internal work running in Sys and Dflt Control Groups.

For example, things like space accounting execute in the internal Virtual Partition, as do other activities not associated directly with user work, but that are important to get done. In addition, some internal activities that used to run in Performance Groups L and R in SLES 10 will now run in the internal Virtual Partition.

The internal Virtual Partition will be given a low share assignment, so as not to impact other user work that is executing in user-defined Virtual Partitions. Overall, it is expected to use a very light level of resources.

Preemption

- The SLES 11 OS supports preemption which is the act of temporarily interrupting a task with the intention of resuming the task later
- It is done by the **preemptive scheduler** which has the power to interrupt and later resume tasks in the system
- It allows a **Tactical** Workload task to get access to the **CPU immediately** upon entering the system if a lower priority task is using the CPU
- Any task with a **SMALLER** Virtual Runtime can take the CPU away from any other task with a **LARGER** Virtual Runtime
- There must be a **notable difference** in the Virtual Runtimes of the two tasks
- Under non-preemptive conditions a task is given an allotment of CPU and will be allowed to consume the CPU allotment and then have its Virtual Runtime updated
- **Under preemption, neither access to the CPU or length of time spent using the CPU is based on timers or a fixed time quantum**

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The SLES 11 operating system supports preemption as a natural part of decision-making functionality. Preemption is most important for tactical work, as it allows a tactical query to get access to the CPU immediately upon entering the system, if a lower priority task is using the CPU at that time.

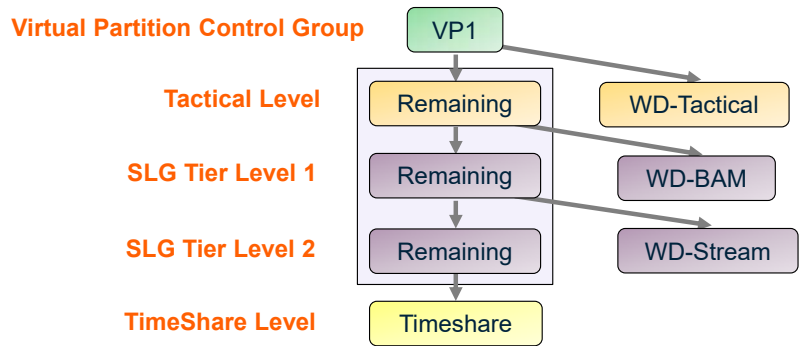
Any task with a smaller virtual runtime (which is seen as more deserving in the red-black tree) can take the CPU away from any other task with a larger virtual runtime (which is seen as less deserving). However, before preemption is allowed to take place, there has to be a notable difference between the two tasks' virtual runtimes. Otherwise, small differences between tasks could lead to constant context switching and unproductive overhead. Tactical queries are expected to be able to consistently preempt the CPU when their neediness is compared to a non-tactical task, due to their extraordinarily high operating system share assignment.

When a task needs CPU and is placed in the red-black tree, the operating system makes a check of its virtual runtime and compares it against the virtual runtime of the task that is currently using CPU. If the running task is much less deserving of the CPU, the operating system scheduler takes the CPU away immediately and gives it to the ready-to-run task that has the much lower virtual runtime (and is therefore considered more deserving).

Under non-preemptive conditions, a task that is given the CPU will be given an allotment of CPU based on the extent of his need (or how deserving he is) compared to the other tasks waiting to run. If there is no preemption while that task is running, then that task will use the amount of CPU it has been given, if it needs to, and when it's complete will have its virtual runtime updated. If the task still needs to run, it is returned to a new position in the red-black tree. Neither access to CPU or length of time spent using CPU is based on timers or a fixed time quantum, as was common in earlier operating system versions.

Remaining control group

At each level below the Virtual Partition level and before the TimeShare level, there will be an internally created Control Group labeled “Remaining”.



- The purpose of the Remaining Control Group is to be a conduit for resources intended for lower workloads and resources that cannot be used at that level
- Any resources at the Tactical Level that cannot be consumed by the Tactical Level workloads will flow down to the SLG Tier Levels
- Any resources not consumed at each SLG Tier Level workloads will flow down to the TimeShare level
- Typically, Remaining will end up with more than its assigned share percent

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Group labeled “Remaining”. This is an internally-created Workload whose sole purpose is to be a conduit for resources that cannot be used at that level and that will flow to the Workloads in the level below.

For example, on the Tactical level there is a Workload named WD-Tactical. A second Workload, Remaining, is automatically defined on the same level, without the administrator having to explicitly define it. All the resources that WD-Tactical Workload cannot consume will flow to the Remaining Control Group at that level. Remaining acts as a parent and passes the resources to the next level below. Without the Remaining Workload, resources below would have no way to receive resources.

By default, the automatically-created Workload called Remaining on each SLG Tier will always have a few percentage points as its Workload Share Percent. This ensures that all levels in the hierarchy will have some amount of resources available to run, even if it is small amount.

Remaining will typically end up with a larger value as its share percent than this minimum, however. When Workloads are added to an SLG Tier, the total of their share percents will be subtracted from 100%, and that is the percent that Remaining will be granted. This happens automatically without the user having to do anything. If additional workloads are added to the Tier later, their Workload Share Percent will further take away from the share percent of Remaining, until such time as the minimum share percent for Remaining is reached, which is likely to be about 5%.

Mechanisms are in place in Viewpoint Workload Designer to prevent the share percent belonging to Remaining from going below this minimum. The sum of this minimum and all of the Workload Share Percents within that Tier will never be allowed to be greater than 100%.

Current topic: Tactical workload management method

- The completely fair scheduler
- Priority scheduler overview
- **Tactical workload management method**
 - **What is the tactical workload management method?**
 - **Tactical workload exceptions**
 - **Reserving AMP work tasks**
 - **Guidelines for reserving AWTs**
- SLG tier workload management method
- Timeshare workload management method
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- Lab activity

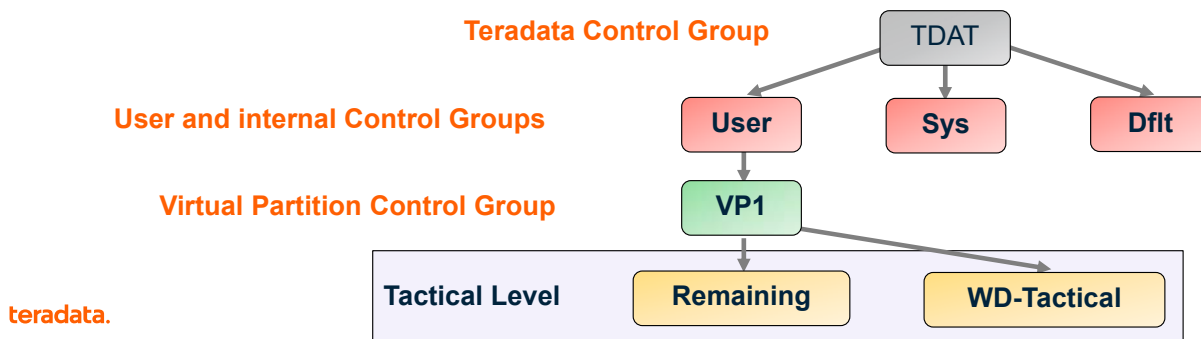
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What is the tactical workload management method?

Tactical management method is the **first level** under the Virtual Partition

- This level is intended for Workloads containing **highly tuned**, very **short** requests
- Workloads identified as tactical will always receive **highest priority** and will be allowed to consume whatever level of CPU resource required within their VP
- **Concurrency** levels **will not dilute** the priority
- Workloads on the Tactical Level will have the following benefits:
 - Automatically run with an **Expedited status** and access to pool of reserved AMP worker tasks
 - Special **internal performance advantages** including a boost in I/O priority
 - Able to more **easily preempt the CPU** from other tasks



Workloads that specify a workload management method of “tactical” will be in the first level under the Virtual Partition. Tactical is intended for Workloads that represent highly-tuned very short requests that have to finish as quickly as possible, even at the expense of other work in the system. An example of a tactical Workload is one that is composed of single-AMP or very short few-step all-AMP queries. Workloads identified by the administrator as tactical will receive the highest priority available to user work and will be allowed to consume whatever level of CPU within their Virtual Partition that they require. Tactical Workloads run with an expedited status which orders them higher in the AMP message processing queues and provides access to reserved AMP worker tasks (AWTs). All tactical workloads are defined with exception criteria to ensure queries exceeding thresholds are reclassified to an appropriate workload.

Workloads on the Tactical level will have several special benefits: Tactical Workloads are automatically run with an expedited status, which will give queries running in the Workload access to special pools of reserved AMP worker tasks if such reserves are defined and provides them with other internal performance boosts. In addition, tasks running in a tactical Workload are able to more easily preempt the CPU from other tasks running in the same or in a different Virtual Partition.

Tactical workload exceptions

Workload Designer TDT1 POWM Starting Ruleset

Workload: Tactical

GENERAL CLASSIFICATION THROTTLES SERVICE LEVEL GOALS EXCEPTIONS TACTICAL EXCEPTION

All tactical workloads have an associated tactical exception. Tactical exceptions apply across all planned environments.

Criteria

Actions are triggered when both thresholds in one of the following two criteria are met.

Tactical CPU Time

CPU (sum over all nodes): 2 CPU-Seconds

CPU per Node: 2 CPU-Seconds

Tactical I/O Physical Byte

I/O (sum over all nodes): 200 MB

I/O per Node: 200 MB

Action

Change Workload: WD-Default

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Tactical workloads require exceptions and are **always active**

- Two workload exceptions are defined on **CPU and on I/O**
- Query is **demoted** to a different, **non-tactical workload** if one or both exceptions are detected
- **Demotes** queries that **exhibit non-tactical behavior**
- **Prevents** tactical workloads from **over-consuming** resources
- Default exception thresholds **may be modified** by the user
- The **workload** that is the target of the demotion **can be modified**
- **Default** CPU per node is **2 seconds** and I/O per node is **200 MB**

Workloads assigned to the Tactical level require a CPU and I/O exception rule be defined.

The default CPU and I/O exceptions can be modified but cannot be removed.

Reserving AMP worker tasks

- Messages dispatched to the AMP **must be assigned an AWT**.
- If **all AWTs are in-use**, messages will be placed into a queue sorted by the work type and priority of the work.
- To avoid queuing tactical queries, you can **reserve AWTs** to support work assigned to the **Tactical Workload Method** and **optionally SLG Tier 1**
- When reserving AWTs, **3 new work types** (work type plus 2 levels of spawned work) will be used, each with a new reserve pool.
 - **WorkType8** for dispatched work and **WorkType9** and **WorkType10** for 1st and 2nd level spawned work.
- Reserving AWTs **removes the reserve number** for WorkType8 and WorkType9 plus 2 for WorkType10 **from the unreserved pool**
 - A **reserve of 3** will **remove 8 AWTs** ($3 + 3 + 2$) **from the unreserved pool**.
 - This will diminish the number of AWTs available for non-expedited work.
- To **compensate** for the reduced AWTs available to the unreserved pool, **increasing** the total number of **AWTs for an AMP** may be an option.
- **Unless there is a shortage of AWTs, causing high priority queries to queue, there is no value to reserving AWTs.**

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Another opportunity for assisting tactical query response times that will be appropriate for some users is to set up a reserve of AMP worker tasks (AWT) specifically for that type of work. Instituting a reserve of AMP worker tasks does not impact other system resources. It does not cause CPU to be held in reserve, for example, or memory or I/O.

AMP worker tasks are execution threads/processes that do the work of executing a query step, once the step is dispatched to the AMP. A fixed number of AWTs are pre-allocated, and most systems will have 80 residing on each AMP.

Each work request coming into an AMP is assigned a work type. The work type indicates when the request should be run relative to other work requests that are waiting to execute. By default, on each AMP there are 8 different types of work requests, with 3 AWTs reserved for each type. These reserved AWTs come from the general pool, so the original 80 are reduced down to 56 unreserved AWTs ($8 * 3 = 24$; $80 - 24 = 56$). These 56 unreserved AWTs can be used for any type of work.

All work waiting for an AWT will be sorted by work type, in descending sequence. That means that MSGWORKCONTROL messages are always first in line to receive an AWT, and MSGWORKNEW work is always last in line.

A shortage of AMP worker tasks can cause tactical queries to wait. A tactical query step being dispatched at a time there are no AWTs free, will be placed in a queue. If that step waits in the queue very long, this wait will cause the query response time to get longer. Creating new reserved work types is a method to increase availability of AWTs when needed by tactical queries.

In order to accommodate one and two levels of spawned work, as is done for work that has not been expedited, three new work types are being made available, but only when a reserve

number is specified. These are:

- MSGWORKEIGHT, a step from the dispatcher for an expedited Allocation Group
- MSGWORKNINE, the first level of spawned work coming from a step running as work type MSGWORKEIGHT
- MSGWORKTEN, the second level of spawned work

Reserving AMP worker tasks (cont.)



Reserving 3 AWTs for expedited work reduces the pool of unreserved AWTs

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The example on the facing page reserves 2 AWT for expedited work. Creating new reserved work types, and selectively expediting allocation groups, is a method to increase availability of AWTs when needed by tactical queries.

In order to accommodate one and two levels of spawned work, as is done for new work that has not been expedited, three new work types are coming into existence. These can be identified as:

- MSGWORKEIGHT, a step from the dispatcher for an expedited allocation group
- MSGWORKNINE, the first level of spawned work coming from a step running as work type MSGWORKEIGHT
- MSGWORKTEN, the second level of spawned work

Whatever number you select for the reserve during Priority Scheduler setup, this number will be tripled internally, and applied individually to the 3 new work types. If your reserve specifies 2, for example, a total of 6 AWTs will be reserved, 2 each for the 3 new work types. As the total number of 80 AWTs is not being increased, your reserve of 2 will cause the general pool of AWTs to be reduced by 6, taking it from 56 to 50.

Guidelines for reserving AWTs

The screenshot shows the 'Workload Designer' interface for a rule set named 'z3FnaRuleset'. The 'LIMITS/RESERVES' tab is selected. Under the 'Limits' section, 'CPU Limit' and 'IO Limit' are both set to '100 % of System'. In the 'Tactical Reserves' section, the 'Reserved AWTs' checkbox is checked and its value is set to '0'. The 'Max AWTs for new tactical work' checkbox is also checked and its value is set to '50'. The 'Planned Environment' is set to 'Always'.

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- Consider Workload Designer **Throttles as an alternative** to reserving AWTs
- Do not reserve AWTs unless you have **identified** that a **shortage is impacting tactical** query performance
- Don't select the reserve number **based on peak processing**, but on **standard usage**.
- Keep the reserve number as **low as possible**
- Set the **limit parameter** for the new reserve pools at **50**, the **same** limit for **MSGWORKNEW**
- Once a set of reserve AWT pools have been established, it becomes **more critical** to **monitor AWT usage**, to ensure that tradeoffs have been appropriately determined

There are several important recommendations related to using this feature. Most importantly, do not set a reserve number to a value greater than zero unless you have identified that a shortage of AMP worker tasks is impacting tactical query performance.

- Because the general pool of unreserved AWTs are available for use by expedited allocation groups, don't select the reserve number based on peak processing, but rather on standard, usual usage
- Whatever number is selected for the number of AWTs to be reserved, that number will be increased by 4. It is advisable to keep the reserve number as low as possible. The number 1 is a good starting point if the work is strictly single-AMP reads. One reserved AWT can service hundreds of single or few AMP queries in a second on most systems. For all-AMP high priority work or single row updates, start with a reserve of 2
- Consider setting the limit parameter for the new reserve pools at 50, which is the same limit used by MSGWORKNEW

Once a set of reserve AWT pools have been established, it becomes more critical to monitor AWT usage, to ensure that tradeoffs have been appropriately determined.

Current topic: SLG tier workload management method

- The completely fair scheduler
- Priority scheduler overview
- Tactical workload management method
- **SLG tier workload management method**
 - **What is the SLG tier workload management method?**
 - **SLG workload share percent**
 - **SLG tier target share percent**
- Timeshare workload management method
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- Lab activity

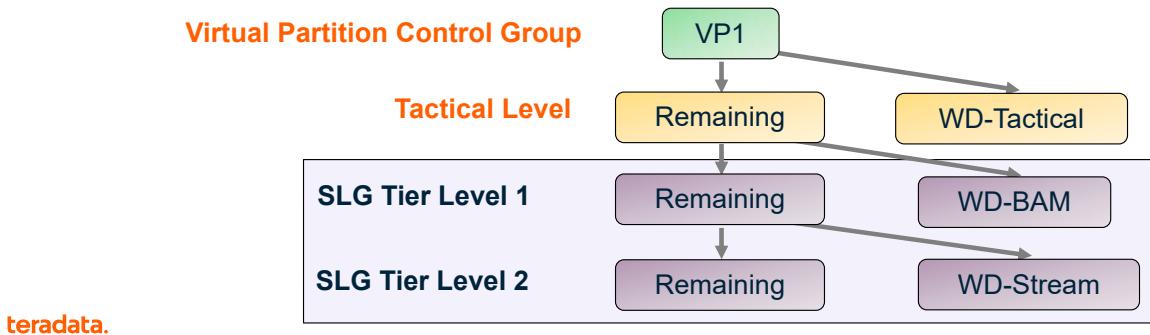
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SLG tier workload management method

SLG Tier workload management method is intended for workloads with **short** Service Level Goals (SLGs)

- **Response time** is **critical** to the business
- More **complex tactical** queries
- The SLG Tier level may consist of up to **5 tiers**
- The **higher** tiers will have their Workload serviced **before lower** tiers
- SLG **Tier 1** workloads may optionally be **expedited**



There may be one or as many as five levels in the hierarchy between the Tactical level and Timeshare level. Resources not consumed by workloads assigned to the Tactical tier flow down to the first SLG tier. These “SLG Tier” levels are intended for Workloads associated with a service level goal, or other non-tactical work whose response time is critical to the business. It may be that only a single SLG Tier will be adequate to satisfy this non-tactical, yet time-dependent work.

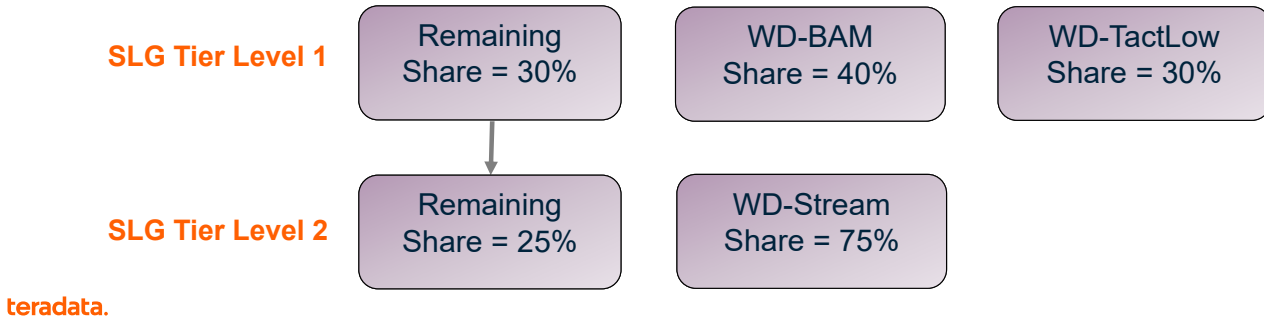
If more than one SLG Tier is assigned Workloads, the higher Tiers will have their Workload requirements met before the lower Tiers. Workloads in Tier 1 will always be serviced ahead of Workloads in Tier 2; Tier 2 will be serviced before Tier 3, and so forth.

Each Tier will automatically be provided with a Remaining Workload to act as a conduit for resources that flow into the Tier but that either is not used or have been set aside for the Tiers below. This Workload is referred to as “Remaining” because it represents the resources remaining after Workloads on that Tier have been provided with their defined percent of Tier resources.

SLG workload share percent

The SLG Workload Shares are specified as a Percentages using Workload Designer

- It will be internally converted into operating system shares that will be assigned to Control Groups and tasks
- The Remaining Control Group will automatically be given a Workload Share Percent and will be equal to the sum of the Workload Share Percent's assigned on that tier minus 100
- The share percent is the percent of resources to be allocated from the resource percent that flows down from the higher-level tiers



Each SLG Tier can support multiple Workloads. When the DBA assigns a workload to a particular SLG Tier, he will be prompted to specify a Workload Share Percent. This represents the percent of the resources that the administrator would like to target to that particular Workload *from within the resources that are made available to that Tier*. In other words, the share percent is a percent of that Tier's available resources, *not* a percent of system resources, and *not* a percent of Virtual Partition resources.

Concurrency within a Workload *will* make a difference to what each task is given. All the requests active within a given Workload will share equally in the Workload Share Percent that Workload is assigned.

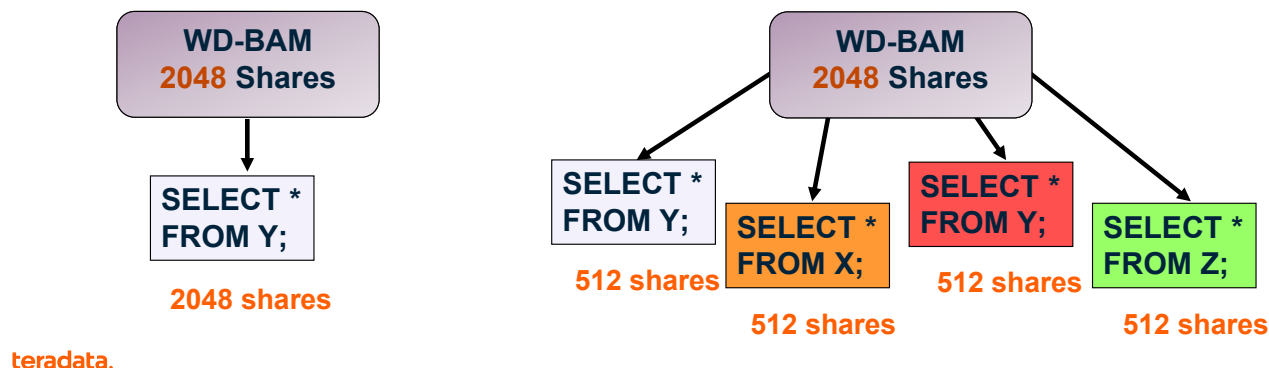
The Workload Share Percent is not an upper limit. If more resources are available after satisfying the share percents of other Workloads on the same Tier and the Tiers below, then a Workload may be offered more resources. Under some conditions, a Workload Throttle may be appropriate for maintaining consistent resource levels for requests within high concurrency Workloads

This Workload Share Percent that the administrator will assign to an SLG Tier Workload is different from the Linux operating system shares. The operating system shares are the actual mechanism to enforce the desired priority the administrator expresses when he selects a Workload Share Percent. For SLG Tiers, the shares that the operating system assigns to tasks and Control Groups, which are invisible to the user, are derived from the Workload Share Percent that the administrator sets and tunes.

Note that Remaining will automatically be given a Workload Share Percent behind the scenes. Remaining's percent will be equal to the sum of the standard Workload Share Percents on that Tier subtracted from 100%. Remaining's percent represents the share of the resources flowing into that Tier that will be always be directed to the Tiers and levels below.

Workload share percent (cont.)

- Mechanisms are in place in Workload Designer to ensure **Remaining will > 0%**
- Higher SLG Tiers will get **higher** priority and **greater access** to resources than **lower** SLG Tiers
- OS shares given to a Workload will be **divided up** among the active tasks within that Workload on that node
- **More active tasks** executing **concurrently** in an SLG Tier Workload, the **fewer shares** an active task will receive



In the case of Workloads placed on an SLG Tier, both the Tier number and the Workload Share Percent will factor into the number of shares that tasks running within that Workload will receive. Higher Tiers mean a larger number of operating system shares at runtime, which means greater access to resources.

The operating system shares given to the Workload will be divided up among the active tasks within the workload on that node. The more requests running concurrently in an SLG Tier Workload, the fewer shares each will receive. If only a single request is active within a given SLG Tier Workload, then that single request will experience a somewhat higher priority because its tasks will be given the entire share that Workload is entitled to.

The CPU seconds already used to get work done by the task will be divided by the combination of the Tier and its operating system shares. The larger the divisor in the virtual runtime calculation, the smaller the result from the virtual runtime formula, and the more needy the task will appear to the operating system. Tasks coming from a high Tier with a high Workload Share Percent will appear the neediest and will tend to run ahead of tasks from lower Tiers, even if they have just as high a Workload Share Percent. However, since CPU time that has already been used by such tasks is in the numerator, as more resources are consumed, the virtual runtimes increase, until at some point those higher priority tasks are no longer seen as the neediest.

SLG tier target share percent

A SLG Tier Workload is **likely to consume** a percentage of resources that is quite **different** than the **targeted percentage**

Contributing **factors** include:

- Other active **Virtual Partitions** have **unused resources**
- Workloads on **higher tiers** don't use all the resources they were allocated
- One or more other **Workloads** on the same tier are **inactive**
- The **Workload** itself **cannot consume all** the resources it was **allocated**
- Workloads in the **Timeshare** level cannot consume all unused resources that flow into that level
- Workloads in **Tactical** consume more or less than expected

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When a Workload on an SLG Tier has no active tasks, its definition and defined Workload Share Percent remain intact, but the Control Group that represents the inactive Workload will temporarily be excluded from the internal calculations of operating system shares.

When a Workload is inactive on a Tier, two things will happen internally:

1. All the Workload Share Percents for Workloads that *do* have active tasks, including Remaining, are summed. Using the figure above, this runtime calculation would look like this:

$$40\% + 25\% + 5\% = 70\%$$

2. Then using this new base, new runtime share percents are calculated for each Workload, including Remaining:

$$\begin{aligned}\text{WD-BAMHigh's new share} &= 40/70 = 57\% \\ \text{WD-DSSHigh's new share} &= 25/70 = 36\% \\ \text{Remaining's new share} &= 5/70 = 7\%\end{aligned}$$

It is important to note that Remaining, which is the conduit for resources flowing to the lower levels, will experience an increase in its runtime allocation as well. When one or more Workloads are inactive on a given Tier, a larger percent of resources will be made available for the Tiers below, if Workloads below are able to use them.

Current topic: Timeshare workload management method

- The completely fair scheduler
- Priority scheduler overview
- Tactical workload management method
- SLG tier workload management method
- **Timeshare workload management method**
 - **What is the SLG tier workload management method?**
 - **Timeshare access rates**
 - **Automatic decay option**
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- Lab activity

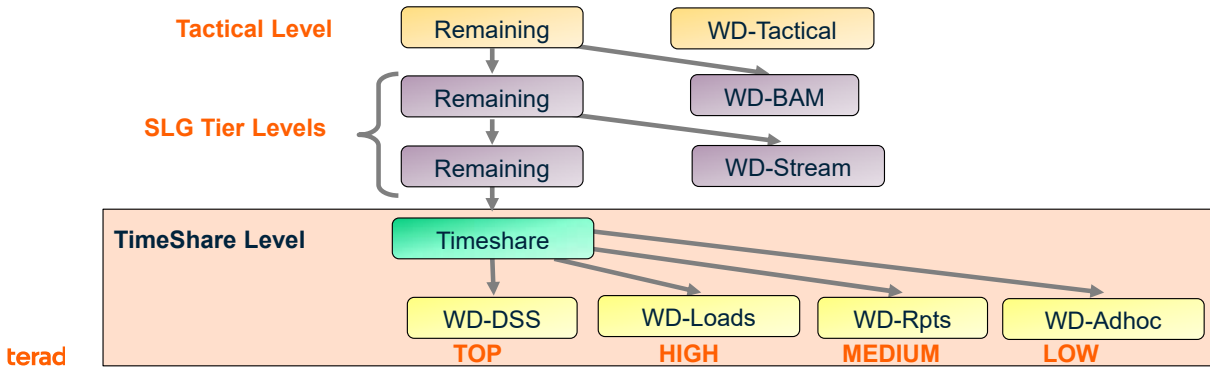
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What is the timeshare workload management method?

Timeshare workload management method is intended for **lower priority non critical** workloads and do not have SLG expectations

- Resources not consumed by the Tactical Level or the SLG Tier levels will **flow down** to the Timeshare level
- It is expected that the **majority of resources** will be consumed by Workloads running in Timeshare
- Timeshare is at the bottom of the hierarchy and has **no remaining** workload



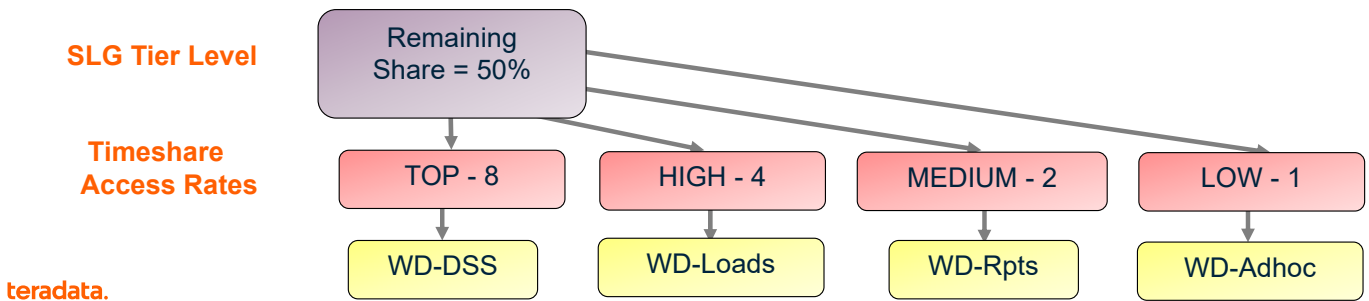
It is expected that the majority of the resources will be consumed by Workloads running in Timeshare. Timeshare is intended for Workloads whose response time is less critical to the business and that do not have a service level expectation, and for background work, sandbox applications, and other generally lower priority work. Resources not able to be used by Tactical or the SLG Tiers, or resources that remain unused due to the presence of the Remaining Workloads above, will flow down to the Timeshare Workloads. This can be a considerable amount of resource, or it can be a slight amount.

- The Timeshare tier workloads are at the bottom of the workload priority hierarchy
- The Timeshare tier has four priority levels to which workloads are assigned
- Each priority level is assigned a relative resource rate:
- This is interpreted as Top requests get eight times the resources of Low requests
- High requests get four times the resources of Low requests
- Medium requests get twice the resources of Low requests
- If not using SLG tiers, place workload in Top if they are very high priority but do not qualify as tactical

Timeshare access rates

Timeshare workload management method comes with 4 fixed Access Rates representing 4 different priorities: **Top, High, Medium and Low**

- A Workload must be associated with one of the 4 Access Levels
- Each of the 4 Access Levels has a **fixed access rate** that cannot be changed:
 - Each request in **Top** will always get **8 times** the resource of a Low Request
 - Each request in **High** will always get **4 times** the resource of a Low Request
 - Each request in **Medium** will always get **2 times** the resource of a Low Request
- The concurrency of active requests will not reduce the priority differentiation between the different Access Levels



The Timeshare Workload Method comes with 4 Access Rates, representing 4 different priorities: Top, High, Medium, and Low. The DBA must associate a Workload to one of those 4 Access Levels when he assigns a Workload to Timeshare. Workloads in the Low Access Level receive the least amount of resources among the Timeshare Workloads, while Workloads in the Top Access Level receive the most.

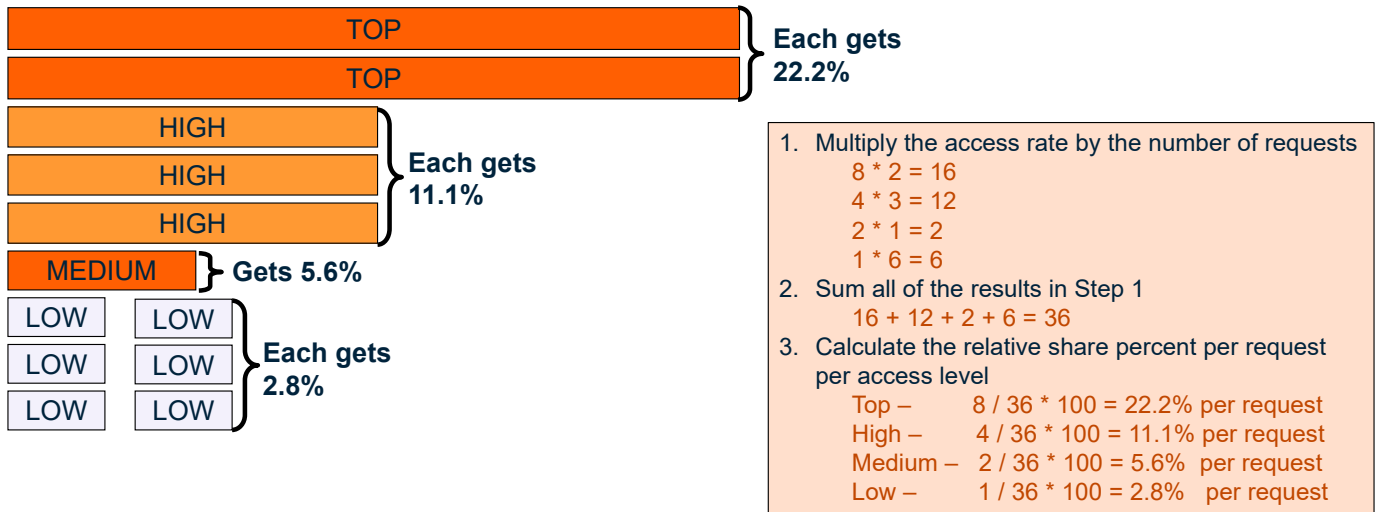
Each of the 4 Access Levels has a different Access Rate:

- Each request in Top gets 8 times the resource as a Low request
- Each request in High gets 4 times the resource as a Low request
- Each request in Medium gets 2 times the resource as a Low request
- Each low request gets a minimum base share, based on what is available from the Tier above

The actual resource distribution will depend on which Access Levels are supporting work at any point in time. However, at each Access Level, the concurrency of active requests will not reduce the priority differentiation between the levels. For example, a query running in Top and a query running in Low will always receive resources in an 8-to-1 ratio whether they are running alone or concurrently with 10 other queries within their Access Level.

There will always be some small percent of resources flowing into Timeshare from the Tiers above. For example, in the starting with the Teradata 14.0 release, the SLG Tiers will only support Workload Share Percents that sum up to a number close to 95% on a given Tier, always allowing some small level of resources to be always available to the Tiers below.

Timeshare access rates concurrency



**Concurrency has no impact on the priority differences between
Top, High, Medium and Low access levels –**
 $(22.2\% * 2) + (11.1\% * 3) + (5.6\% * 1) + (2.8\% * 6) = 100\%$ of Timeshare

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Here are the steps to take to determine the percent of Timeshare resources that each request will receive.

Step 1. First, multiply the Access Rate by the number of requests that are active at each Access Level:

2 request in Top $8 * 2 = 16$
 3 requests in High $4 * 3 = 12$
 1 request in Medium $2 * 1 = 2$
 6 requests in Low $1 * 6 = 6$

Step 2. Then sum all the results from Step 1 (Access Rates x number of requests) for all Access Levels:

$16 + 12 + 2 + 6 = 36$

Step 3. Finally, calculate the relative share percent per request per Access Level, using the sum in Step 2 in the denominator:

Top $8 / 36 * 100 = 22.2\%$ for each of 2 requests
 High $4 / 36 * 100 = 11.1\%$ for each of 3 requests
 Medium $2 / 36 * 100 = 5.6\%$ for one request
 Low $1 / 36 * 100 = 2.8\%$ for each of 6 requests

Notice that each High requests will be allocated $\frac{1}{2}$ of what is allocated to each Top request, and that the sole Medium request receives $\frac{1}{2}$ of what each High request gets, and so forth. No matter what the concurrency within each Access Level, the contrast in what is allocated will maintain the same ratio, based on the Access Level each request runs in.

Timeshare Access Rates are set at 1, 2, 4, and 8.

Automatic decay option

The decay option is intended to **give priority to shorter** requests over longer requests mixed in the same workload

- Timeshare workloads will have an automatic decay option, **off by default**
- If decay option is selected, it will **automatically reduce the access rate** of a running request **based on** either a specified **CPU or I/O** threshold
- Initially, the request is reduced down to an **access rate of half** the original access rate
- If a **second threshold is reached**, the request will be reduced to an access rate a **quarter the original** access rate
- Decay thresholds applies to **all requests running in Timeshare**, it can not be applied to a specific workload or specific access rate
- Workload **classification** based on **estimated** processing time may be **effective without relying** on decay option
- Classify **short** running queries to workloads assigned to **higher** access levels and **long** running queries to workloads in **lower** access levels

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An option is available that will automatically apply a decay mechanism to Timeshare Workloads. This decay option is intended to give priority to shorter requests over longer requests. Only requests running in Timeshare will be impacted by this option. Decay is off by default.

If this option is turned on, the decay mechanism will automatically reduce the Access Rate of a running request, if the request uses a specified threshold of either CPU or I/O. Initially, the request is reduced down to an Access Level that is $\frac{1}{2}$ the original Access Level. If a second threshold is reached, the request will be further reduced to an Access Level that is $\frac{1}{4}$ the original Access Level. This process of Access Rate reduction includes the Low Access Level and means that the Access Rate could be as low as 0.25 (Low typically has an Access Rate of 1) for some requests running in Low.

Decay may be a consideration in cases where there are very short requests mixed into very long requests in a single Workload, and there is a desire to reduce the priority of the long-running queries. Keep in mind, however, that if decay is on, all queries in all Workloads across all Access Levels in Timeshare will be candidates for being decayed if the decay thresholds are met.

Workload classification based on estimated processing time may be effective without relying on the decay option for ensuring that queries expected to be short-running run at a higher Access Level, and queries that are expected to be long-running classify to a Workload in a lower Access.

Decay is not an option than can be applied Workload by Workload or Access Level by Access Level. If most, or all queries experience the automatic decay, then all or most of the active requests will be in the same relationship to each other as they were prior to the decay being applied, in terms of their relative share of resources.

Automatic decay characteristics

- A single request will only undergo a **maximum of two decay actions**
- Decay thresholds are fixed:
 - First decay is by **half** (0.5) after **10 CPU** seconds or **100 MB** of I/O per node
 - Second decay is by a **quarter** (0.25) after **200 CPU** seconds or **10,000 MB** per node
- Decay decisions are made **at the node level**, not the system level
- There is no synchronization of the decay action **between nodes**
- Decayed requests are **not moved to another workload** only the access rate is changed
- Once a decay has taken place for a request, **both** its access to **CPU and I/O** will be **reduced**, not just the resource threshold that was exceeded
- **Workload exception thresholds that can move a request to another workload on a lower access level may be a better alternative to the decay option**

	Access Level	1 st Decay	2 nd Decay
Top	8	4	1
High	4	2	0.5
Medium	2	1	0.25
Low	1	0.5	0.125

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Characteristics of the decay process include:

- A single request will only ever undergo two decay actions, each resulting in a reduction of the request's Access Rate
- Decay decisions are made at the node level, not the system level
- There is no synchronization of the decay action between nodes, so it is possible that a Timeshare request on one node has decayed, but the same request on another node has not
- Decayed requests are not moved to a different workload, the way a workload exception might behave
- Once decay has taken place for a given request, both its access to CPU and to I/O will be reduced, not just the resource whose threshold was exceeded

The decay feature works as follows:

- A request starts with the access rate assigned to its access level
- After the request consumes 10 seconds of CPU or 100 MB of I/O resources, the request's access is decreased by half
- After the request consumes 200 seconds of CPU or 10000 MB of I/O resources, the request's access rate is again decreased by half. This access rate remains constant for the remaining duration of the request's execution
- The decay is performed on each node by Priority Scheduler so the request on different nodes can be running at different decay levels. There is no synchronization of decay levels between nodes

Tradeoffs using the Decay Option

- All the requests that run in timeshare access level workloads will be impacted, decay

cannot be targeted to specific workloads

- The thresholds that trigger decay are the same for all workloads within all access levels; Top is treated the same as Low
- If there are many requests that start in the Low access level that experience decay, they may get so few resources that they hold locks and AMP worker tasks for unreasonably long times

Current topic: Priority scheduler recommendations

- The completely fair scheduler
- Priority scheduler overview
- Tactical workload management method
- SLG tier workload management method
- Timeshare workload management method
- **Priority scheduler recommendations**
 - **Managing resources**
 - **I/O prioritization**
 - **Tactical recommendations**
 - **SLG tier recommendations**
 - **Timeshare recommendations**
- Assigning workloads to workload management methods
- Lab activity

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Managing resources

- Recommended that **Tactical Workloads** contain requests that use **very little resource** so that a majority of the resource flows down to the tiers below
- Putting **heavy resource consuming** queries at the **tactical level** will prevent resources from flowing down potentially **starving lower-level** tiers
- **SLG Tier Workloads** will **receive their share** of resources based on the share percent of the resource flows **from above**
- Workloads belonging to **Timeshare** are at the bottom of the hierarchy and are **most dependent** on resources that cannot be consumed **by the higher levels**
- Under **normal** situations, a **majority** of the resources should flow down to **Timeshare**
- However, there are **internal mechanisms** in place to **ensure** that **some** percentage of resource will always **flow down**
- **Failsafe mechanisms** in the form of automatic **tactical exceptions** can prevent tactical requests from consuming an unreasonable amount of resources

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Resources flow from the top of the Control Group hierarchy downwards, with different workload methods having different usage characteristics.

Tasks originating in Workloads on the Tactical Tier are allowed to consume as much resource as they require, within the allocation of their Virtual Partition. If tactical requests are so resource-intensive that they are able to consume almost all the platform resources, then very little resource will fall to the lower Tiers. It is recommended that Workloads only be placed on the Tactical Tier if they are light consumers of resource, as the architecture of the new Priority Scheduler is built around the concept that the majority of resources will fall through from the Tactical Tier to the tiers below.

Workloads in the SLG Tiers will be associated with a Workload Share Percent. This is different from the other two Workload methods, Tactical and Timeshare. This Workload Share Percent represents a share of the resources that are intended for this Workload among the resources that flow into that tier.

By design, Workloads on an SLG Tier will not be offered a greater level of resources than that specified by their Workload Share Percent. But this is not always the case. If Workloads in the SLG Tiers below cannot use the resources that are intended to flow to them, based on their respective share percents, *and* Timeshare cannot use all of the leftover resources it is entitled to, then Workloads on higher tiers can consume beyond their specified percents.

Essentially, Workloads that belong to the Timeshare access method will be mostly dependent on resources that cannot be used by the higher tiers. However, internal mechanisms are in place to ensure that some small percent of resources will always flow from Tactical and the SLG Tiers to Timeshare. For example, these higher level Tiers will have a remaining Workload in place that will guarantee that some few percent of the resources that flow into the Tier will flow down to the next level in the tree.

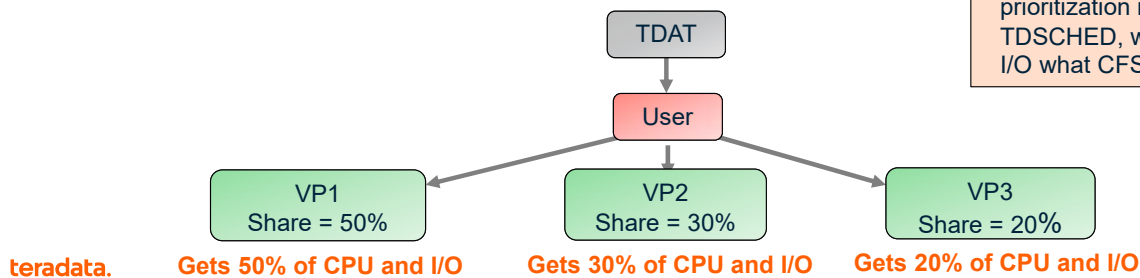
One advantage of Priority Scheduler's hierarchy-based approach to sharing resources is that under normal situations plenty of resources may flow to Timeshare. But when critical work surges at the SLG Tier level, the share percents on those Workloads can act to keep more resource at that level to ensure that adequate resources are available to the more critical work.

I/O prioritization

I/O prioritization is **automatic**

- Based on CPU prioritization parameters, **no additional** I/O prioritization setup or parameters are required
- Actual methods used to enforce I/O and CPU prioritization are independent of each other *
- I/O prioritization is **aware** of the **internal OS shares** that differentiate the Workloads and requests assigned to the Workloads
- Logical I/O's (FSG cache) **will not be charged** with having performed an I/O
- Physical I/O **is charged by the bandwidth (sectors)** not the number
- Makes **priority decisions** on **each disk drive** independently

* CPU is prioritized using the Completely Fair Scheduler (CFS) in SLES11. I/O prioritization is handled by TDSCHED, which enforces for I/O what CFS does for CPU.



With the new SLG Driven Priority Scheduler, a new I/O priority infrastructure has been architected. It is designed to recognize the Tier level and Workload Share Percent in how it treats various I/O requests. The share percents that are assigned to Virtual Partitions and to SLG Tier Workloads have a similar impact on I/O prioritization as they do on CPU prioritization, even though each type of prioritization is working independently.

I/O prioritization knows about the internal operating system shares that different Workloads, and requests under Workloads, have been assigned. It also relies on special red-black tree structures, similar to those used by the operating system's CPU scheduler, to determine which task is the most deserving of I/O at any point in time.

There is no special set up or parameters required for I/O prioritization. It happens automatically, and it responds to changes made in the basic Priority Scheduler resource allocation hierarchy, in terms of hierarchy position or share percent. Such tuning changes translate into a position in the red-black tree that controls which request for I/O will be honored next.

I/O prioritization algorithms rely on physical I/O, the I/O that is incurred when there is actual read or write to disk. If a required data block is found in the FSG cache, the task requesting it will not be charged with having performed an I/O, and may for that reason look more deserving of additional I/Os sooner. Such a task is not charged with an I/O because CPU is the only resource involved in reading a data block from cache.

Physical I/O is not measured by the number performed, but rather by using a bandwidth measurement. Disk usage is expressed in sectors transferred by the task, as the data it requires is either read or written from disk. (A sector is 512 bytes.) Disk usage is maintained for each disk independently, with I/O prioritization software making disk-level decisions about which I/O requests to honor first.

Priority Scheduler does not have insight into different hardware platforms and what their maximum I/O bandwidths are. When assessing I/O usage of a Workload in the new SLES 11 schmon (scheduler monitoring) tool, there is a column called "I/O Usg %." The percent that is displayed in that column is a percent of the total I/O kilobytes transferred, not a percent of the total *potential* I/O kilobytes transferred. In other words, if all the active Workloads at a point in time were doing very little I/O, schmon monitor output could show that a single Workload was consuming 75% of the I/O. That may not be an indication that that Workload is doing I/O-intensive work. It only means that when all the KBs of data transferred across all Workloads were summed during this collection interval, this Workload was responsible for 75% of that total. There may be plenty of spare I/Os on the platform at that time. I/O Wait metrics can be used to assess the degree of pressure on the I/O devices.

More details can be found in the Priority Scheduler for SLES 11 Orange Book here (must login): <https://docs.teradata.com/v/u/Orange-Book/Priority-Scheduler-for-Linux-SLES-11-Orange-Book>

Tactical recommendations

- Only assign workloads that support **highly tuned**, very **short requests**, such as single AMP requests to the **Tactical** level
- **Do not assign** workloads that support **load utilities** to the **Tactical** level
- Rely on **reasonable automatic exception** thresholds to **demote requests** with non-tactical characteristics
- **Monitor** tactical **exceptions** regularly and **adjust** the exception thresholds when necessary
- Don't increase the **AMP Work Tasks reserve count** above zero unless an AWT shortage is impacting tactical performance
- **If setting a reserve for AWTs**, set the reserve count for the worst case for all tactical workloads across all virtual partitions
- **Avoid** placing **tactical workloads** in virtual partitions with an **inadequate** share percent

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The following are some important recommendations to consider before assigning Workloads to the Tactical Workload Method:

- Only assign Workloads that support highly-tuned, very short queries, such as single-AMP, to Tactical
- Do not assign Workloads that support load utilities into Tactical
- Rely on reasonable exception thresholds to demote queries with non-tactical characteristics
- Monitor tactical exceptions regularly, adjust the exception thresholds, when needed
- Don't increase the AMP worker task reserve count above zero unless a shortage of AMP worker tasks is impacting the tactical performance
- If specifying reserved AWTs, set the reserved count for the worst case AMP usage by all tactical query Workloads combined, including tactical Workloads across all Virtual Partitions
- Avoid placing tactical Workloads in Virtual Partitions with an inadequate percent of resources allocated

SLG tier recommendations

- SLG Tiers are intended for **high priority** work that is associated with response time expectations
- **Use only a single level** SLG Tier if only a few workloads fall into this category
- If a large number of workloads fall into this category, place the workloads with **more critical response time** expectations on the **higher SLG Tiers**
- **For more consistent performance** on **lower** level SLG Tiers, keep Workload Share Percents low or moderate on the SLG Tiers above
- Linux **SLES 11 scheduler** supports **higher granularity** in enforcing priorities
- Differences in OS shares that are **fractions of a percent** can be **effective** in managing performance, **large contrasts** in percentages is **not necessary**
- Having a **larger number of workloads** on the SLG Tiers is **not** a cause for **concern**

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Some of the important recommendations for managing workloads on the SLG Tiers include:

- SLG Tiers are intended for high priority work that is associated with response time expectations
- Use only a single SLG Tier if only a few workloads fall into this category
- If a large number of Workloads with widely-varying priorities fall into this category, place the Workloads with the more critical service level expectations on the higher SLG Tiers
- If more than one SLG Tier supports Workloads, attempt to define smaller Workload Share Percents on higher Tiers, to allow a more predictable level of resources to be available for the lower SLG Tiers

The Linux SLES 11 operating system allows Priority Scheduler to enforce priorities in a highly granular way. Differences in operating system shares that are fractions of a percent apart can be effectively managed. Do not be concerned if you have a large number of Workloads that fall into the SLG Tier Workload Method. Do not feel that you have to have large contrast among them to get priority differences.

Timeshare recommendations

- All workloads are appropriate to consider for Timeshare with the exception of tactical workloads
- If SLG Tiers are not used, place high priority workloads in the Top access level
- Reasonably effective priority differences can be achieved using Top, High, Medium and Low access levels
- Concurrency will not dilute the requests priority
- If a penalty box workload is being used, put it on the Low access level
- For more predictable priority differentiation in Timeshare, keep the automatic decay option turned off
- Use workload exception thresholds to demote requests to workloads on lower access levels
- For higher consistency within Timeshare, make sure that remaining in the SLG tier above allows adequate resources to flow into Timeshare

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Below are some of the important considerations for Workloads in Timeshare:

- If SLG Tiers are not being used, place Workloads in the Top Access Level that are very high priority, but do not qualify to be defined as tactical
- If a penalty box Workload exists when migrating to the new Priority Scheduler, add that Workload into the Timeshare Low Access Level
- For more predictable priority differentiation in Timeshare, keep the decay option turned off

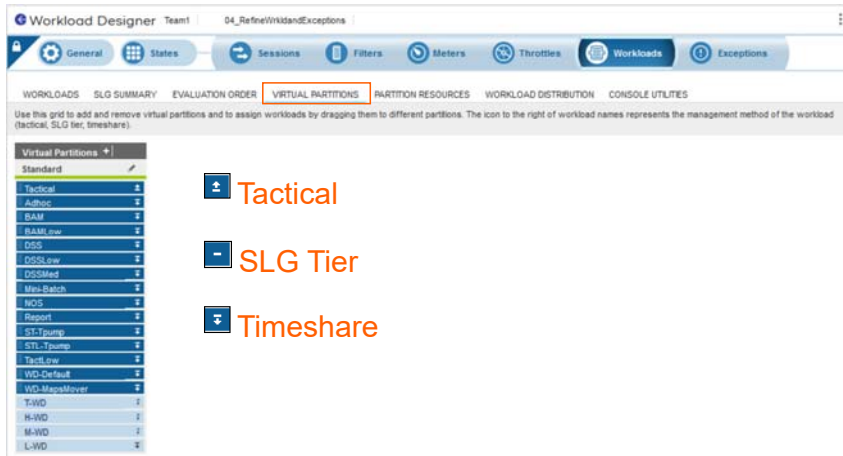
Current topic: Assigning workloads to workload management methods

- The completely fair scheduler
- Priority scheduler overview
- Tactical workload management method
- SLG tier workload management method
- Timeshare workload management method
- Priority scheduler recommendations
- **Assigning workloads to workload management methods**
 - **Adding virtual partitions**
 - **Partition resources**
 - **Workload distribution**
 - **System workload report**
 - **Penalty box workload**
- Lab activity

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Virtual partitions



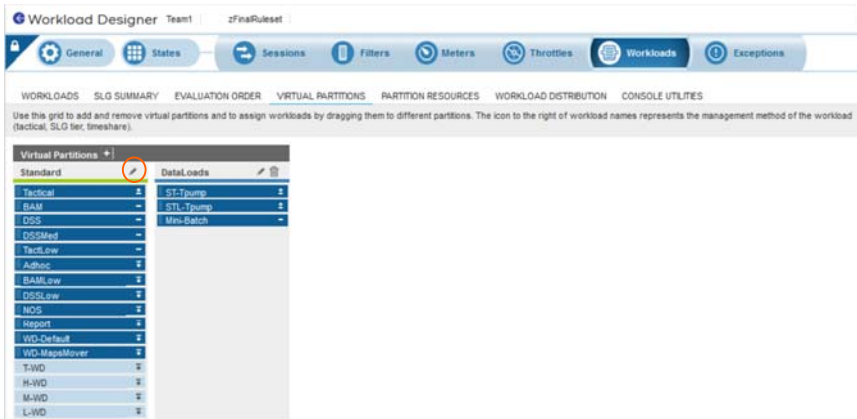
By default, all workloads are assigned to the default **Standard** Virtual Partition.

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The first level in the priority hierarchy that the administrator can interact with is the virtual partition level. A virtual partition represents a collection of workloads. A single virtual partition exists for user work by default, but up to 10 can be defined. A single virtual partition is allocated 100% of the system resources.

A single virtual partition is expected to be adequate to support most priority setups. Multiple virtual partitions are intended for platforms supporting several distinct business units or geographic entities that require strict separation.

Adding virtual partitions



To add additional Virtual Partitions:

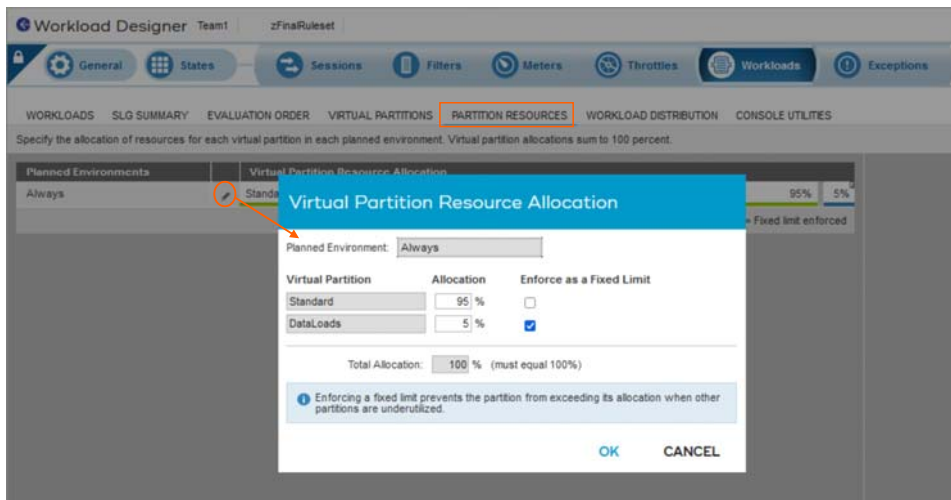
- Click on the + sign and enter the name of the VP
- Drag and drop the Workloads to the new VP

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By clicking on the plus sign next to the “Virtual Partition” label, a new virtual partition may be defined.

During setup and definition time, workloads can be moved from one virtual partition to another by dragging and dropping them, once new virtual partitions have been defined.

Partition resources



The virtual partition **share percent**, for **each Planned Environment**, is set by dragging the boundary line between the defined virtual partitions.

Virtual Partitions can have **hard CPU and I/O limits** enforced.

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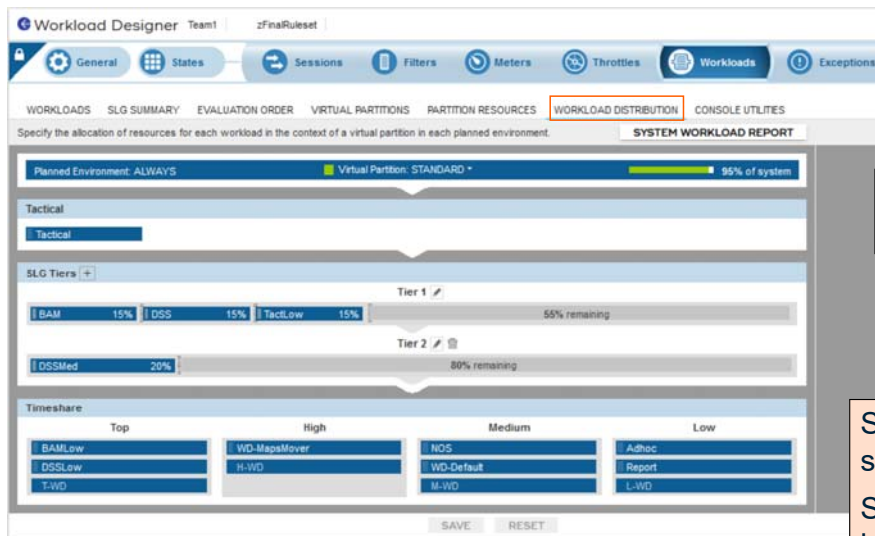
By clicking on the plus sign next to the “Virtual Partition” label, a new virtual partition may be defined.

During setup and definition time, workloads can be moved from one virtual partition to another by dragging and dropping them, once new virtual partitions have been defined.

The system resource allocation percentages for virtual partitions are defined as:

- Dynamic (soft) – If the other virtual partitions are not using their full allocation, then a busy virtual partition can consume more resources than it is allocated
- Fixed – The virtual partition cannot consume more resources than allocated, even if the other virtual partitions do not use their full allocation

Workload distribution



Up to 5 additional SLG Tiers can be added by clicking the + sign.

SLG Tier share percentages can be set by dragging the boundary bar
SLG Tier mapping can be changed by dragging and dropping the workload.

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The Workload Distribution tab is used to set the SLG Tier share percents and to map Workloads.

Workload distribution (cont.)

The screenshot shows the Teradata Workload Designer interface. The main window displays a hierarchy of workloads (DSS, BAM, DSSLow) and their allocation to SLG tiers. Two pop-up windows show the 'Workload Resource Allocation' dialog for Tier 1 and Tier 2.

Workload Resource Allocation (Tier 1):

Workload	Tier Allocation	% of System
DSS	60 %	54 %
(Remaining)	40 %	36 %

Options: ☐ Enable Hard Limit ☐ Enable SLG Expedite

Total Tier Allocation: 100 % (must equal 100%)

† Hard limits are a percentage of overall system resources. Values 3% or greater are recommended.

Workload Resource Allocation (Tier 2):

Workload	Tier Allocation	% of System
BAM	25 %	8 %
BAMLow	25 %	8 %
(Remaining)	50 %	16 %

Options: ☐ Enable Hard Limit

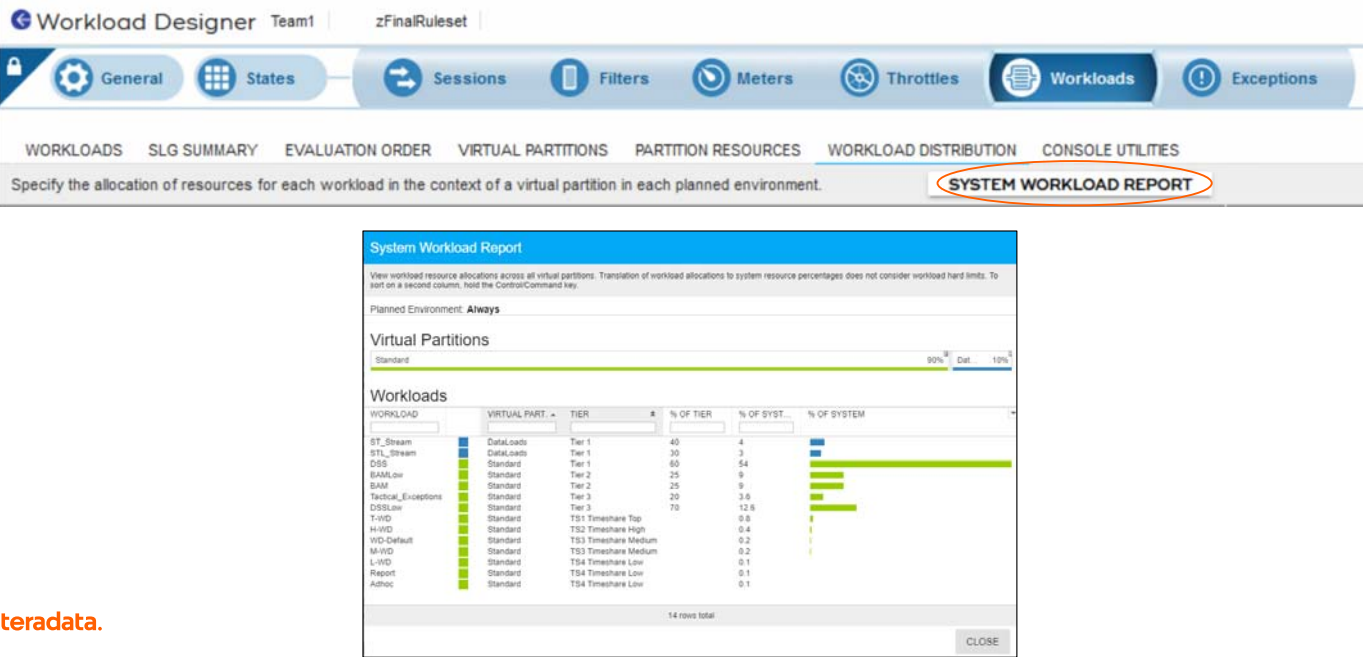
Total Tier Allocation: 100 % (must equal 100%)

† Hard limits are a percentage of overall system resources. Values 3% or greater are recommended.

To add up to 5 additional SLG tier levels, click the plus sign.

SLG Tier 1 can be expedited. All SLG Tiers can have hard CPU and I/O limits enforced.

System workload report



The System Workload Report of the Workload Designer shows the resulting percentage of system resources allocated to each workload (this report assumes no workload is running in the Tactical tier).

Penalty box workload

- To support improved control of resources, it may be useful to create a **special containment workload** sometimes referred to as a **Penalty Box Workload**
- The workload should be **mapped** to a **Low Timeshare Level**
- The workload will typically be used **strictly for demotions**
- To avoid classifying requests directly, the **classification criteria** should be setup to **exclude all users** or **moved after WD_Default** in the Evaluation Order
- The Penalty Box can have value but there are some **negative side effects**
 - It **cannot control other resources** such as I/O, memory, spool, AWTs and locks
 - **Holding** those uncontrolled **resources** for **longer** periods of time can impact other higher priority requests
- **Alternative** is to **use classification** criteria to better detect requests that should be contained and **use throttles** to control concurrency to reduce the number of requests holding critical resources

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To support improved control goals, many customers find it useful to have a Containment workload, also referred to as a penalty box in some situations. A Containment workload is mapped to an Allocation Group that receives a very low priority, sometimes further restricted with a fixed CPU limit on resources it can utilize. This results in the contained requests receiving a very low amount of resources. Typically, 1-5% of system resources are allowed for processing requests assigned to the Containment workload. Generally, requests are assigned to the Containment workload when it is deemed the request needs to run yet it cannot be allowed take significant resources away from the rest of the workloads in the system and risk impacting the ability to meet the Service Level Goals of the other workloads.

An example of Containment Workload usage follows: Classification criteria can be defined for the Containment Workload based on very long estimated processing time. Alternatively or in addition, exception(s) on other workloads can be defined to identify those already-executing requests that should be contained, for example, based on realizing a high CPU to IO ratio, or utilizing too many CPU resources. The automated action of the exception is to change the workload to the Containment workload.

Considerations:

The use of an exception to redirect an already-executing request to the Containment workload does have some negative side-effects. While the associated AG does limit the amount of CPU consumption of its requests, it cannot limit other resource usage such as disk, memory, spool, AMP Worker Tasks (AWTs) and locks. In fact, the release of those resources is often dependent on the requests getting the CPU they need, but the Teradata Priority Scheduler is withholding that CPU at a very low level. As a result, many higher priority requests may be impacted while they wait for availability of the other resources.

For this reason, attempt as much as possible to use classification (applied before the query

begins executing) rather than exceptions to detect queries that should be contained. This allows low concurrency throttles to be enacted on the containment queries. In turn, the number of low-priority requests holding onto critical resources like spool, AWTs and locks are limited, greatly decreasing the chance that those resources will impact the performance of higher priority requests.

Current topic: Lab activity

- The completely fair scheduler
- Priority scheduler overview
- Tactical workload management method
- SLG tier workload management method
- Timeshare workload management method
- Priority scheduler recommendations
- Assigning workloads to workload management methods
- **Lab activity**
 - **Map workloads to priority levels**
 - **Save and activate the ruleset**
 - **Run the Mixed-Workload simulation and capture performance metrics**

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Summary

- Linux SLES 11 offers a completely new scheduler and Teradata's Priority Scheduler is built to leverage that functionality
- The Vantage Analytics Database leverages the Control Group and Resource Shares to manage Workloads priority to CPU and I/O resources
- Priority Scheduler use the concept of Virtual Partitions and the following workload management methods to manage priorities:
 - Tactical
 - SLG Tiers
 - Timeshare
- Resources flow down through the hierarchy
- Ensure that the share percentages are defined so that adequate resource flows down to the lower levels
- Concurrency does not dilute a requests priority in Timeshare

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This slide summarizes this module.

Module 15: Summary

Teradata Vantage optimizing the
Analytics Database through Workload
Management

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Objectives

After completing this module, you will be able to:

- Explain the Workload Optimization Analysis Process
- Identify which Optimization Tools were used in this workshop

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Topics

- Mixed workload review
- A final solution

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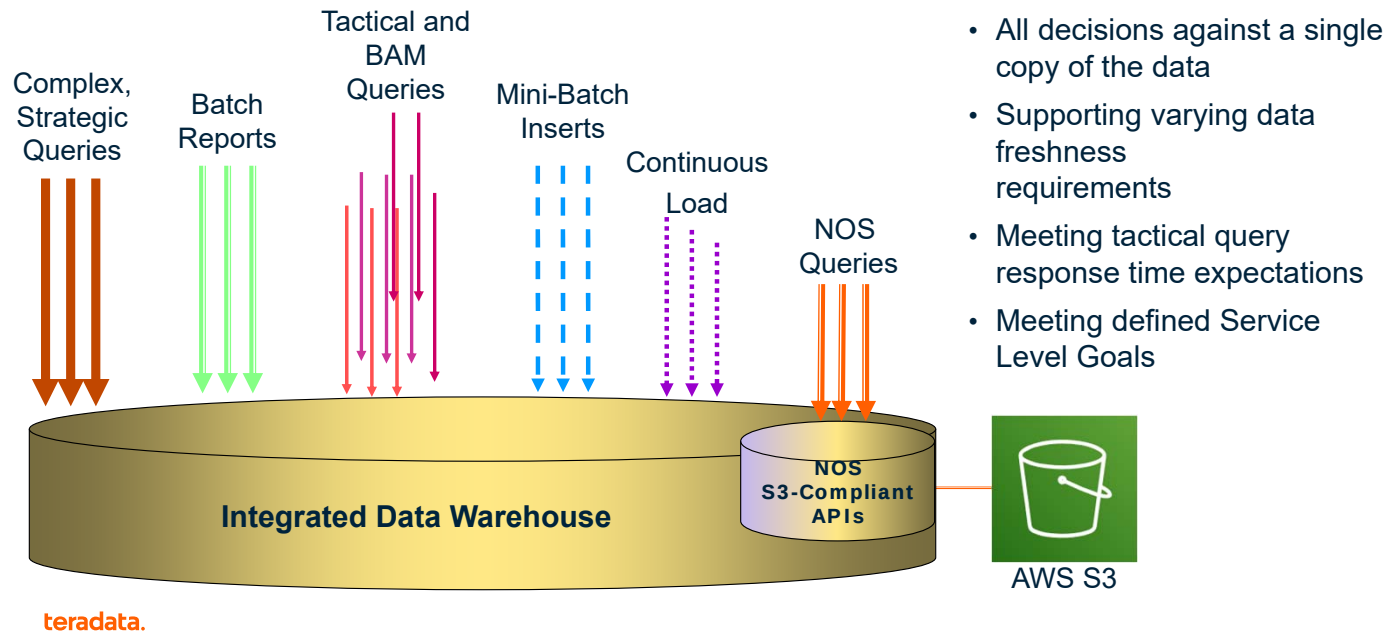
Current topic: Mixed workload review

- **Mixed workload review**
 - **Mixed workloads**
 - **What is workload management?**
 - **What are the advantages of workload definitions?**
- A final solution

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Mixed workloads



Traditionally, data warehouse workloads have been based on drawing strategic advantage from the data. Strategic queries are often complex, sometimes long-running, and usually broad in scope. The parallel architecture of Vantage Analytics Database supports these types of queries by spreading the work across all the parallel units and nodes in the configuration.

Today, data warehouses are being asked to support a diverse set of workloads. These range from the traditional complex strategic queries and batch reporting, which are usually all AMP requests requiring large amounts of I/O and CPU, to tactical queries, which are similar to the traditional OLTP characteristics of single or few AMPs requiring little I/O and CPU.

In addition, the traditional batch window processes of loading data are being replaced with more real-time data freshness requirements.

The ability to support these diverse workloads, with different service level goals, on a single data warehouse is the vision of Teradata's Active DW. However, the challenge for the PS consultants is to implement, manage and monitor an effective mixed workload environment.

What is workload management?

What is Workload Management?

- The Workload Management infrastructure is a **Goal-Oriented**, **Automatic Management** and Advisement technology in support of performance tuning, workload management, capacity management, configuration and system health management
- It consists of several products/tools that assist the DBA or application developer in defining (and refining) the rules that **control the allocation** of resources to workloads running on a system
- It provides for framework for **workload-centric** rather than system-centric database management analysis

Key products that are used to create and manage workloads are:

- Workload Designer portlet
- Workload Monitor portlet
- Workload Health portlet

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Workload Management is made up of several products/tools that assist the DBA or application developer in defining and refining the rules that control the allocation of resources to workloads running on a system. These rules include filters, throttles, and “workload definitions”.

Rules to control the allocation of resources to workloads are effectively represented as workload definitions which are new with Teradata V2R6.1. Tools are also provided to monitor workloads in real time and to produce historical reports of resource utilization by workloads. By analyzing this information, the workload definitions can be adjusted to improve the allocation of system resources.

Workload Management is primarily comprised of the following products to help create and manage “workload definitions”.

- Workload Designer
- Workload Monitor
- Workload Health

Workload Designer is a key supporting product component for Workload Management. The major functions performed by the DBA include:

- Define general Workload Management controls
- Define State Matrix
- Define Session Control
- Define Filters and Throttles
- Define Workloads

The benefit of Workload Management is to automate the allocation of resources to workloads and to assist the DBA or application developer regarding system performance management.

The benefits include:

- Fix and prevent problems before they happen. Seamlessly and automatically manage resource allocation; removes the need for constant setup and adjustment as workload conditions change
- Improved reporting of both real-time and long-term trends – Service Level statistics are now reported for each workload. This helps manage Service Level Goals (SLG) and Service Level Agreements (SLA) – applications can be introduced with known response times
- Automated Exception Handling – queries that are running in an inappropriate manner can be automatically detected and corrected
- Reduced total cost of ownership – one administrator can analyze, tune, and manage a system's performance

What are the advantages of workload definitions?

What are the advantages of Workload Definitions?

- Improved Control of Resource Allocation
 - **Resource priority** is given based on belonging to a particular workload.
 - **Classification rules** permit queries to run at the correct priority from the start.
- Improved Reporting
 - Workload definitions allow you to see **who is using the system** and how much of the various system resources.
 - **Service level statistics** are reported for each workload
 - Real-time and long-term **trends** for workloads are available
- Automatic Exception Detection and Handling
 - After a query has started executing, a query that is running in an **inappropriate** manner can be **automatically detected**. Actions can be taken based on exception criteria that has been defined for the workload

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The reason to create workload definitions is to allow Workload Management to manage and monitor the work executing on a system.

There are two basic reasons for grouping requests into a workload definition.

- Improved Control – some requests need to obtain higher priority to system resources than others. Resource priority is given based on belonging to a particular workload
- Accounting Granularity – workload definitions allow you to see who is using the system and how much of the various system resources. This is useful information for performance tuning efforts
- Automatic Exception Handling – queries can be checked for exceptions while they are executing, and if an exception occurs, a user-defined action can be triggered

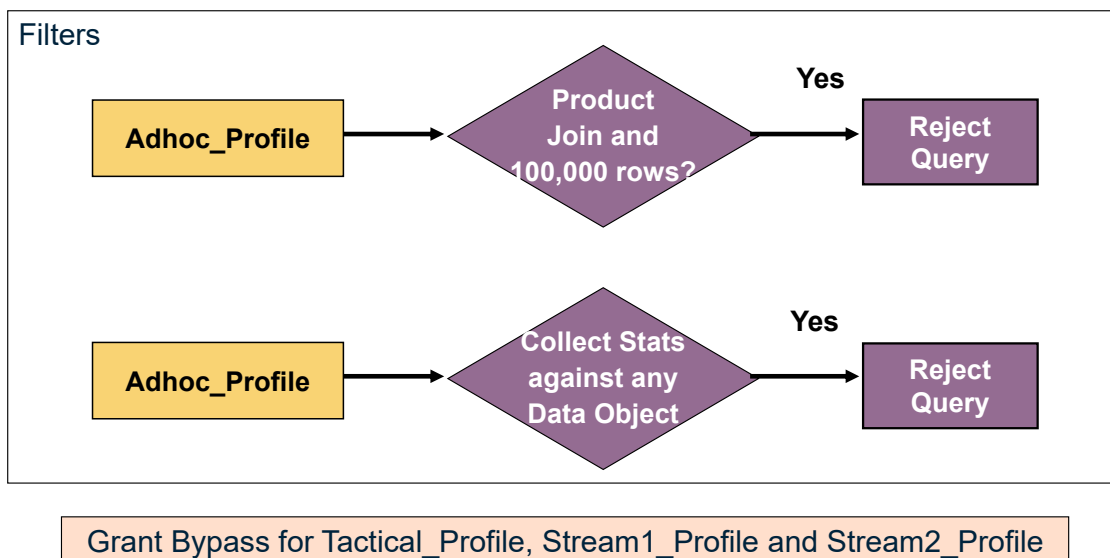
Current topic: A final solution

- Mixed workload review
- **A final solution**
 - **Workload management solution**
 - **Baseline results**
 - **Filters and throttles results**
 - **Refining the workloads and exceptions results**
 - **Workload management final results**

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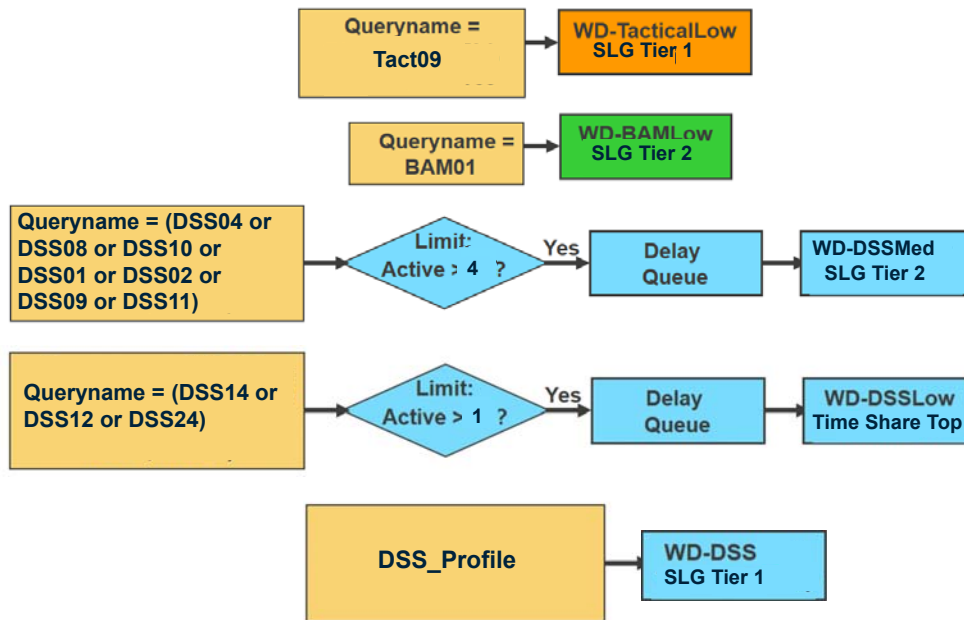
Workload management solution



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This slide has the final solution from our testing.

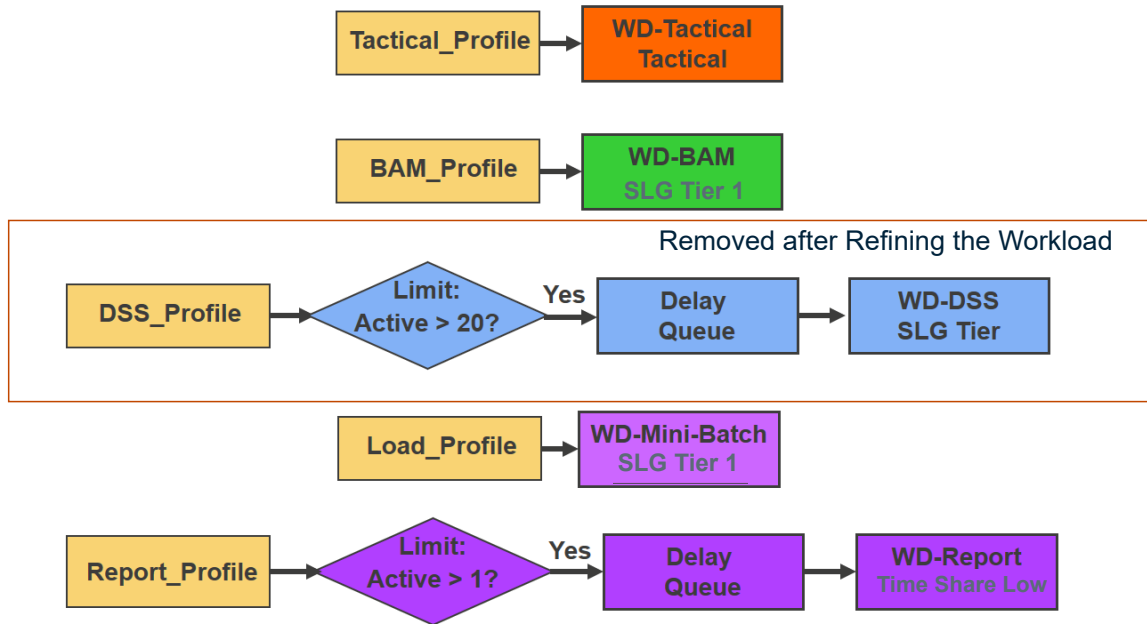
Workload management solution (cont.)



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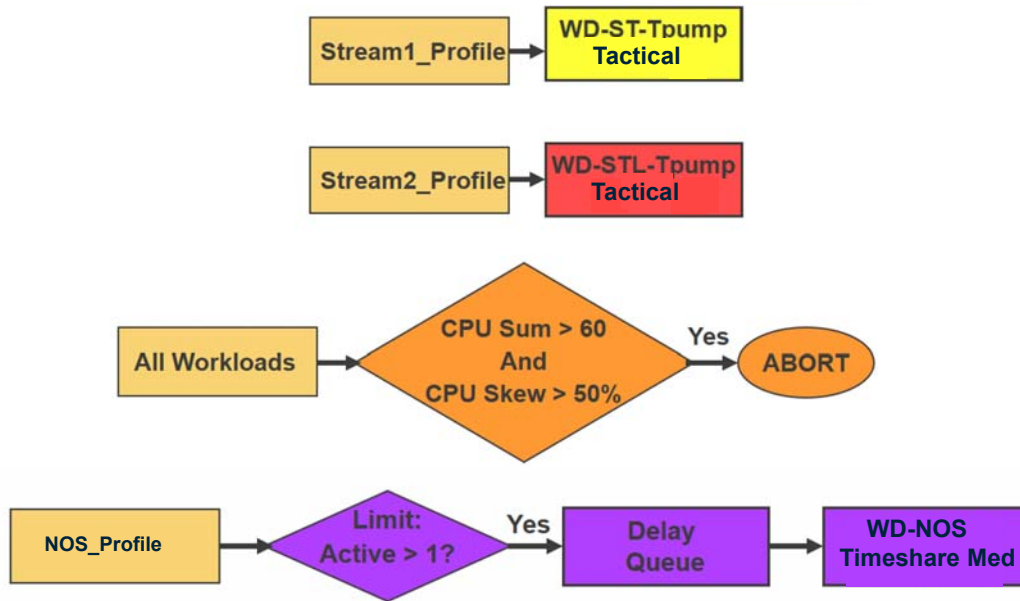
Workload management solution (cont.)



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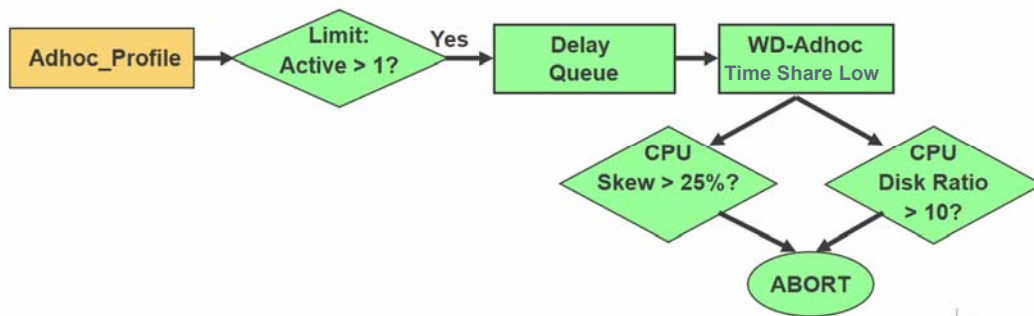
Workload management solution (cont.)



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This slide has the final solution from our testing.

Workload management solution (cont.)



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This slide has the final solution from our testing.

Workload management solution (cont.)

The screenshot shows the Teradata Workload Designer interface. The top navigation bar includes tabs for General, States, Sessions, Filters, Meters, Throttles, Workloads (selected), and Exceptions. Below this, a sub-navigation bar shows various workload management options, with 'VIRTUAL PARTITIONS' selected. A descriptive text states: 'Use this grid to add and remove virtual partitions and to assign workloads by dragging them to different partitions. The icon to the right of workload names represents the management method of the workload (tactical, SLG tier, timeshare).' The main area contains a grid with two columns: 'Standard' and 'DataLoads'. The 'Standard' column lists various partition types like Tactical, BAM, DSS, DSSMed, TactLow, Adhoc, BAMLow, DSSLow, NOS, Report, WD-Default, WD-MapMover, T-WD, H-WD, M-WD, and L-WD. The 'DataLoads' column lists specific workloads: ST-Tpump, STL-Tpump, and Mini-Batch. Each workload name has a small icon to its right indicating its management method.

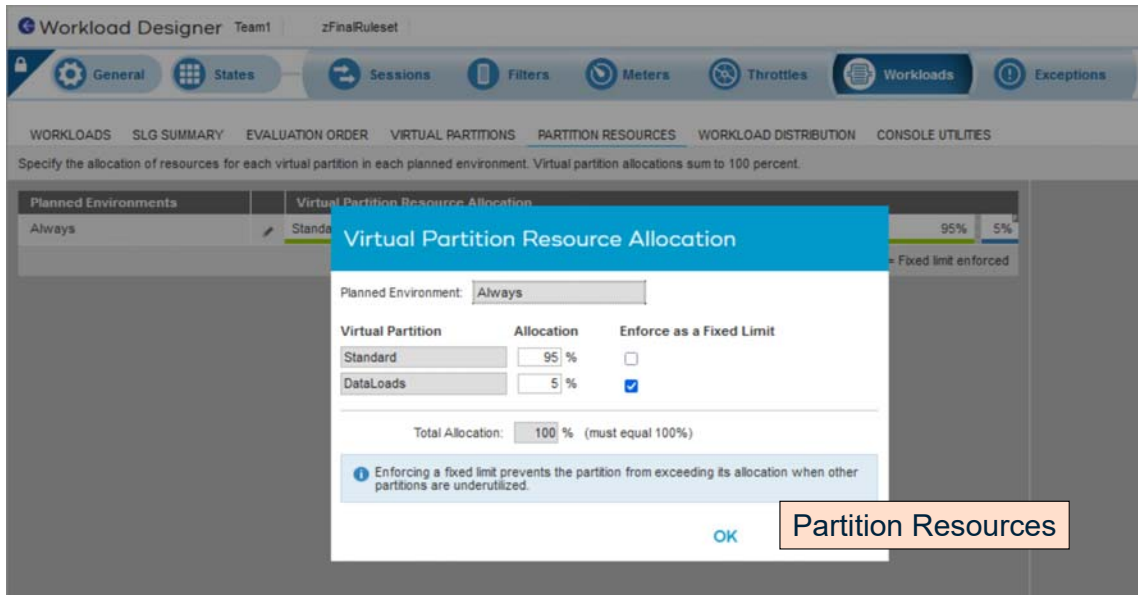
Standard	DataLoads
Tactical	ST-Tpump
BAM	STL-Tpump
DSS	Mini-Batch
DSSMed	
TactLow	
Adhoc	
BAMLow	
DSSLow	
NOS	
Report	
WD-Default	
WD-MapMover	
T-WD	
H-WD	
M-WD	
L-WD	

Virtual Partitions

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This slide has the final solution from our testing.

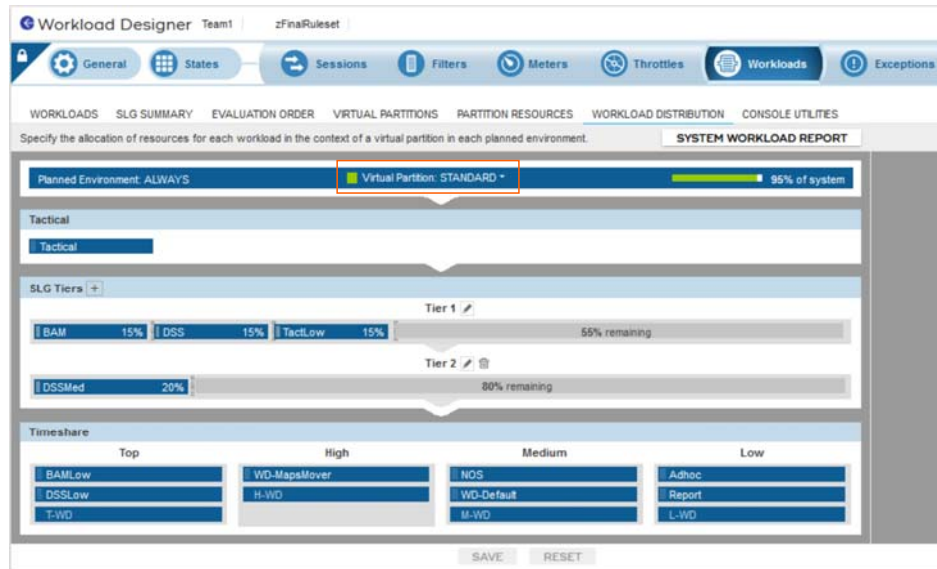
Workload management solution (cont.)



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This slide has the final solution from our testing.

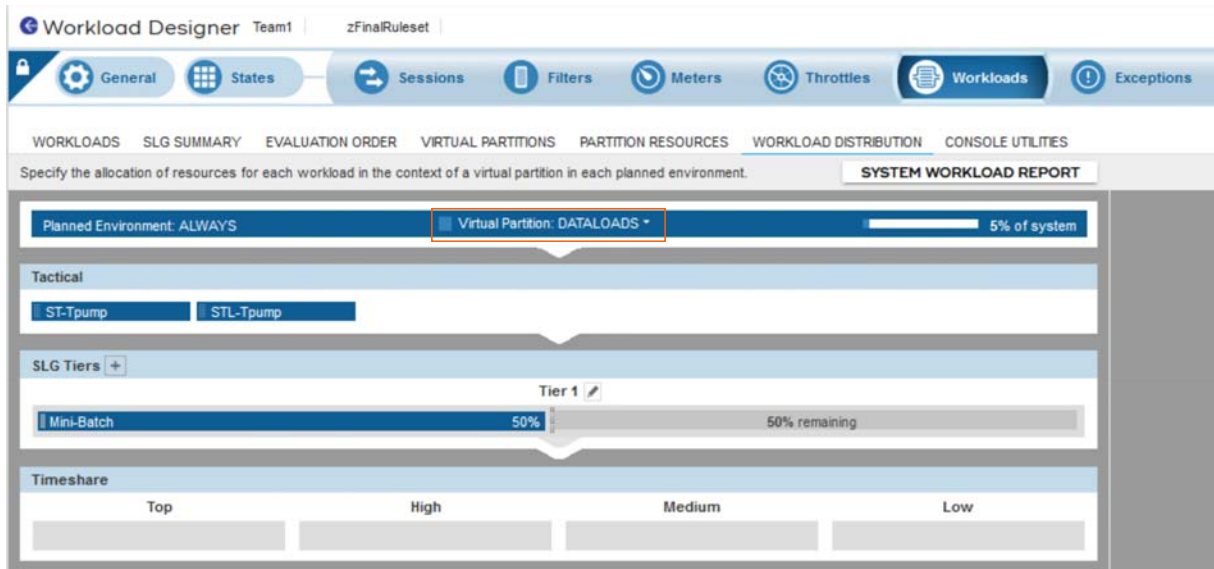
Workload management solution (cont.)



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Workload Distribution

Workload management solution (cont.)



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Workload Distribution

the results after applying our Filters and Throttles in our testing.

Baseline lab exercise results

Avg Response time	DSS	182.39	120
	Tactical	3.57	1
	BAM	94.92	6

Throughput Numbers	DSS	560	1000
	Tactical	7402	45,000
	BAM	66	60
	II Mini-Batch	22.22	60
	ST TPump	22.22	90
	STL T pump	44.44	200

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This slide has the results for our Baseline testing.

Filters and throttles lab exercise results

Avg Response time	DSS	76.4	120
	Tactical	1.17	1
	BAM	21.64	6

Throughput Numbers	DSS	1358	1000
	Tactical	32602	45,000
	BAM	64	60
	II Mini-Batch	55.55	60
	ST TPump	88.88	90
	STL T pump	155.55	200

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This slide has the results after applying our Filters and Throttles in our testing.

Refine workloads and exceptions lab exercise results

Avg Response time	DSS	147.56	120
	Tactical	.44	1
	BAM	4.98	6

Throughput Numbers	DSS	728	1000
	Tactical	54322	45,000
	BAM	66	60
	II Mini-Batch	66.66	60
	ST TPump	128.267	90
	STL T pump	242.37	200

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This slide has the results after refining workloads and applying exception rules in our testing.

Workload management final lab exercise results

Avg Response time	DSS	101.32	120
	Tactical	.62	1
	BAM	5.37	6

Throughput Numbers	DSS	1052	1000
	Tactical	47110	45,000
	BAM	66	60
	II Mini-Batch	66.66	60
	ST TPump	90.75	90
	STL Tpump	198.86	200

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This slide has the workload management lab exercise results from final testing.

Course summary

Workload Management provides a number of **rules** that can be used to **automate** and **manage** a mixed workload environment to meet performance requirements

Some of the Recommendations include:

- Keep the number of **workloads** to a manageable number, usually **10 to 30**
- Keep **classification criteria simple**, leading with Request Source or Queryband for exactness and add additional criteria as necessary
- Exception rules are used to handle **misclassified** queries
- When setting up priorities, start with a **single Virtual Partition** and a **single SLG Tier** and expand based on business need that requires more complexity
- Keep the State Matrix **simple** with a **small** number of **States** in the range of 3 to 5
- Use the State Matrix to **change working values** rather than new rulesets
- Apply **Throttles** to low priority workloads to reduce resource contention
- Apply **Filters** to reject poorly formulated queries

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Workload Management provides a number of rules that can be used to automate and manage a mixed workload environment to meet performance requirements

Some of the Recommendations include:

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- Apply Filters to reject poorly formulated queries

Appendix A: Viewpoint configuration

Vantage: Optimizing the Analytics
Database
through Workload Management

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Objectives

After completing this module, you will be able to:

- Discuss the purpose of various Viewpoint administrative portlets
- Explain how Viewpoint was configured for the Mixed Workload simulation labs

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Topics

- Viewpoint overview
- Administration portlets used in this training

teradata.



Current topic: Viewpoint overview

- Viewpoint overview
 - **What is viewpoint?**
 - **Administration portlets**
- Administration portlets used in this training course

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What is viewpoint?

Viewpoint is the cornerstone Vantage Systems Monitoring and Management.

- Provides systems management via a web browser
- Provides a single operational view (SOV) for multi-system environment
- Highly customizable and can be personalized

Viewpoint Architecture:

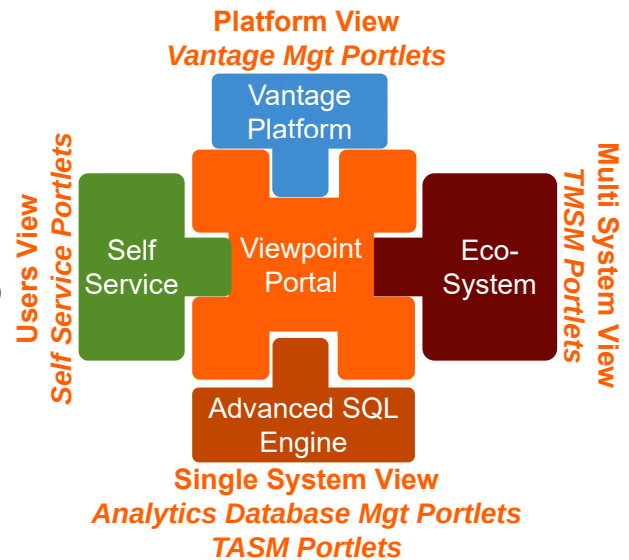
- Viewpoint Server—external server or appliance that executes the Viewpoint application
- Viewpoint Portal built on modern web technologies: AJAX - Web 2.0 application
- Data Collection Service (DCS) is performed by the Viewpoint server
- Postgres database

Viewpoint Supported Browsers:

- Mozilla Firefox 56+
- Google Chrome 61+
- Apple Safari 10.4+
- Microsoft Edge 40+
- Microsoft Internet Explorer 11
- Chrome Android mobile 61+
- Safari iOS mobile 10+



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Viewpoint is the foundation for **monitoring, reporting and management** of Vantage Systems.

Teradata Viewpoint is intended as a Teradata customer's Single Operational View (SOV), meaning it supports various Teradata systems including Teradata Vantage Analytics Database, Teradata Vantage Machine Learning Engine, Teradata Hadoop System, and Teradata QueryGrid.

It provides a web-based interface (a set of portals) for a wide range of capabilities and features, such as monitoring, management, alerting, and others. It serves both system administrators as well as business users. It also serves as a user interface for other Teradata products, for example Teradata Data Lab.

Teradata Viewpoint provides systems management via a web browser which is extensible to Teradata end users and management, allowing them to understand the state of the system and make intelligent decisions about their work day.

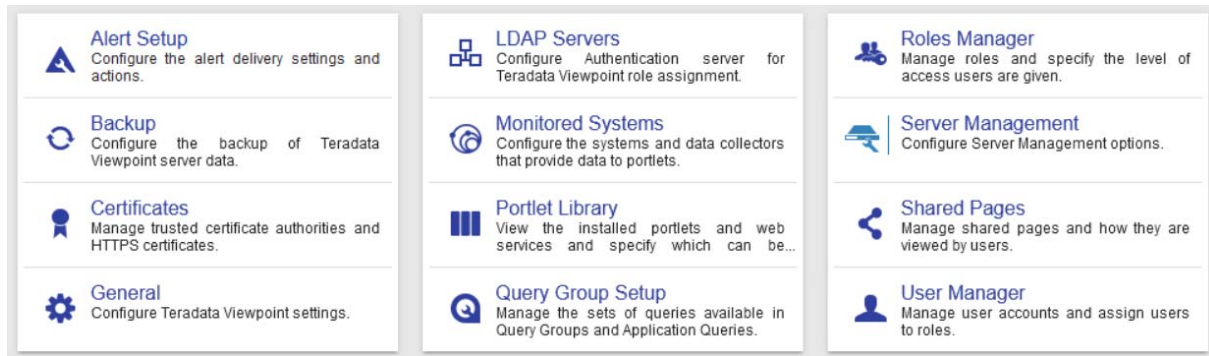
Teradata Viewpoint allows users to view system information, such as query progress, performance data, and system saturation and health through preconfigured portlets displayed from within the Teradata Viewpoint portal. Portlets can also be customized to suit individual user needs. User access to portlets is managed on a per-role basis.

Administrators can use Teradata Viewpoint to determine system status, trends, and individual query status. By observing trends in system usage, system administrators are better able to plan project implementations, batch jobs, and maintenance to avoid peak periods of use. Business users can use Teradata Viewpoint to quickly access the status of reports and queries and drill down into details.

Administration portlets

The Teradata Viewpoint administrative portlets allow the Viewpoint Administrator to **configure access** to Vantage and Viewpoint resources.

The Administrative portlets are available from the **Admin Portlet button**



The Teradata Viewpoint administrative portlets allow the Teradata Viewpoint Administrator to provide access to Teradata Viewpoint resources and information.

You can access these portlets from the Teradata Viewpoint portal page if your role has permission.

Alert Setup

Configure the alert delivery settings and actions.

Backup

Configure the backup of Teradata Viewpoint server data.

Certificates

Manage trusted certificate authorities and HTTPS certificates.

General

Configure Teradata Viewpoint settings.

LDAP Servers

Configure the LDAP servers for Teradata Viewpoint to authenticate users and assign user roles.

Monitored Systems

Add the systems and configure the data collectors that provide data to portlets. You also can add and configure a managed system available to display in the **Viewpoint Monitoring** portlet.

Portlet Library

View the installed portlets and specify which portlets can be enabled.

Query Group Setup

Manage the sets of queries available to users in the **Query Groups** and **Application Queries** portlets. Can also be used to define the criteria that associate a query with a particular application in the **Query Log** portlet.

Roles Manager

Manage roles and specify the level of access users are given.

Server Management

Configure Server Management options.

Shared Pages

Manage shared pages and how they are viewed by users.

User Manager

Manage user accounts and assign users to roles.

Current topic: Administration portlets used in this training

- Viewpoint overview
- **Administration portlets used in this training course**
 - **Monitored systems**
 - **Portlet library**
 - **User manager**
 - **Roles manager**

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Monitored systems portlet

The **MONITORED SYSTEMS** portlet allows the Teradata Viewpoint Administrator to **add, configure, enable, and disable systems**, as well as view the amount of disk space used and set a threshold for a disk usage alert.

- General – Configure the system nickname, **TDPID, login names, passwords**, and account strings (optional)
- Data Collectors – Enable, disable, and **configure data collectors** to capture and retain portlet, disk usage, and resource data
- System Health – Enable **metrics for the SYSTEM HEALTH** portlet. Configure degraded and critical thresholds for each metric
- Canary Queries – Configure canary queries used to **test** Analytics Database **response** times
- Alerts – Add, delete, copy, and **configure alerts**, or migrate existing Teradata Manager alerts
- Monitor Rates – Set Analytics Database **internal sample rates** for Sessions, Node logging, and Vproc logging
- Log Table Clean Up – Select system log tables to **clean up**
- Clean Up Schedule – **Schedule clean up** of system log tables

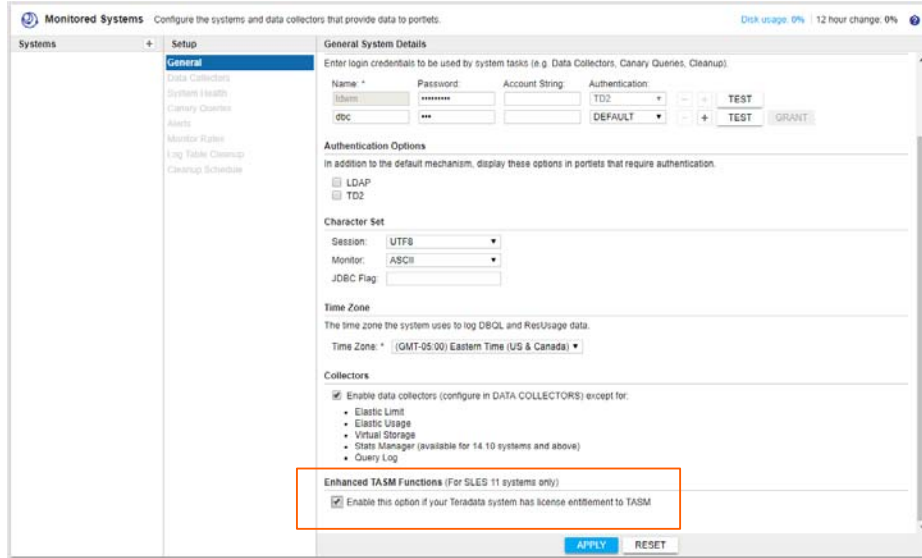
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The **MONITORED SYSTEMS** portlet allows the Teradata Viewpoint Administrator to add, configure, enable, and disable Vantage systems using specific dialog boxes:

- **General** - Configure the system nickname, TDPID, login names, passwords (hidden), and account strings (optional). Test the connection to Analytics Database, and add or delete login names.
- **Data Collectors** - Enable, disable, and configure data collectors to capture and retain portlet, disk usage, and resource data.
- **System Health** - Enable metrics for the **SYSTEM HEALTH** portlet. Configure degraded and critical thresholds for each metric.
- **Canary Queries** Configure canary queries used to test Analytics Database response times. **System Heartbeat** - canary query cannot be removed.
- **Alerts** - Add, delete, copy, and configure alerts, or migrate existing Teradata Manager alerts.
- **Monitor Rates** - Set Analytics Database internal sample rates for sessions, node logging, and vproc logging.
- **Log Table Clean Up** - Select system log tables to clean up.
- **Clean Up Schedule** - Schedule clean up of system log tables.

Monitored systems portlet: General

Button:  (Viewpoint Administration) > Portlet: Monitored Systems > Systems: System Name > Setup: General



Monitored Systems Configure the systems and data collectors that provide data to portlets. Disk usage: 0% | 12 hour change: 0%

Systems + **Setup**

General System Details

Enter login credentials to be used by system tasks (e.g. Data Collectors, Canary Queries, Cleanup).

Name: Password: Account String: Authentication:

Authentication Options

In addition to the default mechanism, display these options in portlets that require authentication.

☐ LDAP ☐ TD2

Character Set

Session: Monitor: JDBC Flag:

Time Zone

The time zone the system uses to log DBQL and ResUsage data.

Time Zone:

Collectors

☒ Enable data collectors (configure in DATA COLLECTORS) except for:

- Elastic Limit
- Elastic Usage
- Virtual Storage
- State Manager (available for 14.10 systems and above)
- Query Log

Enhanced TASM Functions (For SLES 11 systems only)

☒ Enable this option if your Teradata system has license entitlement to TASM

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Configure the system nickname, TDPID, login names, passwords (hidden), and account strings (optional). Test the connection to Analytics Database, and add or delete login names.

The enable full TASM functionality, the Enhanced TASM Functions must be checked.

Monitored systems portlet: Data collectors

Button:  (Viewpoint Administration) > Portlet: **Monitored Systems** >
Systems: **System Name** > Setup: **Data Collectors** > Data Collectors: **Account Info**

Monitored Systems - Configure the systems and data collectors that provide data to portlets. Disk usage: 1% | 12 hour change: +9%

Systems	Setup	Data Collectors	Account Info Collector Details
1011	General Data Collectors System Health Canary Queries Alerts Monitor Rules Log Table Cleanup Cleanup Schedule	Account Info Alert Request AWT Info Database Space Dictionary Disk Space Elastic Limit Elastic Usage Lock Info Query Count Query Log Resource Usage Sessions Stats Manager System Config System Stats Table Space TASM Config TASM Distribution TASM Exception TASM State TASM Summary Virtual Storage	☒ Enable Account Info Collector Login: dbc Sample Rate ☒ Recommended sample rate (43200 seconds) ☐ Custom sample rate: 43200 seconds (3600-86400) * Delete Data ☒ After 1 Week ☐ Over 10 Gigabytes

APPLY RESET

Enable, disable, and configure data collectors to capture and retain portlet, disk usage, and resource data.

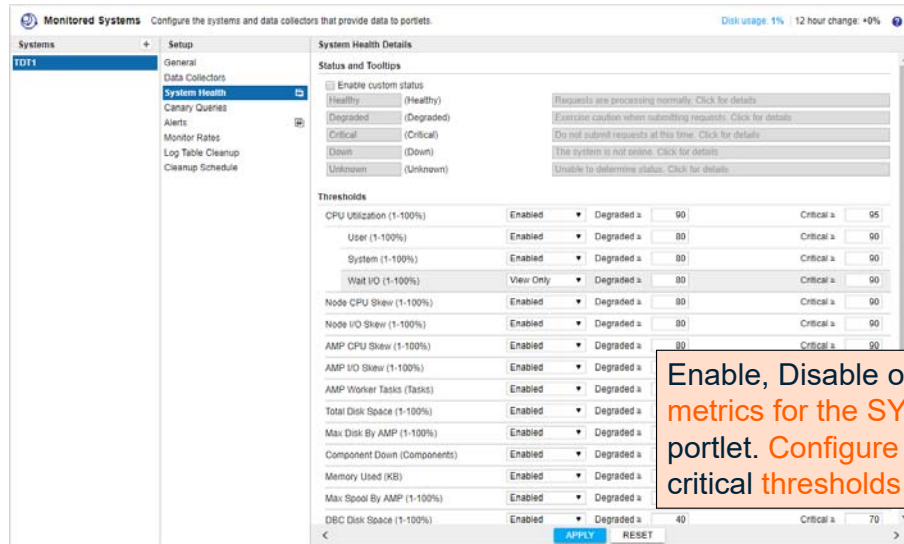
teradata.

Data Collectors are used to monitor systems. After a system has been configured in Teradata Viewpoint, data collectors can be configured to monitor the system. Data collectors gather information from different sources and make the data available to Teradata Viewpoint portlets. Each data collector has a sample rate, or frequency, used to collect data from the system and a retention rate used to keep the collected data for a time period or up to a certain size.

Enable, disable, and configure data collectors to capture and retain portlet, disk usage, and resource data.

Monitored systems portlet: System health

Button:  (Viewpoint Administration) > Portlet: Monitored Systems > Systems: System Name > Setup: System Health



Enable, Disable or View Only metrics for the **SYSTEM HEALTH** portlet. Configure degraded and critical thresholds for each metric

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You can customize system status and tooltips and configure metrics and thresholds. The thresholds are settings for the data collected by canary queries and the disk space, sessions, and system statistics data collectors.

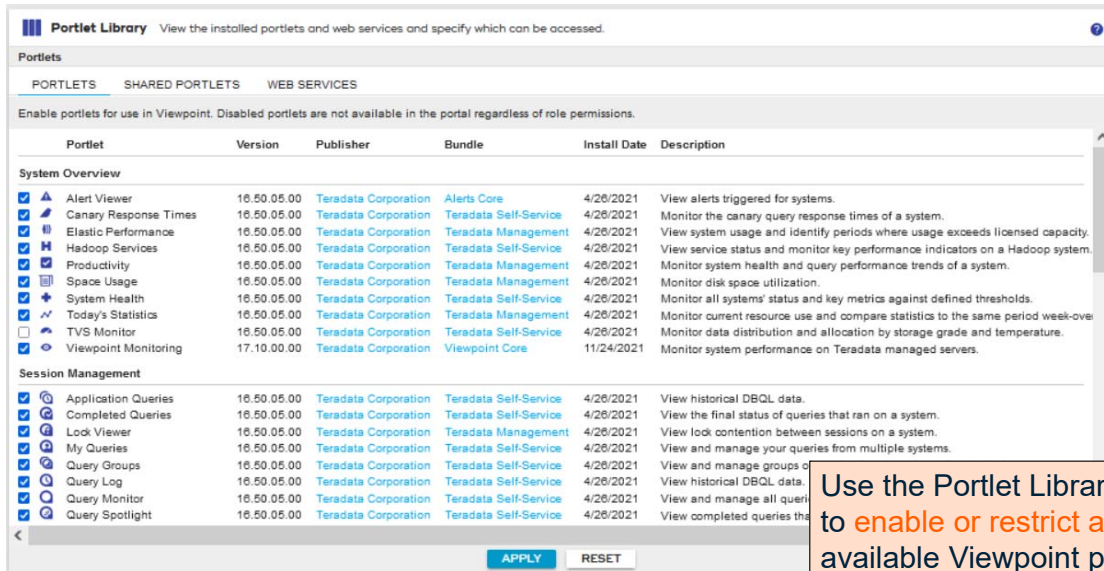
For Vantage Analytics Databases, the system status, tooltips, metrics, and thresholds appear in the System Health and Productivity portlets.

Enable metrics for the **SYSTEM HEALTH** portlet. Configure degraded and critical thresholds for each metric

- Enabled** Makes the metric visible in the System Health portlet. Uses the threshold values in the system status calculation.
- Disabled** Omits the metric in the System Health portlet. Does not use threshold values in the system status calculation.
- View Only** Makes the metric visible in the System Health portlet. Does not use threshold values in the system status calculation.

Portlet library

Button:  (Viewpoint Administration) > Portlet: Portlet Library > Tab: Portlets



Portlet Library View the installed portlets and web services and specify which can be accessed.

PORTLETS SHARED PORTLETS WEB SERVICES

Enable portlets for use in Viewpoint. Disabled portlets are not available in the portal regardless of role permissions.

Portlet	Version	Publisher	Bundle	Install Date	Description
System Overview					
☒ Alert Viewer	16.50.05.00	Teradata Corporation	Alerts Core	4/26/2021	View alerts triggered for systems.
☒ Canary Response Times	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	Monitor the canary query response times of a system.
☒ Elastic Performance	16.50.05.00	Teradata Corporation	Teradata Management	4/26/2021	View system usage and identify periods where usage exceeds licensed capacity.
☒ Hadoop Services	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View service status and monitor key performance indicators on a Hadoop system.
☒ Productivity	16.50.05.00	Teradata Corporation	Teradata Management	4/26/2021	Monitor system health and query performance trends of a system.
☒ Space Usage	16.50.05.00	Teradata Corporation	Teradata Management	4/26/2021	Monitor disk space utilization.
☒ System Health	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	Monitor all systems' status and key metrics against defined thresholds.
☒ Today's Statistics	16.50.05.00	Teradata Corporation	Teradata Management	4/26/2021	Monitor current resource use and compare statistics to the same period week-over-week.
☐ TVS Monitor	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	Monitor data distribution and allocation by storage grade and temperature.
☒ Viewpoint Monitoring	17.10.00.00	Teradata Corporation	Viewpoint Core	11/24/2021	Monitor system performance on Teradata managed servers.
Session Management					
☒ Application Queries	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View historical DBQL data.
☒ Completed Queries	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View the final status of queries that ran on a system.
☒ Lock Viewer	16.50.05.00	Teradata Corporation	Teradata Management	4/26/2021	View lock contention between sessions on a system.
☒ My Queries	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View and manage your queries from multiple systems.
☒ Query Groups	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View and manage groups of queries.
☒ Query Log	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View historical DBQL data.
☒ Query Monitor	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View and manage all queries.
☒ Query Spotlight	16.50.05.00	Teradata Corporation	Teradata Self-Service	4/26/2021	View completed queries that are currently running.

APPLY RESET

Use the Portlet Library portlet to enable or restrict access to available Viewpoint portlets

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The Portlet Library allows you to enable or disable portlets globally. Even if a portlet is enabled for a role, it must be enabled in Portlet Library for a user in the role to have access to it.

The Portlets tab displays installed portlets, grouped by category, and provides the following information:

- Portlet name
- Version number
- Publisher name
- Bundle name
- Installation date
- Portlet description

Select portlets for activation. Using a simple checklist, you can either enable or restrict access to available Teradata Viewpoint portlets.

The Shared Portlets tab displays information about shared portlets. A shared portlet is a user-defined version of a portlet. The Parent Portlet column identifies the original portlet before it was customized as a shared portlet. Portlet names and descriptions can be edited. Portlets can be deleted. Shared portlet permissions can be edited using Roles Manager.

User manager portlet

Button:  (Viewpoint Administration) > Portlet: User Manager

User Manager Manage user accounts and assign users to roles.

Users

LAST NAME	FIRST NAME	PORTAL USERNAME	EMAIL	LAST LOGIN	ROLES
Administrator	Viewpoint	admin	administrator@teradata.com	4/15/19 10:02:56 AM	Administrator, User

User Manager Manage user accounts and assign users to roles.

Administrator, Viewpoint

GENERAL **ROLES**

Portal Username: * admin

First Name: * Viewpoint

Last Name: * Administrator

Email: * administrator@teradata.com

Last Login: 4/15/2019 10:02:56 AM

CHANGE PASSWORD

Assign roles to specify the Viewpoint features this user can access. Users have access to features that are granted to any role to which they are assigned dictate the initial portlet display.

Available Roles:

Selected roles:

User
Administrator

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Use the User Manager portlet to
Add Viewpoint user accounts

The **User Manager** portlet allows the Teradata Viewpoint Administrator to manage Teradata Viewpoint user accounts.

Using this portlet, you can:

- Define or modify a user account.
- Reset forgotten or compromised passwords.
- Assign roles to users.
- Set role precedence.
- Search for existing users.

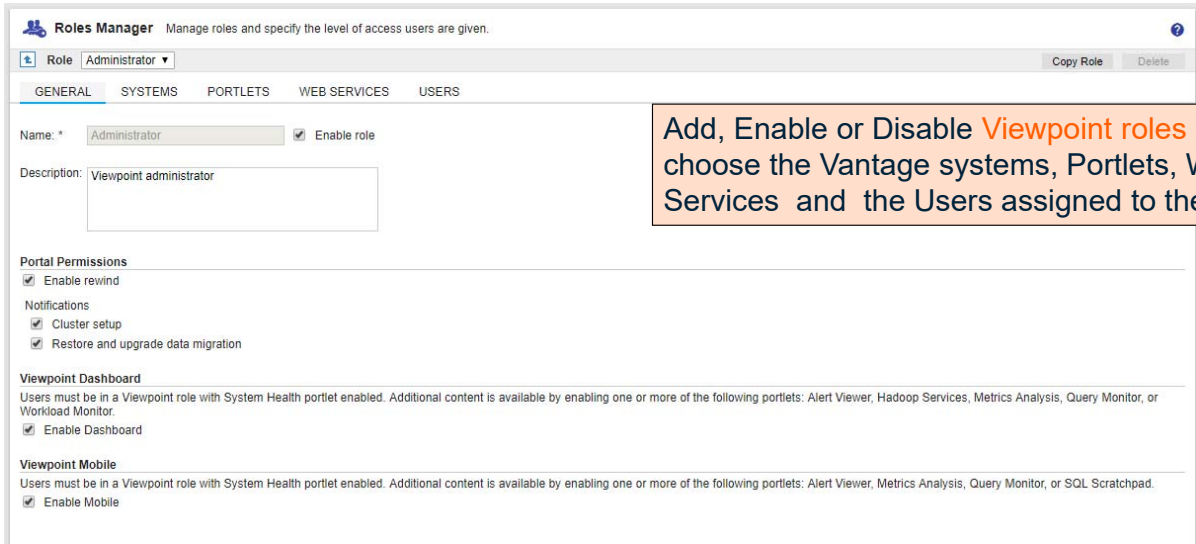
The **User Manager** portlet provides the following views:

USER LIST Allows you to add users or to search for and select an existing user account to modify. A search tool is provided to help locate an individual user or groups of users when the user list is long. It is the default view. **USER DETAILS** Shows details about the selected user. This view includes the following tabs:

- **General** (default): Modify the selected user's account, including name and email address.
 - **Roles**: Assign available roles to the selected user and set role precedence.
- Note:** A role must be defined using the **Roles Manager** portlet before it can be assigned to a user.

Roles manager portlet: General

Button:  (Viewpoint Administration) > Portlet: Roles Manager > Button: Add Role > Tab: General



Roles Manager Manage roles and specify the level of access users are given.

Role: Administrator Copy Role Delete

GENERAL SYSTEMS PORTLETS WEB SERVICES USERS

Name: * Administrator ☒ Enable role

Description: Viewpoint administrator

Portal Permissions

☒ Enable rewind

Notifications

☒ Cluster setup

☒ Restore and upgrade data migration

Viewpoint Dashboard

Users must be in a Viewpoint role with System Health portlet enabled. Additional content is available by enabling one or more of the following portlets: Alert Viewer, Hadoop Services, Metrics Analysis, Query Monitor, or Workload Monitor.

☒ Enable Dashboard

Viewpoint Mobile

Users must be in a Viewpoint role with System Health portlet enabled. Additional content is available by enabling one or more of the following portlets: Alert Viewer, Metrics Analysis, Query Monitor, or SQL Scratchpad.

☒ Enable Mobile

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The **Roles Manager** portlet allows the Teradata Viewpoint Administrator to assign permissions efficiently by creating classes of users called roles.

The Teradata Viewpoint Administrator can perform the following tasks:

- Add and configure new roles
- Edit the configuration and settings of existing and default roles
- Copy roles, saving time in creating new roles
- Enable or disable portlets for a role.
- Delete roles that are no longer needed

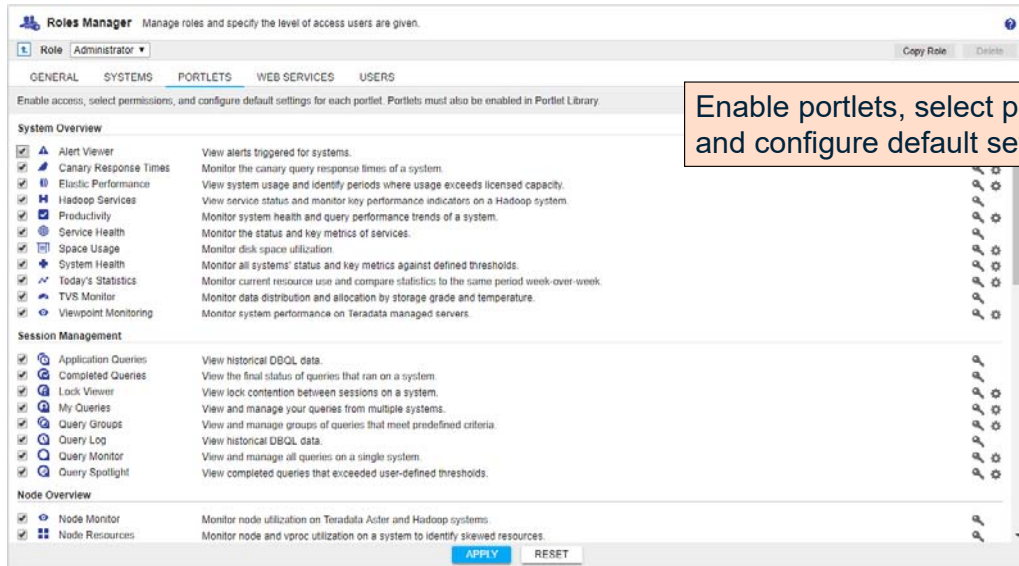
Teradata Viewpoint includes the following preconfigured roles:

- **Administrator** This role has all permissions and can be assigned to any account. It is recommended that this role be used only by the Teradata Viewpoint Administrator.
- **User** This role is assigned to every Teradata Viewpoint user and cannot be removed from Teradata Viewpoint. It is recommended that this role be set with minimum user permissions.

It is recommended that you configure new roles with partial permissions that are appropriate to all users in that role. Each role you create controls access to specific systems, portlets, metrics, preferences, and permissions in portlets.

Roles manager portlet: Portlets

Button:  (Viewpoint Administration) > Portlet: Roles Manager > Button: Add Role > Tab: Portlets



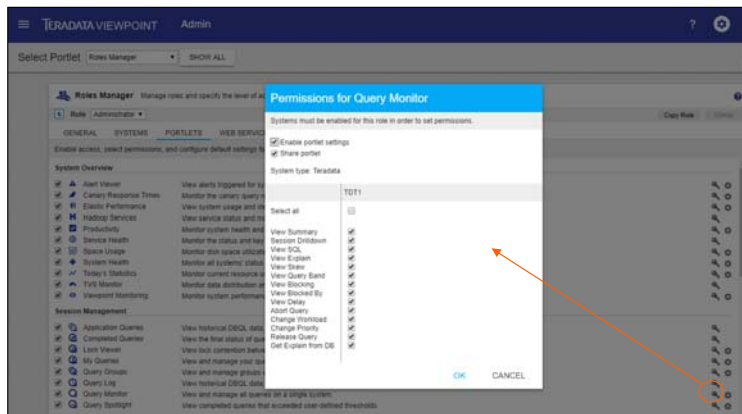
teradata.

Use the Portlets tab to enable or disable portlets for a role.

This tab can also be used to select permissions and configure default settings.

Roles manager portlet: Permissions

Button:  (Viewpoint Administration) > Portlet: Roles Manager > Button: Add Role > Tab: Portlets > Button: Set Portlet Permissions



After choosing a portlet, select the permissions to be granted to users of the portlet

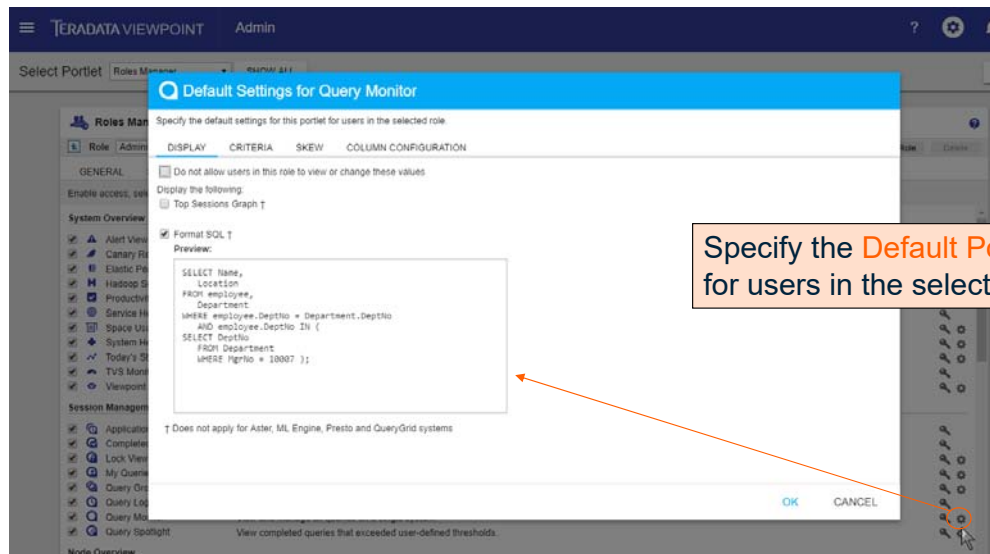
Also, choose if the users are going to be allowed to set their own preferences and if they will be able share customized versions of the portlet with other users

teradata.

After choosing a portlet, select the permissions to be granted to users of the portlet. Also, choose if the users are going to be allowed to set their own preferences and if they will be able share customized versions of the portlet with other users.

Roles manager portlet: Default settings

Button:  (Viewpoint Administration) > Portlet: Roles Manager > Button: Add Role > Tab: Portlets > Button: Set Default Portlet Settings



Specify the **Default Portlet Settings** for users in the selected Role

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Specify the Default Portlet Settings for users in the selected Role.

Summary

The Teradata Viewpoint administrative portlets allow the Viewpoint Administrator to provide access to Teradata Viewpoint resources

- **MONITORED SYSTEMS** – Configure, enable, and disable Viewpoint servers and data collectors. After a server is defined to Viewpoint, you can maintain logins, accounts, passwords, and character set settings.
- **PORTLET LIBRARY** – Select portlets for activation. Using a simple checklist, you can either enable or restrict access to available Teradata Viewpoint portlets.
- **USER MANAGER** – Manage Teradata Viewpoint user accounts by creating user accounts, assigning or resetting passwords, and assigning users to predefined roles.
- **ROLES MANAGER** – Manage roles, assign users, and grant permissions efficiently. After a role is created, you can customize the role by assigning users, enabling portlets, grant permissions for metrics, and granting user permissions for portlets.

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The Teradata Viewpoint administrative portlets allow the Viewpoint Administrator to provide access to Teradata Viewpoint resources

- **MONITORED SYSTEMS** – Configure, enable, and disable Viewpoint servers and data collectors. After a server is defined to Viewpoint, you can maintain logins, accounts, passwords, and character set settings.
- **PORTLET LIBRARY** – Select portlets for activation. Using a simple checklist, you can either enable or restrict access to available Teradata Viewpoint portlets.
- **USER MANAGER** – Manage Teradata Viewpoint user accounts by creating user accounts, assigning or resetting passwords, and assigning users to predefined roles.
- **ROLES MANAGER** – Manage roles, assign users, and grant permissions efficiently. After a role is created, you can customize the role by assigning users, enabling portlets, grant permissions for metrics, and granting user permissions for portlets.

Appendix B: Viewpoint portlets

Vantage: Optimizing the Analytics
Database
through Workload Management

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Objectives

After completing this module, you will be able to:

- Add Pages and Portlets to your Viewpoint screen
- Explain how to view previous information on a Viewpoint Page with the Rewind feature
- Describe the characteristics and purpose of various Viewpoint portlets
- Use various Viewpoint portlets in the mixed workload simulation labs

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Topics

- The viewpoint user interface
- Portlets to use in this training course

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Current topic: The viewpoint user interface

- **The viewpoint user interface**
 - Portal basics
 - Create and access additional pages
 - Add portals to the current page
 - Rewind
- Portlets to use in this training course

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Viewpoint portal basics

The screenshot shows the Teradata Viewpoint portal interface. At the top, there is a navigation bar with the Teradata logo, the text 'VIEWPOINT', and a 'Home' link. A red box labeled 'Current Page' highlights the 'Home' link. On the left side, there is a selector button (a circle with a plus sign) and a red arrow pointing to it with the text 'Access to additional pages are available via this selector button.' On the right side, there is an 'ADD CONTENT' button (a circle with a plus sign) and a red arrow pointing to it with the text 'New portlets are added to a Page via the Add Content.' The main area of the portal is divided into two sections. The left section is titled 'Query Monitor' and displays a table of query execution details. The right section is titled 'System Health' and displays various system metrics and graphs. A red box labeled 'Portlets' points to the 'Query Monitor' section.

Access to additional pages are available via this selector button.

Current Page

ADD CONTENT

Portlets

To help you work efficiently, Teradata Viewpoint uses a page metaphor as the framework for displaying and updating portlets. Each portal page is a virtual work space where you decide which portlets to display and how to arrange them on the page.

Examples of ways to organize your work include defining a page for each system being monitored, or for each type of query or user. As you work, Teradata Viewpoint continually updates the information displayed on the page that currently fills the Teradata Viewpoint portal. This page is called the *active page*.

Manage portal pages using the following guidelines:

- Add portal pages at any time during a Teradata Viewpoint session.
- Access any portal page by clicking its tab; only one page can be active at a time.
- Change the name of any tab, including the **Home** page tab; page names can be duplicated.
- Rearrange pages by dragging and dropping into a new location.
- Remove pages, along with any portlets contained on the page, with a single mouse-click.
- One page (tab) must remain, as well as the **Add Page** tab.

Viewpoint portal basics: Create and access additional pages

Access to additional pages are available via this selector button.

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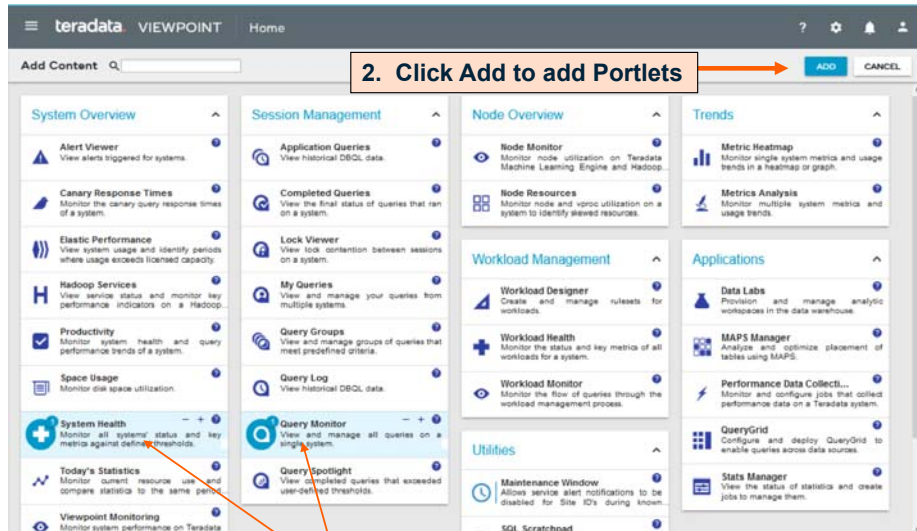
The screenshot shows the Teradata Viewpoint portal interface. At the top, there is a header bar with the Teradata logo, the text "VIEWPOINT", and a "Home" button. Below the header, there is a "REWIND ON" toggle switch set to "OFF". On the left side, there is a sidebar menu with a "Dashboard" button and a "MY PAGES" section. The "MY PAGES" section contains a list of pages: "Home" and "Workload Management". The "Home" page is highlighted. In the main content area, there is a "System Health" section with various metrics and a "Workload" section with a table of data. Annotations with orange boxes and arrows point to specific elements: an arrow points to the "Home" button in the sidebar; an arrow points to the "+" button in the "MY PAGES" section; an arrow points to the "Home" page in the "MY PAGES" list; and an arrow points to the "Workload Management" page in the "MY PAGES" list.

Add New Page button allows you to create additional pages

The Home page is your initial page
Notice that the "selected" page is highlighted in the MY PAGES list

The additional pages that you add will be displayed here
Clicking the name of the page will display its Portlets

Viewpoint portal basics: Add portals to the current page



2. Click Add to add Portlets

ADD CANCEL

New portlets are added to a Page via the **Add Content**.

1. Select portlets to be added to a page by clicking on its name (E.G. Clicking twice on a portlet will put two of the same portlet on your page)

teradata.

Viewpoint rewind

REWIND ON ☒ OFF

4/15/2019 10:55:54 AM 1d 1h 10m 1m 10s 1s 1s 10s 1m 10m 1h 1d

Date/Time Selector **Back** **Forward**

Rewind, replay, fast forward Viewpoint portlets to review Analytics Database operations at past points in time

Query Monitor 10:55:25 AM

SESSION ID	STATE ICON	ΔCPU	ΔI/O	SNAPSHOT CPU SKEW
3767	III	0	0	0
3844	III	0	0	28.4
3845	III	0	0	31.67
3846	III	0	0	31.04
3847	III	0	0	29.37
3848	III	0	0	21.32
3849	III	0	0	26.46
3850	III	0	0	19.21
3735	III	0	0	0
3737	III	0	0	0
3736	III	0	0	0
3734	III	166.5	230K	53.86

27 rows total

System Health 10:55:54 AM

METRIC	VALUE	THRESHOLDS	LAST 60 MINUTES
CPU Utilization	48.96%		
User	42.15%		
Custom	6.913%		
AMP CPU Skew	48.05%		
AMP I/O Skew	1.215%		
AMP Worker Tasks	14		
Total Disk Space	7.925%		
Max Disk By AMP	7.943%		
Component Down	0		
Memory Used	112		
Max Spool By AMP	0.626%		
DBC Disk Space	0.096%		
Active Sessions	12		

The rewind feature allows you to view data that corresponds to dates and times in the past and compare it to data for a different date and time. You can rewind the data for some or all portlets on a portal page to a previous point in time, such as when a job failed. Rewinding portlet data is useful for identifying and resolving issues.

You can rewind data as far back as data is available. The rewind feature is not available for portlets that have portlet-specific methods for reviewing data over time.

Using the rewind toolbar, you can enter a specific date and time as well as scroll through the data in increments of seconds, minutes, hours, or days. All portlets on the page that are participating in rewind activities display data that corresponds to the selected rewind date and time each time a selection is made.

Current topic: Portlets to use in this training course

- The viewpoint user interface
- **Portlets to use in this training course**
 - **Query monitor**
 - **System health**
 - **Alert viewer**

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Viewpoint query monitor summary view

Query Monitor 10:55:25 AM

TDT1 By Session All

27 All 21 Not Idle 12 Active 0 Block 9 Delay 0 Abort 0 Resp 6 Idle 0 Parse 0 Unknown 0 QTDelayed 0 SesDelayed 0 Response...

SESSION ID	STATE ICON	ΔCPU	ΔI/O	SNAPSHOT CPU SKEW	PJI	DURATION	USERNAME	ACCOUNT	CPU USE
3644	☐	0	0			0:00:00	DBC	DBC	0
3653	☐	0	0			0:00:00	DBC	DBC	0
3734	→	166.5	230K	53.86	0.724	0:05:00	RPTUSER1	RPT_&I	8.674
3735	☐	0	0	0	0	0:05:00	RPTUSER1	RPT_&I	0
3736	☐	0	0	0	0	0:05:00	RPTUSER1	RPT_&I	0
3737	☐	0	0	0	0	0:05:00	RPTUSER1	RPT_&I	0
3739	☐	0	0	0	0	0:04:00	ADHOCUSER1	ADHOC_&S	0
3741	☐	0	0	0	0	0:04:00	ADHOCUSER2	ADHOC_&S	0
3743	☐	0	0	0	0	0:04:00	ADHOCUSER2	ADHOC_&S	0
3744	☐	0	0	0	0	0:04:00	ADHOCUSER1	ADHOC_&S	0
3745	☐	0	0	0	0	0:04:00	ADHOCUSER1	ADHOC_&S	0
3746	☐	0	0	0	0	0:04:00	ADHOCUSER2	ADHOC_&S	0
3747	→	718.1	362K				ADHOCUSER2	ADHOC_&S	37.4
3765	☐	0	0				DBC	DBC	0
3767	☐	0	0				DBC	DBC	0
3843	☐	0	0			0:00:00	LOADUSER2	STREAM_&H	0
3844	→	0	0	28.4	0.086	0:00:00	LOADUSER2	STREAM_&H	0
3845	→	0	0	31.67	0.077	0:00:00	LOADUSER2	STREAM_&H	0
3846	→	0	0	34.84	0.084	0:00:00	LOADUSER2	STREAM_&H	0

27 rows total

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The **QUERY MONITOR** portlet allows you to view information about queries running in a Analytics Database so you can spot problem queries. You can analyze and decide whether a query is important, useful, and well written. After you have identified a problem query, you can take action to correct the problem by changing the priority or workload, releasing the query, or aborting the query or session. You can take these actions for one query or session, or multiple queries or sessions at a time.

The summary view contains a table with one row allocated to each of the sessions, account strings, users, or utilities running on the database.

The portlet allows you to filter queries in all of the session views. You can set thresholds for any column and when the threshold is exceeded, the information is highlighted in the sessions table.

Select a row to access session and query information in the details view.

Using Query Monitor, you can also determine the types of utilities that are running most frequently on the system and then set utility limits. You can spot utilities that are using a large number of partition connections and, potentially, a high number of resources.

From the **PREFERENCES** view, you can set the criteria values used to display sessions in the My Criteria view and customize the information displayed in the views. Set criteria values to display only those sessions currently running on the selected system that exceed the specified criteria. For example, you can troubleshoot Analytics Database problems to quickly explore details about queries such as the current state of a query or how long a query has been blocked.

Viewpoint query monitor detail view

Query Monitor 10:55:25 AM	
TDT1 > By Session All	
Session: 3747	
OVERVIEW	SQL EXPLAIN SKEW QUERY BAND
QUERY INFO	
State:	» Active
Time in state:	0:02:00
Total duration:	0:04:00
Spool space:	49,773,457,000
Hot AMP spool:	2,702,163,970
Spool skew:	7.9007506
Temp space:	0
Request CPU:	2350.5
Request I/O:	1,986,571
PJI:	1.18
Unnecessary I/O:	0.85
SESSION INFO	
Name:	Adhoc
Method:	Timeshare
CPU decay:	Level 0
Classification mode:	Auto
Virtual partition:	Standard
I/O decay:	Level 0
User:	ADHOCUSER2
Account:	ADHOC_&S
Partition:	DBC/SQL
Requests:	2
Source:	(TCP/IP) b327 172.32.0.50 TDAWS 15612 EC2-USER ADHOC_B TQ. 01 LSS
SNAPSHOT INFO	
CPU use:	37.4%
Impact CPU:	2430.56
Snapshot CPU...	9.4%
Snapshot I/O s...	21.6%

Depending on Query State, Tabs available include:

- Overview
- SQL
- Explain
- Blocked By
- Delay
- Query Band

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The details view displays statistics and information about the selected session. This view can be accessed by clicking on a session row in the summary view.

When viewing a request, you can see detailed information from the following tabs:

- **Overview** - Key statistics for a session. Any value exceeding the thresholds is highlighted.
- **SQL** - SQL for the selected query.
- **Explain** - Explain steps for the query, including step statistics and explain text.
- **Blocked By** - Details about other queries that are blocking this query.
- **Delay** - Details about rules delaying this query.
- **Query Band** - Displays the query band name and value for the selected query.

Use the **Tools** menu to change the priority or workload, release a query, or abort a query or session for one query or session at a time.

Use the **Next** and **Previous** buttons to move through sessions without returning to the summary view.

Viewpoint query monitor configure columns

Query Monitor10:55:25 AM

TDT1By SessionAll

27

21

12

0

9

0

0

6

0

0

0

0

0

AllNot IdleActiveBlockDelayAbortRespIdleParseUnknownQTDelayedSesDelayedResponse...

SESSION ID	STATE ICON	ΔCPU	ΔI/O	SNAPSHOT CPU SKEW	PJI	DURATION	USERNAME	ACCOUNT	CPU USE
3644		0	0			0:00:00	DBC	DBC	0
3653		0	0			0:00:00	DBC	DBC	0
3734		166.5	230K	53.86	0.724	0:05:00	RPTUSER1	RPT_&i	8.674
3735		0	0	0	0	0:05:00	RPTUSER1	RPT_&i	0
3736		0	0	0	0	0:05:00	RPTUSER1	RPT_&i	0
3737		0	0	0	0	0:05:00	RPTUSER1	RPT_&i	0
3739		0	0	0	0	0:04:00	ADHOCUSER1	ADHOC_&s	0
3741		0	0	0	0	0:04:00	ADHOCUSER2	ADHOC_&s	0
3743		0	0	0	0	0:04:00	ADHOCUSER2	ADHOC_&s	0
3744		0	0	0	0	0:04:00	ADHOCUSER1	ADHOC_&s	0
3745		0	0	0	0	0:04:00	ADHOCUSER1	ADHOC_&s	0
3746		0	0	0	0	0:04:00	ADHOCUSER2	ADHOC_&s	0
3747		718.1	362K	9.357	1.183	0:04:00	ADHOCUSER2	ADHOC_&s	37.4
3765		0	0			0:00:00	DBC	DBC	0
3767		0	0			0:00:00	DBC	DBC	0
3843		0	0			0:00:00	LOADUSER2	STREAM_&h	0
3844		0	0	28.4	0.086	0:00:00	LOADUSER2	STREAM_&h	0
3845		0	0	31.67	0.077	0:00:00	LOADUSER2	STREAM_&h	0
3846		0	0	34.84	0.084	0:00:00	LOADUSER2	STREAM_&h	0

27 rows total

Clear Filters

Configure Columns

Abort...

Change Workload...

Release...

Export...

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From the drop down menu, selecting configure columns, open the dialog box that allows you to choose which columns and the order of columns to display on the summary view.

Query monitor: Configure columns (cont.)

Portlet: **Query Monitor** > Selector: **Table Actions** > **Configure Columns**

Choose which columns to **display**, the **order** to display the columns and set any **thresholds** for **highlighting**

The 1st column can be **locked**. Click the lock icon to keep the 1st column in place when **scrolling horizontally**

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Configure Columns

Drag the rows to reorder the columns.
Click the lock icon to keep the column on the left when scrolling horizontally.

☒ SESSION ID
☒ USERNAME
☐ HOST
☐ START
☒ STATE
☐ STATE ICON
☐ REQ CPU Set Threshold
☒ ΔCPU Set Threshold
☐ REQ I/O Set Threshold
☐ ΔI/O Set Threshold
☒ SNAPSHOT CPU SKEW > 10 % OK
☐ REQ CPU SKEW Set Threshold
☐ SNAPSHOT I/O SKEW Set Threshold
☐ REQ I/O SKEW Set Threshold
☐ SPOOL Set Threshold
☐ HOT AMP SPOOL Set Threshold
☐ SPOOL SKEW Set Threshold
☐ TEMP SPACE Set Threshold
☒ PJI Set Threshold
☒ DURATION Set Threshold

OK CANCEL

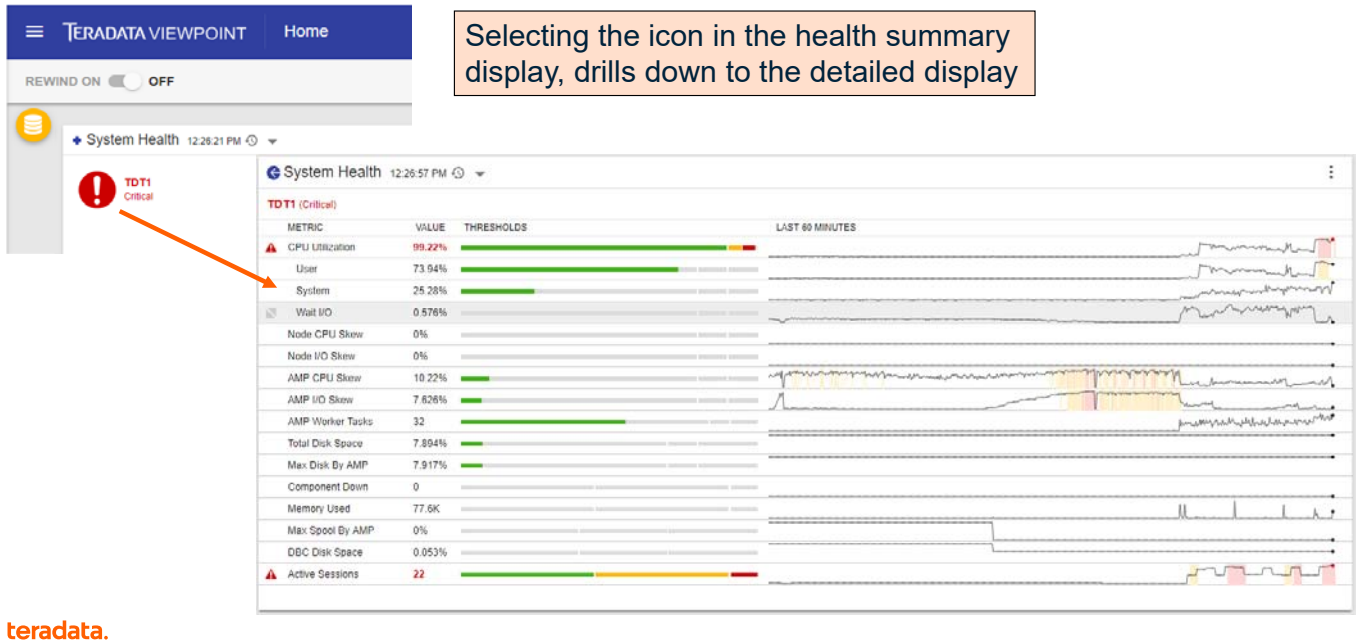
Username has been reordered, **State** name is displayed and queries that have used more than 5 CPU seconds and have a **SNAPSHOT CPU SKEW > 10%** are in **red**

SESSION ID	USERNAME	STATE	ΔCPU	ΔI/O	SNAPSHOT CPU SKEW	PJI	DURATION	ACCOUNT	CPU USE
1644	DBC	Idle	0	0	0.00	0.00	0.00	DBC	0
1653	DBC	Idle	0	0	0.00	0.00	0.00	DBC	0
1734	RPTUSER1	Active	144.9	239K	63.86	9.724	0.05:00	RPT_AJ	0.674
1735	RPTUSER1	Delayed	0	0	0	0	0.05:00	RPT_AJ	0
1736	RPTUSER1	Delayed	0	0	0	0	0.05:00	RPT_AJ	0
1737	RPTUSER1	Delayed	0	0	0	0	0.05:00	RPT_AJ	0
1738	ADHOCUSER1	Delayed	0	0	0	0	0.04:00	ADHOC_AS	0
1741	ADHOCUSER2	Delayed	0	0	0	0	0.04:00	ADHOC_AS	0
1742	ADHOCUSER2	Delayed	0	0	0	0	0.04:00	ADHOC_AS	0
1743	ADHOCUSER1	Delayed	0	0	0	0	0.04:00	ADHOC_AS	0
1745	ADHOCUSER2	Delayed	0	0	0	0	0.04:00	ADHOC_AS	0
1746	ADHOCUSER2	Delayed	0	0	0	0	0.04:00	ADHOC_AS	0
1747	ADHOCUSER2	Active	718.1	32K	9.387	1.183	0.04:00	ADHOC_AS	37.4
1762	DBC	Idle	0	0	0.00	0.00	0.00:00	DBC	0
1767	DBC	Idle	0	0	0.00	0.00	0.00:00	DBC	0
1843	LOADUSER2	Idle	0	0	0.00	0.00	0.00:00	STREAM_AH	0
1844	LOADUSER2	Active	0	0	26.4	0.088	0.00:00	STREAM_AH	0
1845	LOADUSER2	Active	0	0	25.87	0.077	0.00:00	STREAM_AH	0

From the menu provided, choose which columns to display, set thresholds for highlighting metrics and choose the order of the columns that will be displayed in the summary view.

The display now has columns reordered, some columns not displayed and columns meeting specified thresholds are highlighted.

System health



The **SYSTEM HEALTH** portlet monitors and displays the status of the selected Analytics Database using a predefined set of metrics and thresholds. This portlet reports status as one of five states: healthy, warning, critical, down, or unknown, and allows you to investigate metrics exceeding healthy thresholds.

This portlet has two main views:

SYSTEM HEALTH

- Provides status at a glance using color-coded text and icons to indicate overall health of monitored systems. Typically, metrics and thresholds are carefully selected to highlight when there is an unusual load on the system that has the potential to impact overall performance.

SYSTEM HEALTH DETAILS

- Provides details and information about the metrics used to evaluate overall system health. For less-than-healthy systems, metrics exceeding thresholds

Alert viewer

If an Alert Action was defined to write a row into the Alert Log when the event was detected, use the Alert Viewer portlet to display the alert information.

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Alert Viewer10:55:54 AM

19All

0Critical

7High

12Medium

0Low

Last 1 hour

SE...	TIMESTAMP	ALERT TYPE	ALERT NAME	SOURCE	SYSTEM NAME
	12/16/15 1:08:04 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 1:08:04 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 1:05:57 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 1:05:57 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 1:02:44 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 1:02:44 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 1:00:27 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 1:00:27 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:41:58 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:39:54 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 12:39:54 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:37:53 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:37:16 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 12:35:40 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:28:33 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:28:33 PM	System Health	VeryBusy	Viewpoint	Calvin
	12/16/15 12:18:05 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:15:02 PM	System Health	BusyAlert	Viewpoint	Calvin
	12/16/15 12:13:02 PM	System Health	BusyAlert	Viewpoint	Calvin

19 rows total

The **ALERT VIEWER** portlet allows users to view alerts defined for the system. The alert information in the summary view is updated every 30 seconds. Every alert has a timestamp, displaying the date and time at which the alert was issued.

You can filter the alerts by for example severity, time period, type, or name. You can also combine the filters to narrow the results further.

The **ALERT DETAILS** view displays detailed information about what triggered the alert, the source of the alert, and any relevant messages.

An alert is an event that the Vantage System Administrator defines as being significant. The Vantage System Administrator assigns alert severity levels to rank alerts, and can also include an explanatory message. The severity levels are: critical, high, medium, or low. The alerts displayed in the **ALERT VIEWER** portlet are specific to your system.

Summary

The Teradata Viewpoint has a number of portlets to access and monitor Analytics Database resources

- **Alert Viewer** – allows users to view alerts defined for the system
- **Query Monitor** – allows users to view information about requests
- **System Health** – allows users to monitor and display the status of a selected Analytics Database

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The Teradata Viewpoint Management and Self-Service portlets allow the Viewpoint user access Viewpoint and Teradata resources

- Alert Viewer – allows users to view alerts defined for the system
- Query Monitor – allows users to view information about requests
- System Health – allows users to monitor and display the status of a selected Analytics Database